# MITx MicroMaster in Statistics and Data Science

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**Abstract**

These are my personal study notes for the MITx MicroMaster in Statistics and Data Science. The material follows the main courses of the program. The text was drafted with the help of an AI assistant and then adapted and edited for my own use. The goal is to have a concise and easy-to-read reference, useful for quick review, that follows the structure of the original courses.
The sections are organized hierarchically as *course.unit.x*.
Images are generated with Gemini AI

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# 1 Probability - The Science of Uncertainty and Data (6.431x)

## 1.1 Probability models and axioms

A probabilistic model describes the behavior of a random experiment, such as flipping a coin, rolling dice, or throwing a dart. Building such a model has two steps: first, we list all possible outcomes; second, we assign probabilities to them. In this section, we focus on the first step — describing the possible outcomes clearly and systematically.

### 1.1.1 Sample space

All possible outcomes of an experiment form a set, denoted by $\Omega$, called the sample space. Each element of $\Omega$ represents one possible result. The outcomes must be mutually exclusive (only one can happen at a time) and collectively exhaustive (some outcome from $\Omega$ must always happen).
How detailed our sample space is depends on what we care about in the experiment. It should include all relevant distinctions but ignore unimportant ones. Choosing a good sample space is part of the modeling process.

If we flip a coin, a simple sample space is

$$\Omega = \{\text{heads}, \text{tails}\}.$$

This covers all outcomes and they don't overlap.
If we also record whether it's raining, we could define a larger sample space,

$$\Omega' = \{\text{heads and rain}, \text{heads and no rain}, \text{tails and rain}, \text{tails and no rain}\}.$$

This is valid too, but since weather usually doesn't affect a coin toss, it's an unnecessary detail. We include such extra factors only when they matter for the problem.

### 1.1.2 Discrete and continuous sample spaces

Sample spaces can be discrete or continuous.
For example, rolling a four-sided die twice and recording the two results gives

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4\}\},$$

with 16 possible ordered pairs like $(1, 1)$ or $(3, 4)$. When experiments have several stages, tree diagrams can help visualize the structure of outcomes.
Other sample spaces are continuous. For instance, when throwing a dart at a unit square, each landing point is a pair of coordinates $(x, y)$:

$$\Omega = \{(x, y) : 0 \leq x \leq 1,\ 0 \leq y \leq 1\}.$$

Here, every point inside the square is a possible outcome.

### 1.1.3 Events and probability laws

Once we have a sample space, we can talk about events. An event $A$ is any subset of $\Omega$. After an experiment, if the observed outcome $\omega$ belongs to $A$, we say the event occurred.
In discrete cases, we can assign probabilities to single outcomes. In continuous cases, each exact point has probability 0, so we assign probabilities to regions instead (for example, "the dart lands in the upper half of the square"). The probability law gives each event $A$ a value $P(A)$, showing how likely that event is.

### 1.1.4 Probability axioms and basic set notation

Probabilities follow three basic rules, called axioms:

- **Nonnegativity**: For any event $A$, $P(A) \geq 0$.
- **Normalization**: $P(\Omega) = 1$, since something in the sample space must occur.
- **Additivity**: If two events $A$ and $B$ are disjoint (that is, $A \cap B = \emptyset$), then

$$P(A \cup B) = P(A) + P(B).$$

You can think of probability as a kind of "mass" spread over $\Omega$. If two events don't overlap, the total mass on their union is just the sum of their individual masses.

For two events $A$ and $B$:

- The intersection $A \cap B$ is the set of outcomes that belong to both $A$ and $B$.
- The union $A \cup B$ is the set of outcomes that belong to either $A$ or $B$ (or both).
- Two events are disjoint if $A \cap B = \emptyset$, meaning they have no outcomes in common.

### 1.1.5 Simple properties derived from the axioms

The three axioms look simple, but they already imply many useful and intuitive properties. For example, we can prove that probabilities are always at most 1, and that the probability of the empty set is 0, instead of assuming these as extra axioms.
Consider an event $A$ and its complement $A^c$, which contains all outcomes in $\Omega$ that are not in $A$. The set $A$ and its complement $A^c$ are disjoint, and their union is the entire sample space:

$$A \cup A^c = \Omega, \qquad A \cap A^c = \emptyset.$$

Using normalization and additivity, we get

$$P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) = 1.$$

From this relation we also obtain

$$P(A) = 1 - P(A^c).$$

Because probabilities are nonnegative, $P(A^c) \geq 0$, so $P(A) \leq 1$. This shows that probabilities are always bounded above by 1, without adding a separate axiom.
If we apply the same relation to the special case where $A = \Omega$, then the complement $A^c$ is the empty set $\emptyset$. We already know that $P(\Omega) = 1$, so

$$1 = P(\Omega) = P(\Omega) + P(\emptyset),$$

which forces $P(\emptyset) = 0$. This matches the idea that the empty set is "impossible," so its probability should be zero.
Finally, the additivity axiom for two disjoint events extends to any finite collection of disjoint events. If $A_1, A_2, \ldots, A_k$ are pairwise disjoint, then

$$P\Big(\bigcup_{i=1}^{k} A_i\Big) = \sum_{i=1}^{k} P(A_i).$$

A common special case is when we have a finite subset of the sample space consisting of points $s_1, \ldots, s_k$. We can view this set as the union of the singletons $\{s_1\}, \ldots, \{s_k\}$, which are disjoint. Then the probability of the finite set is just the sum of the probabilities of its individual elements. In practice we often write $P(s_i)$ instead of $P(\{s_i\})$ and talk informally about "the probability of an outcome $s_i$" even though probabilities are formally defined on sets.

The axioms also give us some useful monotonicity and union properties. Suppose we have two events $A$ and $B$ with $A \subseteq B$. Intuitively, $B$ is "larger," so its probability should be at least as big as that of $A$. Formally, we can write

$$B = A \cup \big(B \cap A^c\big),$$

where $A$ and $B \cap A^c$ are disjoint. By additivity,

$$P(B) = P(A) + P\big(B \cap A^c\big) \geq P(A),$$

since probabilities are nonnegative.
We can also express the probability of a union when the sets are not disjoint. If $A$ and $B$ overlap, then the union $A \cup B$ can be decomposed into three disjoint parts: the part of $A$ outside $B$, the intersection $A \cap B$, and the part of $B$ outside $A$. If we call their probabilities $a$, $b$, and $c$, then

$$P(A \cup B) = a + b + c.$$

On the other hand,

$$P(A) = a + b, \qquad P(B) = b + c, \qquad P(A \cap B) = b.$$

Putting these together gives the standard identity

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

A direct corollary is the union bound. Since $P(A \cap B) \geq 0$, we get

$$P(A \cup B) \leq P(A) + P(B),$$

which is useful when we only need an upper bound on a probability.
The same idea extends to three sets. For events $A$, $B$, and $C$, we can rewrite their union as a union of three disjoint pieces: $A$ itself, the part of $B$ outside $A$, and the part of $C$ outside both $A$ and $B$. Using additivity on these disjoint parts gives a formula for $P(A \cup B \cup C)$ in terms of simpler probabilities. More generally, we can keep breaking unions into disjoint pieces to get expressions for probabilities of unions of several events.

### 1.1.6 A discrete example and the discrete uniform law

Consider again two rolls of a tetrahedral die. The sample space has 16 outcomes, each an ordered pair $(X, Y)$, where $X$ is the first roll and $Y$ is the second. Assume all 16 outcomes are equally likely, so each has probability $1/16$.
As a first example, look at the event $\{X = 1\}$: the first roll is 1. In the grid of all $(X, Y)$ pairs, this event contains the four outcomes where the first coordinate is 1 and the second is 1, 2, 3, or 4. Since each outcome has probability $1/16$, we get

$$P(X = 1) = 4 \cdot \frac{1}{16} = \frac{1}{4}.$$

Now define $Z$ to be the smaller of the two numbers in the two rolls. For example, if $X = 2$ and $Y = 3$, then $Z = 2$. For $P(Z = 4)$, this happens only if both $X$ and $Y$ are 4, so there is exactly one outcome in the event and

$$P(Z = 4) = \frac{1}{16}.$$

For $\{Z = 2\}$, we need one roll equal to 2 and the other at least 2. This event contains $(2, 2)$, $(2, 3)$, $(2, 4)$, $(3, 2)$, and $(4, 2)$, for a total of five outcomes. Thus

$$P(Z = 2) = 5 \cdot \frac{1}{16}.$$

This is an instance of the discrete uniform law. If the sample space $\Omega$ is finite with $n$ elements and all are equally likely, then, since $P(\Omega) = 1$, we must have

$$P(\{\omega\}) = \frac{1}{n}$$

for each outcome $\omega$. If an event $A$ has exactly $k$ elements, then

$$P(A) = k \cdot \frac{1}{n} = \frac{k}{n}.$$

So in a discrete uniform model, computing probabilities amounts to counting: we count all possible outcomes $n$, and the favorable ones $k$, then take the ratio $k/n$.

### 1.1.7 A continuous example

Now consider a probability calculation in a continuous sample space. The experiment is again throwing a dart at the unit square, and the outcome is the point $(x, y)$ where the dart lands. The sample space is the unit square, and we assume a uniform law: the probability of any region in the square is equal to its area. This is natural if we think the dart is equally likely to land anywhere.
Take the event that the sum of the coordinates is at most $1/2$:

$$A = \{(x, y) : x + y \leq 1/2\}.$$

In the unit square this is a right triangle under the line $x + y = 1/2$. Since probability equals area,

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

As another example, consider the event that the dart hits exactly the point $(0.5, 0.3)$. This event is a single point with area 0, so its probability is 0. The same holds for any single point in the unit square: in a continuous model, individual points have probability 0, while regions with positive area can have positive probability.
This example fits a general four-step recipe for probability problems. Starting from a word description of an experiment, we:

1. Specify the sample space.
2. Choose a probability law (a model for how likely different outcomes are).
3. Describe the event of interest precisely, ideally also with a picture.
4. Use the probability law and the event description to compute the desired probability.

Sometimes the law is given explicitly for all events; in other cases we only know it for some basic events and need extra work to obtain the probability we want.

### 1.1.8 Countable additivity and infinite sample spaces

We now look at a discrete but infinite sample space. Imagine an experiment whose outcome is a positive integer, for example the number of coin tosses until the first heads appears. The sample space is $\{1, 2, 3, \dots\}$. Suppose we are told that

$$P(\{n\}) = \frac{1}{2^n}, \quad n = 1, 2, 3, \dots$$

First we check that this is a valid probability law by summing over all $n$:

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{2} \cdot \frac{1}{1 - 1/2} = 1.$$

Now consider the event that the outcome is even, the infinite set $\{2, 4, 6, \dots\}$. We can write it as a union of singletons:

$$\{2, 4, 6, \dots\} = \{2\} \cup \{4\} \cup \{6\} \cup \dots$$

These sets are disjoint, so we would like to write

$$P(\text{even}) = P(\{2\}) + P(\{4\}) + P(\{6\}) + \dots$$

and evaluate the resulting infinite series. To make this step valid in general, we extend the additivity axiom from finite unions to countably infinite unions.
The extra axiom, called countable additivity, says: if $A_1, A_2, A_3, \dots$ is a sequence of pairwise disjoint events, then

$$P\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} P(A_n).$$

The word "sequence" matters: the events can be listed in order $A_1, A_2, \dots$, so we can form a countable sum. This axiom lets us compute probabilities of many events in infinite sample spaces by summing over simpler disjoint pieces.
It is also important to see what countable additivity does not say. In the continuous unit square model, the square is the union of singletons (one per point), and each singleton has probability 0. If we tried to add their probabilities, we would get a sum of zeros, but the probability of the whole square should be 1. The catch is that the unit square is uncountable: its points cannot be arranged as a sequence $A_1, A_2, \dots$, so the axiom does not apply. In short, some infinite sets are countable (like the integers) and others are uncountable (like the real line or the unit square), and countable additivity only covers the countable case.
There are deeper measure-theoretic issues in the background, especially when we try to define probability on all subsets of a continuous space. For this course, the key point is that standard constructions, like defining probability on the unit square via area, can be made rigorous and do satisfy countable additivity for all "nice" sets we will use. The strange sets where problems arise are mostly of theoretical interest and do not appear in typical applications.

### 1.1.9 Interpretations and uses of probability

To close, it is useful to think about what probability really means. In the narrow sense, probability theory is a branch of mathematics: we start from axioms, study models that satisfy them, and prove theorems. This view does not fix any particular real-world interpretation.
One common interpretation is in terms of long-run frequencies. For example, if we have a fair coin and toss it many times, the fraction of heads is expected to approach 1/2. In this view, the probability of an event is the limiting frequency with which it occurs in a long sequence of repeated trials. This fits well for repeatable experiments like coin tosses.
But not all probability statements are like that. A sentence such as "the current president of my country will be reelected with probability 0.7" does not describe an experiment we can repeat infinitely many times. In such cases it is more natural to see probabilities as measuring degrees of belief: they summarize how strongly we believe an event will happen. Thinking in terms of bets, a probability expresses how much we are willing to stake on an outcome.
This raises the worry that probabilities are just subjective. The answer is that probability theory at least gives consistent rules for reasoning under uncertainty. If our probability model is a reasonable description of the situation, those rules help us make predictions and decisions in a systematic way.
Whether our decisions are actually good depends on whether we chose a good model. Building such models from data is the task of statistics and statistical inference. The real world generates data; statistics uses these data to construct probabilistic models; and probability theory then analyzes these models to produce predictions or decisions that we can apply back to the real world.

### 1.1.10 Sets, complement, empty set, and subsets

A set is a collection of distinct elements, usually denoted by a capital letter like $S$. We can specify a set by listing its elements inside braces, for example $\{a, b, c, d\}$, which is a finite set. Other sets, such as the set of all real numbers, are infinite. We write $x \in S$ if $x$ is an element of $S$, and $x \notin S$ if $x$ does not belong to $S$.
Sometimes it is convenient to define a set by a property instead of listing elements. For example, starting from the real numbers, we can consider all $x$ such that $\cos(x) > 1/2$; the collection of all such $x$ is a set. Often we also fix a larger "universe" of objects, called the universal set and denoted by $\Omega$, and only look at sets contained in this universe.
Given a universal set $\Omega$ and a subset $S \subseteq \Omega$, the complement of $S$, denoted $S^c$, is the set of all elements in $\Omega$ that are not in $S$:

$$S^c = \{x \in \Omega : x \notin S\}.$$

Taking complements twice brings us back to the original set: $(S^c)^c = S$. A special set is the empty set $\emptyset$, which contains no elements. For example, the complement of $\Omega$ is $\emptyset$, since there is nothing outside the universal set.
If every element of a set $S$ also belongs to a set $T$, we say that $S$ is a subset of $T$ and write $S \subseteq T$. This includes the case $S = T$. Intuitively, $S$ is "contained" inside $T$.

### 1.1.11 Union, intersection, and infinite collections

Given two sets $S$ and $T$, their union $S \cup T$ is the set of all elements that belong to at least one of them:

$$x \in S \cup T \quad \text{iff} \quad x \in S \text{ or } x \in T.$$

Their intersection $S \cap T$ is the set of elements that belong to both:

$$x \in S \cap T \quad \text{iff} \quad x \in S \text{ and } x \in T.$$

These notions extend to infinite families of sets. If we have sets $S_1, S_2, S_3, \ldots$, their union is

$$\bigcup_{n=1}^{\infty} S_n = \{x : x \in S_n \text{ for at least one } n\},$$

and their intersection is

$$\bigcap_{n=1}^{\infty} S_n = \{x : x \in S_n \text{ for all } n\}.$$

Informally, the union collects elements that appear in at least one of the sets, while the intersection keeps only the elements common to all of them.

### 1.1.12 Basic algebra of sets

Set operations satisfy some familiar algebraic properties. Union and intersection are both commutative:

$$S \cup T = T \cup S, \qquad S \cap T = T \cap S,$$

and associative:

$$(S \cup T) \cup U = S \cup (T \cup U), \qquad (S \cap T) \cap U = S \cap (T \cap U).$$

Because of associativity, we usually drop parentheses when writing unions or intersections of three or more sets.
The universal set $\Omega$ behaves like an identity: $S \cup \Omega = \Omega$ and $S \cap \Omega = S$. There are also distributive laws describing how union and intersection interact, for example

$$S \cap (T \cup U) = (S \cap T) \cup (S \cap U),$$

and a similar law with $\cup$ and $\cap$ swapped. A standard way to check such identities is to argue element-wise: assume $x$ belongs to one side, unpack the definitions, and show that this forces $x$ to belong to the other side as well. If each side is a subset of the other, then the two sets are equal.

### 1.1.13 De Morgan's laws

De Morgan's laws relate unions, intersections, and complements. For two sets $S$ and $T$, the first law says that the complement of their intersection is the union of their complements:

$$(S \cap T)^c = S^c \cup T^c.$$

Intuitively, if a point is not in both $S$ and $T$ at the same time, then it must be missing from at least one of them, so it lies in $S^c$ or in $T^c$. A formal proof rewrites the membership condition step by step: $x \notin S \cap T$ means "$x$ is not in $S$ or not in $T$," which is the same as "$x \in S^c$ or $x \in T^c$," that is, $x \in S^c \cup T^c$.
The second law is symmetric and says that the complement of a union is the intersection of the complements:

$$(S \cup T)^c = S^c \cap T^c.$$

Here the idea is that if a point is not in $S \cup T$, then it cannot be in $S$ and cannot be in $T$, so it must belong to both complements. Both identities extend to families of sets $\{S_n\}$:

$$\Big(\bigcap_n S_n\Big)^c = \bigcup_n S_n^c, \qquad \Big(\bigcup_n S_n\Big)^c = \bigcap_n S_n^c.$$

These laws are very useful when we want to switch between unions and intersections, especially in probability.

### 1.1.14 Sequences and their limits

A sequence is a collection of elements indexed by the positive integers. We usually write it as $\{a_i\}_{i=1}^{\infty}$, or simply $a_i$, where each $a_i$ belongs to some set $S$. Often $S = R$ and we have a sequence of real numbers, but $S$ could also be $R^n$ (a sequence of vectors) or some other set. Formally, a sequence is just a function $f$ from the natural numbers to $S$, and the $i$-th term is $a_i = f(i)$.
We say that a sequence $a_i$ converges to a real number $a$ if, as $i$ becomes large, the terms of the sequence get arbitrarily close to $a$ and then stay close. Formally, $a_i \to a$ as $i \to \infty$ if for every $\varepsilon > 0$ there exists an index $i_0$ such that for all $i \geq i_0$,

$$|a_i - a| < \varepsilon.$$

Geometrically, if you draw a horizontal band of width $2\varepsilon$ around the limit $a$, then from some point on all the sequence terms lie inside that band. The definition may look technical, but it just encodes the idea that the sequence eventually stays as close to $a$ as we like.
Convergence behaves well under basic operations. If $a_i \to a$ and $b_i \to b$, then the sum sequence $a_i + b_i$ converges to $a+b$, and the product sequence $a_i b_i$ converges to $ab$. If $g$ is a continuous function and $a_i \to a$, then the sequence $g(a_i)$ converges to $g(a)$; for example, if $a_i \to a$, then $a_i^2 \to a^2$. These properties let us manipulate limits in a straightforward way.

### 1.1.15 Practical convergence criteria

In practice, it is helpful to have quick tests for convergence. One common situation is a monotone sequence. If $a_i$ is increasing (or at least never decreases) and is bounded above, then it converges to some finite limit. If it keeps increasing without any bound, we say that it "converges to infinity" rather than to a finite number.
Another strategy is to bound the distance $|a_i - a|$ from a candidate limit $a$. If we can find an upper bound on $|a_i - a|$ that goes to 0 as $i$ grows, then $a_i$ must converge to $a$. A related idea is the sandwich (or squeeze) argument: if a sequence $a_i$ is always between two other sequences $b_i$ and $c_i$, and both $b_i$ and $c_i$ converge to the same limit $a$, then $a_i$ also converges to $a$. These tools will appear often, sometimes implicitly, when analyzing limits in the course.

### 1.1.16 Infinite Series

Consider a sequence $a_i$ and the sum:

$$\sum_{i=1}^{\infty} a_i.$$

This sum is defined as the limit of the partial sums:

$$S_n = \sum_{i=1}^{n} a_i,$$

if the limit exists and is finite as $n \to \infty$. The series converges if this limit exists; otherwise, it diverges.
For non-negative terms, partial sums are increasing and bounded, ensuring convergence (finite or infinite). For alternating signs, the limit may depend on the order of terms, so absolute convergence is preferred. If

$$\sum_{i=1}^{\infty} |a_i|$$

converges, then the series $\sum a_i$ converges as well.

A key example is the geometric series. For $|\alpha| < 1$, consider:

$$\sum_{i=0}^{\infty} \alpha^i = 1 + \alpha + \alpha^2 + \dots$$

The sum is:

$$\sum_{i=0}^{\infty} \alpha^i = \frac{1}{1-\alpha}.$$

Alternatively, using the trick $s = \sum_{i=0}^{\infty} \alpha^i$, we get:

$$s = \frac{1}{1-\alpha}, \quad |\alpha| < 1.$$

### 1.1.17 Series with Multiple Indices and Order of Summation

For series with two indices, like $a_{ij}$ with $i, j = 1, 2, \dots$, we sum over a grid of points $(i, j)$. The sum can be written as:

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} \quad \text{or} \quad \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

Different summation orders may yield different results, but absolute convergence ensures the sum is well-defined, and the order does not matter:

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{ij}|.$$

Without absolute convergence, changing the order can lead to different sums, such as 0 or 1, due to cancellations between positive and negative terms.
We can also sum over restricted sets, e.g., $\sum_{i=1}^{\infty} \sum_{j=1}^{i} a_{ij}$, and the order of summation can vary. With absolute convergence, the result is independent of the summation order.

### 1.1.18 Countable and Uncountable Sets

In probability theory, not all infinite sets are the same. Some are countable, others uncountable. A set is countable if its elements can be mapped one-to-one to natural numbers, i.e., they can be listed in a sequence.
Positive integers are countable, as are all integers, which can be listed as $0, 1, -1, 2, -2, 3, -3, \dots$. The same applies to pairs $(i, j)$ of positive integers, which can be arranged in a zigzag pattern.
A less obvious example is the set of rationals in $[0, 1]$. Since every rational can be written as $p/q$, we can list them by varying $q$ and $p$, skipping duplicates, resulting in a sequence containing all rationals in $[0, 1]$.
A set is uncountable if it cannot be listed in this way. Typical examples include intervals of real numbers, the real line, and spaces like the plane or 3D space. These sets are too large to be listed. The distinction between countable and uncountable sets is key to understanding the difference between discrete and continuous probability models.

### 1.1.19 Cantor's Diagonal Argument

Cantor's diagonal argument proves that the set of real numbers in $[0, 1]$ is uncountable. The idea is to assume that we can list all real numbers in this interval, each represented by an infinite decimal expansion. If we arrange them in a sequence, like:

$$r_1 = 0.a_{11}a_{12}a_{13}\dots, \quad r_2 = 0.a_{21}a_{22}a_{23}\dots, \quad r_3 = 0.a_{31}a_{32}a_{33}\dots, \dots$$

Then, by constructing a new number $r$ whose $n$-th decimal place is different from the $n$-th decimal of $r_n$, we ensure that $r$ is not on the list. This contradiction shows that the real numbers in $[0, 1]$ cannot be counted, proving they are uncountable.
Thus, Cantor's argument demonstrates that no list can exhaust all real numbers, highlighting the distinction between countable and uncountable sets.

## 1.2 Conditioning and independence

### 1.2.1 Conditional probabilities

Conditional probabilities formalize how we update a probability model once we learn that some event has occurred. Conditional probabilities are probabilities associated with a revised model that takes into account extra information about the outcome of a probabilistic experiment. The question is how to systematically revise our model when we are told that a certain event has occurred.

Consider a probability model with a sample space $\Omega$ and a probability law $P$. Let $A$ and $B$ be two events, that is, two subsets of $\Omega$. We will be interested in the probability of $A$ under the information that $B$ has occurred.

To build intuition, start with a finite sample space with equally likely outcomes. Suppose that $\Omega$ has 12 possible outcomes, each with probability $1/12$. Let $A$ be an event consisting of 5 outcomes, so that $P(A) = 5/12$, and let $B$ be an event consisting of 6 outcomes, so that $P(B) = 6/12$. Assume that $A$ and $B$ overlap in exactly 2 outcomes, so that

$$P(A \cap B) = \frac{2}{12}.$$

Now suppose we are told that $B$ has occurred, but nothing more. How should we revise the model? First, all outcomes outside $B$ are no longer possible. In the updated model they must have probability 0. Second, inside $B$ we still have 6 outcomes. In the original model they were equally likely, and the new information does not change their relative likelihoods, so they should remain equally likely. Since the total probability inside $B$ must now be 1, each of the 6 outcomes in $B$ should have probability $1/6$ in the revised model.

In this revised model the probability of $A$ is just the fraction of the outcomes in $B$ that also belong to $A$. There are 2 such outcomes, so

$$P(A \mid B) = \frac{2}{6} = \frac{1}{3}.$$

Here we have introduced the notation $P(A \mid B)$ for the probability of $A$ given $B$, read as "the probability of $A$ given $B$". Similarly, since $B$ is now certain in the revised model, we have $P(B \mid B) = 1$.

The same idea works even if the outcomes of $\Omega$ are not equally likely. Suppose we only know the probabilities of certain regions of the sample space without assuming any uniformity inside them. For instance, we might still have

$$P(A) = \frac{5}{12}, \qquad P(B) = \frac{6}{12}, \qquad P(A \cap B) = \frac{2}{12},$$

but now these numbers come directly from the probability law $P$ without referring to equally likely elementary outcomes. Inside $B$ the subset where $A$ also occurs, namely $A \cap B$, has total probability $P(A \cap B) = 2/12$. Relative to the total probability of $B$, which is $P(B) = 6/12$, this corresponds to a fraction

$$\frac{P(A \cap B)}{P(B)} = \frac{2/12}{6/12} = \frac{1}{3}.$$

It is therefore natural to define the conditional probability of $A$ given $B$ to be this fraction. Intuitively, once we restrict attention to $B$ we renormalize the probability law by dividing by $P(B)$ and then ask what portion of the "mass" inside $B$ lies in $A$.

This leads to the formal definition. Given two events $A$ and $B$ with $P(B) > 0$, the conditional probability of $A$ given $B$ is defined by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The numerator $P(A \cap B)$ is the probability that both $A$ and $B$ occur, while the denominator $P(B)$ is the probability that $B$ occurs. Their ratio measures the fraction of the probability of $B$ that also belongs to $A$. This definition only makes sense when $P(B) > 0$. If $P(B) = 0$, the expression $P(A \cap B)/P(B)$ is undefined, and conditional probabilities given $B$ are not defined.

It is important to emphasize that this is a definition, not a theorem. There is no question of "proving" that

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

is correct. Rather, we adopt this formula because it captures the intuitive procedure of restricting the model to $B$ and renormalizing the probabilities. The earlier examples simply motivate why this is the most natural way to construct a conditional probability law that reflects the information that the event $B$ has occurred.

Conditional probabilities obey the same axioms as ordinary probabilities and therefore behave as an ordinary probability law on a new sample space.

Conditional probabilities are defined from an underlying probability model $(\Omega, P)$ and an event $B$ with $P(B) > 0$. For any event $A$, the conditional probability of $A$ given $B$ is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This definition suggests that we can think of a new probabilistic model in which the outcomes are restricted to the event $B$ and the probabilities are renormalized. In this new model the function $A \mapsto P(A \mid B)$ should behave like a probability law. We now check that it satisfies the usual axioms.
First, conditional probabilities are nonnegative. Indeed, both $P(A \cap B)$ and $P(B)$ are nonnegative, and $P(B) > 0$ by assumption, so the ratio

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

is always greater than or equal to 0. Thus the nonnegativity axiom remains valid under conditioning.
Second, we verify that the whole sample space has probability 1 in the conditional model. Consider the conditional probability of $\Omega$ given $B$:

$$P(\Omega \mid B) = \frac{P(\Omega \cap B)}{P(B)}.$$

Since $B \subseteq \Omega$, the intersection $\Omega \cap B$ is just $B$, so the numerator is $P(B)$. Therefore

$$P(\Omega \mid B) = \frac{P(B)}{P(B)} = 1.$$

We can also view the new sample space as $B$ itself, because outcomes outside $B$ are considered impossible once we know that $B$ has occurred. In that view the "entire sample space" is $B$, and its conditional probability is

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

So the normalization axiom $P(\text{whole space}) = 1$ holds whether we regard $\Omega$ or $B$ as the relevant sample space under conditioning.
Third, we check the additivity axiom. Let $A$ and $C$ be two disjoint events, so that $A \cap C = \emptyset$. We want to show that

$$P(A \cup C \mid B) = P(A \mid B) + P(C \mid B).$$

Using the definition of conditional probability, we have

$$P(A \cup C \mid B) = \frac{P\big((A \cup C) \cap B\big)}{P(B)}.$$

By a basic set identity,

$$(A \cup C) \cap B = (A \cap B) \cup (C \cap B).$$

Now note that $A$ and $C$ are disjoint, so their intersections with $B$ are also disjoint. Indeed, any point in $(A \cap B) \cap (C \cap B)$ would belong to both $A$ and $C$, which is impossible. Therefore $(A \cap B)$ and $(C \cap B)$ are disjoint sets, and by the additivity axiom for the original probability measure $P$,

$$P\big((A \cup C) \cap B\big) = P(A \cap B) + P(C \cap B).$$

Substituting into the expression for the conditional probability gives

$$P(A \cup C \mid B) = \frac{P(A \cap B) + P(C \cap B)}{P(B)} = \frac{P(A \cap B)}{P(B)} + \frac{P(C \cap B)}{P(B)} = P(A \mid B) + P(C \mid B).$$

This shows that conditional probabilities preserve finite additivity for disjoint events. The same reasoning extends, without essential changes, to finite unions of pairwise disjoint events and even to countable unions when countable additivity holds for the original probability measure $P$. Thus, conditional probabilities satisfy nonnegativity, normalization, and (finite or countable) additivity, and therefore they obey all the standard axioms of probability theory.
An important consequence is that conditional probabilities can be treated as ordinary probabilities. Any identity or theorem that we derive for ordinary probabilities continues to hold if we replace $P(\cdot)$ by $P(\cdot \mid B)$. In later sections we will repeatedly use this idea to apply general probability rules, such as the product rule or Bayes' rule, in conditional form.

EXAMPLE: RADAR DETECTION PROBLEM

Conditional probabilities are useful both for updating a model when new information arrives and for building multi-stage probabilistic models.

Consider a simple radar detection problem. Either there is an airplane in the sky or there is not. Let $A$ be the event that an airplane is present and $A^c$ the event that there is no airplane. Suppose prior information tells us that

$$P(A) = 0.05, \qquad P(A^c) = 0.95.$$

A radar looks at the sky and either reports a detection or not. Let $B$ be the event that the radar reports a detection and $B^c$ the event that it does not. The performance of the radar is summarized by conditional probabilities that describe how it behaves in the two possible worlds $A$ and $A^c$.

If an airplane is present, the radar should detect it most of the time, but it can also miss it. For example, we might have

$$P(B \mid A) = 0.99, \qquad P(B^c \mid A) = 0.01,$$

so that conditionally on $A$ the radar detects the plane with probability 99% and misses it with probability 1%. If no airplane is present, the radar can still produce a false alarm or remain silent. For instance,

$$P(B \mid A^c) = 0.10, \qquad P(B^c \mid A^c) = 0.90,$$

so there is a 10% chance of a false alarm and a 90% chance that the radar correctly reports no detection when nothing is flying.

These four conditional probabilities, together with the prior probabilities of $A$ and $A^c$, define a complete probabilistic model. We can represent it with a tree diagram: first choose between $A$ and $A^c$, then from each of these nodes choose between $B$ and $B^c$ using the corresponding conditional probabilities. The probability of each "leaf" of the tree is obtained by multiplying along the path that leads to it. This is an instance of the product rule

$$P(A \cap B) = P(A)\, P(B \mid A),$$

which follows directly from the definition of conditional probability.

As a first example, let us compute the probability that there is an airplane and that the radar detects it, that is, $P(A \cap B)$. Using the product rule,

$$P(A \cap B) = P(A)\, P(B \mid A) = 0.05 \times 0.99 = 0.0495.$$

In the tree picture, this is the probability of the path where $A$ occurs at the first stage and $B$ occurs at the second stage.

Next, consider the total probability that the radar reports a detection, $P(B)$. There are two disjoint ways in which this can happen: either there is an airplane and it is detected, or there is no airplane and the radar produces a false alarm. In symbols,

$$B = (A \cap B) \cup (A^c \cap B),$$

and the two events on the right-hand side are disjoint. Therefore

$$P(B) = P(A \cap B) + P(A^c \cap B).$$

Using the product rule for each term, we obtain

$$P(B) = P(A)\, P(B \mid A) + P(A^c)\, P(B \mid A^c) = 0.05 \times 0.99 + 0.95 \times 0.10.$$

Numerically,

$$0.05 \times 0.99 = 0.0495, \qquad 0.95 \times 0.10 = 0.095,$$

so

$$P(B) = 0.0495 + 0.095 = 0.1445.$$

This calculation is an example of the law of total probability: we express the probability of $B$ as a sum over different scenarios ($A$ and $A^c$) weighted by their prior probabilities and the corresponding conditional probabilities of $B$.

Finally, suppose the radar has reported a detection. We would like to know how likely it is that there really is an airplane in the sky, that is, we want $P(A \mid B)$. By the definition of conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

We already computed both the numerator and the denominator:

$$P(A \cap B) = 0.0495, \qquad P(B) = 0.1445,$$

so

$$P(A \mid B) = \frac{0.0495}{0.1445} \approx 0.34.$$

Thus, even though the radar is quite reliable in both scenarios, the probability that there is actually an airplane given that the radar reports a detection is only about 34%.
This example illustrates three basic tools that arise from conditional probabilities: the product rule, which expresses joint probabilities in terms of conditional probabilities; the law of total probability, which decomposes a probability into contributions from different scenarios; and the use of conditional probability to update our beliefs about $A$ when we observe $B$. Variants of these ideas will appear repeatedly in more complex models.

### 1.2.2 The multiplication rule

Conditional probabilities lead to a general multiplication rule that expresses the probability of several events occurring together as a product of ordinary and conditional probabilities.
We start from the definition of conditional probability. For events $A$ and $B$ with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Rearranging this equality gives

$$P(A \cap B) = P(B)\, P(A \mid B).$$

This says that the probability that both $A$ and $B$ occur is equal to the probability that $B$ occurs, times the conditional probability that $A$ occurs given that $B$ has occurred. Since we are free to choose which event we treat as "first", we can also write

$$P(A \cap B) = P(A)\, P(B \mid A).$$

These are the simplest forms of the multiplication rule and they are exactly what we used in the radar example when we computed the probability of a particular path in the tree diagram.
The same idea extends to three events. Consider events $A$, $B$, and $C$. We can write their intersection as

$$A \cap B \cap C = (A \cap B) \cap C.$$

Applying the multiplication rule to the "composite" event $A \cap B$ and the event $C$ gives

$$P(A \cap B \cap C) = P(A \cap B)\, P\big(C \mid A \cap B\big).$$

We can then expand $P(A \cap B)$ using the two-event rule:

$$P(A \cap B) = P(A)\, P(B \mid A).$$

Substituting into the previous expression yields

$$P(A \cap B \cap C) = P(A)\, P(B \mid A)\, P\big(C \mid A \cap B\big).$$

This formula has a natural interpretation in terms of a three-stage tree: first we pick $A$ or $A^c$, then from there we pick $B$ or $B^c$, and finally we pick $C$ or $C^c$. The probability of a leaf is the product of the probability of the first branch, the conditional probability of the second branch given the first, and the conditional probability of the third branch given the previous two.
The same reasoning works for more than three events. Let $A_1, A_2, \ldots, A_n$ be events and assume that all the conditional probabilities we write below are well-defined. We can write

$$A_1 \cap A_2 \cap \cdots \cap A_n = (A_1 \cap \cdots \cap A_{n-1}) \cap A_n,$$

and apply the multiplication rule step by step. The result is the general multiplication rule:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \prod_{i=2}^{n} P\big(A_i \mid A_1 \cap \cdots \cap A_{i-1}\big).$$

In words, the probability that all of $A_1, \ldots, A_n$ occur is equal to the probability that $A_1$ occurs, times the probability that $A_2$ occurs given that $A_1$ has occurred, times the probability that $A_3$ occurs given that $A_1$ and $A_2$ have occurred, and so on. This is the most general form of the multiplication rule and it underlies the computation of path probabilities in multi-stage models.

### 1.2.3 Total probability theorem

The total probability theorem formalizes the "divide and conquer" idea: to find the probability of an event that can occur under several disjoint scenarios, we add its probabilities across those scenarios.
Consider a sample space $\Omega$ and events $A_1, A_2, A_3$ that form a **partition** of $\Omega$. This means: - the events are pairwise disjoint, - their union is the whole sample space, - and each has positive probability:

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \qquad A_1 \cup A_2 \cup A_3 = \Omega, \qquad P(A_i) > 0.$$

We can think of $A_1, A_2, A_3$ as three mutually exclusive "scenarios" under which the experiment may unfold.
Now take an event $B$. The event $B$ can only occur through one of these scenarios. In set notation,

$$B = (B \cap A_1) \cup (B \cap A_2) \cup (B \cap A_3).$$

The three sets on the right-hand side are disjoint, because the $A_i$ are disjoint. Therefore, by the additivity axiom,

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3).$$

For each term we can use the multiplication rule:

$$P(B \cap A_i) = P(A_i)\, P(B \mid A_i), \qquad i = 1, 2, 3.$$

Substituting these expressions into the previous equation gives

$$P(B) = P(A_1)\, P(B \mid A_1) + P(A_2)\, P(B \mid A_2) + P(A_3)\, P(B \mid A_3).$$

This is the **total probability theorem** for three scenarios. It says that the total probability of $B$ is the sum over all scenarios of:

(probability of the scenario) $\times$ (probability of $B$ within that scenario).

Equivalently, $P(B)$ is a weighted average of the conditional probabilities $P(B \mid A_i)$, with weights $P(A_i)$ that sum to 1.
The same reasoning applies to any finite number of scenarios. If $A_1, \ldots, A_n$ form a partition of $\Omega$ and $P(A_i) > 0$ for all $i$, then for any event $B$,

$$P(B) = \sum_{i=1}^{n} P(A_i \cap B) = \sum_{i=1}^{n} P(A_i)\, P(B \mid A_i).$$

This is the general statement of the total probability theorem. If we partition $\Omega$ into countably many events $A_1, A_2, \ldots$, the same derivation goes through, except that we use countable additivity instead of finite additivity, and we obtain an infinite series

$$P(B) = \sum_{i=1}^{\infty} P(A_i)\, P(B \mid A_i).$$

In all cases, the key idea is to decompose $B$ into disjoint pieces corresponding to different scenarios and then sum their probabilities.

### 1.2.4 Bayes' rule

Bayes' rule describes how to update the probabilities of different scenarios after observing an event.
We keep the same setting as for the total probability theorem. The sample space $\Omega$ is partitioned into disjoint events $A_1, A_2, \ldots, A_n$, which we think of as *scenarios*. They satisfy

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \qquad \bigcup_{i=1}^{n} A_i = \Omega, \qquad P(A_i) > 0.$$

The numbers $P(A_i)$ represent our *initial beliefs* about how likely each scenario is before seeing any additional information. There is also an event $B$ that may occur under each scenario, and for each $i$ we know the conditional probability $P(B \mid A_i)$, that is, how likely $B$ is to occur if scenario $A_i$ is the true one.
Once we perform the experiment, we may observe that $B$ has actually occurred. This new information should cause us to revise our beliefs about which scenario $A_i$ is more likely. The updated belief about scenario $A_i$ is the conditional probability $P(A_i \mid B)$.
To compute $P(A_i \mid B)$, we start from the definition of conditional probability:

$$P(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)}, \qquad P(B) > 0.$$

The numerator $P(A_i \cap B)$ can be written using the multiplication rule:

$$P(A_i \cap B) = P(A_i)\, P(B \mid A_i).$$

The denominator $P(B)$ is given by the total probability theorem:

$$P(B) = \sum_{j=1}^{n} P(A_j \cap B) = \sum_{j=1}^{n} P(A_j)\, P(B \mid A_j).$$

Substituting these expressions into the definition yields

$$P(A_i \mid B) = \frac{P(A_i)\, P(B \mid A_i)}{\sum_{j=1}^{n} P(A_j)\, P(B \mid A_j)}.$$

This formula is **Bayes' rule**. It combines three ingredients: - the prior probabilities $P(A_i)$, which encode our initial beliefs about each scenario; - the likelihoods $P(B \mid A_i)$, which tell us how compatible the observation $B$ is with each scenario; - the normalizing denominator $\sum_j P(A_j)\, P(B \mid A_j)$, which ensures that the updated probabilities $P(A_i \mid B)$ sum to 1.
Conceptually, Bayes' rule starts from conditional probabilities "in one direction", $P(B \mid A_i)$, provided by the model, and uses them to compute conditional probabilities in the **reverse direction**, $P(A_i \mid B)$, which represent our revised beliefs about the scenarios after observing $B$. This is the basic mechanism of statistical inference: we begin with a probabilistic model and prior weights on possible explanations, observe data $B$, and then use Bayes' rule to update the probabilities of the different scenarios in light of the new evidence.

### 1.2.5 Independence

Independence formalizes the idea that learning one event has occurred does not change our beliefs about another event.
We return to the example of three tosses of a biased coin. The coin lands heads with probability $p$, where $p$ is not necessarily $1/2$. The experiment can be represented by a tree with three stages, one for each toss. Each branch corresponds to a possible result at a given toss, and each leaf corresponds to a complete outcome of the three tosses. Probabilities and conditional probabilities are attached to the branches: for example, the probability of heads on the first toss is $p$, and the probability of heads on the second toss given that the first toss was heads is also $p$. Similarly, the probability of heads on the second toss given that the first toss was tails is $p$ as well. More generally, in this model every toss has probability $p$ of being heads, regardless of the results of previous tosses.
We can use the multiplication rule to compute the probability of any specific outcome. For instance, consider the outcome "tails, then heads, then tails". Its probability is obtained by multiplying along the corresponding path in the tree:

$$P(T, H, T) = (1-p)\, p\, (1-p).$$

We can also compute the probability of more complicated events. Take the event that there is *exactly one head* in the three tosses. This event consists of three distinct outcomes:

$$(H, T, T), \quad (T, H, T), \quad (T, T, H).$$

Using the tree, we find that each of these outcomes has probability $p(1-p)^2$. Therefore, the total probability of exactly one head is

$$P(\text{exactly one head}) = 3\, p(1-p)^2.$$

All three outcomes in this event are equally likely.
Now consider a conditional probability. Suppose we are told that there was exactly one head in the three tosses. We want the probability that the *first* toss was heads under this information. In other words, we want

$$P(H_1 \mid \text{exactly one head}).$$

Intuitively, within the event "exactly one head" the three possible outcomes are equally likely, so each should have conditional probability $1/3$. Only one of them—$(H, T, T)$—has heads on the first toss, so we expect

$$P(H_1 \mid \text{exactly one head}) = \frac{1}{3}.$$

We can verify this formally using the definition of conditional probability. Let $E$ be the event "exactly one head" and $G$ be the event "first toss is heads". Then

$$P(G \mid E) = \frac{P(G \cap E)}{P(E)}.$$

The event $G \cap E$ corresponds to the single outcome $(H, T, T)$, which has probability $p(1-p)^2$. The denominator $P(E)$ is the probability of exactly one head, which we already computed as $3p(1-p)^2$. Thus

$$P(G \mid E) = \frac{p(1-p)^2}{3p(1-p)^2} = \frac{1}{3}.$$

This example has a special property: knowing the result of one toss does not change the probabilities for future tosses. For instance, the conditional probability of heads on the second toss given heads on the first toss is

$$P(H_2 \mid H_1) = p,$$

and the conditional probability of heads on the second toss given tails on the first toss is

$$P(H_2 \mid T_1) = p.$$

So the probability of heads on the second toss remains $p$, regardless of what happened on the first toss. We can also compute the unconditional probability of heads on the second toss using the total probability theorem:

$$P(H_2) = P(H_1)\, P(H_2 \mid H_1) + P(T_1)\, P(H_2 \mid T_1) = p \cdot p + (1-p) \cdot p = p.$$

Therefore, the unconditional probability of heads on the second toss is equal to both conditional probabilities. Learning the outcome of the first toss does not change our belief about the second toss.
This motivates the notion of independence. Let $A$ and $B$ be two events. Informally, $B$ is independent of $A$ if knowing that $A$ has occurred does not change our belief about $B$, in the sense that

$$P(B \mid A) = P(B).$$

Using the multiplication rule,

$$P(A \cap B) = P(A)\, P(B \mid A).$$

If we also have $P(B \mid A) = P(B)$, then this becomes

$$P(A \cap B) = P(A)\, P(B).$$

This identity turns out to be the most convenient formal definition of independence. We say that two events $A$ and $B$ are *independent* if

$$P(A \cap B) = P(A)\, P(B).$$

This definition has several advantages. It is symmetric in $A$ and $B$, so it does not distinguish between "$A$ is independent of $B$" and "$B$ is independent of $A$". It is also consistent with the intuitive definition: if $P(A) > 0$ and $P(B) > 0$, then

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = P(B),$$

and similarly $P(A \mid B) = P(A)$ whenever $P(B) > 0$. Finally, the product definition makes sense even when one of the events has probability zero, whereas conditional probabilities are not defined in that case.
It is important to distinguish independence from disjointness. Suppose $A$ and $B$ are disjoint events with strictly positive probabilities. Then

$$P(A \cap B) = 0,$$

but

$$P(A)\, P(B) > 0.$$

So the defining equality for independence does *not* hold, and $A$ and $B$ are not independent. Intuitively, disjoint events are as dependent as possible: if you know that $A$ has occurred, you are certain that $B$ has not, and vice versa. In contrast, independent events are such that the occurrence of one does not affect the probability of the other.
In many real-world situations, independence is a reasonable modeling assumption when two events are determined by physically separate and noninteracting processes. For example, the result of a coin toss and the weather on a specific future day can often be modeled as independent. In other cases, apparent independence may arise from a numerical coincidence, so one should always think carefully about whether the independence assumption is justified in a given context.

### 1.2.6 Independence of event complements

Independence between events automatically extends to their complements in a natural way.
Suppose that events $A$ and $B$ are independent. Intuitively, if learning that $A$ occurred does not change our belief about whether $B$ will occur, it also should not change our belief about whether $B$ will *not* occur. In other words, if $A$ is independent of $B$, then it is reasonable to expect that $A$ is also independent of $B^c$. We now show this formally.
Start with the event $A$. We can decompose $A$ into two disjoint pieces:

$$A = (A \cap B) \cup (A \cap B^c).$$

These two sets are disjoint, and together they make up the whole of $A$. By additivity,

$$P(A) = P(A \cap B) + P(A \cap B^c).$$

Since $A$ and $B$ are independent, their joint probability factorizes:

$$P(A \cap B) = P(A)\,P(B).$$

Substituting this into the previous equation gives

$$P(A) = P(A)\,P(B) + P(A \cap B^c).$$

Solving for $P(A \cap B^c)$ yields

$$P(A \cap B^c) = P(A) - P(A)\,P(B) = P(A)\big(1 - P(B)\big).$$

Using the relation $P(B^c) = 1 - P(B)$, we can rewrite this as

$$P(A \cap B^c) = P(A)\,P(B^c).$$

This is exactly the condition for $A$ and $B^c$ to be independent. Thus, we have shown that if $A$ and $B$ are independent, then $A$ and $B^c$ are independent as well.

### 1.2.7 Conditional independence

Conditional independence extends the idea of independence to a setting where we already condition on some additional information.
Conditional probabilities are like ordinary probabilities, except that they apply in a new situation where we are told that a certain event has occurred. Because of this, any notion we define for ordinary probabilities has an analogue for conditional probabilities. In particular, we can talk about conditional independence, which is simply the notion of independence applied to a conditional model.
More concretely, suppose we have events $A$, $B$, and $C$ in a probability space, with $P(C) > 0$. Once we are told that $C$ has occurred, we work with the conditional probabilities $P(\cdot \mid C)$. In the original (unconditional) model, independence of $A$ and $B$ is defined by

$$P(A \cap B) = P(A)\,P(B).$$

In the conditional model given $C$, we use exactly the same pattern but with conditional probabilities in place of ordinary ones. We say that $A$ and $B$ are *conditionally independent given* $C$ if

$$P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C),$$

provided all these conditional probabilities are defined. Intuitively, this means that after we learn that $C$ has occurred, learning whether $A$ occurs does not change our belief about $B$, and learning whether $B$ occurs does not change our belief about $A$, within the "universe" where $C$ holds.
A natural question is whether ordinary independence implies conditional independence. The answer is no in general. It may happen that $A$ and $B$ are independent in the unconditional model, so that $P(A \cap B) = P(A)\,P(B)$, but once we restrict attention to a particular event $C$, the geometry of the sets inside $C$ changes. For example, it could be that within $C$ the events $A$ and $B$ have no overlap at all, so that

$$P(A \cap B \mid C) = 0,$$

while both $P(A \mid C)$ and $P(B \mid C)$ are positive. In that case,

$$P(A \cap B \mid C) \neq P(A \mid C)\,P(B \mid C),$$

and $A$ and $B$ are not conditionally independent given $C$. In this way, conditioning on $C$ can introduce dependencies even when none were present before.
The main takeaway is that conditional independence is defined by the same product relation as ordinary independence, but using conditional probabilities. However, unconditional independence does not guarantee conditional independence once we restrict to a particular event $C$.

### 1.2.8 Independence vs conditional independence

Independence and conditional independence can behave very differently; independent events may become dependent after conditioning.
Consider two possible coins, coin A and coin B. If we know that coin A has been chosen, then in this conditional universe every toss is heads with probability 0.9, and different tosses are independent. Formally, given "coin A", the coin flips are *conditionally independent* and identically distributed with $P(H) = 0.9$. Similarly, given "coin B", the flips are conditionally independent with $P(H) = 0.1$ at each toss.
Now suppose that, before tossing, we pick one of the two coins at random, with

$$P(\text{coin A}) = 0.5, \qquad P(\text{coin B}) = 0.5,$$

and then we keep tossing that same coin. We consider the *overall* model where we do not know in advance which coin was selected, and we ask whether the different tosses are independent in this unconditional model.
First compute the unconditional probability of heads on the 11th toss. Using the total probability theorem,

$$P(H_{11}) = P(\text{coin A})\, P(H_{11} \mid \text{coin A}) + P(\text{coin B})\, P(H_{11} \mid \text{coin B}) = 0.5 \cdot 0.9 + 0.5 \cdot 0.1 = 0.5.$$

So before seeing any outcomes, each toss has probability 0.5 of being heads on average.
Next, suppose we are told that the first 10 tosses all resulted in heads. Intuitively, this makes coin A extremely likely: observing 10 heads in a row is very plausible under coin A, but extremely unlikely under coin B. In fact, by Bayes' rule one can show that

$$P(\text{coin A} \mid H_1, \ldots, H_{10})$$

is very close to 1, and the conditional probability of heads on the 11th toss satisfies

$$P(H_{11} \mid H_1, \ldots, H_{10}) \approx P(H_{11} \mid \text{coin A}) = 0.9.$$

Thus

$$P(H_{11} \mid H_1, \ldots, H_{10}) \approx 0.9 \neq 0.5 = P(H_{11}).$$

Information about the first ten tosses changes our belief about the 11th toss, so the 11th toss is *not* independent of the first ten in the overall model.
This example shows that even if the tosses are conditionally independent given the coin (they are independent within each conditional model), they need not be independent in the unconditional model where the underlying coin is unknown. Conditional independence does not imply unconditional independence, and unconditional independence does not imply conditional independence.

### 1.2.9 Independence of a collection of events

When we talk about independence for many events, we want a notion that captures "no information flow" between any subset and the rest of the family.
Suppose we have a collection of events $A_1, A_2, A_3, \ldots$. Intuitively, we would like to say that this family is independent if knowing what happened for some of the events does not change our beliefs about the others. For example, if $A_1, A_2, A_3$ are independent, then learning that $A_1$ occurred and $A_2$ did not occur should not change the probability we assign to $A_3$; the conditional probability of $A_3$ under that information should still be $P(A_3)$. More generally, any information about which of $A_1, A_2$ occurred should leave our probability for $A_3$ unchanged.
Stating independence directly in terms of "any information about some events does not affect the others" would lead to a very complicated definition. Instead, we use a cleaner condition that parallels the two-event case. Recall that two events $A$ and $B$ are independent exactly when

$$P(A \cap B) = P(A)\, P(B).$$

We extend this pattern to collections by requiring that we can always compute probabilities of intersections by multiplying individual probabilities.
For concreteness, consider three events $A_1, A_2, A_3$. We say that they are *independent* if the following relations all hold:

$$P(A_1 \cap A_2) = P(A_1)\, P(A_2),$$

$$P(A_1 \cap A_3) = P(A_1)\, P(A_3),$$

$$P(A_2 \cap A_3) = P(A_2)\, P(A_3),$$

and, in addition,

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)\, P(A_2)\, P(A_3).$$

The first three conditions mean that every pair of events is independent; this is called *pairwise independence*. The last condition is extra and does not follow automatically from pairwise independence. It enforces that the three-way intersection probability also factors as a product of individual probabilities. For three events, our definition requires both pairwise independence and the three-way product condition.
This definition extends to any finite family $A_1, \ldots, A_n$. We say that these events are independent if for every choice of distinct indices $i_1, \ldots, i_k$ (with $k \geq 2$) we have

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k}).$$

In other words, the intersection of any subcollection of events has probability equal to the product of the individual probabilities. One can prove that, when these conditions are satisfied, all the intuitive "no-information" properties follow. For example, if $A_1, A_2, A_3$ are independent in this sense, then for any pattern of occurrences of $A_1$ and $A_2$ we have

$$P(A_3 \mid \text{any specification of } A_1, A_2) = P(A_3),$$

so knowing which of $A_1$ and $A_2$ occurred does not change the probability of $A_3$. The formal proof that the product conditions imply all such conditional equalities is somewhat tedious, so it is usually omitted, but the key point is that the product definition compactly captures the idea that information about any subfamily of events does not affect the probabilities of the remaining events.

### 1.2.10 Independence vs pairwise independence

Independence of a collection of events is stronger than pairwise independence; events can be pairwise independent without being independent as a family.
Consider two independent tosses of a fair coin. Let $H_1$ be the event that the first toss is heads, and $H_2$ the event that the second toss is heads. Each toss has probability 1/2 of heads, and the independence assumption gives

$$P(H_1 \cap H_2) = P(H_1)\, P(H_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

By symmetry, each of the four outcomes $(H, H)$, $(H, T)$, $(T, H)$, and $(T, T)$ has probability 1/4.
Now define a new event $C$ as "the two tosses have the same result", which corresponds to the outcomes $(H, H)$ and $(T, T)$. Thus

$$P(C) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

We first check pairwise independence. Take $H_1$ and $C$:

$$H_1 \cap C$$

is the event that the first toss is heads and the two tosses are the same, which happens only for $(H, H)$, so

$$P(H_1 \cap C) = \frac{1}{4}.$$

On the other hand,

$$P(H_1)\, P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

So $P(H_1 \cap C) = P(H_1)\, P(C)$, and $H_1$ and $C$ are independent. By symmetry, $H_2$ and $C$ are also independent, and we already know that $H_1$ and $H_2$ are independent. Thus all three pairs $(H_1, H_2)$, $(H_1, C)$, and $(H_2, C)$ are pairwise independent.
To test whether $H_1, H_2$, and $C$ are independent as a collection, we must also look at the three-way intersection. The event

$$H_1 \cap H_2 \cap C$$

means "first toss heads, second toss heads, and the two tosses are the same", which again corresponds only to the outcome $(H, H)$, so

$$P(H_1 \cap H_2 \cap C) = \frac{1}{4}.$$

The product of individual probabilities is

$$P(H_1)\, P(H_2)\, P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $\frac{1}{4} \neq \frac{1}{8}$, the factorization condition for three events fails. Therefore $H_1, H_2, C$ are not independent as a collection, even though every pair among them is independent. This illustrates that pairwise independence does not imply independence of the whole family.

The intuition is that knowing only $H_1$ does not help us decide whether $C$ occurs: given that the first toss is heads, the second toss is equally likely to be heads or tails, so

$$P(C \mid H_1) = \frac{1}{2} = P(C).$$

Similarly $P(C \mid H_2) = P(C)$. However, if we know both $H_1$ and $H_2$ occurred, then we know the results of the two tosses are the same, so

$$P(C \mid H_1 \cap H_2) = 1 \neq P(C).$$

Thus the pair $(H_1, H_2)$ together carries information about $C$, even though each event separately does not. This is exactly the situation where pairwise independence holds but full independence of the collection fails.

### 1.2.11 Reliability

Independence lets us analyze complex systems by breaking them into simpler independent pieces, which is especially useful in reliability calculations.
Consider a system with $n$ components (or units). The $i$-th unit can be "up" (working) or "down" (failed). Let $U_i$ be the event that unit $i$ is up, with

$$P(U_i) = p_i,$$

and assume the events $U_1, \ldots, U_n$ are independent. Intuitively, this means that whether some units are up or down does not change the probabilities for the others. Equivalently, if we define $F_i = U_i^c$ as the event that unit $i$ has failed, then the failure events $F_1, \ldots, F_n$ are also independent: information about which units have failed does not affect the failure probabilities of the remaining units.
Let us look at a system with three components and define the event "system up" to mean that there exists a path from left to right consisting only of working units.
First, consider a series system where all three units lie on the same path. In this case, the system is up if and only if all three units are up. The event "system up" is

$$U_1 \cap U_2 \cap U_3,$$

and by independence its probability is

$$P(\text{system up}) = P(U_1 \cap U_2 \cap U_3) = P(U_1)\, P(U_2)\, P(U_3) = p_1 p_2 p_3.$$

Next, consider a parallel system where three units provide alternative paths from left to right. The system is up as long as at least one unit is up. The event "system up" is

$$U_1 \cup U_2 \cup U_3.$$

Independence is most convenient for intersections, so we rewrite this probability using complements. The complement of "at least one unit is up" is "all units are down", i.e.

$$(U_1 \cup U_2 \cup U_3)^c = F_1 \cap F_2 \cap F_3.$$

Therefore,

$$P(\text{system up}) = P(U_1 \cup U_2 \cup U_3) = 1 - P(F_1 \cap F_2 \cap F_3).$$

By independence of the failure events, this intersection probability factors:

$$P(F_1 \cap F_2 \cap F_3) = P(F_1)\, P(F_2)\, P(F_3).$$

Since $P(F_i) = 1 - p_i$, we obtain

$$P(\text{system up}) = 1 - (1 - p_1)(1 - p_2)(1 - p_3).$$

These formulas illustrate how independence allows us to compute the reliability (probability of correct operation) of systems built from multiple components. For series configurations we multiply the success probabilities of all components, and for parallel configurations we use complements and multiply failure probabilities. More complex networks can be analyzed by combining these ideas.

EXAMPLE: THE KING'S SIBLING

The probability that the king's sibling is female depends on how we model the royal family's childbearing decision. Assume first that the royal couple decided in advance to have exactly two children, that each child is independently a boy or a girl with probability 1/2, and that a boy always becomes king if there is at least one boy. Under these assumptions there are four equally likely gender outcomes for the two children:

$$(B, B),\ (B, G),\ (G, B),\ (G, G),$$

each with probability 1/4. Since there is a king, there must be at least one boy, so the outcome $(G, G)$ is impossible. Conditioning on "at least one boy", we restrict to

$$(B, B),\ (B, G),\ (G, B),$$

which are now equally likely and each has conditional probability 1/3. In two of these three outcomes the sibling is a girl, so

$$P(\text{sibling is female} \mid \text{there is a king, with exactly two children}) = \frac{2}{3}.$$

However, this answer relies crucially on the assumption that "two children" was fixed in advance. If we change the family rule, the answer changes. For example, if the royal family "has children until the first boy is born," then the fact that there are two children tells us the first was a girl and the second a boy, so the sibling is certainly a girl and the probability is 1. If instead they "have children until they get two boys," then being told there are exactly two children implies both are boys, so the sibling cannot be a girl and the probability is 0.

Thus 2/3 is the correct answer only under the model where the number of children (two) is fixed beforehand, genders are independent and equally likely, and a boy always becomes king. Different modeling assumptions lead to different probabilities, which illustrates how important it is to clarify the probabilistic model before computing.

## 1.3 Counting

### 1.3.1 The counting principle

We often construct an outcome through a sequence of stages, making a choice at each stage. The *counting* principle says: if there are $r$ stages and at stage $i$ we always have $n_i$ options (independent of previous choices), then the total number of possible outcomes is

$$n_1 n_2 \cdots n_r.$$

EXAMPLE: OUTFITS

Suppose you have 4 shirts, 3 ties, and 2 jackets. Think of getting dressed as a 3-stage process:

shirt : 4 choices, tie : 3 choices, jacket : 2 choices.

By the counting principle, the number of different outfits is

$$4 \cdot 3 \cdot 2 = 24.$$

Figure 1: Enter Caption

EXAMPLE: LICENSE PLATES
Consider license plates of the form "2 letters followed by 3 digits." We construct the plate sequentially.
*With repetition allowed.* At each stage:

$$\underbrace{26}_{\text{1st letter}} \cdot \underbrace{26}_{\text{2nd letter}} \cdot \underbrace{10}_{\text{1st digit}} \cdot \underbrace{10}_{\text{2nd digit}} \cdot \underbrace{10}_{\text{3rd digit}}.$$

So the number of such plates is

$$26^2 \cdot 10^3.$$

*Without repetition.* Now require that no letter and no digit can be used more than once. Then:

$$\underbrace{26}_{\text{1st letter}} \cdot \underbrace{25}_{\text{2nd letter}} \cdot \underbrace{10}_{\text{1st digit}} \cdot \underbrace{9}_{\text{2nd digit}} \cdot \underbrace{8}_{\text{3rd digit}}.$$

Thus, the number of such plates is

$$26 \cdot 25 \cdot 10 \cdot 9 \cdot 8.$$

EXAMPLE: PERMUTATIONS AND FACTORIAL
Let a set have $n$ distinct elements; we want to order them (form a permutation). Think of $n$ slots and fill them one at a time:

$$\underbrace{n}_{\text{1st slot}} \cdot \underbrace{(n-1)}_{\text{2nd slot}} \cdot \underbrace{(n-2)}_{\text{3rd slot}} \cdots \underbrace{1}_{\text{last slot}}.$$

The total number of permutations is

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1.$$

EXAMPLE: NUMBER OF SUBSETS
Let a set have $n$ elements; we want to count its subsets. Consider the elements one by one. For each element, we have 2 options: include it in the subset or not. By the counting principle, the total number of subsets is

$$2 \cdot 2 \cdot \cdots \cdot 2 = 2^n,$$

with $n$ factors.
*Sanity check for* $n = 1$. If the set is $\{a\}$, its subsets are

$$\{a\}, \quad \varnothing,$$

so there are 2 subsets, in agreement with $2^1 = 2$. Note that when we count subsets we include both the whole set and the empty set.

### 1.3.2 A die-roll example

We roll an ordinary six-sided die six times, and we are interested in the probability that all six results are different; denote this event by $A$. We assume that all $6^6$ possible sequences of six rolls are equally likely, so we are in a discrete uniform model and probabilities can be computed by counting.

Sample space
A typical outcome is a sequence such as $(2, 3, 4, 3, 6, 2)$. At each of the 6 rolls we have 6 possible results, so by the counting principle the total number of outcomes is

$$|\Omega| = 6^6.$$

Under the equally likely assumption, each sequence has probability $1/6^6$. This is consistent with viewing the experiment as 6 independent rolls of a fair die, each outcome having probability $(1/6)^6 = 1/6^6$.

Counting the favorable outcomes
Event $A$ consists of all sequences in which no number is repeated, i.e., each of the numbers $1, \ldots, 6$ appears exactly once. Such sequences are precisely the permutations of $\{1, 2, 3, 4, 5, 6\}$, and there are

$$|A| = 6!$$

of them.

Probability of all distinct rolls
By the discrete uniform model,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{6!}{6^6}.$$

### 1.3.3 Combinations

We start with a set of $n$ elements and a non-negative integer $k$. A *combination* is a subset of the original set that has exactly $k$ elements. The number of such subsets is denoted by the binomial coefficient

$$\binom{n}{k},$$

read as "$n$-choose-$k$".

Two ways to count ordered $k$-tuples
Consider a related problem: construct an ordered list of $k$ *distinct* elements chosen from the $n$-element set.
*First method (direct counting).* Think of $k$ slots and fill them one by one with distinct elements:

$$n \cdot (n-1) \cdot (n-2) \cdots (n-k+1).$$

This product can be written as

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

*Second method (choose then order).* First choose which $k$ elements will appear (ignore order), then order them.

$$\underbrace{\binom{n}{k}}_{\text{choose the set of } k \text{ elements}} \cdot \underbrace{k!}_{\text{permute these } k \text{ elements}}.$$

Both methods count the same set of ordered lists, so

$$\binom{n}{k} k! = \frac{n!}{(n-k)!}.$$

Solving for $\binom{n}{k}$ gives the standard formula

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$

valid for integers $n \geq 0$ and $0 \leq k \leq n$.

Extreme cases and the convention $0! = 1$
Consider $\binom{n}{n}$. From the combinatorial definition, choosing $n$ elements from an $n$-element set leaves only one possibility: take all elements, so $\binom{n}{n} = 1$. From the formula,

$$\binom{n}{n} = \frac{n!}{n!\,0!},$$

so this equals 1 if we adopt the convention $0! = 1$.
Now consider $\binom{n}{0}$. Combinatorially, there is exactly one subset with zero elements: the empty set, so $\binom{n}{0} = 1$. From the formula,

$$\binom{n}{0} = \frac{n!}{0!\,n!} = 1,$$

again consistent with $0! = 1$.

A useful binomial identity
The sum

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$$

can be given a combinatorial interpretation. Here, $\binom{n}{k}$ counts the subsets of size $k$ of an $n$-element set. Summing over all $k$ from 0 to $n$ counts all subsets of the set. But the total number of subsets of an $n$-element set is $2^n$. Therefore,

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

### 1.3.4 Binomial probabilities

The binomial coefficients $\binom{n}{k}$ are closely related to probabilities in repeated coin-tossing experiments. Consider a coin tossed $n$ times independently. Let $p$ be the probability of heads on a single toss, so the probability of tails is $1 - p$. We want the probability of observing exactly $k$ heads in these $n$ tosses.

Probability of a particular sequence
Take a concrete example with $n = 6$, and consider a specific sequence of heads and tails, e.g. H T T H H H. By independence, the probability of this sequence is

$$p \cdot (1-p) \cdot (1-p) \cdot p \cdot p \cdot p = p^4(1-p)^2.$$

In general, for any *given* sequence with $k$ heads and $n - k$ tails, the probability of that sequence is

$$p^k(1-p)^{n-k},$$

because each head contributes a factor $p$ and each tail a factor $1 - p$.

Number of $k$-head sequences
We now want the probability of the event "exactly $k$ heads in $n$ tosses". This event consists of all sequences of length $n$ that contain exactly $k$ heads. Every such sequence has the same probability $p^k(1-p)^{n-k}$. Therefore, the desired probability equals

$$\big(\text{number of sequences with exactly } k \text{ heads}\big) \times p^k(1-p)^{n-k}.$$

To count the sequences with exactly $k$ heads, think of the $n$ tosses as $n$ time slots. To specify a $k$-head sequence, we must choose which $k$ of the $n$ slots contain heads; the remaining $n - k$ slots will contain tails. Thus, the number of such sequences is

$$\binom{n}{k}.$$

The binomial formula
Putting everything together, the probability of exactly $k$ heads in $n$ independent tosses of a coin with head probability $p$ is

$$P(\text{exactly } k \text{ heads}) = \binom{n}{k} p^k(1-p)^{n-k},$$

which is called the *binomial* probability formula.

### 1.3.5 Example: Each person gets an ace

We have a standard 52-card deck, dealt fairly to four players, 13 cards each (as in bridge). Outcomes are all partitions of the 52 cards into four hands of size 13, and all such partitions are assumed equally likely. We want the probability that each player gets *exactly one ace*.

Counting via multinomial coefficients
The sample space size is the number of ways to partition 52 distinct cards into hands of sizes $13, 13, 13, 13$, namely the multinomial coefficient

$$|\Omega| = \binom{52}{13, 13, 13, 13} = \frac{52!}{13!\,13!\,13!\,13!}.$$

Now count the favorable outcomes where each person gets exactly one ace.
First distribute the four aces: there are

$$4 \cdot 3 \cdot 2 \cdot 1 = 4!$$

ways to assign the four distinct aces to the four distinct players (each player gets one).
Then distribute the remaining 48 non-aces arbitrarily, giving 12 additional cards to each player. The number of such partitions is

$$\binom{48}{12, 12, 12, 12} = \frac{48!}{12!\,12!\,12!\,12!}.$$

By the counting principle, the number of favorable outcomes is

$$|A| = 4! \cdot \binom{48}{12, 12, 12, 12} = 4! \cdot \frac{48!}{12!\,12!\,12!\,12!}.$$

Thus, the desired probability is

$$P(A) = \frac{|A|}{|\Omega|} = \frac{4! \cdot \dfrac{48!}{12!\,12!\,12!\,12!}}{\dfrac{52!}{13!\,13!\,13!\,13!}}.$$

Alternative "aces first" viewpoint
There is a faster way to obtain the same probability. Imagine the 52 slots corresponding to the four players' 13-card hands. Place the four aces *first*, one at a time, uniformly at random among the 52 slots.
The first ace can go anywhere.
For the second ace, there are 51 remaining slots, of which 39 belong to players who do not yet have an ace (3 players $\times$ 13 slots each). So the conditional probability that the second ace goes to a different player than the first is

$$\frac{39}{51}.$$

Assuming this has happened, two players now have aces. For the third ace there are 50 remaining slots; the "good" slots are those belonging to the two players without aces, i.e. $2 \times 13 = 26$ slots. So the corresponding probability is

$$\frac{26}{50}.$$

Assuming all three aces have gone to different players, three players now have aces. For the fourth ace, there are 49 remaining slots, of which 13 belong to the last player without an ace, giving probability

$$\frac{13}{49}.$$

Multiplying these conditional probabilities, the overall probability that the four aces land in slots belonging to four different players is

$$P(A) = \frac{39}{51} \cdot \frac{26}{50} \cdot \frac{13}{49}.$$

One can check algebraically that this expression is equal to the earlier ratio of multinomial coefficients.

## 1.4 Discrete random variables

### 1.4.1 Definition of random variables

A random variable is a function from the sample space to the real numbers. It assigns to each outcome of the experiment a numerical value, determined once the outcome is known.

Example: Weight of a student
Let the sample space be $\Omega = \{a, b, c, d\}$, where each element is a student. The experiment is to pick one student at random and record their weight in kilograms. Define $W(\omega)$ as the weight of student $\omega$. Then $W$ is a random variable, because for each outcome $\omega \in \Omega$ it produces a real number $W(\omega)$.

Another random variable and a function of them
On the same experiment, define $H(\omega)$ as the height (in meters) of student $\omega$. This $H$ is also a random variable. We can form a new random variable by combining them, for instance the body mass index

$$B(\omega) = \frac{W(\omega)}{H(\omega)^2}.$$

Once the outcome $\omega$ is known, both $W(\omega)$ and $H(\omega)$ are known, so $B(\omega)$ is determined as well.

Discrete and continuous random variables
Random variables can be discrete or continuous. For example, the number of heads in 10 coin tosses is a discrete random variable taking values in $\{0, 1, \ldots, 10\}$. The time at which an event occurs, measured with infinite accuracy, is modeled as a continuous random variable taking real values in some interval.

Notation
We use uppercase letters such as $X, Y, W, H$ for random variables (the functions) and lowercase letters such as $x, y, w, h$ for their numerical values. If $X$ and $Y$ are random variables, then $X + Y$ is also a random variable, defined by

$$(X + Y)(\omega) = X(\omega) + Y(\omega).$$

In general, any function of random variables is again a random variable.

### 1.4.2 Probability mass function

A discrete random variable can take different numerical values, and some of these values are more likely than others. These relative likelihoods are described by the *probability mass function* (PMF), which gives the probability of each possible numerical value.

Definition of the PMF
Let $X$ be a discrete random variable. For any real number $x$, we write

$$p_X(x) = \mathbb{P}(X = x).$$

This is the probability of the event that $X$ takes the value $x$. The function $p_X$ is called the PMF (or probability law, or probability distribution) of $X$.
The argument $x$ ranges over the possible values that $X$ can take. For each such $x$, $p_X(x)$ is a number between 0 and 1. If we had another random variable $Y$ on the same sample space, its PMF would be denoted $p_Y(y) = \mathbb{P}(Y = y)$.

Basic properties
The PMF satisfies two fundamental properties:

$$p_X(x) \geq 0 \quad \text{for all } x,$$
$$\sum_x p_X(x) = 1,$$

where the sum is over all possible values of $X$. The first property holds because probabilities are never negative. The second holds because the events $\{X = x\}$ for all possible $x$ are disjoint and together cover the entire sample space, so their probabilities must add up to 1.

Example: A simple random variable
Consider an experiment with four equally likely outcomes $a, b, c, d$. Suppose a random variable $X$ is defined by

$$X(a) = 5, \quad X(b) = 5, \quad X(c) = 4, \quad X(d) = 3.$$

Each outcome has probability 1/4.
The event $\{X = 5\}$ is $\{a, b\}$, so

$$p_X(5) = \mathbb{P}(X = 5) = \mathbb{P}(\{a, b\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Similarly,

$$p_X(4) = \mathbb{P}(X = 4) = \mathbb{P}(\{c\}) = \frac{1}{4},$$
$$p_X(3) = \mathbb{P}(X = 3) = \mathbb{P}(\{d\}) = \frac{1}{4}.$$

For any other $x$, we have $p_X(x) = 0$. These values satisfy $p_X(3) + p_X(4) + p_X(5) = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$, as required.

Example: Sum of two tetrahedral dice
Consider rolling two independent tetrahedral dice, each with faces $\{1, 2, 3, 4\}$. Let $X$ be the result of the first roll and $Y$ the result of the second roll. All 16 pairs $(i, j)$ with $i, j \in \{1, 2, 3, 4\}$ are assumed equally likely, each with probability 1/16.
Define a new random variable

$$Z = X + Y.$$

The possible values of $Z$ are $2, 3, 4, 5, 6, 7, 8$. To compute the PMF of $Z$, we proceed value by value, each time collecting the outcomes that give that sum and adding their probabilities.
For example, the event $\{Z = 2\}$ is $\{(1, 1)\}$, so

$$p_Z(2) = \mathbb{P}(Z = 2) = \frac{1}{16}.$$

The event $\{Z = 3\}$ is $\{(1, 2), (2, 1)\}$, so

$$p_Z(3) = \mathbb{P}(Z = 3) = \frac{2}{16}.$$

The event $\{Z = 4\}$ is $\{(1, 3), (2, 2), (3, 1)\}$, so

$$p_Z(4) = \mathbb{P}(Z = 4) = \frac{3}{16}.$$

Continuing in the same way, we find

$$p_Z(5) = \frac{4}{16}, \quad p_Z(6) = \frac{3}{16}, \quad p_Z(7) = \frac{2}{16}, \quad p_Z(8) = \frac{1}{16}.$$

These numbers are all nonnegative and they add up to 1:

$$\frac{1}{16} + \frac{2}{16} + \frac{3}{16} + \frac{4}{16} + \frac{3}{16} + \frac{2}{16} + \frac{1}{16} = 1.$$

Thus we have completely determined the probability mass function of $Z$.

### 1.4.3 Bernoulli and indicator random variables

A *Bernoulli* random variable is a discrete random variable that takes only the values 0 and 1. Its distribution is determined by a parameter $p \in [0, 1]$, interpreted as the probability of the outcome 1.
Using PMF notation, a Bernoulli random variable $X$ satisfies

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p.$$

If we plot its PMF, we obtain two bars: one at $x = 0$ of height $1 - p$, and one at $x = 1$ of height $p$. Bernoulli random variables model experiments with two possible outcomes, such as success/failure or heads/tails.

Indicator random variables
Let $\Omega$ be a sample space and let $A \subseteq \Omega$ be an event. The *indicator* random variable of $A$ is defined by

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Thus $I_A = 1$ exactly when the event $A$ occurs, and $I_A = 0$ otherwise.
Since $I_A$ takes only the values 0 and 1, it is a Bernoulli random variable. Moreover,

$$\mathbb{P}(I_A = 1) = \mathbb{P}(A), \qquad \mathbb{P}(I_A = 0) = 1 - \mathbb{P}(A),$$

so $I_A$ is Bernoulli with parameter $p = \mathbb{P}(A)$. Indicator random variables are useful because they convert questions about events into questions about random variables, where algebraic manipulations are often easier.

### 1.4.4 Uniform random variables

A *discrete uniform* random variable is one whose possible values all have the same probability. It takes integer values in some finite interval and assigns equal mass to each of them.

Definition
Let $a$ and $b$ be integers with $a \leq b$. We consider the experiment of choosing an integer at random from the set

$$\{a, a+1, a+2, \ldots, b\}.$$

There are $b - a + 1$ possible values.
Let $X$ be the random variable that equals the chosen integer. Then $X$ has PMF

$$\mathbb{P}(X = x) = \begin{cases} \dfrac{1}{b-a+1} & \text{if } x \in \{a, a+1, \ldots, b\}, \\ 0 & \text{otherwise.} \end{cases}$$

In this setting, the outcome of the experiment is already a number, and this number is the value of the random variable, so there is no distinction between outcome and numerical value.

Example: Seconds on a digital clock
Consider the seconds reading on a digital clock. The possible values are the integers from 0 to 59, so there are 60 possibilities. If you glance at the clock at a time chosen completely at random, there is no reason to think one second value is more likely than another.
Let $S$ be the seconds reading. Then $S$ is (ideally) discrete uniform on $\{0, 1, \ldots, 59\}$, with

$$\mathbb{P}(S = s) = \frac{1}{60}, \quad s = 0, 1, \ldots, 59.$$

Degenerate case
If $a = b$, then the range of $X$ consists of a single value $a$. In that case,

$$\mathbb{P}(X = a) = 1,$$

and $X$ is a *constant* or *degenerate* random variable. It has no randomness in the usual sense, but it still fits the formal definition of a random variable as a function on the sample space.

### 1.4.5 Binomial random variables

A *binomial* random variable models the number of successes in a fixed number of independent, identical trials, where each trial has two outcomes (success/failure) and the same success probability $p$. Typical examples are repeated coin tosses or repeated yes/no experiments.

Definition
Consider $n$ independent trials. On each trial, the probability of success is $p$ and the probability of failure is $1 - p$. Define the random variable $X$ to be the number of successes observed in these $n$ trials.
Then $X$ is called a *binomial* random variable with parameters $n$ and $p$. Its possible values are $0, 1, \ldots, n$.

PMF of a binomial random variable
We now recall the probability that $X$ takes a specific value $k$. The event $\{X = k\}$ consists of all sequences of $n$ trials that contain exactly $k$ successes and $n - k$ failures. Each particular sequence with $k$ successes has probability

$$p^k (1-p)^{n-k},$$

because we multiply a factor $p$ for each success and a factor $1 - p$ for each failure.
The number of such sequences is

$$\binom{n}{k},$$

since we must choose which $k$ of the $n$ positions are successes. Therefore, the PMF of $X$ is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Interpretation and shape

When the coin is fair, $p = 1/2$, the PMF is largest near $k = n/2$. For example, with $n = 10$, the most likely outcome is $X = 5$, and probabilities decrease as $k$ moves away from 5 in either direction. For larger $n$, such as $n = 100$, the most likely values of $X$ form a band around $np$; values far from $np$ have very small probability.

When $p$ is small (for instance $p = 0.2$), small values of $X$ are more likely. If $n$ is moderate, such as $n = 10$, we typically see a few successes, often between 0 and 4. If $n$ is large, such as $n = 100$ with $p = 0.1$, we say we expect "about" $np = 10$ successes, meaning that values near 10 (roughly in some neighborhood such as from 0 up to around 20) have noticeable probability, while much larger values are extremely unlikely.

### 1.4.6 Geometric random variables

A *geometric* random variable models the number of trials needed until the first success occurs, in a sequence of independent, identical trials with success probability $p$ at each trial. Typical examples are repeated coin tosses until the first head appears, or repeated attempts at an experiment until it succeeds.

Definition and PMF

Consider an infinite sequence of independent trials, each with probability $p$ of success and $1 - p$ of failure, where $0 < p \leq 1$. Let $X$ be the number of the trial on which the first success occurs. Then $X$ is called a *geometric* random variable. Its possible values are the positive integers $1, 2, 3, \ldots$.

To compute the PMF, fix $k \geq 1$. The event $\{X = k\}$ means that the first $k - 1$ trials are failures and the $k$-th trial is a success. By independence,

$$\mathbb{P}(X = k) = (1 - p)^{k-1} p, \qquad k = 1, 2, 3, \ldots$$

As $k$ increases, the factor $(1 - p)^{k-1}$ makes these probabilities decrease geometrically.

Remark on "never seeing a success"

There is a logically possible outcome where every trial is a failure (for example, all tails when tossing a coin), so there is no first success and $X$ would be "infinite". However, the probability of this event is

$$\lim_{k \to \infty} (1 - p)^k = 0,$$

because $0 < 1 - p < 1$. So with probability 1 the random variable $X$ takes a finite value in $\{1, 2, 3, \ldots\}$, and the probabilities $(1 - p)^{k-1} p$ sum to 1.

### 1.4.7 Expectation

The *expected value* (or expectation, or mean) of a random variable is a single number that describes what value we see "on average" in many independent repetitions of the experiment. For a discrete random variable, it is defined as a weighted average of its possible values, where the weights are the corresponding probabilities.

Definition for a discrete random variable

Let $X$ be a discrete random variable with PMF $p_X(x) = \mathbb{P}(X = x)$. The expected value of $X$, denoted $\mathbb{E}[X]$, is

$$\mathbb{E}[X] = \sum_x x \, p_X(x),$$

where the sum runs over all possible values of $X$. Each term is "value $\times$ probability of that value".

If $X$ takes infinitely many (discrete) values, this becomes an infinite series. We then assume that

$$\sum_x |x| \, p_X(x) < \infty,$$

so that $\mathbb{E}[X]$ is well defined and finite.

Example: Game of chance

Suppose a random variable $X$ represents your gain in a single play of a game, with

$$\mathbb{P}(X = 1) = \frac{2}{10}, \quad \mathbb{P}(X = 2) = \frac{5}{10}, \quad \mathbb{P}(X = 4) = \frac{3}{10}.$$

Then

$$\mathbb{E}[X] = 1 \cdot \frac{2}{10} + 2 \cdot \frac{5}{10} + 4 \cdot \frac{3}{10} = \frac{2}{10} + \frac{10}{10} + \frac{12}{10} = \frac{24}{10} = 2.4.$$

If you play many times, your average gain per game will be close to 2.4.

Bernoulli and indicator random variables
Let $X$ be a Bernoulli random variable with parameter $p$, so

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p.$$

Then

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

As a special case, let $X = I_A$ be the indicator of an event $A$, so that $X = 1$ if $A$ occurs and $X = 0$ otherwise. Then

$$\mathbb{P}(I_A = 1) = \mathbb{P}(A),$$

and

$$\mathbb{E}[I_A] = \mathbb{P}(A).$$

So the expected value of an indicator random variable equals the probability of the corresponding event.

Uniform random variable on $\{0, 1, \ldots, n\}$
Let $X$ be discrete uniform on $\{0, 1, \ldots, n\}$. Then

$$\mathbb{P}(X = k) = \frac{1}{n+1}, \quad k = 0, 1, \ldots, n.$$

Using the definition,

$$\mathbb{E}[X] = \sum_{k=0}^{n} k \cdot \frac{1}{n+1} = \frac{1}{n+1} \sum_{k=0}^{n} k = \frac{1}{n+1} \cdot \frac{n(n+1)}{2} = \frac{n}{2}.$$

Thus the expectation is the midpoint of the interval $\{0, \ldots, n\}$. More generally, when a PMF is symmetric, the expected value lies at the center of symmetry.

Population-average interpretation
Consider a class of $n$ students, and let $x_i$ be the weight of the $i$-th student. You perform an experiment where you pick one student uniformly at random, and let $X$ be the weight of the selected student.
The PMF of $X$ is

$$\mathbb{P}(X = x_i) = \frac{1}{n}, \quad i = 1, \ldots, n.$$

Then

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \cdot \frac{1}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

So in this example, the expected value of $X$ is exactly the usual average (arithmetic mean) of the students' weights. This illustrates that expectation can be seen both as the long-run average over many repetitions, and as an average over a finite population when all individuals are equally likely to be chosen.

Elementary properties of expectation
A random variable $X$ can be summarized by its expected value $\mathbb{E}[X]$. Some basic properties of expectation are very natural and are worth stating explicitly.

Nonnegativity
If $X(\omega) \geq 0$ for every outcome $\omega$ in the sample space, then

$$\mathbb{E}[X] \geq 0.$$

In words, if the random variable never takes negative values, its average value cannot be negative. This follows because $\mathbb{E}[X]$ is a sum of terms of the form $x\, p_X(x)$, and both factors are nonnegative.

Bounds from the range
Suppose $X(\omega)$ always lies between two constants $a$ and $b$, that is

$$a \leq X(\omega) \leq b \quad \text{for all } \omega.$$

Then the expectation also lies between $a$ and $b$:

$$a \leq \mathbb{E}[X] \leq b.$$

To see why $\mathbb{E}[X] \geq a$, write

$$\mathbb{E}[X] = \sum_x x\, p_X(x).$$

Each possible value $x$ of $X$ satisfies $x \geq a$, so

$$\mathbb{E}[X] = \sum_x x\, p_X(x) \geq \sum_x a\, p_X(x) = a \sum_x p_X(x) = a,$$

because the probabilities $p_X(x)$ sum to 1. A symmetric argument with $x \leq b$ gives $\mathbb{E}[X] \leq b$.

Expectation of a constant

A constant $c$ can be viewed as a (degenerate) random variable that always takes the value $c$. Its PMF assigns probability 1 to the value $c$ and probability 0 to all other values. Then

$$\mathbb{E}[c] = c \cdot 1 = c.$$

So if a quantity is always equal to $c$, its average is also $c$.

Expected value rule

A random variable $Y$ is often defined as a function of another random variable $X$, say

$$Y = g(X).$$

We want a convenient way to compute $\mathbb{E}[Y]$ without first finding the PMF of $Y$. The *expected value rule* gives exactly this.

Let $X$ be a discrete random variable with PMF $p_X(x) = \mathbb{P}(X = x)$, and let

$$Y = g(X)$$

for some function $g$. Then

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x),$$

where the sum is over all possible values of $X$.

Conceptually, we sum over the values of $X$, and for each possible $x$ we take the corresponding value $g(x)$ of $Y$, weighted by the probability that $X$ equals $x$.

Example

Suppose $X$ takes values $2, 3, 4, 5$ with probabilities

$$\mathbb{P}(X = 2) = 0.1, \quad \mathbb{P}(X = 3) = 0.2, \quad \mathbb{P}(X = 4) = 0.3, \quad \mathbb{P}(X = 5) = 0.4.$$

Define $Y = g(X)$ by

$$g(2) = 3, \quad g(3) = 3, \quad g(4) = 4, \quad g(5) = 4.$$

Then

$$\mathbb{E}[Y] = \sum_x g(x)\, p_X(x) = 3 \cdot 0.1 + 3 \cdot 0.2 + 4 \cdot 0.3 + 4 \cdot 0.4.$$

This matches the intuitive reasoning: 10

Why the rule works (idea)

By definition,

$$\mathbb{E}[Y] = \sum_y y\, \mathbb{P}(Y = y).$$

For each fixed $y$, the event $\{Y = y\}$ is the union of all outcomes for which $X$ takes some value $x$ with $g(x) = y$. Thus

$$\mathbb{P}(Y = y) = \sum_{x : g(x) = y} \mathbb{P}(X = x).$$

Substituting this into the definition and rearranging the order of summation leads to

$$\mathbb{E}[Y] = \sum_y y \sum_{x : g(x) = y} \mathbb{P}(X = x) = \sum_x g(x)\, \mathbb{P}(X = x),$$

which is the expected value rule.

A USEFUL SPECIAL CASE
If $g(x) = x^2$, then

$$\mathbb{E}[X^2] = \sum_x x^2 \, p_X(x).$$

In general,

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

for most functions $g$. So, for example, typically

$$\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2.$$

LINEARITY FOR ONE RANDOM VARIABLE
Linearity of expectation describes how expectation behaves under multiplication by constants and addition. It says that taking expectations is a linear operation.
Let $X$ be a random variable and let $a, b$ be constants. Define a new random variable

$$Y = aX + b.$$

Then

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

DERIVATION
We view $Y$ as a function of $X$, namely $Y = g(X)$ with $g(x) = ax + b$. By the expected value rule,

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x) = \sum_x (ax + b)\, p_X(x).$$

We split the sum:

$$\mathbb{E}[Y] = \sum_x ax\, p_X(x) + \sum_x b\, p_X(x) = a \sum_x x\, p_X(x) + b \sum_x p_X(x).$$

The first sum is $\mathbb{E}[X]$. The second sum is 1, since the probabilities $p_X(x)$ add up to 1. Thus

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

SPECIAL CASE: AVERAGE SALARY
If $X$ is the salary of a random person in a population, then $\mathbb{E}[X]$ is the average salary. If everyone's salary is updated to

$$Y = 2X + 100,$$

then the new average is

$$\mathbb{E}[Y] = 2\,\mathbb{E}[X] + 100.$$

Doubling all salaries and adding 100 to each one doubles the average and adds 100.

COMMENT
For linear functions $g(x) = ax + b$, we have

$$\mathbb{E}[g(X)] = g(\mathbb{E}[X]).$$

For general, nonlinear functions $g$, this identity usually does *not* hold.

### 1.4.8 Variance

Variance measures how much a random variable tends to deviate (in squared distance) from its mean. It complements the expectation by quantifying the spread of the distribution around the average.

DEFINITION
Let $X$ be a random variable with mean $\mu = \mathbb{E}[X]$. The *variance* of $X$ is defined as

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big].$$

This is the expected value of the squared distance from the mean. Since $(X - \mu)^2 \geq 0$ always, the variance is always nonnegative.

Using the expected value rule, for a discrete random variable with PMF $p_X(x)$, we can write

$$\mathrm{Var}(X) = \sum_x (x-\mu)^2 \, p_X(x),$$

where the sum is over all possible values of $X$. For each value $x$, we compute its squared distance from $\mu$ and weight it by the probability that $X$ takes that value.

Standard deviation

The *standard deviation* of $X$ is defined as

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

It has the same units as $X$ (unlike the variance, which has squared units), and is often easier to interpret as a measure of typical deviation from the mean.

Effect of adding and scaling

Let $Y = aX + b$, where $a$ and $b$ are constants. Then

$$\mathrm{Var}(Y) = a^2 \, \mathrm{Var}(X).$$

In particular,

$$\mathrm{Var}(X+b) = \mathrm{Var}(X), \qquad \mathrm{Var}(aX) = a^2 \, \mathrm{Var}(X)[][].$$

Intuitively, adding a constant $b$ just shifts the distribution right or left without changing its spread, so the variance stays the same. Multiplying by $a$ stretches or shrinks the distribution horizontally by a factor $|a|$, so the spread (variance) scales by $a^2$

Example

If $\mathrm{Var}(X) = 2$, then

$$\mathrm{Var}(3-4X) = (-4)^2 \, \mathrm{Var}(X) = 16 \cdot 2 = 32.$$

The additive constant 3 does not affect the variance, while the factor $-4$ multiplies the variance by 16.

Alternative formula

There is a useful identity that often simplifies calculations:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2[][].$$

It comes from expanding

$$\mathrm{Var}(X) = \mathbb{E}\big[(X-\mu)^2\big] = \mathbb{E}\big[X^2 - 2\mu X + \mu^2\big] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2,$$

since $\mu = \mathbb{E}[X]$. This formula allows us to compute the variance by first finding $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$, and then taking their difference.

Variance of a Bernoulli random variable

A *Bernoulli* random variable and a *discrete uniform* random variable are simple examples where the variance can be computed explicitly. We first recall the Bernoulli case, then the uniform on $\{0, \ldots, n\}$, and finally a general uniform on $\{a, \ldots, b\}$.

Let $X$ be Bernoulli with parameter $p$, so

$$\mathbb{P}(X=1) = p, \qquad \mathbb{P}(X=0) = 1-p.$$

We know that

$$\mathbb{E}[X] = p.$$

Using the alternative variance formula,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2[][].$$

For a Bernoulli random variable, $X^2 = X$, since $0^2 = 0$ and $1^2 = 1$. Hence

$$\mathbb{E}[X^2] = \mathbb{E}[X] = p,$$

and

$$\mathrm{Var}(X) = p - p^2 = p(1-p).$$

As a function of $p$, $p(1-p)$ is a parabola on $[0,1]$ with maximum at $p=\frac{1}{2}$ and zeros at $p=0$ and $p=1$. This matches the intuition that a fair coin ($p=\frac{1}{2}$) is "most random", while a deterministic coin ($p=0$ or $p=1$) has variance 0.

Variance of a uniform on $\{0,1,\dots,n\}$
Let $X$ be discrete uniform on $\{0,1,\dots,n\}$. Then

$$\mathbb{P}(X=k)=\frac{1}{n+1},\quad k=0,1,\dots,n.$$

We first compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
For the mean,

$$\mathbb{E}[X]=\sum_{k=0}^{n}k\cdot\frac{1}{n+1}=\frac{1}{n+1}\sum_{k=0}^{n}k=\frac{1}{n+1}\cdot\frac{n(n+1)}{2}=\frac{n}{2}[][].$$

For $\mathbb{E}[X^2]$,

$$\mathbb{E}[X^2]=\sum_{k=0}^{n}k^2\cdot\frac{1}{n+1}=\frac{1}{n+1}\sum_{k=0}^{n}k^2.$$

The sum of squares satisfies

$$\sum_{k=0}^{n}k^2=\frac{n(n+1)(2n+1)}{6}[][],$$

so

$$\mathbb{E}[X^2]=\frac{1}{n+1}\cdot\frac{n(n+1)(2n+1)}{6}=\frac{n(2n+1)}{6}.$$

Now apply $\mathrm{Var}(X)=\mathbb{E}[X^2]-\big(\mathbb{E}[X]\big)^2$:

$$\mathrm{Var}(X)=\frac{n(2n+1)}{6}-\left(\frac{n}{2}\right)^2=\frac{n(2n+1)}{6}-\frac{n^2}{4}.$$

Bringing to a common denominator,

$$\mathrm{Var}(X)=\frac{4n(2n+1)-3n^2}{12}=\frac{8n^2+4n-3n^2}{12}=\frac{5n^2+4n}{12}.$$

This is the variance of the uniform distribution on $\{0,1,\dots,n\}$ written via the sum-of-squares formula.
In many references, one instead writes the variance of the uniform on $\{0,1,\dots,n\}$ as

$$\mathrm{Var}(X)=\frac{n(n+2)}{12},$$

which is algebraically equivalent after simplifying the sum carefully.

Uniform on $\{a,a+1,\dots,b\}$
Now consider a general discrete uniform random variable $Y$ on $\{a,a+1,\dots,b\}$. Let

$$n=b-a.$$

We can write

$$Y=a+X,$$

where $X$ is uniform on $\{0,1,\dots,n\}$. Adding the constant $a$ does not change the variance, so

$$\mathrm{Var}(Y)=\mathrm{Var}(X)[][].$$

Thus, substituting $n=b-a$,

$$\mathrm{Var}(Y)=\frac{(b-a)\big((b-a)+2\big)}{12},$$

which expresses the variance in terms of the endpoints of the range.

Conditional PMF given an event

Let $X$ be a discrete random variable with PMF $p_X(x) = \mathbb{P}(X = x)$. Let $A$ be an event with $\mathbb{P}(A) > 0$. The *conditional PMF* of $X$ given $A$ is

$$p_{X|A}(x) = \mathbb{P}(X = x \mid A) = \frac{\mathbb{P}(X = x \cap A)}{\mathbb{P}(A)}.$$

This is an ordinary PMF on the same set of values as $X$, but computed in the conditional model where $A$ is known to have occurred. Its entries satisfy

$$\sum_x p_{X|A}(x) = 1.$$

Conditional expectation given an event

The *conditional expectation* of $X$ given $A$ is the mean of $X$ in the conditional model. It is defined by

$$\mathbb{E}[X \mid A] = \sum_x x\,\mathbb{P}(X = x \mid A) = \sum_x x\,p_{X|A}(x).$$

This is the same formula as for an ordinary expectation, but with conditional probabilities throughout.

Example: Conditioning a uniform

Let $X$ be uniform on $\{1, 2, 3, 4\}$. Then

$$\mathbb{P}(X = x) = \frac{1}{4}, \quad x = 1, 2, 3, 4.$$

By symmetry,

$$\mathbb{E}[X] = \frac{1+4}{2} = \frac{5}{2},$$

and using the uniform-variance formula on $\{1, 2, 3, 4\}$,

$$\mathrm{Var}(X) = \frac{1}{12}(4-1)\big((4-1)+2\big) = \frac{1}{12} \cdot 3 \cdot 5 = \frac{5}{4}.$$

Now condition on the event

$$A = \{X \in \{2, 3, 4\}\}.$$

In the conditional model, outcome 1 is impossible, so

$$\mathbb{P}(X = 1 \mid A) = 0.$$

The remaining values $2, 3, 4$ had equal probabilities in the original model, so they remain equally likely given $A$. Since their probabilities must sum to 1,

$$\mathbb{P}(X = 2 \mid A) = \mathbb{P}(X = 3 \mid A) = \mathbb{P}(X = 4 \mid A) = \frac{1}{3}.$$

Thus the conditional PMF is uniform on $\{2, 3, 4\}$.
The conditional expectation is the mean of this conditional uniform:

$$\mathbb{E}[X \mid A] = \frac{2+4}{2} = 3.$$

For the conditional variance $\mathrm{Var}(X \mid A)$, we can compute directly. The mean is 3, so

$$\mathrm{Var}(X \mid A) = \mathbb{E}\big[(X-3)^2 \mid A\big] = (2-3)^2 \cdot \frac{1}{3} + (3-3)^2 \cdot \frac{1}{3} + (4-3)^2 \cdot \frac{1}{3} = 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}.$$

Notice that

$$\mathrm{Var}(X \mid A) = \frac{2}{3} < \mathrm{Var}(X) = \frac{5}{4}.$$

In the conditional model we have more information (we know that $X$ is not 1), so there is less uncertainty, and the variance decreases.

### 1.4.9 Total expectation theorem

The Total Expectation Theorem provides a way to compute an expectation by splitting the probability space into scenarios and averaging the conditional expectations over those scenarios. It is the expectation analogue of the Total Probability Theorem.
Let $X$ be a random variable and let $A_1, A_2, A_3$ be events that form a partition of the sample space, that is:

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \quad A_1 \cup A_2 \cup A_3 = \Omega, \quad \mathbb{P}(A_i) > 0.$$

Then the *Total Expectation Theorem* says that

$$\mathbb{E}[X] = \mathbb{P}(A_1)\,\mathbb{E}[X \mid A_1] + \mathbb{P}(A_2)\,\mathbb{E}[X \mid A_2] + \mathbb{P}(A_3)\,\mathbb{E}[X \mid A_3].$$

More generally, for a finite or countable partition $\{A_i\}$,

$$\mathbb{E}[X] = \sum_i \mathbb{P}(A_i)\,\mathbb{E}[X \mid A_i].$$

From total probability to total expectation
For a fixed value $x$, the Total Probability Theorem applied to the event $\{X = x\}$ gives

$$\mathbb{P}(X = x) = \sum_i \mathbb{P}(X = x \mid A_i)\,\mathbb{P}(A_i).$$

This holds for every possible value $x$. Multiply both sides by $x$ and sum over all $x$:

$$\sum_x x\,\mathbb{P}(X = x) = \sum_x x \sum_i \mathbb{P}(X = x \mid A_i)\,\mathbb{P}(A_i).$$

The left-hand side is $\mathbb{E}[X]$. On the right-hand side, we can interchange the order of summation:

$$\mathbb{E}[X] = \sum_i \mathbb{P}(A_i) \sum_x x\,\mathbb{P}(X = x \mid A_i).$$

But for each $i$,

$$\sum_x x\,\mathbb{P}(X = x \mid A_i) = \mathbb{E}[X \mid A_i].$$

Substituting this into the previous expression gives the Total Expectation Theorem.

Interpretation
We can think of $\{A_1, A_2, A_3\}$ as different scenarios. Under each scenario $A_i$, the random variable $X$ has a conditional distribution and a conditional expectation $\mathbb{E}[X \mid A_i]$. The theorem says that the overall expectation $\mathbb{E}[X]$ is the weighted average of these conditional expectations, where the weights are the scenario probabilities $\mathbb{P}(A_i)$.

Example: Splitting a PMF into two scenarios
Consider a discrete random variable $X$ with PMF having six mass points: three "small" values and three "large" values. Define

$$A_1 = \{\text{“small” values of } X\}, \quad A_2 = \{\text{“large” values of } X\},$$

so that $\{A_1, A_2\}$ is a partition of the sample space.
Suppose the probabilities of the three small values are all $1/9$, so

$$\mathbb{P}(A_1) = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3},$$

and the probabilities of the three large values are all $2/9$, so

$$\mathbb{P}(A_2) = \frac{2}{9} + \frac{2}{9} + \frac{2}{9} = \frac{2}{3}.$$

In the conditional model given $A_1$, the three small values are equally likely, so $X$ is uniform on $\{0, 1, 2\}$ and

$$\mathbb{E}[X \mid A_1] = 1$$

(the midpoint). Similarly, given $A_2$, $X$ is uniform on $\{6, 7, 8\}$, so

$$\mathbb{E}[X \mid A_2] = 7.$$

By the Total Expectation Theorem,

$$\mathbb{E}[X] = \mathbb{P}(A_1)\,\mathbb{E}[X \mid A_1] + \mathbb{P}(A_2)\,\mathbb{E}[X \mid A_2] = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 7 = \frac{1}{3} + \frac{14}{3} = 5.$$

Thus, instead of computing $\mathbb{E}[X]$ directly from all six probabilities at once, we divided the problem into two simpler conditional expectations and combined them with appropriate weights.

### 1.4.10 Geometric PMF and memorylessness

Let $X$ be the number of tosses until the first heads. Then $X$ takes values $1, 2, 3, \ldots$, and

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots$$

The key property is *memorylessness*. Suppose we know that the first toss was tails, so $X > 1$. From that point on, we are again waiting for the first heads in a sequence of independent tosses with success probability $p$. Thus, the number of *additional* tosses until heads, namely $X - 1$, has the same geometric distribution as $X$, when we condition on $X > 1$:

$$X - 1 \mid (X > 1) \sim \text{Geometric}(p).$$

More generally, for any integer $n \geq 1$,

$$X - n \mid (X > n) \sim \text{Geometric}(p).$$

Concretely, for example

$$\mathbb{P}(X - 1 = 3 \mid X > 1) = \mathbb{P}(\text{T,T,T,H on tosses } \not\models, \not\models, \not\models \mid \text{T on toss } \not\models).$$

Because tosses are independent, conditioning on tails on the first toss does not change the probabilities of future sequences, so this equals

$$\mathbb{P}(\text{T,T,T,H on tosses } 2, 3, 4) = (1-p)^3 p,$$

which is exactly $\mathbb{P}(X = 4)$, the geometric PMF at $k = 4$. This illustrates that the conditional distribution of the remaining waiting time is geometric with parameter $p$.

Using memorylessness to find the expectation

Directly computing

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

is possible but involves an infinite series. Instead, we use memorylessness and the total expectation idea.

Write

$$X = 1 + (X - 1).$$

Taking expectations,

$$\mathbb{E}[X] = 1 + \mathbb{E}[X - 1].$$

Now split $\mathbb{E}[X - 1]$ into two scenarios according to the outcome of the first toss:

$$\mathbb{E}[X - 1] = \mathbb{E}[X - 1 \mid X = 1]\,\mathbb{P}(X = 1) + \mathbb{E}[X - 1 \mid X > 1]\,\mathbb{P}(X > 1).$$

If $X = 1$, then $X - 1 = 0$, so

$$\mathbb{E}[X - 1 \mid X = 1] = 0, \quad \mathbb{P}(X = 1) = p.$$

Thus the first term is zero. The second term is

$$\mathbb{E}[X - 1 \mid X > 1]\,\mathbb{P}(X > 1) = \mathbb{E}[X - 1 \mid X > 1](1 - p).$$

By memorylessness,

$$X - 1 \mid (X > 1) \sim \text{Geometric}(p),$$

so

$$\mathbb{E}[X - 1 \mid X > 1] = \mathbb{E}[X].$$

Therefore

$$\mathbb{E}[X - 1] = (1 - p)\,\mathbb{E}[X].$$

Substitute back into $\mathbb{E}[X] = 1 + \mathbb{E}[X - 1]$:

$$\mathbb{E}[X] = 1 + (1 - p)\,\mathbb{E}[X].$$

Rearrange:

$$\begin{aligned} \mathbb{E}[X] - (1-p)\,\mathbb{E}[X] &= 1, \\ p\,\mathbb{E}[X] &= 1, \\ \mathbb{E}[X] &= \frac{1}{p}. \end{aligned}$$

This makes intuitive sense: if the success probability $p$ is small, we expect to wait longer on average until the first heads.

### 1.4.11 Joint PMFs and the expected value rule

When dealing with several discrete random variables together, their joint behaviour is described by a *joint* PMF. From the joint PMF we can recover the individual (marginal) PMFs and compute expectations of functions of several variables.

Joint and marginal PMFs
Let $X$ and $Y$ be discrete random variables. Their *joint* PMF is

$$p_{X,Y}(x,y) = \mathbb{P}(X = x,\, Y = y).$$

This is a function of two arguments: for each pair $(x, y)$ it gives the probability that $X$ takes value $x$ and $Y$ takes value $y$ simultaneously. It satisfies

$$\sum_x \sum_y p_{X,Y}(x,y) = 1.$$

The *marginal* PMFs of $X$ and $Y$ are obtained by summing out the other variable:

$$p_X(x) = \mathbb{P}(X = x) = \sum_y p_{X,Y}(x,y),$$

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_x p_{X,Y}(x,y).$$

For example, if the joint PMF is given in a table, then $p_X(x)$ is the sum of the entries in the column corresponding to $x$, and $p_Y(y)$ is the sum of the entries in the row corresponding to $y$.

Example: reading a joint PMF table
Suppose the joint PMF of $X$ and $Y$, for $x \in \{0, 1, 2\}$ and $y \in \{1, 2, 3\}$, is given in a $3 \times 3$ table of probabilities. Then: - The probability that $X = 1$ and $Y = 3$ is the entry in the row $y = 3$, column $x = 1$. - The marginal $\mathbb{P}(X = 2)$ is the sum of the entries in the column $x = 2$. - The marginal $\mathbb{P}(Y = 2)$ is the sum of the entries in the row $y = 2$.
If we want $\mathbb{P}(X = Y)$, we add the probabilities of all diagonal pairs $(0, 0)$, $(1, 1)$, $(2, 2)$ (or whichever pairs satisfy $x = y$) in the joint table.

Joint PMF for more variables
For three discrete random variables $X, Y, Z$, the joint PMF is

$$p_{X,Y,Z}(x,y,z) = \mathbb{P}(X = x, Y = y, Z = z),$$

and

$$\sum_x \sum_y \sum_z p_{X,Y,Z}(x,y,z) = 1.$$

From this we can obtain marginals by summing out the variables we are not interested in, for example

$$p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x,y,z),$$

$$p_{X,Y}(x,y) = \sum_z p_{X,Y,Z}(x,y,z).$$

Functions of two random variables and their PMF
Given two random variables $X$ and $Y$, a function

$$Z = g(X, Y)$$

is itself a random variable. To find the PMF of $Z$, we use the joint PMF of $(X, Y)$:

$$p_Z(z) = \mathbb{P}(Z = z) = \mathbb{P}\big(g(X,Y) = z\big) = \sum_{(x,y):\, g(x,y)=z} p_{X,Y}(x,y).$$

That is, we sum the probabilities of all pairs $(x, y)$ for which $g(x, y)$ equals the value $z$.

Expected value rule for functions of two variables
For a function $g(X, Y)$, there is an expected value rule analogous to the one-variable case:

$$\mathbb{E}[g(X,Y)] = \sum_x \sum_y g(x,y)\, p_{X,Y}(x,y).$$

Interpretation: for each possible pair $(x, y)$, the random variables take this pair with probability $p_{X,Y}(x, y)$, and then the function has value $g(x, y)$; the product $g(x, y)\, p_{X,Y}(x, y)$ is the contribution of that pair to the overall average, and we sum over all pairs.

### 1.4.12 Linearity of expectations

Linearity of expectation extends to sums of multiple random variables and lets us compute the binomial mean very easily.

Linearity for several random variables
For any discrete random variables $X_1, \ldots, X_n$,

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

For example, for three variables,

$$\mathbb{E}[2X + 3Y - Z] = 2\,\mathbb{E}[X] + 3\,\mathbb{E}[Y] - \mathbb{E}[Z].$$

Derivation for two variables
For two variables $X$ and $Y$,

$$\mathbb{E}[X+Y] = \sum_x \sum_y (x+y)\, p_{X,Y}(x,y).$$

Split the sum:

$$\mathbb{E}[X+Y] = \sum_x \sum_y x\, p_{X,Y}(x,y) + \sum_x \sum_y y\, p_{X,Y}(x,y).$$

In the first term, for fixed $x$,

$$\sum_y x\, p_{X,Y}(x,y) = x \sum_y p_{X,Y}(x,y) = x\, p_X(x),$$

so

$$\sum_x \sum_y x\, p_{X,Y}(x,y) = \sum_x x\, p_X(x) = \mathbb{E}[X].$$

A symmetric argument shows that the second term equals $\mathbb{E}[Y]$. Thus

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

The general $n$-variable case follows similarly.

Mean of a binomial via indicators
Let $X \sim \text{Binomial}(n,p)$. Interpret $X$ as the number of successes in $n$ independent trials, each with success probability $p$.
Define indicator variables

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success,} \\ 0 & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, n.$$

Each $X_i$ is Bernoulli($p$), so

$$\mathbb{P}(X_i = 1) = p, \quad \mathbb{P}(X_i = 0) = 1 - p, \quad \mathbb{E}[X_i] = p.$$

The total number of successes is the sum of these indicators:

$$X = X_1 + X_2 + \cdots + X_n.$$

By linearity of expectation,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] = p + \cdots + p = np.$$

So a binomial random variable with parameters $n$ and $p$ has mean

$$\mathbb{E}[X] = np,$$

which matches the intuition that in $n$ trials with success probability $p$, we expect $np$ successes on average.

### 1.4.13 Conditional PMFs

When we condition on the value of another random variable, we obtain a family of conditional PMFs for $X$, one for each value of $Y$.

Conditional PMF given another variable
Let $X$ and $Y$ be discrete random variables with joint PMF $p_{X,Y}(x, y)$ and marginal $p_Y(y) = \mathbb{P}(Y = y)$. For any $y$ with $p_Y(y) > 0$, the *conditional PMF* of $X$ given $Y = y$ is

$$p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

If we fix $y$, this is a PMF in the variable $x$, and its values satisfy

$$\sum_x p_{X|Y}(x \mid y) = 1.$$

Changing $y$ changes the conditional PMF, so $\{p_{X|Y}(\cdot \mid y)\}$ is a family of PMFs indexed by $y$.

Example from a joint table
Suppose the joint PMF of $(X, Y)$ is given by a table, and we condition on $Y = 2$. First compute

$$p_Y(2) = \sum_x p_{X,Y}(x, 2),$$

by adding the entries in the row $y = 2$. Then, for each $x$,

$$p_{X|Y}(x \mid 2) = \frac{p_{X,Y}(x, 2)}{p_Y(2)}.$$

For instance, if the row $y = 2$ has entries proportional to $0, 1, 3, 1$, their sum is 5, so the conditional PMF values are

$$0, \ \frac{1}{5}, \ \frac{3}{5}, \ \frac{1}{5}.$$

Geometrically, the conditional PMF for $X \mid Y = 2$ is the corresponding row of the joint PMF, rescaled so that its entries add to 1.

Multiplication rule in PMF form
From the definition

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)},$$

we can rewrite

$$p_{X,Y}(x, y) = p_Y(y)\, p_{X|Y}(x \mid y),$$

which is the multiplication rule in PMF notation. Symmetrically,

$$p_{X,Y}(x, y) = p_X(x)\, p_{Y|X}(y \mid x).$$

Thus we can specify a joint distribution either directly via $p_{X,Y}$ or indirectly by giving a marginal (say $p_Y$) and a conditional PMF (say $p_{X|Y}$).

More than two variables
For three random variables $X, Y, Z$, the conditional PMF of $X$ given $Y = y$ and $Z = z$ is

$$p_{X|Y,Z}(x \mid y, z) = \mathbb{P}(X = x \mid Y = y, Z = z) = \frac{p_{X,Y,Z}(x, y, z)}{p_{Y,Z}(y, z)},$$

where

$$p_{Y,Z}(y, z) = \sum_x p_{X,Y,Z}(x, y, z).$$

We also have a multiplication rule for three variables. Let

$$A = \{X = x\}, \quad B = \{Y = y\}, \quad C = \{Z = z\}.$$

Then

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\, \mathbb{P}(B \mid A)\, \mathbb{P}(C \mid A \cap B),$$

which in PMF notation becomes

$$p_{X,Y,Z}(x, y, z) = p_X(x)\, p_{Y|X}(y \mid x)\, p_{Z|X,Y}(z \mid x, y).$$

CONDITIONAL EXPECTATION
A conditional expectation is just the expectation of $X$ computed in the conditional model "given $Y = y$".
Let $X$ and $Y$ be discrete random variables, and suppose $\mathbb{P}(Y = y) > 0$. The conditional expectation of $X$ given $Y = y$ is

$$\mathbb{E}[X \mid Y = y] = \sum_x x\, \mathbb{P}(X = x \mid Y = y).$$

Here we use the conditional PMF $p_{X|Y}(x \mid y)$ instead of the original PMF of $X$.

EXPECTED VALUE RULE (CONDITIONAL FORM)
If $g$ is a function and $Z = g(X)$, then in the conditional model "given $Y = y$" we have

$$\mathbb{E}[g(X) \mid Y = y] = \sum_x g(x)\, \mathbb{P}(X = x \mid Y = y).$$

This is the same expected value rule as before, but with conditional probabilities.

TOTAL EXPECTATION THEOREM (VERSION WITH $Y$)
Let $X$ and $Y$ be discrete random variables, and suppose $Y$ takes values in some countable set $\mathcal{Y}$. Then the total expectation theorem says

$$\mathbb{E}[X] = \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y)\, \mathbb{E}[X \mid Y = y].$$

So we can view the possible values of $Y$ as "scenarios": for each scenario $Y = y$, we compute the conditional mean $\mathbb{E}[X \mid Y = y]$, and then average these means using $\mathbb{P}(Y = y)$ as weights.

REMARK ON INFINITE RANGES
If $Y$ can take infinitely many discrete values, the sum

$$\sum_y \mathbb{P}(Y = y)\, \mathbb{E}[X \mid Y = y]$$

becomes an infinite series. Under the usual assumption that $\mathbb{E}[X]$ is well defined (finite), this identity still holds, and we can safely use the total expectation theorem even when $Y$ ranges over an infinite discrete set.

### 1.4.14 Independence of random variables

Independence for random variables extends the idea of independence for events to the level of entire distributions.
Let $X$ be a random variable and $A$ an event. We say that $X$ and $A$ are *independent* if, for every real number $x$,

$$\mathbb{P}(X = x \text{ and } A) = \mathbb{P}(X = x)\, \mathbb{P}(A).$$

Equivalently, whenever $\mathbb{P}(A) > 0$,

$$\mathbb{P}(X = x \mid A) = \mathbb{P}(X = x) \quad \text{for all } x.$$

Intuition: knowing that $A$ occurred does not change the distribution of $X$.
Symmetrically, for any $x$ with $\mathbb{P}(X = x) > 0$,

$$\mathbb{P}(A \mid X = x) = \mathbb{P}(A),$$

so learning the value of $X$ does not change your beliefs about $A$.

INDEPENDENCE OF TWO RANDOM VARIABLES
Let $X$ and $Y$ be discrete random variables. We say that $X$ and $Y$ are *independent* if, for all real $x$ and $y$,

$$\mathbb{P}(X = x,\, Y = y) = \mathbb{P}(X = x)\, \mathbb{P}(Y = y).$$

In terms of PMFs,

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y) \quad \text{for all } x, y.$$

So the joint PMF factorizes as the product of the marginals.

Equivalently, for any $y$ with $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \mathbb{P}(X = x) \quad \text{for all } x,$$

that is,

$$p_{X|Y}(x \mid y) = p_X(x).$$

So knowing $Y = y$ does not change the distribution of $X$. Symmetrically,

$$p_{Y|X}(y \mid x) = p_Y(y),$$

so knowing $X = x$ does not change the distribution of $Y$.

Independence of multiple random variables

For three random variables $X, Y, Z$, we say they are *mutually independent* if, for all $x, y, z$,

$$p_{X,Y,Z}(x, y, z) = p_X(x)\, p_Y(y)\, p_Z(z).$$

More generally, a finite collection $X_1, \ldots, X_n$ is mutually independent if the joint PMF factors as the product of all the marginals:

$$p_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} p_{X_i}(x_i)$$

for all choices of $(x_1, \ldots, x_n)$.

Intuitively, this means that information about some of the variables does not change the distribution of any of the others: all conditional PMFs coincide with the corresponding unconditional PMFs. In practical terms, independence models situations where each random variable is generated by its own separate probabilistic experiment, with no interaction and no shared source of randomness.

Independence gives one more important rule for expectations: products of independent random variables have factorizing expectations.

Product of independent random variables

Let $X$ and $Y$ be independent discrete random variables. Then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

*Derivation.* Using the expected value rule with the joint PMF:

$$\mathbb{E}[XY] = \sum_x \sum_y xy\, p_{X,Y}(x, y).$$

Independence means

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y),$$

so

$$\mathbb{E}[XY] = \sum_x \sum_y xy\, p_X(x)\, p_Y(y).$$

For fixed $x$, the factor $x\, p_X(x)$ does not depend on $y$, hence

$$\mathbb{E}[XY] = \sum_x x\, p_X(x) \sum_y y\, p_Y(y) = \Big(\sum_x x\, p_X(x)\Big)\Big(\sum_y y\, p_Y(y)\Big) = \mathbb{E}[X]\,\mathbb{E}[Y].$$

Products of functions of independent variables

If $X$ and $Y$ are independent, and $g$ and $h$ are functions, then the random variables $g(X)$ and $h(Y)$ are also independent. Therefore,

$$\mathbb{E}[g(X)\, h(Y)] = \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)].$$

We can view this either as a consequence of independence of $g(X)$ and $h(Y)$, or prove it directly by repeating the previous derivation with $g(x)$ in place of $x$ and $h(y)$ in place of $y$.

We want to understand how variance behaves for sums, and then use this to find the variance of a binomial random variable.

Variance of a sum under independence

In general, $\mathrm{Var}(X+Y)$ is not equal to $\mathrm{Var}(X)+\mathrm{Var}(Y)$. However, if $X$ and $Y$ are independent, we do have

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Assume first that $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, just to simplify the algebra. Then

$$\mathrm{Var}(X+Y) = \mathbb{E}\big[(X+Y)^2\big] = \mathbb{E}\big[X^2 + 2XY + Y^2\big].$$

Using linearity of expectation,

$$\mathrm{Var}(X+Y) = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2].$$

Since the means are zero, $\mathbb{E}[X^2] = \mathrm{Var}(X)$ and $\mathbb{E}[Y^2] = \mathrm{Var}(Y)$.
If $X$ and $Y$ are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] = 0 \cdot 0 = 0.$$

So

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Without the zero-mean assumption, one can expand $\mathrm{Var}(X+Y)$ using the general formula $\mathrm{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2$ and obtain the same result; the cross terms again cancel because $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ under independence.

Examples showing dependence matters

If $Y = X$, then $X + Y = 2X$, so

$$\mathrm{Var}(X+Y) = \mathrm{Var}(2X) = 4\ \mathrm{Var}(X),$$

which is generally not equal to $\mathrm{Var}(X) + \mathrm{Var}(Y) = 2\ \mathrm{Var}(X)$. Here $X$ and $Y$ are completely dependent.
If $Y = -X$, then $X + Y = 0$ always, so

$$\mathrm{Var}(X+Y) = \mathrm{Var}(0) = 0,$$

again very different from $\mathrm{Var}(X) + \mathrm{Var}(Y)$.
These examples show that knowing just the individual variances is not enough; the dependence structure between $X$ and $Y$ is crucial.

Example with independence and scaling

If $X$ and $Y$ are independent, then $X$ is also independent of any function of $Y$, such as $-3Y$. Thus

$$\mathrm{Var}(X - 3Y) = \mathrm{Var}(X) + \mathrm{Var}(-3Y) = \mathrm{Var}(X) + 9\ \mathrm{Var}(Y),$$

using the rule $\mathrm{Var}(aZ) = a^2\,\mathrm{Var}(Z)$.

Variance of a binomial random variable

Let $X$ be binomial with parameters $n$ and $p$. Interpret $X$ as the number of successes in $n$ independent trials, each with success probability $p$.
Define indicator random variables

$$X_i = \begin{cases} 1 & \text{if the } i\text{-th trial is a success,} \\ 0 & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, n.$$

Each $X_i$ is Bernoulli($p$), and the total number of successes is

$$X = X_1 + X_2 + \cdots + X_n.$$

The independence of the trials means the indicators $X_1, \ldots, X_n$ are independent.
For each $X_i$,

$$\mathbb{E}[X_i] = p, \qquad \mathrm{Var}(X_i) = p(1-p).$$

Using independence and the variance-of-sum rule,

$$\mathrm{Var}(X) = \mathrm{Var}\big(X_1 + \cdots + X_n\big) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = \sum_{i=1}^{n} p(1-p) = n\,p(1-p).$$

Thus a binomial random variable with parameters $n$ and $p$ has variance

$$\mathrm{Var}(X) = np(1-p).$$

### 1.4.15 The hat problem

There are $n$ people and $n$ hats, all permutations equally likely. Let $X$ be the number of people who get their own hat, and $X_i$ the indicator for person $i$.

Definition of the indicators
For each $i = 1, \ldots, n$,

$$X_i = \begin{cases} 1 & \text{if person } i \text{ gets their own hat,} \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$X = X_1 + X_2 + \cdots + X_n.$$

Expectation of each $X_i$
In a random permutation, person $i$ is equally likely to receive any of the $n$ hats. Only one of these is their own hat, so

$$\mathbb{P}(X_i = 1) = \frac{1}{n}.$$

Thus

$$\mathbb{E}[X_i] = 1 \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right) = \frac{1}{n}.$$

Expectation of $X$
By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{1}{n} = 1.$$

So in the hat problem, the expected number of people who get their own hat back is always

$$\mathbb{E}[X] = 1,$$

independent of $n$.

## 1.5 Continuous random variables

A continuous random variable takes values over a continuous range, such as the real line or an interval on the real line. In the discrete case, one unit of probability is distributed over the possible values as point masses, and the PMF tells us how much probability sits on top of each value. In the continuous case, one unit of probability is still assigned to the possible values, but now it is spread continuously over the real line.
Some parts of the real line receive more probability mass per unit length, others receive less. The *probability density function* (PDF), denoted $f_X(x)$, describes how densely probability is distributed along the real line. The letter $f$ indicates we are dealing with a PDF, and the subscript indicates which random variable we are considering.

Interpretation of the PDF
Think of probability as snowfall. One pound of snow has fallen on the real line, and the PDF tells us the height of the snow accumulated over each point. The weight of snow sitting on top of an interval $[a, b]$ is found by calculating the area under the PDF curve over that interval. Mathematically, area under the curve is an integral, so

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

This is the probability that $X$ takes values in the interval $[a, b]$.

Properties of a PDF
By analogy with the discrete case, a PDF must satisfy two conditions:

$$f_X(x) \geq 0 \quad \text{for all } x,$$

$$\int_{-\infty}^{+\infty} f_X(x)\, dx = 1.$$

The first condition ensures we do not obtain negative probabilities. The second condition ensures that the total probability over the entire real line equals one, since $X$ is certain to lie somewhere on the real line. These two conditions are all that we need for a legitimate PDF.

Formal definition of a continuous random variable
A continuous random variable is a random variable whose probabilities can be described by a PDF. That is, there exists a function $f_X$ such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\, dx$$

for any interval $[a, b]$. It is not enough for a random variable to take values on a continuous set. A continuous random variable must also be describable by a PDF according to this formula.

Probabilities of more complicated sets
Once we have the probabilities of intervals, we can use additivity to calculate the probabilities of more complicated sets. For example, suppose we want the probability that $X$ lies between 1 and 3 or between 4 and 5. This is the probability of a region consisting of two disjoint intervals. By additivity,

$$\mathbb{P}(1 \leq X \leq 3 \text{ or } 4 \leq X \leq 5) = \mathbb{P}(1 \leq X \leq 3) + \mathbb{P}(4 \leq X \leq 5).$$

Each of the probabilities on the right is calculated using the PDF:

$$= \int_1^3 f_X(x)\, dx + \int_4^5 f_X(x)\, dx.$$

The role of the sample space
There is still an underlying sample space lurking in the background. Different outcomes in the sample space result in different numerical values for the random variable $X$. When we talk about the probability that $X$ takes values in an interval $[a, b]$, we mean the probability of those outcomes for which the resulting value of $X$ lies in that interval.
However, once we have a PDF in our hands, we can completely forget about the sample space and carry out any calculations just by working with the PDF. As we move on in this course, the sample space will appear less and less, and we will work directly with PDFs or PMFs.

Probabilities of small intervals
Let us look at an interval that starts at some point $a$ and goes up to $a+\delta$, where $\delta$ is a small positive number. The probability that $X$ falls in this interval is the area under the PDF over the interval $[a, a+\delta]$. As long as $f_X$ does not change too much over this little interval, which will be the case if $f_X$ is continuous, we can approximate the area by the area of a rectangle with constant height. The area of this rectangle is

$$\mathbb{P}(a \leq X \leq a+\delta) \approx f_X(a) \cdot \delta.$$

This gives us an interpretation of PDFs in terms of probabilities of small intervals.
If we rearrange this approximation, we see that

$$f_X(a) \approx \frac{\mathbb{P}(a \leq X \leq a+\delta)}{\delta}.$$

So the value of the PDF can be interpreted as *probability per unit length.* PDFs are not probabilities, they are densities, and their units are probability per unit length.

Probability of a single point
If the probability per unit length is finite and the length $\delta$ is sent to zero, we obtain zero probability. More formally, if we let $b = a$ in the integral

$$\mathbb{P}(a \leq X \leq a) = \int_a^a f_X(x)\, dx,$$

we get an integral over a zero-length interval, and that integral is zero. Therefore,

$$\mathbb{P}(X = a) = 0$$

for any particular point $a$.
So for a continuous random variable, any particular point has zero probability. Yet, collectively, the infinitely many points in an interval together have positive probability. This is not a puzzle, it is exactly what happens with the ordinary notion of length. Single points have zero length, yet by putting together lots of points we can create a set with positive length.

Open and closed intervals
A final consequence of the fact that individual points have zero probability is that it does not matter whether we include the endpoints of an interval or not. Using additivity,

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(X = a) + \mathbb{P}(a < X < b) + \mathbb{P}(X = b).$$

Since $\mathbb{P}(X = a) = 0$ and $\mathbb{P}(X = b) = 0$, we conclude that

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X < b).$$

When calculating probabilities of intervals, it does not matter whether we include the endpoints or not.

### 1.5.1 Uniform and piecewise constant PDFs

The uniform random variable is a basic example of a continuous random variable. In the discrete uniform case, the possible values are integers between $a$ and $b$, all equally likely. In the continuous case, any real number between $a$ and $b$ is possible, and the PDF has the same height at all points in this interval.
Intuitively, a uniform random variable models a situation where we know that the value lies between $a$ and $b$, but we have no reason to believe that any location inside the interval is more likely than another. In this sense, it represents a situation of complete ignorance.

PDF of a uniform random variable
Let $X$ be uniform on $[a, b]$. Its PDF is constant on $[a, b]$ and zero elsewhere:

$$f_X(x) = \begin{cases} c, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

The total probability must be 1, so the area under the PDF is

$$1 = \int_a^b c\, dx = c(b-a),$$

which gives $c = \frac{1}{b-a}$. Thus

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Any two intervals of the same length inside $[a, b]$ have the same probability, because their probabilities are the base (length) times the same constant height.

Piecewise constant PDFs
A more general kind of PDF is piecewise constant. In this case, the PDF is constant on each of several subintervals, possibly with different heights on different intervals. This shows that PDFs need not be continuous; they may have jumps and discontinuities.
For such a PDF to be valid, the total area under the curve must be 1. This total area is the sum of the areas of the rectangles corresponding to each constant piece.

Computing probabilities
With a piecewise constant PDF, probabilities are obtained by adding areas of rectangles. If a certain event corresponds to an interval that intersects several constant pieces, its probability is the sum of the areas over each part of that interval. For example, if over two subintervals the heights are $h_1$ and $h_2$ and the lengths are $\ell_1$ and $\ell_2$, then the probability of their union is

$$h_1\ell_1 + h_2\ell_2.$$

### 1.5.2 Means and variances

We now develop continuous counterparts of the concepts of expectation and variance. In the discrete case, expectation is a weighted average of the possible values of a random variable, weighted by their probabilities. In the continuous case, we use the same idea, but the sum is replaced by an integral and the weights come from the PDF.

Expectation of a continuous random variable
Let $X$ be a continuous random variable with PDF $f_X(x)$. Its expectation (or mean) is defined by

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\, dx.$$

This is a weighted average of the possible values of $X$, where points with higher density $f_X(x)$ contribute more. More generally, for any function $g$,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x)\, dx.$$

This is the continuous analogue of the discrete expected value rule.
To ensure that the expectation is well defined, we usually assume that

$$\int_{-\infty}^{+\infty} |x| f_X(x)\, dx < \infty.$$

Under this assumption, $\mathbb{E}[X]$ exists.
The intuition of expectation remains the same as in the discrete case. It represents the average value we would see if we repeated the experiment many times independently. It can also be viewed as the center of gravity of the distribution. If the PDF is symmetric around a point $c$, then the expectation is $c$.

Basic properties of expectation
Expectations of continuous random variables satisfy the same properties as in the discrete case. If $X \ge 0$ always, then $\mathbb{E}[X] \ge 0$. If $X$ always lies in an interval $[a, b]$, then $\mathbb{E}[X]$ also lies in $[a, b]$.
A key property is linearity of expectation. For constants $a$ and $b$,

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

Linearity holds regardless of whether $X$ is discrete or continuous.

Variance and standard deviation
The variance of a continuous random variable is defined exactly as in the discrete case:

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big],$$

where $\mu = \mathbb{E}[X]$. Using the expected value rule, this becomes

$$\mathrm{Var}(X) = \int_{-\infty}^{+\infty} (x-\mu)^2 f_X(x)\, dx.$$

The standard deviation is

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

Variance measures how spread out the values of $X$ are around the mean. A small variance means that most of the probability mass is concentrated close to $\mu$. A large variance means that values far from $\mu$ occur more often.

Alternative formula for the variance
It is often convenient to use the formula

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

Here

$$\mathbb{E}[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x)\, dx.$$

This alternative expression is derived exactly as in the discrete case and is usually easier to use in calculations.

Effect of linear transformations
If we transform a random variable $X$ by a linear function $aX + b$, the variance behaves as follows:

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X).$$

Adding a constant $b$ does not change the variance. Multiplying by a constant $a$ scales the variance by $a^2$.

### 1.5.3 Mean and variance of the uniform

We now illustrate mean and variance calculations using the continuous uniform random variable. This is the continuous analogue of the discrete uniform, for which we already know formulas for the mean and variance.

Mean of the uniform
Let $X$ be uniform on $[a, b]$, with density

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The mean is

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\, dx.$$

Since $f_X(x) = 0$ outside $[a, b]$, this reduces to

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a} \int_a^b x\, dx.$$

We have

$$\int_a^b x\, dx = \frac{b^2}{2} - \frac{a^2}{2},$$

so

$$\mathbb{E}[X] = \frac{1}{b-a}\left(\frac{b^2}{2} - \frac{a^2}{2}\right) = \frac{a+b}{2}.$$

This is the same as in the discrete case and also matches the midpoint of the interval $[a, b]$, reflecting the symmetry of the PDF around $\frac{a+b}{2}$.

Computing $\mathbb{E}[X^2]$
To use the alternative variance formula, we first compute $\mathbb{E}[X^2]$. By the expected value rule,

$$\mathbb{E}[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x)\, dx = \int_a^b x^2 \cdot \frac{1}{b-a}\, dx.$$

Thus

$$\mathbb{E}[X^2] = \frac{1}{b-a}\int_a^b x^2\,dx.$$

Since

$$\int_a^b x^2\,dx = \frac{b^3}{3} - \frac{a^3}{3},$$

we obtain

$$\mathbb{E}[X^2] = \frac{1}{b-a}\left(\frac{b^3}{3} - \frac{a^3}{3}\right).$$

The cubic terms appear because the antiderivative of $x^2$ is $\frac{x^3}{3}$.

Variance and standard deviation

Using the alternative formula

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2,$$

we substitute

$$\mathbb{E}[X^2] = \frac{1}{b-a}\left(\frac{b^3}{3} - \frac{a^3}{3}\right), \quad \mathbb{E}[X] = \frac{a+b}{2}.$$

After some algebra, this simplifies to

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

The standard deviation is then

$$\sigma = \sqrt{\mathrm{Var}(X)} = \frac{b-a}{\sqrt{12}}.$$

A useful observation is that the standard deviation is proportional to the width $b-a$ of the interval. The wider the uniform distribution, the larger the standard deviation, which matches the intuition that standard deviation measures the spread or width of the distribution. The variance grows with the square of the length $b-a$, reflecting that doubling the width multiplies the variance by four.

### 1.5.4 Exponential random variables

We now introduce a new continuous random variable, the exponential random variable. It is determined by a single positive parameter $\lambda > 0$, and its PDF is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Since the PDF is zero for negative $x$, the exponential random variable is always non-negative.
For $x \geq 0$, the PDF starts at value $\lambda$ when $x = 0$ and then decays exponentially. When $\lambda$ is small, the initial height is small but the decay is slow, so the distribution extends over a wide range. When $\lambda$ is large, the PDF starts high near zero and decays quickly, so most of the probability is concentrated near 0.
The shape of the exponential PDF is analogous to that of the geometric distribution in the discrete case. There, probabilities decrease geometrically with the value; here they decrease exponentially.

Tail probabilities

Fix $a > 0$. We want to compute

$$\mathbb{P}(X \geq a) = \int_a^\infty \lambda e^{-\lambda x}\,dx.$$

Using the antiderivative $\int e^{-\lambda x}dx = -\frac{1}{\lambda}e^{-\lambda x}$, we get

$$\mathbb{P}(X \geq a) = \lambda\left[-\frac{1}{\lambda}e^{-\lambda x}\right]_{x=a}^{x=\infty} = -e^{-\lambda x}\Big|_{x=a}^{x=\infty} = 0 - \big(-e^{-\lambda a}\big) = e^{-\lambda a}.$$

Thus the tail probability decays exponentially in $a$.
If we set $a = 0$, we obtain

$$\mathbb{P}(X \geq 0) = e^{-\lambda \cdot 0} = 1,$$

which confirms that the total integral of the PDF over its support is 1.

Mean and second moment

To compute the mean, we use the definition

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x}\, dx.$$

This integral can be evaluated by integration by parts and the result is

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

For the second moment, we similarly write

$$\mathbb{E}[X^2] = \int_0^\infty x^2\lambda e^{-\lambda x}\, dx,$$

which is again computed by integration by parts (a bit more tedious). The result is

$$\mathbb{E}[X^2] = \frac{2}{\lambda^2}.$$

Variance
Using the alternative variance formula,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2,$$

we substitute

$$\mathbb{E}[X^2] = \frac{2}{\lambda^2}, \quad \mathbb{E}[X] = \frac{1}{\lambda}.$$

Thus

$$\mathrm{Var}(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

When $\lambda$ is small, the PDF decays slowly, so large values of $X$ are more likely. In that case, both the mean $\frac{1}{\lambda}$ and the variance $\frac{1}{\lambda^2}$ are large, reflecting a wide and spread-out distribution.

Interpretation and relation to geometric
The exponential distribution is closely related to the geometric distribution. The mean $\mathbb{E}[X] = 1/\lambda$ parallels the geometric mean $1/p$, where $p$ is the success probability. The exponential is often used to model waiting times until an event occurs, such as the time until a customer arrives, a machine fails, a light bulb burns out, or an email arrives.

### 1.5.5 Cumulative distribution functions

We now introduce a way to describe the distribution of any random variable, discrete or continuous, using the *cumulative distribution function* (CDF). For a random variable $X$, the CDF is the function

$$F_X(x) = \mathbb{P}(X \le x).$$

It gives the probability that $X$ takes a value less than or equal to $x$.
The CDF is defined the same way for all types of random variables. If $X$ is continuous with PDF $f_X$, then

$$F_X(x) = \int_{-\infty}^x f_X(t)\, dt.$$

Example: uniform random variable
Let $X$ be uniform on $[a, b]$. The CDF $F_X(x)$ has three cases:

$$F_X(x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x < b, \\ 1, & x \ge b. \end{cases}$$

For $x \le a$, there is no probability mass to the left, so the CDF is 0. For $a < x < b$, the probability up to $x$ is the area of a rectangle with base $x - a$ and height $\frac{1}{b-a}$, giving $\frac{x-a}{b-a}$. For $x \ge b$, we have already covered all the probability, so the CDF is 1.
The graph starts at 0, rises linearly from $a$ to $b$, and then stays at 1.

Using the CDF to get probabilities
Once we know the CDF, we can compute probabilities of intervals. For example,

$$\mathbb{P}(3 < X \leq 4) = F_X(4) - F_X(3).$$

More generally, for any $a < b$,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

Thus the CDF contains all probabilistic information about the distribution.
For continuous random variables, we can recover the PDF from the CDF wherever the derivative exists:

$$f_X(x) = \frac{d}{dx} F_X(x).$$

Conversely, given the PDF, we obtain the CDF by integrating.

CDFs for discrete random variables
For a discrete random variable $X$, the CDF is still

$$F_X(x) = \mathbb{P}(X \leq x),$$

but it is computed as a sum of PMF values for all outcomes $k$ with $k \leq x$. The CDF has the shape of a staircase: it is constant between the possible values of $X$ and jumps upward at each value where $X$ has positive probability.
At a point $x_0$ where $X$ can take the value $x_0$, the jump size is

$$F_X(x_0) - \lim_{x \uparrow x_0} F_X(x) = \mathbb{P}(X = x_0).$$

The value of the CDF at a jump is the value *including* that point, because the definition uses $\leq x$.

General properties of CDFs
CDFs satisfy some general properties for any random variable:

$$\text{(i) } F_X(x) \text{ is non-decreasing in } x,$$

$$\text{(ii) } \lim_{x \to -\infty} F_X(x) = 0,$$

$$\text{(iii) } \lim_{x \to +\infty} F_X(x) = 1.$$

The non-decreasing property reflects that larger $x$ correspond to larger events $\{X \leq x\}$. As we move far to the left, the probability of being $\leq x$ must go to 0, and as we move far to the right, the probability must approach 1.

### 1.5.6 Normal random variables

Normal random variables, also called Gaussian random variables, are among the most important in probability theory. They play a key role in theoretical results such as the central limit theorem and are widely used in applications as models of random noise.

Standard normal distribution
The *standard normal* random variable is denoted by

$$X \sim \mathcal{N}(0, 1).$$

Its PDF is defined for all real $x$ by

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

This random variable can take any real value.
The factor $e^{-x^2/2}$ has a bell shape centered at 0: it is equal to 1 at $x = 0$ and decreases as $|x|$ increases. The constant $\frac{1}{\sqrt{2\pi}}$ is chosen so that the total area under the curve is 1.
Because the PDF is symmetric around 0, the mean of the standard normal is

$$\mathbb{E}[X] = 0.$$

A calculus computation shows that its variance is

$$\mathrm{Var}(X) = 1.$$

General normal distributions
A general normal random variable with mean $\mu$ and variance $\sigma^2$ is denoted by

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

where $\sigma > 0$. Its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

The exponent $-\frac{(x-\mu)^2}{2\sigma^2}$ is a quadratic centered at $x = \mu$, so the PDF has a bell shape centered at $\mu$. By symmetry, the mean is

$$\mathbb{E}[X] = \mu.$$

A calculus exercise shows that the variance is

$$\mathrm{Var}(X) = \sigma^2.$$

The parameter $\sigma$ controls the spread. When $\sigma$ is small, the quadratic in the exponent rises quickly, the bell is narrow, and the variance is small. When $\sigma$ is large, the bell is wide and the variance is large.

Linear transformations of normal variables
Normal random variables behave very nicely under linear transformations. Let

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

and define

$$Y = aX + b$$

for constants $a$ and $b$.
From linearity of expectation,

$$\mathbb{E}[Y] = a\,\mathbb{E}[X] + b = a\mu + b.$$

From the variance rule for linear functions,

$$\mathrm{Var}(Y) = a^2\mathrm{Var}(X) = a^2\sigma^2.$$

The key additional fact is that $Y$ is itself normal:

$$Y \sim \mathcal{N}(a\mu + b,\, a^2\sigma^2).$$

There is one special case. If $a = 0$, then $Y = b$ is a constant (a degenerate discrete random variable with variance 0). By convention, we can view this as a degenerate normal with mean $b$ and variance 0, so that any linear function of a normal random variable is still considered normal.

### 1.5.7 Calculation of normal probabilities

Normal random variables are very important, so we would like to calculate probabilities such as $\mathbb{P}(X \leq 5)$ when $X$ is normal. In general, the CDF of a normal random variable does not have a closed form expression in terms of elementary functions.

Standard normal CDF and its table
For the standard normal random variable $Y \sim \mathcal{N}(0, 1)$, we denote the CDF by

$$\phi(y) = \mathbb{P}(Y \leq y).$$

Graphically, $\phi(y)$ is the area under the standard normal PDF to the left of $y$. Values of $\phi(y)$ are tabulated in standard normal tables.
To use such a table, we locate the row and column corresponding to the first two decimal digits of $y$. For example, $\phi(0) = 0.5$ by symmetry of the standard normal around 0. If we want $\phi(1.16)$, we find the row for 1.1 and the column for 0.06, and read off the corresponding entry. For $y = 2.9$, we find the row for 2.9 and the column for 0.00, which gives $\phi(2.9)$.
Typical tables list $\phi(y)$ only for $y \geq 0$. To handle negative arguments, we use symmetry of the standard normal:

$$\phi(-y) = \mathbb{P}(Y \leq -y) = 1 - \mathbb{P}(Y \leq y) = 1 - \phi(y), \quad y \geq 0.$$

Thus, to find $\phi(-2)$, we compute $1 - \phi(2)$.

### Standardization of a general normal

Let $X$ be a normal random variable with mean $\mu$ and standard deviation $\sigma > 0$, that is $X \sim \mathcal{N}(\mu, \sigma^2)$. We define the standardized random variable

$$Y = \frac{X - \mu}{\sigma}.$$

The quantity $Y$ measures how many standard deviations $X$ is away from its mean.
The mean of $Y$ is

$$\mathbb{E}[Y] = \frac{\mathbb{E}[X] - \mu}{\sigma} = 0,$$

and the variance of $Y$ is

$$\mathrm{Var}(Y) = \frac{1}{\sigma^2}\mathrm{Var}(X) = 1.$$

If $X$ is normal, then $Y$ is a standard normal random variable. Equivalently,

$$X = \mu + \sigma Y, \quad Y \sim \mathcal{N}(0, 1).$$

### Using standardization to compute probabilities

To compute probabilities for $X$, we transform the event into one involving $Y$. For example, for any real $a$ and $b$ with $a < b$,

$$\mathbb{P}(a \le X \le b) = \mathbb{P}(a \le \mu + \sigma Y \le b) = \mathbb{P}\left(\frac{a - \mu}{\sigma} \le Y \le \frac{b - \mu}{\sigma}\right).$$

If we write

$$\alpha = \frac{a - \mu}{\sigma}, \quad \beta = \frac{b - \mu}{\sigma},$$

then

$$\mathbb{P}(a \le X \le b) = \mathbb{P}(\alpha \le Y \le \beta) = \phi(\beta) - \phi(\alpha).$$

We can then use the standard normal table to find $\phi(\beta)$ and $\phi(\alpha)$, handling negative arguments via symmetry if needed.

### Example of the method

Suppose $X \sim \mathcal{N}(6, 4)$, so $\mu = 6$ and $\sigma = 2$. We want

$$\mathbb{P}(2 \le X \le 8).$$

We standardize:

$$\frac{2 - 6}{2} = -2, \quad \frac{8 - 6}{2} = 1.$$

So

$$\mathbb{P}(2 \le X \le 8) = \mathbb{P}(-2 \le Y \le 1),$$

where $Y \sim \mathcal{N}(0, 1)$. This equals

$$\mathbb{P}(Y \le 1) - \mathbb{P}(Y \le -2) = \phi(1) - \phi(-2).$$

Using symmetry,

$$\phi(-2) = 1 - \phi(2),$$

so

$$\mathbb{P}(2 \le X \le 8) = \phi(1) - \big(1 - \phi(2)\big) = \phi(1) + \phi(2) - 1.$$

Finally, we read $\phi(1)$ and $\phi(2)$ from the standard normal table and substitute the corresponding numerical values.

### 1.5.8 Conditioning a continuous random variable on an event

We extend conditioning to continuous random variables, by replacing sums with integrals and PMFs with PDFs.

Conditional PDF on an event
Let $X$ be continuous with PDF $f_X$ and let $A$ be an event with $\mathbb{P}(A) > 0$. In the conditional world where $A$ is known to have occurred, $X$ has a conditional PDF $f_{X|A}$ such that, for any set $B$,

$$\mathbb{P}(X \in B \mid A) = \int_B f_{X|A}(x)\, dx.$$

This $f_{X|A}$ is a genuine PDF:

$$f_{X|A}(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_{X|A}(x)\, dx = 1.$$

Special case: conditioning on $\{X \in S\}$
Now assume

$$A = \{X \in S\}, \quad S \subseteq \mathbb{R}, \quad \mathbb{P}(X \in S) > 0.$$

We want a formula for $f_{X|X\in S}$.
Think in terms of small intervals: for small $\Delta > 0$,

$$\mathbb{P}(x \leq X \leq x + \Delta) \approx f_X(x)\, \Delta,$$

$$\mathbb{P}(x \leq X \leq x + \Delta \mid X \in S) \approx f_{X|X\in S}(x)\, \Delta.$$

Case 1: $x \notin S$. For small $\Delta$, the interval $[x, x + \Delta]$ does not intersect $S$, so the conditional probability is 0, hence

$$f_{X|X\in S}(x) = 0 \quad \text{if } x \notin S.$$

Case 2: $x \in S$. For small $\Delta$, $[x, x + \Delta] \subseteq S$, so

$$\mathbb{P}(x \leq X \leq x + \Delta \mid X \in S) = \frac{\mathbb{P}(x \leq X \leq x + \Delta)}{\mathbb{P}(X \in S)} \approx \frac{f_X(x)\, \Delta}{\mathbb{P}(X \in S)}.$$

Comparing with

$$\mathbb{P}(x \leq X \leq x + \Delta \mid X \in S) \approx f_{X|X\in S}(x)\, \Delta$$

and cancelling $\Delta$, we get

$$f_{X|X\in S}(x) = \frac{f_X(x)}{\mathbb{P}(X \in S)} \quad \text{if } x \in S,$$

and 0 otherwise.
In compact form,

$$f_{X|X\in S}(x) = \frac{f_X(x)}{\int_S f_X(u)\, du}\, \mathbf{1}_{\{x\in S\}}.$$

*The conditional PDF keeps the same shape as $f_X$ on $S$, but is rescaled so that the total area over $S$ becomes 1.*

Conditional expectation (brief)
Given $A$ with $\mathbb{P}(A) > 0$, the conditional expectation of $g(X)$ given $A$ is

$$\mathbb{E}[g(X) \mid A] = \int_{-\infty}^{\infty} g(x)\, f_{X|A}(x)\, dx.$$

For the special case $A = \{X \in S\}$,

$$\mathbb{E}[g(X) \mid X \in S] = \frac{1}{\mathbb{P}(X \in S)} \int_S g(x)\, f_X(x)\, dx.$$

This is the continuous analogue of the discrete conditional expectation formula, with integrals instead of sums.

### 1.5.9 Conditioning example

We illustrate conditioning of a continuous random variable on an event using a simple piecewise constant PDF.

Setup
Let $X$ have a uniform PDF on $[a, b]$,

$$f_X(x) = \frac{1}{b-a}, \quad a \le x \le b,$$

and $f_X(x) = 0$ otherwise. We condition on the event

$$A = \left\{ \frac{a+b}{2} \le X \le b \right\}.$$

Conditional PDF
The conditional PDF is

$$f_{X|A}(x) = 0 \quad \text{for } x < \frac{a+b}{2} \text{ or } x > b.$$

On the interval $\left[\frac{a+b}{2}, b\right]$, it must keep the same *shape* as $f_X$, which is constant, so $f_{X|A}$ is also constant there. The length of the conditioning interval is

$$b - \frac{a+b}{2} = \frac{b-a}{2}.$$

Since the total area must be 1,

$$\text{height} \times \text{length} = 1 \quad \Rightarrow \quad \text{height} \cdot \frac{b-a}{2} = 1,$$

so the conditional PDF height is

$$\text{height} = \frac{2}{b-a}.$$

Therefore

$$f_{X|A}(x) = \begin{cases} \dfrac{2}{b-a}, & \dfrac{a+b}{2} \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Conditional expectation
Because $X \mid A$ is uniform on $\left[\frac{a+b}{2}, b\right]$, its expectation is the midpoint of that interval:

$$\mathbb{E}[X \mid A] = \frac{1}{2}\left(\frac{a+b}{2}\right) + \frac{1}{2}b = \frac{1}{4}a + \frac{3}{4}b.$$

Conditional expectation of $X^2$
Using the expected value rule with the conditional PDF,

$$\mathbb{E}[X^2 \mid A] = \int_{-\infty}^{\infty} x^2 f_{X|A}(x)\, dx = \int_{(a+b)/2}^{b} x^2 \cdot \frac{2}{b-a}\, dx.$$

That is,

$$\mathbb{E}[X^2 \mid A] = \frac{2}{b-a} \int_{(a+b)/2}^{b} x^2\, dx,$$

an elementary integral that can be evaluated directly if needed.

### 1.5.10 Exponential PDF and memorylessness

Let $T$ be the lifetime of a light bulb and assume $T \sim \text{Exponential}(\lambda)$, so

$$\mathbb{P}(T > x) = e^{-\lambda x}, \quad x \ge 0.$$

Memorylessness
Suppose the bulb has already worked for time $t$ and has not failed, so $T > t$. Let $X = T - t$ be the remaining lifetime. Then

$$\mathbb{P}(X > x \mid T > t) = \frac{\mathbb{P}(T > t + x)}{\mathbb{P}(T > t)} = \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x} = \mathbb{P}(T > x).$$

So the remaining lifetime has the same distribution as a new bulb. *An exponential lifetime does not "remember" how long it has already lasted.*

Small-interval interpretation
For small $\Delta > 0$,

$$\mathbb{P}(0 \le T \le \Delta) \approx \lambda\Delta.$$

By memorylessness, the same is true conditionally at any time $t$:

$$\mathbb{P}(\text{failure in } [t, t+\Delta) \mid T > t) \approx \lambda\Delta.$$

*At each tiny time step, the failure chance is always about $\lambda\Delta$, independent of the past.*

### 1.5.11 Total probability and expectation theorems (continuous case)

We now state continuous versions of the total probability and total expectation theorems, and illustrate them with a simple scenario model.

Total probability for PDFs
Let $A_1, \ldots, A_n$ be a partition of the sample space, that is, disjoint events whose union is the whole space, and let $X$ be a continuous random variable with density $f_X$. For each scenario $A_i$, let $f_{X|A_i}$ be the conditional PDF of $X$ given $A_i$.
Then the PDF of $X$ is

$$f_X(x) = \sum_{i=1}^{n} \mathbb{P}(A_i)\, f_{X|A_i}(x).$$

*The overall density at a point $x$ is a weighted sum of the conditional densities at $x$, with weights given by the scenario probabilities.*

Total expectation (continuous)
Under the same setup, the total expectation theorem says

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{P}(A_i)\, \mathbb{E}[X \mid A_i].$$

*The overall mean is the average of the conditional means, weighted by the probabilities of the scenarios.*

Example: Bill and the supermarket
Bill will go to the supermarket at a random time $X$ (in hours from now). There are two scenarios:
Scenario 1 $A_1$: with probability $\mathbb{P}(A_1) = \frac{1}{3}$, Bill goes between 0 and 2 hours, uniformly:

$$f_{X|A_1}(x) = \begin{cases} \frac{1}{2}, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Scenario 2 $A_2$: with probability $\mathbb{P}(A_2) = \frac{2}{3}$, Bill goes between 6 and 8 hours, uniformly:

$$f_{X|A_2}(x) = \begin{cases} \frac{1}{2}, & 6 \le x \le 8, \\ 0, & \text{otherwise.} \end{cases}$$

Using the total probability formula,

$$f_X(x) = \mathbb{P}(A_1) f_{X|A_1}(x) + \mathbb{P}(A_2) f_{X|A_2}(x),$$

so

$$f_X(x) = \begin{cases} \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}, & 0 \le x \le 2, \\ \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}, & 6 \le x \le 8, \\ 0, & \text{otherwise.} \end{cases}$$

For the expectation, use total expectation:

$$\mathbb{E}[X] = \mathbb{P}(A_1)\, \mathbb{E}[X \mid A_1] + \mathbb{P}(A_2)\, \mathbb{E}[X \mid A_2].$$

Under $A_1$, $X$ is uniform on $[0, 2]$, so $\mathbb{E}[X \mid A_1] = 1$. Under $A_2$, $X$ is uniform on $[6, 8]$, so $\mathbb{E}[X \mid A_2] = 7$. Thus

$$\mathbb{E}[X] = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 7 = \frac{1}{3} + \frac{14}{3} = 5.$$

This example shows how scenario-wise conditional PDFs and expectations combine to give the overall distribution and mean.

### 1.5.12 Mixed random variables

We now consider random variables that are neither purely discrete nor purely continuous.

Motivating example
You have 1 dollar and can choose whether to play a lottery.
- With probability 1/2, you do nothing and keep your 1 dollar. In this scenario, $X = 1$ for sure. - With probability 1/2, you play the lottery and get a random return $U$ that is uniform on $[0, 2]$. In this scenario, $X = U$.
The resulting $X$ can take any value in $[0, 2]$, but

$$\mathbb{P}(X = 1) = \frac{1}{2} > 0.$$

So:
- $X$ is *not* discrete, because it is not supported on a countable set only - $X$ is *not* continuous, because a continuous random variable must satisfy $\mathbb{P}(X = x) = 0$ for all $x$, which fails at $x = 1$
We call such an $X$ a *mixed random variable.*

General construction
More generally, suppose:
- With probability $p$, $X$ behaves like a discrete random variable $Y$ - With probability $1-p$, $X$ behaves like a continuous random variable $Z$
Then $X$ has no PMF (not purely discrete) and no PDF (not purely continuous), but its CDF is always well defined. Let $F_Y$ and $F_Z$ be the CDFs of $Y$ and $Z$. Using the total probability theorem for CDFs,

$$F_X(x) = \mathbb{P}(X \le x) = p\,\mathbb{P}(Y \le x) + (1-p)\,\mathbb{P}(Z \le x) = p\,F_Y(x) + (1-p)\,F_Z(x).$$

*The CDF of a mixed random variable is a weighted sum of the CDFs of its discrete and continuous parts.*
For the expectation, we define

$$\mathbb{E}[X] = p\,\mathbb{E}[Y] + (1-p)\,\mathbb{E}[Z],$$

consistent with the total expectation theorem.

CDF in the lottery example
In the lottery example:
- Scenario $A_1$ (play): probability 1/2, $X$ is uniform on $[0, 2]$

$$f_{X|A_1}(x) = \begin{cases} \frac{1}{2}, & 0 \le x \le 2, \\ 0, & \text{otherwise}, \end{cases}$$

so

$$F_{X|A_1}(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{2}, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

- Scenario $A_2$ (do nothing): probability 1/2, $X \equiv 1$ Then

$$F_{X|A_2}(x) = \begin{cases} 0, & x < 1, \\ 1, & x \ge 1. \end{cases}$$

Using

$$F_X(x) = \frac{1}{2}F_{X|A_1}(x) + \frac{1}{2}F_{X|A_2}(x),$$

we get:
- For $x < 0$: $F_X(x) = 0$ - For $0 \le x < 1$:

$$F_X(x) = \frac{1}{2} \cdot \frac{x}{2} + \frac{1}{2} \cdot 0 = \frac{x}{4},$$

a line of slope 1/4 - At $x = 1$: we add the jump from the discrete part

$$\lim_{x \to 1^-} F_X(x) = \frac{1}{4}, \qquad F_X(1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4},$$

so the jump size at $x = 1$ is $3/4 - 1/4 = 1/2$, which equals $\mathbb{P}(X = 1)$ - For $1 < x \leq 2$:

$$F_X(x) = \frac{1}{2} \cdot \frac{x}{2} + \frac{1}{2} \cdot 1 = \frac{x}{4} + \frac{1}{2},$$

again a line of slope 1/4 until it reaches 1 at $x = 2$ - For $x > 2$: $F_X(x) = 1$
The resulting CDF has:
- Linear growth with slope 1/4 on $[0, 1)$ - A jump of size 1/2 at $x = 1$ - Linear growth with slope 1/4 on $(1, 2]$, then flat
*This illustrates that a mixed random variable may not have a single PDF or PMF, but its CDF is easy to construct from the component distributions.*

### 1.5.13 Joint PDFs

We now move to *multiple* continuous random variables and introduce the joint PDF, the continuous analogue of the joint PMF.

Definition and probability of a set
Let $X$ and $Y$ be random variables. A *joint PDF* is a function $f_{X,Y}(x, y)$ such that

$$f_{X,Y}(x, y) \geq 0 \quad \text{for all } (x, y),$$
$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y)\, dx\, dy = 1.$$

If $X$ and $Y$ are *jointly continuous*, probabilities are computed by

$$\mathbb{P}\big((X, Y) \in B\big) = \iint_B f_{X,Y}(x, y)\, dx\, dy,$$

for any region $B \subseteq \mathbb{R}^2$. *Geometrically, this is the volume under the surface $z = f_{X,Y}(x, y)$ above the set $B$.*

Rectangles and small rectangles
For a rectangle

$$B = \{(x, y) \colon a \leq x \leq b,\ c \leq y \leq d\},$$

we write

$$\mathbb{P}(a \leq X \leq b,\ c \leq Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x, y)\, dx\, dy.$$

If the rectangle is very small, say $[x, x + \Delta] \times [y, y + \Delta]$ with small $\Delta > 0$, then

$$\mathbb{P}\big(x \leq X \leq x + \Delta,\ y \leq Y \leq y + \Delta\big) \approx f_{X,Y}(x, y)\, \Delta^2.$$

So *the joint PDF is a probability density per unit area*: probability $\approx$ density $\times$ area.

Zero-area sets
For jointly continuous random variables, any set of area 0 has probability 0. In particular:
- $\mathbb{P}(X = x_0,\ Y = y_0) = 0$ for any point $(x_0, y_0)$ - $\mathbb{P}\big((X, Y) \text{ lies on a specific curve or line}\big) = 0$

When continuity fails jointly
Consider $X$ continuous and define $Y = X$. Each variable alone is continuous, but the pair $(X, Y)$ always lies on the line $y = x$. All probability is concentrated on this one-dimensional line, which has area 0.
This contradicts the *jointly continuous* picture where zero-area sets must have probability 0. So $X$ and $Y$ here are *not* jointly continuous, even though each is continuous by itself.

### 1.5.14 From the joint to the marginal

We now see how to recover the individual (marginal) PDFs of $X$ and $Y$ from their joint PDF.

Deriving the marginal of $X$
Let $X, Y$ be jointly continuous with joint PDF $f_{X,Y}(x, y)$. The CDF of $X$ is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Geometrically, this is the probability that the point $(X, Y)$ falls in the half-plane $\{(s, t) \colon s \leq x\}$. Using the joint PDF,

$$F_X(x) = \iint_{\{(s,t)\colon s \leq x\}} f_{X,Y}(s, t)\, ds\, dt.$$

We can write this as an iterated integral (use $s$ for the $X$-coordinate and $t$ for the $Y$-coordinate):

$$F_X(x) = \int_{-\infty}^{\infty} \left( \int_{-\infty}^{x} f_{X,Y}(s, t)\, ds \right) dt.$$

To get the PDF of $X$, we differentiate:

$$f_X(x) = \frac{d}{dx} F_X(x) = \frac{d}{dx} \int_{-\infty}^{\infty} \left( \int_{-\infty}^{x} f_{X,Y}(s, t)\, ds \right) dt.$$

Differentiating w.r.t. the upper limit $x$ of the inner integral gives

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, t)\, dt.$$

So the *marginal PDF of* $X$ is obtained by integrating the joint PDF over all possible $y$-values:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy.$$

Similarly,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx.$$

Example: uniform joint PDF on a region
Suppose $(X, Y)$ has a joint PDF that is uniform on a set $S$ in the plane, and zero outside. If $\text{area}(S) = A$, then

$$f_{X,Y}(x, y) = \begin{cases} \dfrac{1}{A}, & (x, y) \in S, \\ 0, & \text{otherwise.} \end{cases}$$

For any set $B$,

$$\mathbb{P}\big((X, Y) \in B\big) = \iint_B f_{X,Y}(x, y)\, dx\, dy = \frac{1}{A} \text{area}(B \cap S).$$

Now fix some $x$. To find $f_X(x)$, integrate along the vertical line at that $x$:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy = \frac{1}{A} \times \big(\text{length of the segment of this line inside } S\big).$$

If that vertical slice inside $S$ has length $L(x)$, then

$$f_X(x) = \frac{L(x)}{A}.$$

Similarly, for $f_Y(y)$, we integrate horizontally; if the horizontal slice at $y$ has length $M(y)$ inside $S$, then

$$f_Y(y) = \frac{M(y)}{A}.$$

*In words: the marginal at a given point is the joint density times the "thickness" of $S$ along the corresponding line, i.e., the length of the slice through $S$.*

### 1.5.15 Continuous analogs of various properties

We now briefly summarize how standard discrete results extend to jointly continuous random variables.

Joint PDFs for many variables
For more than two continuous random variables $X_1, \ldots, X_n$, we can define a joint PDF

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n),$$

which is nonnegative and integrates to 1 over $\mathbb{R}^n$. Probabilities of events in $\mathbb{R}^n$ are obtained by integrating this joint PDF over the corresponding region.

From joint to marginals
The marginal PDF of any subset of the variables is obtained by integrating out the others. For example, for the first variable,

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)\, dx_2 \cdots dx_n.$$

Functions of jointly continuous variables
If $X_1, \ldots, X_n$ are jointly continuous and $g$ is a real function, we can define

$$Y = g(X_1, \ldots, X_n).$$

Then the expected value rule has the same form as in the discrete case:

$$\mathbb{E}[Y] = \mathbb{E}\big[g(X_1, \ldots, X_n)\big] = \int \cdots \int g(x_1, \ldots, x_n)\, f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n.$$

Linearity of expectation
From this rule we recover the usual linearity properties. For constants $a, b$ and random variables $X, Y$,

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y],$$

with no extra assumptions needed.
*In summary, all familiar discrete laws carry over to the continuous case by the usual recipe: replace sums by integrals and PMFs by PDFs.*

### 1.5.16 Joint CDFs

We can also describe the distribution of multiple random variables using a joint CDF, in complete analogy with the one-dimensional case.

Definition and relation to the joint PDF
For a pair $(X, Y)$, the *joint CDF* is

$$F_{X,Y}(x, y) = \mathbb{P}(X \le x,\ Y \le y).$$

Geometrically, this is the probability that $(X, Y)$ falls in the rectangle $(-\infty, x] \times (-\infty, y]$.
If $X, Y$ are jointly continuous with joint PDF $f_{X,Y}$, then

$$F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(s, t)\, ds\, dt.$$

Conversely, we can recover the joint PDF by differentiating:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x, y),$$

assuming the derivatives exist.

Example: uniform on the unit square
Suppose $(X, Y)$ is uniform on the unit square $[0, 1] \times [0, 1]$. Then

$$f_{X,Y}(x, y) = \begin{cases} 1, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 \le x \le 1,\ 0 \le y \le 1$, the joint CDF is the probability of the rectangle $[0, x] \times [0, y]$, which is just its area:

$$F_{X,Y}(x, y) = xy.$$

Differentiating,

$$\frac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x, y) = \frac{\partial}{\partial y}(y) = 1,$$

which recovers the joint PDF on the unit square, as expected.

### 1.5.17 Conditional PDFs of one variable given another

Let $X$ and $Y$ be jointly continuous with joint PDF $f_{X,Y}(x, y)$ and marginal $f_Y(y)$. For $y$ such that $f_Y(y) > 0$, the *conditional PDF* of $X$ given $Y = y$ is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Small-interval interpretation
For small $\delta > 0$, we interpret

$$\mathbb{P}(x \le X \le x + \delta \mid Y \approx y) \approx f_{X|Y}(x \mid y)\, \delta.$$

So, given that $Y$ is (approximately) $y$, the function $x \mapsto f_{X|Y}(x \mid y)$ plays the same role as an ordinary PDF of $X$.
Conditional probabilities of sets
For any set $B \subseteq \mathbb{R}$, we define

$$\mathbb{P}(X \in B \mid Y = y) \triangleq \int_B f_{X|Y}(x \mid y)\, dx,$$

and treat this as the conditional probability of $X$ lying in $B$, given that $Y$ took value $y$.
Example
Suppose

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 \le x \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then for $0 \le y \le 1$,

$$f_Y(y) = \int_0^y 2\, dx = 2y,$$

and, for $0 < y \le 1$,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{2}{2y} = \frac{1}{y}, \qquad 0 \le x \le y,$$

so conditionally on $Y = y$, the random variable $X$ is uniform on $[0, y]$.

### 1.5.18 Comments on conditional PDFs

Basic properties
For jointly continuous $X, Y$ with joint PDF $f_{X,Y}(x, y)$ and marginal $f_Y(y)$, recall

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \qquad \text{for } f_Y(y) > 0.$$

This conditional PDF is always nonnegative, because the joint PDF is nonnegative and the denominator is positive.
Slices of the joint PDF
Fix a value $y$ with $f_Y(y) > 0$. As a function of $x$, the numerator $f_{X,Y}(x, y)$ describes how the joint density varies along the horizontal line $\{(x, y) \colon x \in \mathbb{R}\}$. Thus $x \mapsto f_{X,Y}(x, y)$ is a *slice* of the joint surface taken at height $y$.
The conditional PDF $f_{X|Y}(x \mid y)$ is obtained from this slice by multiplying by the constant factor $1/f_Y(y)$. So, for fixed $y$, the conditional PDF has the *same shape* in $x$ as the corresponding slice of the joint PDF, just rescaled vertically.
Why it integrates to 1
For any fixed $y$ with $f_Y(y) > 0$,

$$\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)\, dx = \int_{-\infty}^{\infty} \frac{f_{X,Y}(x, y)}{f_Y(y)}\, dx = \frac{1}{f_Y(y)} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx.$$

By the marginalization rule,

$$\int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx = f_Y(y),$$

so the ratio is equal to 1. This shows that, for each fixed $y$, the conditional PDF behaves like an ordinary PDF in the variable $x$: it is nonnegative and integrates to 1.

Multiplication rule for densities

Starting again from

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

we can rearrange to obtain

$$f_{X,Y}(x,y) = f_{X|Y}(x \mid y)\, f_Y(y).$$

This has the same form as the multiplication rule for probabilities, where we write $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B)$. The only difference is that here we are dealing with densities, not probabilities, but the algebraic structure is identical.

By symmetry, we also have

$$f_{X,Y}(x,y) = f_{Y|X}(y \mid x)\, f_X(x).$$

So, at the level of formulas, conditional PDFs mirror all the familiar rules from the discrete case, with PMFs replaced by PDFs.

### 1.5.19 Total probability and total expectation theorems

Continuous total probability

For jointly continuous $X, Y$ and any set $A \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A) = \int_{-\infty}^{\infty} \mathbb{P}(X \in A \mid Y = y)\, f_Y(y)\, dy.$$

Conditional expectation

For $f_Y(y) > 0$, the conditional mean of $X$ given $Y = y$ is

$$\mathbb{E}[X \mid Y = y] = \int_{-\infty}^{\infty} x\, f_{X|Y}(x \mid y)\, dx.$$

Total expectation

The continuous total expectation theorem (law of total expectation) is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y]\, f_Y(y)\, dy.$$

Expected value rule in the conditional universe

For any function $g$,

$$\mathbb{E}\big[g(X) \mid Y = y\big] = \int_{-\infty}^{\infty} g(x)\, f_{X|Y}(x \mid y)\, dx.$$

### 1.5.20 Independence

Definition

Let $X$ and $Y$ be jointly continuous with joint PDF $f_{X,Y}(x,y)$ and marginals $f_X(x)$, $f_Y(y)$. They are *independent* if

$$f_{X,Y}(x,y) = f_X(x)\, f_Y(y) \quad \text{for all } x, y.$$

Relation to conditional PDFs

Whenever $f_Y(y) > 0$, the multiplication rule gives

$$f_{X,Y}(x,y) = f_{X|Y}(x \mid y)\, f_Y(y).$$

If $X$ and $Y$ are independent, comparing with

$$f_{X,Y}(x,y) = f_X(x)\, f_Y(y)$$

yields

$$f_{X|Y}(x \mid y) = f_X(x) \quad \text{for all } x \text{ and all } y \text{ with } f_Y(y) > 0.$$

So independence means that conditioning on $Y = y$ does not change the distribution of $X$.

Slice picture

For fixed $y$, the conditional PDF $f_{X|Y}(x \mid y)$ is (up to scaling) a slice of the joint PDF along the horizontal line at height $y$. Independence requires that all these slices, as functions of $x$, have exactly the same shape, namely the shape of the marginal PDF $f_X$.

Intuition
Under independence, no matter what value of $Y$ we observe, the distribution of $X$ remains the same. In this sense, $Y$ carries no information about $X$, and by symmetry, $X$ carries no information about $Y$.
More than two variables
Random variables $X_1, \ldots, X_n$ are independent if their joint PDF factors as

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i) \quad \text{for all } (x_1, \ldots, x_n).$$

Intuitively, knowing some of the values $X_i$ does not affect our beliefs about the others.
Consequences
If $X$ and $Y$ are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y],$$

and, for a sum,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

More generally, if $X$ and $Y$ are independent and $g$ and $h$ are measurable functions, then $g(X)$ and $h(Y)$ are independent, and

$$\mathbb{E}\big[g(X)\,h(Y)\big] = \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)].$$

Example
Suppose $X$ and $Y$ are independent, each uniform on $[0, 1]$. Then

$$f_X(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases} \qquad f_Y(y) = \begin{cases} 1, & 0 \le y \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$f_{X,Y}(x, y) = f_X(x)\,f_Y(y) = \begin{cases} 1, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Here every horizontal slice in $x$ has the same flat shape, matching the marginal density of $X$, which reflects the independence of $X$ and $Y$.

### 1.5.21 Stick-breaking example

Setup
A stick has length $l$. First break: the location $X$ is uniform on $[0, l]$, so

$$f_X(x) = \begin{cases} \frac{1}{l}, & 0 \le x \le l, \\ 0, & \text{otherwise}. \end{cases}$$

Second break: given $X = x$, the location $Y$ is uniform on $[0, x]$, so

$$f_{Y|X}(y \mid x) = \begin{cases} \frac{1}{x}, & 0 \le y \le x, \\ 0, & \text{otherwise}. \end{cases}$$

Dependence
The conditional PDF of $Y$ changes when $x$ changes (its support is $[0, x]$ and its height is $1/x$), so $Y$ clearly depends on $X$. Equivalently, knowing $X = x$ tells us that $0 \le Y \le x$, so $X$ provides information about $Y$, hence $X$ and $Y$ are not independent.
Joint and marginal PDFs
Using the multiplication rule,

$$f_{X,Y}(x, y) = f_X(x)\,f_{Y|X}(y \mid x) = \frac{1}{l} \cdot \frac{1}{x} = \frac{1}{lx},$$

for the *possible* pairs $(x, y)$, namely

$$0 \le y \le x \le l,$$

and $f_{X,Y}(x, y) = 0$ otherwise. This corresponds to a joint density supported on the triangle $\{(x, y) \colon 0 \le y \le x \le l\}$.
To find the marginal of $Y$, fix $y$ and integrate over the admissible $x$:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,dx = \int_y^l \frac{1}{lx}\,dx = \frac{1}{l}\big[\log x\big]_{x=y}^{x=l} = \frac{1}{l}\log\Big(\frac{l}{y}\Big),$$

for $0 < y \le l$, and $f_Y(y) = 0$ otherwise. This density blows up as $y \to 0$, but it is still valid because its total area is 1.
Expected value of $Y$
A direct computation uses

$$\mathbb{E}[Y] = \int_0^l y\, f_Y(y)\, dy = \int_0^l y \cdot \frac{1}{l} \log\Big(\frac{l}{y}\Big)\, dy,$$

but this integral is messy. A simpler route is to use the total expectation theorem and condition on $X$:

$$\mathbb{E}[Y] = \int_0^l \mathbb{E}[Y \mid X = x]\, f_X(x)\, dx.$$

Given $X = x$, the variable $Y$ is uniform on $[0, x]$, so

$$\mathbb{E}[Y \mid X = x] = \frac{x}{2}.$$

Thus

$$\mathbb{E}[Y] = \int_0^l \frac{x}{2} \cdot \frac{1}{l}\, dx = \frac{1}{2l} \int_0^l x\, dx = \frac{1}{2l} \cdot \frac{l^2}{2} = \frac{l}{4}.$$

*Intuitively, each break cuts the expected length by a factor of* $1/2$*, so after two breaks the expected remaining length is* $l/4$.

### 1.5.22 Independent normals

Standard independent normals
Let $X$ and $Y$ be independent standard normal random variables. Their joint PDF is the product of the marginals:

$$f_{X,Y}(x, y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} \exp\Big(-\frac{1}{2}(x^2 + y^2)\Big).$$

For any fixed radius $r$, the circle $\{(x, y) \colon x^2 + y^2 = r^2\}$ is a contour on which the joint PDF is constant, so the contours are circles centered at the origin.
General independent normals
Now let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$ be independent. Their joint PDF is

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{1}{2}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2}\right]\right).$$

The prefactor $1/(2\pi\sigma_x\sigma_y)$ is just the normalizing constant. Contours correspond to level sets of the quadratic form

$$\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} = \text{constant},$$

which are ellipses centered at $(\mu_x, \mu_y)$ with principal axes aligned with the coordinate axes.
Shape and stretching
The joint PDF attains its maximum at $(x, y) = (\mu_x, \mu_y)$, where the exponent is zero. Moving to ellipses farther from this center decreases the value of the joint PDF, but it remains constant along each ellipse.
If $\sigma_y^2 > \sigma_x^2$, then the ellipses are more stretched in the $y$-direction than in the $x$-direction, reflecting that $Y$ typically varies over a wider range than $X$. In the special case $\sigma_x^2 = \sigma_y^2$, these ellipses become circles (after appropriate rescaling), recovering a circularly symmetric bell.
Independence versus dependence
For independent normals, the stretching of the bell is always along the coordinate axes. If the contours were ellipses tilted away from the axes (with principal axes along some diagonal direction), this would indicate dependence between $X$ and $Y$, leading to a general bivariate normal distribution rather than a product of independent one-dimensional normals.

### 1.5.23 Bayes rule variations

Discrete case
In the discrete setting we have a hidden random variable $X$ with a known prior PMF $p_X(x)$, and an observable random variable $Y$ related to $X$ through the conditional PMF $p_{Y|X}(y \mid x)$. After observing $Y = y$, Bayes rule gives the posterior PMF of $X$:

$$p_{X|Y}(x \mid y) = \frac{p_{Y|X}(y \mid x)\, p_X(x)}{p_Y(y)},$$

where the marginal PMF of $Y$ is

$$p_Y(y) = \sum_x p_{Y|X}(y \mid x)\, p_X(x).$$

Continuous case

In the continuous setting the same structure carries over with PMFs replaced by PDFs and sums by integrals. If $X$ has prior PDF $f_X(x)$ and $Y$ has conditional PDF $f_{Y|X}(y \mid x)$, then after observing $Y = y$ the posterior PDF of $X$ is

$$f_{X|Y}(x \mid y) = \frac{f_{Y|X}(y \mid x)\, f_X(x)}{f_Y(y)},$$

with

$$f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x)\, f_X(x)\, dx.$$

Interpretation

In both cases, Bayes rule updates prior beliefs about $X$ using the likelihood of the observation $Y = y$. The numerator combines prior and likelihood into a joint model, while the denominator normalizes over all possible values of $X$ via the total probability theorem, ensuring the posterior is a valid distribution.

### 1.5.24 Detection of a binary signal

Model

A binary random variable $K$ takes values $\pm 1$ with equal prior probabilities

$$\mathbb{P}(K = 1) = \mathbb{P}(K = -1) = \frac{1}{2}.$$

The observation is

$$Y = K + W,$$

where $W$ is standard normal, independent of $K$. Given $K = k$, the random variable $Y$ is normal with mean $k$ and variance 1, so

$$f_{Y|K}(y \mid k) = \frac{1}{\sqrt{2\pi}} \exp\Big( -\frac{1}{2}(y-k)^2 \Big), \qquad k \in \{-1, 1\}.$$

Posterior probability via Bayes rule

We want the posterior probability that $K = 1$ given an observed value $Y = y$:

$$\mathbb{P}(K = 1 \mid Y = y) = \frac{f_{Y|K}(y \mid 1)\, \mathbb{P}(K = 1)}{f_{Y|K}(y \mid 1)\, \mathbb{P}(K = 1) + f_{Y|K}(y \mid -1)\, \mathbb{P}(K = -1)}.$$

Using the equal priors, this becomes

$$\mathbb{P}(K = 1 \mid Y = y) = \frac{f_{Y|K}(y \mid 1)}{f_{Y|K}(y \mid 1) + f_{Y|K}(y \mid -1)}.$$

Substituting the Gaussian PDFs and simplifying the exponentials yields

$$\mathbb{P}(K = 1 \mid Y = y) = \frac{1}{1 + \exp(-2y)}.$$

Interpretation

As a function of $y$, the posterior $\mathbb{P}(K = 1 \mid Y = y)$ is a smooth S-shaped curve:

$$\mathbb{P}(K = 1 \mid Y = y) \to 1 \text{ as } y \to +\infty, \qquad \mathbb{P}(K = 1 \mid Y = y) \to 0 \text{ as } y \to -\infty,$$

and

$$\mathbb{P}(K = 1 \mid Y = 0) = \frac{1}{2}.$$

Large positive $y$ makes it very likely that $K = 1$, large negative $y$ makes it very likely that $K = -1$, and $y = 0$ leaves both possibilities equally likely.

### 1.5.25 Inference of the bias of a coin

Model
We observe a Bernoulli random variable $K \in \{0, 1\}$ with

$$\mathbb{P}(K = 1 \mid Y = y) = y, \qquad \mathbb{P}(K = 0 \mid Y = y) = 1 - y,$$

where the (unknown) bias $Y$ itself is random and modeled as uniform on $[0, 1]$, so

$$f_Y(y) = \begin{cases} 1, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Goal
Given a single observation $K = 1$, we want the posterior density $f_{Y|K}(y \mid 1)$.
Bayes rule
Using Bayes rule with a continuous $Y$ and discrete $K$,

$$f_{Y|K}(y \mid 1) = \frac{\mathbb{P}(K = 1 \mid Y = y)\, f_Y(y)}{\mathbb{P}(K = 1)}.$$

Here

$$\mathbb{P}(K = 1 \mid Y = y) = y, \qquad f_Y(y) = 1 \text{ on } [0, 1],$$

and

$$\mathbb{P}(K = 1) = \int_0^1 \mathbb{P}(K = 1 \mid Y = u)\, f_Y(u)\, du = \int_0^1 u\, du = \frac{1}{2}.$$

Therefore, for $0 \le y \le 1$,

$$f_{Y|K}(y \mid 1) = \frac{y \cdot 1}{1/2} = 2y,$$

and $f_{Y|K}(y \mid 1) = 0$ otherwise.
Interpretation
Initially, $Y$ was uniform, so all biases in $[0, 1]$ were equally plausible. After observing $K = 1$, the posterior density $f_{Y|K}(y \mid 1) = 2y$ tilts toward larger $y$, reflecting that higher biases make the outcome $K = 1$ more likely.

## 1.6 Further topics on random variables

### 1.6.1 Functions of a discrete random variable

PMF of a general function
Let $X$ be a discrete random variable with PMF $p_X(x)$, and let $Y$ be defined as a function of $X$, say

$$Y = g(X).$$

We wish to find the PMF $p_Y(y)$ of $Y$. For any value $y$ in the range of $Y$, the event $\{Y = y\}$ is the same as the event that $X$ takes any value $x$ such that $g(x) = y$. Therefore,

$$p_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}\big(g(X) = y\big) = \sum_{x:\, g(x)=y} \mathbb{P}(X = x) = \sum_{x:\, g(x)=y} p_X(x).$$

In words, to obtain $p_Y(y)$, we sum the probabilities of all values of $X$ that are mapped to $y$ by the function $g$.
Example: many-to-one mapping
Suppose that $X$ takes the values $2, 3, 4, 5$ with probabilities

$$p_X(2) = 0.1, \quad p_X(3) = 0.2, \quad p_X(4) = 0.3, \quad p_X(5) = 0.4,$$

and define $Y = g(X)$ by specifying $g(2), g(3), g(4), g(5)$. If, for instance, $g(4) = 4$ and $g(5) = 4$, then the probability that $Y$ takes the value 4 is

$$p_Y(4) = \mathbb{P}(Y = 4) = \mathbb{P}(X = 4) + \mathbb{P}(X = 5) = p_X(4) + p_X(5) = 0.3 + 0.4 = 0.7.$$

This illustrates the general rule that whenever several values of $X$ are mapped to the same value $y$, the corresponding probability masses add up in the PMF of $Y$.
Linear functions: stretching and shifting
Consider now the special case of a linear function

$$Y = aX + b,$$

where $a$ and $b$ are constants and $a \neq 0$. For any value $y$, the event $\{Y = y\}$ is the same as the event that $X$ takes the unique value

$$x = \frac{y - b}{a}$$

for which $ax + b = y$, provided that this $x$ lies in the support of $X$. Thus,

$$p_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}\Big(X = \frac{y - b}{a}\Big) = p_X\Big(\frac{y - b}{a}\Big),$$

whenever $(y - b)/a$ is a possible value of $X$, and $p_Y(y) = 0$ otherwise.
Example: scaling
Let $X$ be a discrete random variable taking the values $-1, 1, 2$ with

$$p_X(-1) = \frac{2}{6}, \quad p_X(1) = \frac{1}{6}, \quad p_X(2) = \frac{3}{6}.$$

Define

$$Z = 2X.$$

Then the possible values of $Z$ are

$$-2 \text{ when } X = -1, \quad 2 \text{ when } X = 1, \quad 4 \text{ when } X = 2.$$

Using the general rule,

$$p_Z(-2) = \mathbb{P}(Z = -2) = \mathbb{P}(X = -1) = \frac{2}{6},$$

$$p_Z(2) = \mathbb{P}(Z = 2) = \mathbb{P}(X = 1) = \frac{1}{6},$$

$$p_Z(4) = \mathbb{P}(Z = 4) = \mathbb{P}(X = 2) = \frac{3}{6}.$$

If we plot the PMF of $Z$ as a function of $z$, it has exactly the same shape as the PMF of $X$, except that the mass points have been stretched horizontally by a factor of 2. No probabilities change, only the locations of the mass points on the horizontal axis.

Example: scaling and shifting
Using the previous example, define

$$Y = 2X + 3 = Z + 3.$$

The possible values of $Y$ are obtained by applying this transformation to the values of $X$:

$$X = -1 \Rightarrow Y = 1, \quad X = 1 \Rightarrow Y = 5, \quad X = 2 \Rightarrow Y = 7.$$

The corresponding PMF values are

$$p_Y(1) = \mathbb{P}(Y = 1) = \mathbb{P}(X = -1) = \frac{2}{6},$$
$$p_Y(5) = \mathbb{P}(Y = 5) = \mathbb{P}(X = 1) = \frac{1}{6},$$
$$p_Y(7) = \mathbb{P}(Y = 7) = \mathbb{P}(X = 2) = \frac{3}{6}.$$

Thus the PMF of $Y$ has the same shape as the PMF of $Z$, but all mass points have been shifted to the right by 3. In general, for $Y = aX + b$, the factor $a$ stretches or compresses the horizontal axis, and the constant $b$ shifts the PMF left or right, while the probabilities attached to corresponding mass points remain unchanged.
General formula for a linear function
For a discrete random variable $X$ with PMF $p_X$ and a linear function

$$Y = aX + b, \quad a \neq 0,$$

the PMF of $Y$ is given by

$$p_Y(y) = p_X\Big(\frac{y-b}{a}\Big),$$

for those values of $y$ such that $(y - b)/a$ is in the support of $X$, and $p_Y(y) = 0$ for all other values of $y$. This formula captures algebraically the graphical operations of stretching the PMF of $X$ by the factor $a$ and then shifting it horizontally by $b$.

### 1.6.2 A linear function of a continuous random variable

Heuristic picture
Let $X$ be a continuous random variable with PDF $f_X$. Consider a linear transformation

$$Y = aX + b,$$

where $a$ and $b$ are constants and $a \neq 0$. Geometrically, this transformation takes the horizontal axis, stretches it by a factor of $a$, and then shifts it by $b$. If we think of the PDF as a curve whose area under it must stay equal to 1, then stretching the horizontal axis by a factor $|a|$ forces us to compress the vertical scale by a factor $|a|$ so that the total area remains 1.
Intuitive example
Suppose that $X$ has a piecewise constant PDF supported on the interval $[-1, 1]$. Define

$$Z = 2X.$$

Then $Z$ takes values in $[-2, 2]$. The subinterval $[-1, 0]$ of $X$ corresponds to $[-2, 0]$ for $Z$, and if $f_X$ is constant on $[-1, 0]$, then the values of $X$ in this interval are equally likely in the sense of density, so the corresponding values of $Z$ in $[-2, 0]$ should also have a constant density. Similarly, the subinterval $[0, 1]$ of $X$ corresponds to $[0, 2]$ of $Z$, and if $f_X$ is constant on $[0, 1]$, then $f_Z$ is constant on $[0, 2]$.
Assume, for instance, that the probability that $X$ is positive equals $2/3$, which is the area of a rectangle of width 1 and height $2/3$. When we pass to $Z$, the corresponding probability that $Z$ is positive must still be $2/3$, but now this probability is the area of a rectangle of width 2, so its height must be $1/3$. The same reasoning applied to the negative side shows that if the area for negative $X$ was $1/3$, then the rectangle for negative $Z$ now has width 2, so its height must be $1/6$. Thus, the shape of the PDF is stretched horizontally by a factor of 2, but the heights are scaled down by a factor of 2 to preserve total area 1.
Now add a constant and define

$$Y = Z + 3 = 2X + 3.$$

Then the interval $[-2, 0]$ for $Z$ becomes $[1, 3]$ for $Y$, and $[0, 2]$ becomes $[3, 5]$. The areas of the corresponding rectangles must remain the same, so their heights are unchanged. In other words, adding a constant simply shifts the PDF horizontally without altering its shape or vertical scale.
CDF-based derivation for $a > 0$

We now derive a general formula. Assume first that $a > 0$. The CDF of $Y$ is

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(aX + b \leq y).$$

Solving the inequality for $X$ gives

$$aX + b \leq y \quad \Longleftrightarrow \quad X \leq \frac{y-b}{a}.$$

Therefore,

$$F_Y(y) = \mathbb{P}\Big(X \leq \frac{y-b}{a}\Big) = F_X\Big(\frac{y-b}{a}\Big),$$

where $F_X$ is the CDF of $X$. To obtain the PDF, we differentiate both sides with respect to $y$. Using the chain rule,

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}F_X\Big(\frac{y-b}{a}\Big) = f_X\Big(\frac{y-b}{a}\Big) \cdot \frac{d}{dy}\Big(\frac{y-b}{a}\Big) = \frac{1}{a}\, f_X\Big(\frac{y-b}{a}\Big),$$

for all $y$ such that $(y-b)/a$ lies in the support of $X$, and $f_Y(y) = 0$ otherwise.

CDF-based derivation for $a < 0$

Assume now that $a < 0$. We again start from

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(aX + b \leq y).$$

Solving the inequality for $X$, the sign of the inequality reverses when we divide by the negative constant $a$:

$$aX + b \leq y \quad \Longleftrightarrow \quad X \geq \frac{y-b}{a}.$$

Thus,

$$F_Y(y) = \mathbb{P}\Big(X \geq \frac{y-b}{a}\Big) = 1 - \mathbb{P}\Big(X < \frac{y-b}{a}\Big).$$

Since $X$ is continuous, $\mathbb{P}(X < c) = \mathbb{P}(X \leq c)$ for any constant $c$, so

$$F_Y(y) = 1 - F_X\Big(\frac{y-b}{a}\Big).$$

Differentiating both sides with respect to $y$ and again using the chain rule gives

$$f_Y(y) = \frac{d}{dy}F_Y(y) = -f_X\Big(\frac{y-b}{a}\Big) \cdot \frac{d}{dy}\Big(\frac{y-b}{a}\Big) = -f_X\Big(\frac{y-b}{a}\Big) \cdot \frac{1}{a} = \frac{1}{|a|}\, f_X\Big(\frac{y-b}{a}\Big),$$

because when $a < 0$ we have $-1/a = 1/|a|$.

Unified formula and interpretation

Combining the two cases $a > 0$ and $a < 0$, we obtain the general formula for any $a \neq 0$:

$$f_Y(y) = \frac{1}{|a|}\, f_X\Big(\frac{y-b}{a}\Big),$$

for all $y$ such that $(y-b)/a$ lies in the support of $X$, and $f_Y(y) = 0$ otherwise. This formula encodes three geometric effects of the transformation $Y = aX + b$:

- The argument $(y-b)/a$ inside $f_X$ corresponds to a horizontal stretching or compression by the factor $a$.
- The term $b$ corresponds to a horizontal shift of the PDF by $b$.
- The factor $1/|a|$ corresponds to a vertical scaling that ensures the area under $f_Y$ is equal to 1.

Comparison with the discrete case

In the discrete case, for $Y = aX + b$ we had

$$p_Y(y) = p_X\Big(\frac{y-b}{a}\Big),$$

without any extra scaling factor. This reflects the fact that in the discrete setting we preserve the individual point probabilities exactly, and there is no notion of stretching intervals and adjusting heights to keep total probability equal to 1. In the continuous setting, probabilities are represented by areas under the PDF, so when we stretch the horizontal axis by $|a|$, we must divide the height of the curve by $|a|$ to keep the total area unchanged.

### 1.6.3 A linear function of a normal random variable

Setup

Let $X$ be a normal random variable with mean $\mu$ and variance $\sigma^2$, so

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

and its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left( -\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 \right).$$

Consider the linear transformation

$$Y = aX + b,$$

where $a$ and $b$ are constants with $a \neq 0$.

Using the general transformation formula

From the general result for linear functions of a continuous random variable, the PDF of $Y$ is given by

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).$$

Substituting the normal PDF of $X$ into this expression gives

$$f_Y(y) = \frac{1}{|a|} \cdot \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left( -\frac{1}{2}\left(\frac{\frac{y-b}{a} - \mu}{\sigma}\right)^2 \right).$$

We now simplify the expression inside the exponential and the prefactor. First rewrite

$$\frac{\frac{y-b}{a} - \mu}{\sigma} = \frac{y - b - a\mu}{a\sigma},$$

so the exponent becomes

$$-\frac{1}{2}\left(\frac{y-b-a\mu}{a\sigma}\right)^2 = -\frac{1}{2}\left(\frac{y-(a\mu+b)}{|a|\sigma}\right)^2,$$

because the square absorbs the sign of $a$.

For the prefactor, we have

$$\frac{1}{|a|} \cdot \frac{1}{\sqrt{2\pi}\,\sigma} = \frac{1}{\sqrt{2\pi}\,|a|\sigma} = \frac{1}{\sqrt{2\pi}\,\sqrt{a^2\sigma^2}}.$$

Putting these together,

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sqrt{a^2\sigma^2}} \exp\left( -\frac{1}{2}\left(\frac{y-(a\mu+b)}{\sqrt{a^2\sigma^2}}\right)^2 \right).$$

Identification of the resulting distribution

The last expression has exactly the standard normal PDF form in the variable $y$, with mean

$$\mu_Y = a\mu + b$$

and variance

$$\sigma_Y^2 = a^2\sigma^2.$$

Therefore,

$$Y = aX + b \sim \mathcal{N}(a\mu + b,\ a^2\sigma^2).$$

The fact that the mean and variance transform as $E[Y] = a\mu + b$ and $\mathrm{Var}(Y) = a^2\sigma^2$ agrees with the general formulas for linear functions of any random variable. The special feature here is that the entire distributional shape is preserved: the bell-shaped normal PDF of $X$ is stretched and shifted, but remains bell shaped, so the resulting random variable $Y$ is again normal.

### 1.6.4 The PDF of a general function of a continuous r.v.

TWO-STEP METHOD
Let $X$ be a continuous random variable with known PDF $f_X$ and CDF $F_X$, and let

$$Y = g(X)$$

for some (possibly nonlinear) function $g$. To find the PDF of $Y$, we use a two-step procedure:

1. First find the CDF of $Y$,
$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\big(g(X) \le y\big),$$
by rewriting the event $\{g(X) \le y\}$ in terms of inequalities involving $X$ and then computing that probability using the known distribution of $X$.

2. Then obtain the PDF by differentiation,
$$f_Y(y) = \frac{d}{dy} F_Y(y),$$
for those values of $y$ in the range of $Y$.

Most of the work lies in the first step, where we translate the condition $g(X) \le y$ into a suitable interval (or union of intervals) for $X$ and then evaluate the corresponding probability.
EXAMPLE: $Y = X^3$ WITH $X \sim$ UNIFORM$(0, 2)$
Suppose that $X$ is uniform on $[0, 2]$, so its PDF is

$$f_X(x) = \begin{cases} \frac{1}{2}, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$Y = X^3.$$

Since $X$ takes values in $[0, 2]$, the range of $Y$ is $[0, 8]$. For $y < 0$ we have $F_Y(y) = 0$, and for $y > 8$ we have $F_Y(y) = 1$, so we focus on $0 \le y \le 8$.
For such $y$,

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}(X^3 \le y).$$

On $[0, 2]$, the function $x \mapsto x^3$ is increasing, so the inequality $X^3 \le y$ is equivalent to

$$X \le y^{1/3}.$$

Thus,

$$F_Y(y) = \mathbb{P}\big(X \le y^{1/3}\big) = \int_0^{y^{1/3}} \frac{1}{2}\, dx = \frac{1}{2}\, y^{1/3}, \quad 0 \le y \le 8.$$

This completes the first step. For the second step, we differentiate:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy}\Big(\frac{1}{2} y^{1/3}\Big) = \frac{1}{2} \cdot \frac{1}{3} y^{-2/3} = \frac{1}{6}\, y^{-2/3}, \quad 0 < y \le 8.$$

For $y$ outside $[0, 8]$, we have $f_Y(y) = 0$. So the PDF of $Y$ is

$$f_Y(y) = \begin{cases} \frac{1}{6}\, y^{-2/3}, & 0 < y \le 8, \\ 0, & \text{otherwise.} \end{cases}$$

This PDF is not constant: it becomes larger as $y$ approaches 0, and in fact $f_Y(y)$ tends to infinity as $y \downarrow 0$, while still integrating to 1 over $(0, 8]$.
EXAMPLE: $Y = 10/X$ WITH $X \sim$ UNIFORM$(5, 10)$
Now let $X$ be uniform on $[5, 10]$, so that

$$f_X(x) = \begin{cases} \frac{1}{5}, & 5 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

Define the random variable

$$Y = \frac{10}{X}.$$

Since $X \in [5, 10]$, the smallest value of $Y$ occurs when $X$ is largest, $X = 10$, giving $Y = 1$, and the largest value of $Y$ occurs when $X$ is smallest, $X = 5$, giving $Y = 2$. Thus, the range of $Y$ is $[1, 2]$, and $f_Y(y) = 0$ outside this interval. For $1 \le y \le 2$,

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\Big(\frac{10}{X} \le y\Big).$$

Because $X > 0$, the inequality $\frac{10}{X} \le y$ is equivalent to

$$10 \le yX \quad \Longleftrightarrow \quad X \ge \frac{10}{y}.$$

So

$$F_Y(y) = \mathbb{P}\Big(X \ge \frac{10}{y}\Big).$$

For $1 \le y \le 2$, the point $10/y$ lies in $[5, 10]$, so we can express this probability as the area under the uniform density from $10/y$ to 10:

$$F_Y(y) = \int_{10/y}^{10} \frac{1}{5}\, dx = \frac{1}{5}\Big(10 - \frac{10}{y}\Big) = \frac{10}{5}\Big(1 - \frac{1}{y}\Big) = 2\Big(1 - \frac{1}{y}\Big), \quad 1 \le y \le 2.$$

Outside this interval, $F_Y(y)$ is 0 for $y < 1$ and 1 for $y > 2$.
To obtain the PDF, we differentiate on $(1, 2)$:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy}\Big(2 - \frac{2}{y}\Big) = 0 - 2 \cdot \Big(-\frac{1}{y^2}\Big) = \frac{2}{y^2}, \quad 1 < y < 2.$$

Thus,

$$f_Y(y) = \begin{cases} \dfrac{2}{y^2}, & 1 < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

The PDF starts at a value of 2 when $y$ is close to 1 and decreases as $y$ increases, reflecting that shorter times (corresponding to higher speeds) are more likely than longer times in this setup.
General strategy
These examples illustrate a general and systematic approach to finding the PDF of a function $Y = g(X)$ of a continuous random variable $X$:

- Identify the range of $Y$, so that you know where $f_Y(y)$ can be nonzero.
- For $y$ in this range, express the event $\{Y \le y\}$ as an event involving $X$, typically of the form $\{X \le h(y)\}$, $\{X \ge h(y)\}$, or a union of such intervals.
- Compute $F_Y(y)$ by integrating the known PDF $f_X$ over the appropriate interval(s) for $X$.
- Differentiate $F_Y(y)$ with respect to $y$ to obtain $f_Y(y)$.

In more complicated situations where $g$ is not monotone on the entire support of $X$, one splits the support of $X$ into regions on which $g$ is monotone, computes the contribution from each region, and then adds them up to obtain the CDF or PDF of $Y$.

### 1.6.5 The monotonic case

Setup and inverse function
Let $X$ be a continuous random variable with PDF $f_X$, and let

$$Y = g(X),$$

where $g$ is a strictly monotonic and differentiable function on the range of interest of $X$. Because $g$ is strictly monotonic, every value $y$ in the range of $Y$ comes from a unique value $x$, so $g$ has an inverse function $h$ satisfying

$$x = h(y) \quad \Longleftrightarrow \quad y = g(x).$$

The function $h$ maps values of $y$ back to the corresponding values of $x$.
Strictly increasing case
Assume first that $g$ is strictly increasing. Fix some $y$ in the range of $Y$. The CDF of $Y$ is

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\big(g(X) \le y\big).$$

Since $g$ is increasing, the inequality $g(X) \leq y$ is equivalent to

$$X \leq h(y),$$

because $h(y)$ is the unique value of $x$ that produces $y$. Therefore,

$$F_Y(y) = \mathbb{P}\big(X \leq h(y)\big) = F_X\big(h(y)\big),$$

where $F_X$ is the CDF of $X$. To obtain the PDF, we differentiate both sides with respect to $y$ and apply the chain rule:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X\big(h(y)\big) = f_X\big(h(y)\big)\, h'(y),$$

for $y$ in the range of $Y$, and $f_Y(y) = 0$ otherwise. When $g$ is increasing, the inverse $h$ is also increasing, so $h'(y) \geq 0$.

Strictly decreasing case

Assume now that $g$ is strictly decreasing. Again fix $y$ in the range of $Y$, and start from

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}\big(g(X) \leq y\big).$$

Because $g$ is decreasing, the inequality $g(X) \leq y$ holds when $X$ is larger than or equal to the value $h(y)$ that produces $y$, so

$$F_Y(y) = \mathbb{P}\big(X \geq h(y)\big).$$

This can be written in terms of the CDF of $X$ as

$$F_Y(y) = 1 - \mathbb{P}\big(X < h(y)\big).$$

Since $X$ is continuous, $\mathbb{P}(X < h(y)) = \mathbb{P}(X \leq h(y)) = F_X\big(h(y)\big)$, and thus

$$F_Y(y) = 1 - F_X\big(h(y)\big).$$

Differentiating both sides with respect to $y$ gives

$$f_Y(y) = \frac{d}{dy} F_Y(y) = -f_X\big(h(y)\big)\, h'(y).$$

In the decreasing case, $h$ is also decreasing, so $h'(y) \leq 0$. Thus the product $-h'(y)$ is nonnegative, as a PDF must be.

Unified formula

In the increasing case we have

$$f_Y(y) = f_X\big(h(y)\big)\, h'(y),$$

and in the decreasing case we have

$$f_Y(y) = -f_X\big(h(y)\big)\, h'(y).$$

These two expressions can be combined into a single formula using absolute values:

$$f_Y(y) = f_X\big(h(y)\big)\, \big|h'(y)\big|,$$

for $y$ in the range of $Y$, and $f_Y(y) = 0$ otherwise. This is the general transformation rule for the PDF of a strictly monotonic and differentiable function of a continuous random variable.

Example: $Y = X^2$ with $X \sim$ Uniform$(0, 1)$

Let $X$ be uniform on $[0, 1]$, so

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise}, \end{cases}$$

and consider

$$Y = g(X) = X^2.$$

The function $g(x) = x^2$ is not monotonic on the entire real line, but it is strictly increasing on $[0, 1]$, which is the support of $X$. Therefore we can apply the monotonic transformation formula on this interval.

First identify the inverse function. On $[0, 1]$,

$$y = x^2 \quad \Longleftrightarrow \quad x = \sqrt{y},$$

so

$$h(y) = \sqrt{y}.$$

Since $X \in [0,1]$, $Y = X^2$ also takes values in $[0,1]$, and outside this interval we must have $f_Y(y) = 0$. For $0 < y < 1$,

$$f_Y(y) = f_X\big(h(y)\big)\,\big|h'(y)\big| = f_X(\sqrt{y}) \cdot \Big|\frac{d}{dy}\sqrt{y}\Big|.$$

On the support of $X$, we have $f_X(\sqrt{y}) = 1$, and

$$\frac{d}{dy}\sqrt{y} = \frac{1}{2\sqrt{y}}.$$

Thus,

$$f_Y(y) = 1 \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1.$$

Including the behavior outside $[0,1]$, the PDF of $Y$ is

$$f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

This result is obtained with much less work than going through the full two-step CDF method, because we only need to identify the inverse function $h$ and its derivative.

Matching small intervals

Let $Y = g(X)$, where $g$ is a strictly monotonic and differentiable function, and let $h$ be its inverse, so that

$$x = h(y) \quad \Longleftrightarrow \quad y = g(x).$$

Fix a particular value $x$ and the corresponding value $y = g(x)$. Consider a small interval around $x$, say $[x, x + \Delta_1]$, and look at the corresponding interval for $Y$, which is approximately $[y, y + \Delta_2]$, where $\Delta_2$ is the small change in $y$ that results from the change $\Delta_1$ in $x$. Because the mapping $y = g(x)$ is one-to-one, the event

$$\{X \in [x, x + \Delta_1]\}$$

is the same as the event

$$\{Y \in [y, y + \Delta_2]\},$$

so these two events have the same probability:

$$\mathbb{P}\big(y \le Y \le y + \Delta_2\big) = \mathbb{P}\big(x \le X \le x + \Delta_1\big).$$

Approximating probabilities with PDFs

For small intervals, probabilities can be approximated using the PDFs. On the $y$-side,

$$\mathbb{P}\big(y \le Y \le y + \Delta_2\big) \approx f_Y(y)\,\Delta_2,$$

and on the $x$-side,

$$\mathbb{P}\big(x \le X \le x + \Delta_1\big) \approx f_X(x)\,\Delta_1.$$

Equating these two approximations gives

$$f_Y(y)\,\Delta_2 \approx f_X(x)\,\Delta_1.$$

To turn this into a relation between $f_Y$ and $f_X$, we need to relate $\Delta_1$ and $\Delta_2$.

Relating the interval lengths via the inverse

The change $\Delta_2$ in $y$ is caused by the change $\Delta_1$ in $x$ through the function $g$. Equivalently, we can look at how a small change $\Delta_2$ in $y$ produces a change $\Delta_1$ in $x$ through the inverse function $h$. When $y$ advances from $y$ to $y + \Delta_2$, the corresponding $x$ advances from $x$ to $x + \Delta_1$, so

$$\Delta_1 \approx h'(y)\,\Delta_2,$$

where $h'(y)$ is the derivative (slope) of the inverse function at $y$.

Substituting this into the earlier approximate equality yields

$$f_Y(y)\,\Delta_2 \approx f_X(x)\,h'(y)\,\Delta_2.$$

Canceling $\Delta_2$ from both sides gives

$$f_Y(y) \approx f_X(x)\,h'(y),$$

and in the limit of vanishingly small intervals this becomes the exact relation

$$f_Y(y) = f_X\big(h(y)\big)\, h'(y),$$

where $x = h(y)$ is the unique value of $X$ that maps to $y$.

Absolute value and decreasing functions

If $g$ is decreasing instead of increasing, the same small-interval argument still applies. The difference is that in this case the inverse function $h$ is also decreasing, so $h'(y) \le 0$. The probability of a small interval in $y$ must still be equal to the probability of the corresponding interval in $x$, so the same algebra leads to

$$f_Y(y) = f_X\big(h(y)\big)\, \big|h'(y)\big|.$$

This absolute value ensures that the PDF $f_Y$ is always nonnegative, regardless of whether $g$ (and hence $h$) is increasing or decreasing. Thus, the intuitive picture is that small intervals on the $x$-axis are stretched or compressed when mapped to the $y$-axis, and the PDFs adjust inversely to this stretching so that the probabilities of corresponding small intervals remain the same.

### 1.6.6 A function of multiple random variables

General methodology

Let $X$ and $Y$ be jointly continuous random variables with known joint PDF $f_{X,Y}(x, y)$. Suppose we define a new random variable

$$Z = g(X, Y),$$

for some function $g$. To find the distribution of $Z$, we follow the same two-step procedure used for functions of a single random variable:

1. First find the CDF of $Z$,

$$F_Z(z) = \mathbb{P}(Z \le z) = \mathbb{P}\big(g(X, Y) \le z\big) = \iint_{\{(x,y):\, g(x,y) \le z\}} f_{X,Y}(x, y)\, dx\, dy.$$

2. Then obtain the PDF by differentiating,

$$f_Z(z) = \frac{d}{dz} F_Z(z),$$

for those values of $z$ in the range of $Z$.

In many examples, step one becomes a geometric problem: describe the region $\{(x, y) : g(x, y) \le z\}$ and integrate $f_{X,Y}$ over that region.

Example: $Z = Y/X$ with $(X, Y)$ uniform on the unit square

Consider $X$ and $Y$ independent and each uniform on $[0, 1]$. Their joint PDF is

$$f_{X,Y}(x, y) = \begin{cases} 1, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

We define

$$Z = \frac{Y}{X}$$

and seek the CDF and PDF of $Z$.

CDF for negative and small positive $z$

The CDF is

$$F_Z(z) = \mathbb{P}(Z \le z) = \mathbb{P}\Big(\frac{Y}{X} \le z\Big).$$

If $z < 0$, then $\frac{Y}{X} \ge 0$ always, because $X$ and $Y$ are nonnegative, so

$$F_Z(z) = 0, \quad z < 0.$$

Now consider $0 \le z \le 1$. The inequality $\frac{Y}{X} \le z$ is equivalent to

$$Y \le zX.$$

On the $(x, y)$-plane, this is the region below the line $y = zx$. When $0 \le z \le 1$, this line intersects the top edge $y = 1$ at $x = 1/z \ge 1$, so over the unit square it intersects only the right edge at $(1, z)$. Thus, the region $\{(x, y) : 0 \le x \le 1,\ 0 \le y \le zx\}$ is a right triangle with base 1 and height $z$, so its area is

$$\text{area} = \frac{1}{2} \cdot 1 \cdot z = \frac{z}{2}.$$

Since the joint density is 1 on the unit square, this area equals the probability, and therefore

$$F_Z(z) = \frac{z}{2}, \quad 0 \le z \le 1.$$

CDF FOR $z > 1$
Now consider $z > 1$. Again, the event $\frac{Y}{X} \le z$ corresponds to the region below the line $y = zx$. When $z > 1$, this line intersects the top edge $y = 1$ at $x = 1/z < 1$, so it cuts off a small triangle near the origin where $y > zx$. The region where $Y/X \le z$ is then the whole unit square minus that triangle.
The small excluded triangle has base $1/z$ on the $x$-axis and height 1 on the $y$-axis, so its area is

$$\text{area}_{\text{triangle}} = \frac{1}{2} \cdot \frac{1}{z} \cdot 1 = \frac{1}{2z}.$$

Thus, the area of the region where $Y/X \le z$ is

$$1 - \frac{1}{2z},$$

and consequently

$$F_Z(z) = 1 - \frac{1}{2z}, \quad z > 1.$$

Putting all cases together, the CDF of $Z$ is

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ \frac{z}{2}, & 0 \le z \le 1, \\ 1 - \frac{1}{2z}, & z > 1. \end{cases}$$

PDF OF $Z$
To obtain the PDF, we differentiate $F_Z(z)$ piecewise.
For $z < 0$, $F_Z(z)$ is constant, so

$$f_Z(z) = 0, \quad z < 0.$$

For $0 < z < 1$,

$$F_Z(z) = \frac{z}{2} \quad \Longrightarrow \quad f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{1}{2}.$$

For $z > 1$,

$$F_Z(z) = 1 - \frac{1}{2z} \quad \Longrightarrow \quad f_Z(z) = \frac{d}{dz}\left(1 - \frac{1}{2z}\right) = \frac{1}{2z^2}.$$

At the transition points $z = 0$ and $z = 1$, the PDF can be defined by continuity on either side, but this does not affect probabilities because these are single points. Thus, the PDF of $Z = Y/X$ is

$$f_Z(z) = \begin{cases} 0, & z < 0, \\ \frac{1}{2}, & 0 < z < 1, \\ \frac{1}{2z^2}, & z > 1. \end{cases}$$

INTERPRETATION
The distribution of $Z = Y/X$ arises from the geometry of the uniform joint distribution on the unit square. For small positive $z$, the allowed region where $Y/X \le z$ is a triangle whose area grows linearly with $z$, giving a constant PDF on $(0, 1)$. For larger $z$, the allowed region is the unit square minus a shrinking triangle near the origin, whose area decreases like $1/z$, and the corresponding PDF decays like $1/z^2$.

### 1.6.7 Convolution of PMFs for sums of independent random variables

Let $X$ and $Y$ be independent discrete random variables with known PMFs $p_X(x)$ and $p_Y(y)$. We define a new random variable

$$Z = X + Y$$

and we want to find the PMF $p_Z(z) = \mathbb{P}(Z = z)$.
BASIC IDEA
The event $\{Z = z\}$ means that the pair $(X, Y)$ takes values that add up to $z$. There are many ways for this to happen; for example, if $z = 3$ then we could have $(X, Y) = (0, 3)$, or $(1, 2)$, or $(2, 1)$, and so on. The probability that $Z$ equals

$z$ is obtained by adding the probabilities of all such pairs. Because $X$ and $Y$ are independent, the probability of any particular pair $(x, y)$ is the product $p_X(x)\, p_Y(y)$. This leads to a systematic formula.

Convolution formula

Fix an integer $z$. For each possible value $x$ of $X$, the value of $Y$ must be $z - x$ in order for the sum to equal $z$. Thus

$$\mathbb{P}(Z = z) = \sum_x \mathbb{P}\big(X = x,\ Y = z - x\big).$$

Independence gives

$$\mathbb{P}\big(X = x,\ Y = z - x\big) = \mathbb{P}(X = x)\, \mathbb{P}(Y = z - x) = p_X(x)\, p_Y(z - x).$$

Therefore the PMF of $Z$ is

$$p_Z(z) = \sum_x p_X(x)\, p_Y(z - x),$$

where the sum runs over all values of $x$ for which both terms in the product are defined and nonzero. This operation on PMFs is called *convolution*. Given $p_X$ and $p_Y$, the convolution formula produces a new PMF $p_Z$ corresponding to the sum $Z = X + Y$.

Example

Suppose $X$ takes values 1 and 4 with

$$p_X(1) = \frac{1}{3}, \qquad p_X(4) = \frac{2}{3},$$

and $Y$ takes values $-2, -1, 1$ with

$$p_Y(-2) = \frac{3}{6}, \qquad p_Y(-1) = \frac{2}{6}, \qquad p_Y(1) = \frac{1}{6},$$

and $X$ and $Y$ are independent. To compute $p_Z(3)$ where $Z = X + Y$, list all pairs $(x, y)$ with $x + y = 3$. These are

$$(x, y) = (1, 2) \ \textit{(impossible here)}, \quad (4, -1).$$

In this particular example the only feasible pair is $(4, -1)$ because $Y$ never takes the value 2. Thus

$$p_Z(3) = \mathbb{P}(X = 4,\ Y = -1) = p_X(4)\, p_Y(-1) = \frac{2}{3} \cdot \frac{2}{6} = \frac{4}{18} = \frac{2}{9}.$$

If instead $Y$ could also take the value 2 with some probability, say $p_Y(2)$, then

$$p_Z(3) = p_X(1)\, p_Y(2) + p_X(4)\, p_Y(-1) = \frac{1}{3} p_Y(2) + \frac{2}{3} \cdot \frac{2}{6}.$$

Graphical interpretation

A useful graphical method to remember the convolution formula is to work with the PMFs drawn as stem plots on the number line. One can reflect the PMF of $Y$ about the vertical axis to obtain a plot of $p_Y(-y)$. Then shift this reflected plot by an amount $z$ so that the point corresponding to $y$ lines up with the point $x = z - y$. For a given shift $z$, the probability $p_Z(z)$ is obtained by multiplying corresponding vertical stems that line up and summing all these products. Changing $z$ corresponds to sliding the reflected PMF left or right and repeating the same multiplication and summation. This picture mirrors exactly the algebraic convolution sum $p_Z(z) = \sum_x p_X(x)\, p_Y(z - x)$.

### 1.6.8 Convolution for sums of independent continuous random variables

Let $X$ and $Y$ be independent continuous random variables with PDFs $f_X(x)$ and $f_Y(y)$. Define

$$Z = X + Y.$$

We want to find the PDF $f_Z(z)$ of the sum $Z$.

Heuristic derivation

For a fixed value $z$, the event $\{Z = z\}$ corresponds to all pairs $(x, y)$ satisfying $x + y = z$. In the discrete case, to obtain the PMF we add the probabilities of all pairs on that line and get a sum (the discrete convolution). In the continuous case, there are infinitely many pairs on the line $x + y = z$, so the "sum over ways" becomes an integral and PMFs are replaced by PDFs. This leads to the continuous convolution formula

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx,$$

where we integrate over all possible values of $x$ and for each such $x$ the corresponding $y$ is $z - x$.

Conditional-density derivation

A more formal justification starts by conditioning on $X$. Fix a value $x$ of $X$; then

$$Z = X + Y = x + Y,$$

so given $X = x$, the random variable $Z$ is just $Y$ shifted by $x$. For a linear transformation $U = Y + b$, the PDF satisfies

$$f_U(u) = f_Y(u - b),$$

because shifting a continuous random variable by a constant simply shifts its density. Thus, for each fixed $x$,

$$f_{Z|X}(z \mid x) = f_Y(z - x).$$

Now use the multiplication rule for densities to get the joint PDF of $(X, Z)$. We have

$$f_{X,Z}(x, z) = f_X(x)\, f_{Z|X}(z \mid x) = f_X(x)\, f_Y(z - x).$$

To obtain the marginal density of $Z$, integrate out $x$ from the joint density:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Z}(x, z)\, dx = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx.$$

This is exactly the continuous convolution formula.

Graphical interpretation

The mechanics of using this formula mirror those of the discrete case. To visualize the integral, one may take the graph of $f_Y(y)$, reflect it horizontally to obtain $f_Y(-y)$, and then shift it by an amount $z$ to form the function $y \mapsto f_Y(z - y)$. For a fixed $z$, the integrand $f_X(x)\, f_Y(z - x)$ multiplies the height of the $f_X$ graph at $x$ by the height of the shifted-reflected $f_Y$ graph at the same $x$. Integrating over all $x$ adds up these products and yields $f_Z(z)$.

### 1.6.9 Sum of independent normal random variables

Let $X$ and $Y$ be independent normal random variables. Assume

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \qquad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2).$$

We define

$$Z = X + Y$$

and we want to find the PDF and distribution of $Z$.

Using the convolution formula

Because $X$ and $Y$ are independent and continuous, the PDF of $Z$ is given by the convolution

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx.$$

For normal $X$ and $Y$,

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma_X} \exp\left(-\frac{(x - \mu_X)^2}{2\sigma_X^2}\right),$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma_Y} \exp\left(-\frac{(y - \mu_Y)^2}{2\sigma_Y^2}\right).$$

In the convolution, $y$ is replaced by $z - x$, so

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\sigma_X} \exp\left(-\frac{(x - \mu_X)^2}{2\sigma_X^2}\right) \cdot \frac{1}{\sqrt{2\pi}\,\sigma_Y} \exp\left(-\frac{(z - x - \mu_Y)^2}{2\sigma_Y^2}\right) dx.$$

The integrand is a product of two exponentials in $x$. By expanding the squares in the exponents and collecting terms in $x$, one can rewrite the product as a single Gaussian in $x$ times a factor that depends only on $z$. Integrating the Gaussian in $x$ over the real line yields 1, leaving a remaining factor in $z$ that has the form of a normal PDF.

Resulting distribution
The final result is that $Z$ is again normal with

$$Z \sim \mathcal{N}(\mu_X + \mu_Y,\ \sigma_X^2 + \sigma_Y^2).$$

Equivalently, the PDF of $Z$ can be written as

$$f_Z(z) = \frac{1}{\sqrt{2\pi(\sigma_X^2 + \sigma_Y^2)}} \exp\left( - \frac{\big(z - (\mu_X + \mu_Y)\big)^2}{2(\sigma_X^2 + \sigma_Y^2)} \right).$$

The mean of the sum is the sum of the means and the variance of the sum is the sum of the variances.

Extension to more than two variables
Now consider independent normal random variables

$$X_1, X_2, \ldots, X_n, \qquad X_i \sim \mathcal{N}(\mu_i, \sigma_i^2).$$

From the two-variable result, the sum $X_1 + X_2$ is normal. Because $X_3$ is independent of $(X_1 + X_2)$ and normal, the sum $(X_1 + X_2) + X_3$ is also normal. Continuing inductively, any finite sum

$$S_n = X_1 + X_2 + \cdots + X_n$$

is normal with

$$S_n \sim \mathcal{N}\left( \sum_{i=1}^n \mu_i,\ \sum_{i=1}^n \sigma_i^2 \right).$$

Thus the class of normal distributions is closed under finite sums of independent variables, which makes normal models particularly convenient when studying linear combinations of random signals or errors.

### 1.6.10 Covariance and dependence

Consider two random variables $X$ and $Y$ with means 0. If $X$ and $Y$ are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Now imagine a discrete joint distribution where points with $X > 0$ tend to occur together with $Y > 0$, and points with $X < 0$ tend to occur together with $Y < 0$. Most outcomes then fall in the quadrants where the product $XY$ is positive, so the average value $\mathbb{E}[XY]$ is positive. In contrast, if typical outcomes lie in quadrants where $X$ and $Y$ have opposite signs, then $XY$ is usually negative and $\mathbb{E}[XY]$ becomes negative. Thus, in the zero-mean case, the sign of $\mathbb{E}[XY]$ tells us whether $X$ and $Y$ tend to move in the same direction or in opposite directions.

General definition
For general random variables $X$ and $Y$ (discrete or continuous) with possibly nonzero means, the covariance is defined as

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])\,(Y - \mathbb{E}[Y])\big].$$

Here $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$ are the deviations from their respective mean values. If the covariance is positive, then when $X$ is typically above its mean, $Y$ also tends to be above its mean, and when $X$ is below its mean, $Y$ tends to be below its mean. If the covariance is negative, then $X$ being above its mean is usually accompanied by $Y$ being below its mean, and vice versa. In summary, the covariance measures whether two random variables tend to be simultaneously high or low in an average sense.

Independence and zero covariance
If $X$ and $Y$ are independent, then

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])\,(Y - \mathbb{E}[Y])\big] = \mathbb{E}[X - \mathbb{E}[X]]\ \mathbb{E}[Y - \mathbb{E}[Y]] = 0,$$

because $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$ are independent and each has expectation 0. Thus independence implies zero covariance. However, the converse is not true: there exist dependent random variables whose covariance is zero.

Example of dependence with zero covariance
Consider a discrete example with four possible outcomes arranged so that at every outcome either $X = 0$ or $Y = 0$. In this construction, the product $XY$ is always equal to 0, so

$$\mathbb{E}[XY] = 0.$$

By symmetry, the means of $X$ and $Y$ are also 0, so

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Nevertheless, $X$ and $Y$ are not independent. For instance, if you are told that $X = 1$, you immediately know which outcome occurred and can deduce that $Y = 0$ with certainty, so knowledge of $X$ reveals information about $Y$.

### 1.6.11 Covariance: basic algebraic properties

For any random variable $X$, the covariance of $X$ with itself is

$$\mathrm{Cov}(X, X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \mathrm{Var}(X).$$

Thus variance is a special case of covariance.

Alternative formula
Starting from the definition

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])\,(Y - \mathbb{E}[Y])\big]$$

and expanding the product gives

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[X]\,\mathbb{E}[Y] + \mathbb{E}[X]\,\mathbb{E}[Y].$$

Using linearity of expectation and pulling constants outside, the middle two terms cancel, leaving

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].$$

This identity often simplifies covariance calculations.

Effect of linear transformations
Consider a linear function of $X$,

$$U = aX + b$$

and another random variable $Y$. Using the definition of covariance and linearity of expectation, one can show

$$\mathrm{Cov}(aX + b,\ Y) = a\ \mathrm{Cov}(X, Y).$$

Multiplying $X$ by $a$ scales the covariance by $a$, while adding a constant $b$ does not change the covariance.

Linearity in one argument
For random variables $X$, $Y$ and $Z$ (assume for simplicity that they have finite second moments),

$$\mathrm{Cov}\big(X,\ Y + Z\big) = \mathbb{E}\big[X(Y + Z)\big] - \mathbb{E}[X]\,\mathbb{E}[Y + Z].$$

Expanding both terms and collecting gives

$$\mathrm{Cov}\big(X,\ Y + Z\big) = \mathrm{Cov}(X, Y) + \mathrm{Cov}(X, Z).$$

So covariance is linear in each argument when the other is held fixed.

### 1.6.12 Variance of a sum of random variables

Let $X_1$ and $X_2$ be random variables and consider their sum

$$S = X_1 + X_2.$$

By definition,

$$\mathrm{Var}(S) = \mathbb{E}\Big[\big(S - \mathbb{E}[S]\big)^2\Big] = \mathbb{E}\Big[\big((X_1 + X_2) - (\mathbb{E}[X_1] + \mathbb{E}[X_2])\big)^2\Big].$$

Rewriting inside the square,

$$S - \mathbb{E}[S] = (X_1 - \mathbb{E}[X_1]) + (X_2 - \mathbb{E}[X_2]).$$

Expanding the square gives

$$\big(S - \mathbb{E}[S]\big)^2 = (X_1 - \mathbb{E}[X_1])^2 + (X_2 - \mathbb{E}[X_2])^2 + 2\,(X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2]).$$

Taking expectations and using linearity,

$$\mathrm{Var}(S) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\ \mathrm{Cov}(X_1, X_2).$$

If $X_1$ and $X_2$ are independent, then $\mathrm{Cov}(X_1, X_2) = 0$ and

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2).$$

Many random variables

Now consider $n$ random variables $X_1, \ldots, X_n$ and their sum

$$S_n = X_1 + X_2 + \cdots + X_n.$$

For simplicity assume $\mathbb{E}[X_i] = 0$, so

$$\mathrm{Var}(S_n) = \mathbb{E}\big[S_n^2\big] = \mathbb{E}\Big[\Big(\sum_{i=1}^{n} X_i\Big)^2\Big].$$

Expanding the square,

$$S_n^2 = \sum_{i=1}^{n} X_i^2 + \sum_{\substack{i,j=1 \\ i\neq j}}^{n} X_i X_j.$$

Taking expectations,

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathbb{E}[X_i^2] + \sum_{\substack{i,j=1 \\ i\neq j}}^{n} \mathbb{E}[X_i X_j].$$

Because $\mathbb{E}[X_i] = 0$, each $\mathbb{E}[X_i^2] = \mathrm{Var}(X_i)$ and each $\mathbb{E}[X_i X_j] = \mathrm{Cov}(X_i, X_j)$. Thus

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + \sum_{\substack{i,j=1 \\ i\neq j}}^{n} \mathrm{Cov}(X_i, X_j).$$

Equivalently, one can write

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \sum_{j=1}^{n} \mathrm{Cov}(X_i, X_j),$$

since the diagonal terms $\mathrm{Cov}(X_i, X_i)$ are exactly the variances. If all covariances are zero, then the variance of the sum reduces to the sum of the individual variances. This happens in particular when the $X_i$ are independent, but the formula itself holds regardless of whether they are independent or have nonzero covariances.

### 1.6.13 The correlation coefficient

The covariance $\mathrm{Cov}(X, Y)$ indicates whether two random variables tend to move together, but its numerical value is hard to interpret and it depends on the measurement units. If $X$ and $Y$ are measured in meters, then the covariance has units of meters$^2$, which makes comparisons across problems or unit systems difficult. To obtain a dimensionless and more interpretable measure of linear dependence, we normalize the covariance by the standard deviations of the two variables.

Definition and basic properties

For random variables $X$ and $Y$ with standard deviations $\sigma_X > 0$ and $\sigma_Y > 0$, the correlation coefficient is defined as

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \, \sigma_Y}.$$

Because the numerator has the same units as the product $\sigma_X \sigma_Y$, the ratio $\rho_{X,Y}$ is dimensionless. If either standard deviation is zero, then the corresponding variable is almost surely constant and the correlation coefficient is not defined because the denominator would be zero. A fundamental fact is that

$$-1 \leq \rho_{X,Y} \leq 1,$$

which provides an absolute scale for judging the strength of linear association between $X$ and $Y$.

Interpretation and extreme cases

If $X$ and $Y$ are independent and have finite variances, then $\mathrm{Cov}(X, Y) = 0$ and

$$\rho_{X,Y} = 0,$$

and in this case the variables are said to be *uncorrelated*. The converse is not generally true: there exist dependent random variables with zero covariance and therefore zero correlation. At the other extreme, consider the correlation coefficient of a random variable with itself:

$$\rho_{X,X} = \frac{\mathrm{Cov}(X, X)}{\sigma_X^2} = \frac{\mathrm{Var}(X)}{\sigma_X^2} = 1.$$

Similarly, the correlation coefficient of $X$ with $-X$ is $-1$. More generally, if there is a deterministic linear relationship between $X$ and $Y$ of the form

$$Y = aX + b$$

with probability one, then

$$|\rho_{X,Y}| = 1,$$

with $\rho_{X,Y} = 1$ when $a > 0$ and $\rho_{X,Y} = -1$ when $a < 0$. Conversely, if $|\rho_{X,Y}| = 1$, then $X$ and $Y$ are related by such a deterministic linear formula, so extreme correlations $\pm 1$ correspond exactly to perfect linear dependence.

Effect of linear transformations

Let $U = aX + b$ for real constants $a$ and $b$ and consider the correlation between $U$ and $Y$. From covariance properties,

$$\mathrm{Cov}(aX + b,\ Y) = a\ \mathrm{Cov}(X, Y).$$

The standard deviation of $aX + b$ is $|a|\,\sigma_X$, because shifting by $b$ does not change the spread and scaling by $a$ multiplies the standard deviation by $|a|$. Therefore,

$$\rho_{aX+b,\,Y} = \frac{\mathrm{Cov}(aX + b,\ Y)}{\sigma_{aX+b}\,\sigma_Y} = \frac{a\ \mathrm{Cov}(X, Y)}{|a|\sigma_X\,\sigma_Y} = \mathrm{sign}(a)\,\rho_{X,Y},$$

where

$$\mathrm{sign}(a) = \begin{cases} 1, & a > 0, \\ -1, & a < 0. \end{cases}$$

The magnitude $|\rho_{aX+b,\,Y}|$ is the same as $|\rho_{X,Y}|$; only the sign may change if $a < 0$. Thus changing units for a variable (for example, converting a temperature from degrees Celsius to degrees Fahrenheit) does not change the absolute value of the correlation with another variable, reflecting the fact that correlation is unitless and invariant under positive linear rescalings.

### 1.6.14 Key properties of the correlation coefficient

Bounded between $-1$ and $1$

To justify the basic bound on the correlation coefficient, consider a simplified setting in which

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0, \qquad \mathrm{Var}(X) = \mathrm{Var}(Y) = 1.$$

In this case the standard deviations are 1, so the correlation coefficient reduces to

$$\rho = \rho_{X,Y} = \mathbb{E}[XY].$$

Define the random variable

$$W = X - \rho Y$$

and examine the expectation of its square:

$$\mathbb{E}[W^2] = \mathbb{E}\big[(X - \rho Y)^2\big].$$

Expanding the square gives

$$\mathbb{E}[W^2] = \mathbb{E}[X^2] - 2\rho\,\mathbb{E}[XY] + \rho^2\,\mathbb{E}[Y^2].$$

Under our assumptions $\mathbb{E}[X^2] = \mathrm{Var}(X) = 1$ and $\mathbb{E}[Y^2] = 1$. Also $\mathbb{E}[XY] = \rho$. Substituting,

$$\mathbb{E}[W^2] = 1 - 2\rho^2 + \rho^2 = 1 - \rho^2.$$

Since $W^2$ is always nonnegative, its expectation must be nonnegative:

$$\mathbb{E}[W^2] \geq 0 \quad \Longrightarrow \quad 1 - \rho^2 \geq 0.$$

Therefore $\rho^2 \leq 1$, which implies

$$-1 \leq \rho \leq 1.$$

With a bit more algebra the same conclusion holds for general random variables $X$ and $Y$ with arbitrary means and nonzero variances.

Characterizing the extreme case $|\rho| = 1$

Suppose now that the absolute value of the correlation coefficient is equal to 1. In the setting above, this means $\rho^2 = 1$, so from the previous result

$$\mathbb{E}[W^2] = 1 - \rho^2 = 0.$$

Again $W^2$ is nonnegative, and a nonnegative random variable with zero expectation must be zero with probability 1. Hence

$$W^2 = 0 \text{ with probability 1}, \quad \text{so} \quad X - \rho Y = 0 \text{ with probability 1}.$$

This shows that

$$X = \rho Y \quad \text{with probability 1},$$

which is a deterministic linear relation between $X$ and $Y$. Because $|\rho| = 1$, there are two possibilities:

$$\rho = 1 \quad \Longrightarrow \quad X = Y \text{ almost surely},$$

$$\rho = -1 \quad \Longrightarrow \quad X = -Y \text{ almost surely}.$$

Thus $|\rho| = 1$ corresponds to an extreme form of dependence, in which one random variable is an exact linear function of the other.

### 1.6.15 Interpreting the correlation coefficient

A correlation coefficient measures the strength of a *linear association* between two random variables, not a causal relationship. A value such as $\rho_{X,Y} = 0.5$ indicates that high values of $X$ tend to be accompanied by high values of $Y$ and low values of $X$ tend to be accompanied by low values of $Y$, in an average sense. However, this does not mean that changing $X$ will directly cause a change in $Y$, or vice versa. In many situations a nonzero correlation arises because both variables are influenced by some common underlying factor, rather than one variable directly affecting the other. For example, mathematical aptitude and musical skill may be positively correlated because both are influenced by certain shared cognitive traits, even though studying one does not necessarily cause improvement in the other.

A simple model with a hidden common factor

To make this idea concrete, consider independent random variables $Z$, $V$, and $W$ with

$$\mathbb{E}[Z] = \mathbb{E}[V] = \mathbb{E}[W] = 0, \qquad \mathrm{Var}(Z) = \mathrm{Var}(V) = \mathrm{Var}(W) = 1.$$

Define

$$X = Z + V, \qquad Y = Z + W.$$

Here the underlying factor $Z$ affects both $X$ and $Y$, while $V$ and $W$ are independent "noise" terms specific to each variable. There is no direct causal influence from $X$ to $Y$ or from $Y$ to $X$; the dependence comes entirely from the shared component $Z$.

Computing the correlation in the example

First compute the variances. Because $Z$ and $V$ are independent,

$$\mathrm{Var}(X) = \mathrm{Var}(Z) + \mathrm{Var}(V) = 1 + 1 = 2,$$

and similarly

$$\mathrm{Var}(Y) = \mathrm{Var}(Z) + \mathrm{Var}(W) = 2.$$

Thus the standard deviations are

$$\sigma_X = \sigma_Y = \sqrt{2}.$$

Next compute the covariance. Since $\mathbb{E}[X] = \mathbb{E}[Y] = 0$,

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] = \mathbb{E}[(Z + V)(Z + W)] = \mathbb{E}[Z^2] + \mathbb{E}[ZW] + \mathbb{E}[VZ] + \mathbb{E}[VW].$$

Because $Z$, $V$, and $W$ are independent and have zero means,

$$\mathbb{E}[Z^2] = 1, \qquad \mathbb{E}[ZW] = \mathbb{E}[Z]\mathbb{E}[W] = 0, \qquad \mathbb{E}[VZ] = 0, \qquad \mathbb{E}[VW] = 0,$$

so

$$\mathrm{Cov}(X, Y) = 1.$$

The correlation coefficient is then

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}.$$

Qualitative meaning of $\rho = \frac{1}{2}$
In this model, the common factor $Z$ and the idiosyncratic parts $V$ and $W$ all have the same variance, so the "shared" part and the "individual noise" parts have comparable strength. The resulting correlation $\rho = \frac{1}{2}$ reflects a moderate linear association: $X$ and $Y$ tend to move together because of their common component $Z$, but not perfectly, since the independent noises $V$ and $W$ partly obscure that relation. If the noise terms $V$ and $W$ were absent, we would have $X = Y = Z$ and the correlation would be 1. If instead $V$ and $W$ had very large variances compared to $Z$, their idiosyncratic fluctuations would dominate, the influence of $Z$ would be hard to detect, and the correlation would be much smaller, approaching 0 in the limit.

### 1.6.16 Why correlations matter in practice

Consider an investment company with capital of 100 million that is allocated equally across 10 different states. Let $X_i$ denote the random profit (in millions) from state $i$, so each investment is 10 million and

$$\mathbb{E}[X_i] = 1, \qquad \mathrm{SD}(X_i) = 1.3, \qquad \mathrm{Var}(X_i) = 1.3^2 = 1.69.$$

The total profit is

$$S = X_1 + X_2 + \cdots + X_{10}.$$

Its expectation is

$$\mathbb{E}[S] = \sum_{i=1}^{10} \mathbb{E}[X_i] = 10.$$

If the state markets are truly unrelated so that the $X_i$ are uncorrelated, then all covariances are zero and

$$\mathrm{Var}(S) = \sum_{i=1}^{10} \mathrm{Var}(X_i) = 10 \times 1.69 = 16.9.$$

Thus

$$\mathrm{SD}(S) = \sqrt{16.9} \approx 4.1.$$

Since the mean profit is 10, breaking even corresponds to an outcome 2.5 standard deviations below the mean (because $10 - 2.5 \times 4.1 \approx 0$). Outcomes this far below the mean have relatively small probability for many light-tailed distributions, so in the uncorrelated case diversification gives a high chance of a positive total return.

Effect of strong positive correlations
Suppose instead that the state markets are strongly linked so that

$$\rho_{X_i,X_j} = 0.9 \quad \text{for all } i \neq j.$$

Each covariance is then

$$\mathrm{Cov}(X_i, X_j) = \rho_{X_i,X_j}\ \mathrm{SD}(X_i)\ \mathrm{SD}(X_j) = 0.9 \times 1.3 \times 1.3 = 1.52.$$

The variance of the sum becomes

$$\mathrm{Var}(S) = \sum_{i=1}^{10} \mathrm{Var}(X_i) + \sum_{\substack{i,j=1 \\ i\neq j}}^{10} \mathrm{Cov}(X_i, X_j) = 10 \times 1.69 + 90 \times 1.52.$$

Here the first term accounts for the 10 individual variances, and the second term has 90 covariance entries because there are $10 \cdot 9$ ordered pairs $(i, j)$ with $i \neq j$. Carrying out the arithmetic,

$$\mathrm{Var}(S) = 16.9 + 136.8 = 153.7 \approx 154,$$

so

$$\mathrm{SD}(S) \approx \sqrt{154} \approx 12.4.$$

The expected profit is still 10, but now a one-standard-deviation drop from the mean $(10 - 12.4)$ already leads to a negative total return. Such a deviation has substantial probability, so the overall investment becomes much riskier when returns are highly positively correlated.

Lessons for risk and diversification
This example illustrates that diversification reduces risk only when the underlying positions are not strongly positively correlated. If the component returns move largely together because of common economic forces, the covariance terms dominate the variance of the aggregate portfolio and the overall risk can remain large even when the portfolio includes many different assets. In the context of real estate, it is not enough to invest in multiple regions if housing markets across regions are driven by shared macroeconomic factors that create high correlations. When such correlations are underestimated or ignored, as in parts of the financial sector before the global financial crisis, portfolios that appear well diversified can still suffer large simultaneous losses.

### 1.6.17 Conditional expectation as a random variable

Consider a random variable $X$ and a function $h$ that maps real numbers to real numbers. For example, if $h(x) = x^2$, then $h(X)$ is defined to be the random variable that takes the value $x^2$ whenever $X = x$. Thus, $h(X) = X^2$ is itself a random variable.
Now let us move to conditional expectations. For a discrete random variable $X$ and another random variable $Y$, the conditional expectation of $X$ given that $Y = y$ is defined as

$$\mathbb{E}[X \mid Y = y] = \sum_x x\,\mathbb{P}(X = x \mid Y = y).$$

In the continuous case, the summation would be replaced by an integral and the conditional probability mass function by a conditional probability density function.
If we fix a specific value $y$, this expression gives a number. However, if $y$ changes, the result changes too, so this quantity depends on $y$. Let us define the function

$$g(y) = \mathbb{E}[X \mid Y = y].$$

This is a function of $y$. If we now apply it to the random variable $Y$ itself, we obtain a new random variable:

$$\mathbb{E}[X \mid Y] = g(Y).$$

This means that when $Y = y$, the random variable $\mathbb{E}[X \mid Y]$ takes the value $g(y) = \mathbb{E}[X \mid Y = y]$. Hence, $\mathbb{E}[X \mid Y]$ is itself a random variable that depends only on $Y$.

Example
Let $X$ represent a student's exam score and $Y$ the number of hours studied. For each fixed $y$, the term $\mathbb{E}[X \mid Y = y]$ represents the expected exam score for students who studied $y$ hours. We can think of $g(y) = \mathbb{E}[X \mid Y = y]$ as a function that increases with $y$. Then, for each student, the random variable $\mathbb{E}[X \mid Y]$ gives the expected score given their own study time $Y$.

Interpretation
The conditional expectation $\mathbb{E}[X \mid Y]$ is therefore an abstract object—a random variable whose value becomes known once $Y$ is known. It has all the properties of a random variable: it has a distribution, a mean, and a variance. Understanding conditional expectation in this way provides the foundation for concepts such as variance decomposition and regression analysis.

### 1.6.18 The law of iterated expectations

We have previously introduced the conditional expectation of one random variable given another and established that it is itself a random variable. Because it is a random variable, it has an expectation of its own. Our goal is to determine what this expectation equals.
Let $X$ and $Y$ be random variables. For each possible value $y$ of $Y$, the conditional expectation $\mathbb{E}[X \mid Y = y]$ is a specific number. Defining

$$g(y) = \mathbb{E}[X \mid Y = y],$$

we obtain a function $g$ of $y$. Then the abstract conditional expectation can be written as

$$\mathbb{E}[X \mid Y] = g(Y),$$

which is a random variable that takes the value $g(y)$ whenever $Y = y$.
We now wish to compute its expected value:

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[g(Y)].$$

How do we find the expectation of a function of a random variable? By the Expected Value Rule, if $Y$ is discrete, we have

$$\mathbb{E}[g(Y)] = \sum_y g(y)\,\mathbb{P}(Y = y).$$

Substituting the definition of $g(y)$ gives

$$\mathbb{E}[g(Y)] = \sum_y \mathbb{E}[X \mid Y = y]\,\mathbb{P}(Y = y).$$

This expression matches the form of the Total Expectation Theorem: we take conditional expectations under each possible scenario and weight them by the corresponding probabilities. Hence,

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

This fundamental identity is called the *Law of Iterated Expectations.*

Interpretation
The Law of Iterated Expectations states that the overall expected value of a random variable can be obtained by first conditioning on another variable and then averaging over all possible outcomes of that conditioning variable. Formally, it generalizes the Total Expectation Theorem and holds for both discrete and continuous random variables (with integrals replacing sums in the continuous case). It provides a unified and abstract perspective: the expectation of a conditional expectation equals the unconditional expectation.

### 1.6.19 Stick-breaking revisited

Consider a stick of length $\ell$ that we first break at a point chosen uniformly at random along its length. Let $Y$ denote the position of the first break, measured from the left end of the stick. Then $Y$ is uniformly distributed on the interval $[0, \ell]$, with density

$$f_Y(y) = \frac{1}{\ell}, \quad 0 \le y \le \ell.$$

Next, we take the remaining piece of length $Y$ and break it again at a point chosen uniformly at random along that remaining piece. Let $X$ denote the position of this second break measured from the left end of the original stick. Conditioned on $Y = y$, the second break point $X$ is uniform on the interval $[0, y]$, so the conditional density is

$$f_{X|Y}(x \mid y) = \frac{1}{y}, \quad 0 \le x \le y.$$

For a fixed numerical value $y$, the conditional distribution of $X$ given $Y = y$ is uniform on $[0, y]$. Therefore, its conditional expectation is the midpoint of that interval:

$$\mathbb{E}[X \mid Y = y] = \frac{y}{2}.$$

This is an equality between numbers, valid for each specific value $y$.
We now rewrite this as an equality between random variables. The abstract conditional expectation $\mathbb{E}[X \mid Y]$ is the random variable that takes the value $\mathbb{E}[X \mid Y = y]$ whenever $Y = y$. From the previous relation, it follows that

$$\mathbb{E}[X \mid Y] = \frac{Y}{2}.$$

Both sides here are random variables: the value of each side is not known until the outcome of $Y$ is revealed.
Using the Law of Iterated Expectations, we can now compute $\mathbb{E}[X]$ as follows:

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}\Big[\frac{Y}{2}\Big] = \frac{1}{2}\,\mathbb{E}[Y].$$

Since $Y$ is uniform on $[0, \ell]$, its expectation is

$$\mathbb{E}[Y] = \frac{\ell}{2},$$

and therefore

$$\mathbb{E}[X] = \frac{1}{2} \cdot \frac{\ell}{2} = \frac{\ell}{4}.$$

Key distinction
The relation

$$\mathbb{E}[X \mid Y = y] = \frac{y}{2}$$

is an equality between numbers: for each fixed $y$, it gives a specific numerical value. In contrast, the relation

$$\mathbb{E}[X \mid Y] = \frac{Y}{2}$$

is an equality between random variables: both sides are random, and their realized values become known only after the experiment is performed and the outcome of $Y$ is observed.

### 1.6.20 The conditional variance

We have defined the conditional expectation of a random variable given another as an abstract object, which is itself a random variable. We now do something analogous for the notion of conditional variance.
Recall that the variance of a random variable $X$ is defined as

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big].$$

If we live in a conditional universe where we are told that another random variable $Y$ has taken the value $y$, then the conditional variance of $X$ given $Y = y$ is defined by

$$\mathrm{Var}(X \mid Y = y) = \mathbb{E}\big[(X - \mathbb{E}[X \mid Y = y])^2 \,\big|\, Y = y\big].$$

Here $\mathbb{E}[X \mid Y = y]$ is the mean of $X$ in that conditional universe, so $X - \mathbb{E}[X \mid Y = y]$ is the deviation of $X$ from its conditional mean. We square this deviation and then take its conditional expectation. For each fixed numerical $y$, the quantity $\mathrm{Var}(X \mid Y = y)$ is just a number.
As with conditional expectation, we can view this as defining a function of $y$. Define

$$v(y) = \mathrm{Var}(X \mid Y = y).$$

This is now a deterministic function of the numerical value $y$. Using this function, we define the abstract conditional variance as the random variable

$$\mathrm{Var}(X \mid Y) = v(Y),$$

which is the random variable that takes the value $\mathrm{Var}(X \mid Y = y)$ whenever $Y = y$. Before the value of $Y$ is revealed, $\mathrm{Var}(X \mid Y)$ is unknown and therefore random.

Example: uniform on $[0, Y]$
Suppose that $Y$ is a nonnegative random variable and that, conditioned on $Y = y$, the random variable $X$ is uniform on the interval $[0, y]$. Then, for a fixed numerical value $y$, the conditional distribution of $X$ given $Y = y$ is $\mathrm{Unif}(0, y)$. A uniform random variable on an interval of length $y$ has variance

$$\mathrm{Var}(X \mid Y = y) = \frac{y^2}{12}.$$

This is an equality between numbers: for each specific $y$, the conditional variance takes the value $y^2/12$.
Turning this into an equality between random variables, we obtain

$$\mathrm{Var}(X \mid Y) = \frac{Y^2}{12},$$

because the random variable $\mathrm{Var}(X \mid Y)$ takes the value $y^2/12$ whenever $Y = y$. Thus the conditional variance is itself a random variable, determined by the value of $Y$.

Total variance decomposition
For conditional expectations we had the Law of Iterated Expectations,

$$\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X].$$

A natural question is whether an analogous identity holds for variances, that is, whether $\mathbb{E}[\mathrm{Var}(X \mid Y)]$ equals $\mathrm{Var}(X)$. In general this is not true. Instead, the variance of $X$ can be decomposed as

$$\mathrm{Var}(X) = \mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big).$$

Here $\mathrm{Var}(X \mid Y)$ is a random variable, so the term $\mathbb{E}[\mathrm{Var}(X \mid Y)]$ is its mean. Likewise, $\mathbb{E}[X \mid Y]$ is a random variable, so $\mathrm{Var}(\mathbb{E}[X \mid Y])$ is the variance of that random variable. This identity is often called the *law of total variance* and expresses the total variability of $X$ as the sum of the average conditional variability plus the variability of the conditional means.

### 1.6.21 Derivation of the law of total variance

We start from the standard identity for the variance of a random variable $X$,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

We apply the same identity inside a conditional universe where we are told that $Y = y$. For each fixed $y$, the conditional variance is

$$\mathrm{Var}(X \mid Y = y) = \mathbb{E}\big[X^2 \mid Y = y\big] - \big(\mathbb{E}[X \mid Y = y]\big)^2.$$

This is an equality between numbers, valid for each specific value $y$.
We now reinterpret this as an equality between random variables. The abstract conditional variance $\mathrm{Var}(X \mid Y)$ is defined as the random variable that takes the numerical value $\mathrm{Var}(X \mid Y = y)$ whenever $Y = y$. Similarly, the random variable $\mathbb{E}[X^2 \mid Y]$ takes the value $\mathbb{E}[X^2 \mid Y = y]$ whenever $Y = y$, and $\mathbb{E}[X \mid Y]$ takes the value $\mathbb{E}[X \mid Y = y]$ whenever $Y = y$. Thus, for all outcomes of $Y$, the corresponding numerical values match, and we have the equality of random variables

$$\mathrm{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - \big(\mathbb{E}[X \mid Y]\big)^2.$$

Next, we take expectations of both sides in order to obtain the first term in the law of total variance:

$$\mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] = \mathbb{E}\big[\mathbb{E}[X^2 \mid Y]\big] - \mathbb{E}\big[\big(\mathbb{E}[X \mid Y]\big)^2\big].$$

The law of iterated expectations gives

$$\mathbb{E}\big[\mathbb{E}[X^2 \mid Y]\big] = \mathbb{E}[X^2],$$

so we obtain

$$\mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] = \mathbb{E}[X^2] - \mathbb{E}\big[\big(\mathbb{E}[X \mid Y]\big)^2\big].$$

We now turn to the second component of the law of total variance, the variance of the conditional expectation. Applying the variance identity to the random variable $\mathbb{E}[X \mid Y]$, we get

$$\mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) = \mathbb{E}\big[\big(\mathbb{E}[X \mid Y]\big)^2\big] - \big(\mathbb{E}[\mathbb{E}[X \mid Y]]\big)^2.$$

By the law of iterated expectations,

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X],$$

so

$$\mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) = \mathbb{E}\big[\big(\mathbb{E}[X \mid Y]\big)^2\big] - \big(\mathbb{E}[X]\big)^2.$$

We now add the two expressions

$$\mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] \quad \text{and} \quad \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big].$$

From the previous formulas we have

$$\mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) = \Big(\mathbb{E}[X^2] - \mathbb{E}\big[\big(\mathbb{E}[X \mid Y]\big)^2\big]\Big) + \Big(\mathbb{E}\big[\big(\mathbb{E}[X \mid Y]\big)^2\big] - \big(\mathbb{E}[X]\big)^2\Big).$$

The middle terms cancel, leaving

$$\mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2 = \mathrm{Var}(X).$$

Therefore, we obtain the law of total variance,

$$\mathrm{Var}(X) = \mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big).$$

### 1.6.22 Section means and variances

Consider a class of 30 students divided into two sections. Let $x_i$ be the score of student $i$. We perform the following experiment: pick one student uniformly at random from the class.
Define two random variables. The numerical random variable $X$ is the score of the selected student, so if student $i$ is picked we have $X = x_i$. The random variable $Y$ is the section of the selected student, so that $Y \in \{1, 2\}$.
Assume that the average score in section 1 is 90 and the average in section 2 is 60. Section 1 has 10 students and section 2 has 20 students. The expected score of the randomly selected student can be computed directly by

$$\mathbb{E}[X] = \frac{1}{30}\sum_{i=1}^{30} x_i = \frac{1}{30}\Big(10 \cdot 90 + 20 \cdot 60\Big) = 70.$$

Now consider conditional expectations. If $Y = 1$, we know the student comes from section 1, and within that section each of the 10 students is equally likely with probability 1/10. Thus

$$\mathbb{E}[X \mid Y = 1] = 90.$$

Similarly, if $Y = 2$, the student is from section 2 and each of the 20 students is equally likely with probability 1/20, so

$$\mathbb{E}[X \mid Y = 2] = 60.$$

The abstract conditional expectation $\mathbb{E}[X \mid Y]$ is then a random variable that takes the value 90 if $Y = 1$ and 60 if $Y = 2$. Since section 1 contains 10 of the 30 students, we have

$$\mathbb{P}(Y = 1) = \frac{1}{3}, \qquad \mathbb{P}(Y = 2) = \frac{2}{3}.$$

Hence $\mathbb{E}[X \mid Y]$ takes value 90 with probability 1/3 and 60 with probability 2/3. Its expectation is

$$\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \frac{1}{3} \cdot 90 + \frac{2}{3} \cdot 60 = 70,$$

which matches the overall class average. This is an instance of the law of iterated expectations: the overall mean can be obtained by taking section means and weighting them by the section probabilities.

Variance of the conditional expectation

We now compute the variance of the random variable $\mathbb{E}[X \mid Y]$. Its mean is 70, and it takes value 90 with probability 1/3 and 60 with probability 2/3. Thus

$$\mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) = \frac{1}{3}(90 - 70)^2 + \frac{2}{3}(60 - 70)^2 = \frac{1}{3} \cdot 20^2 + \frac{2}{3} \cdot (-10)^2 = \frac{400}{3} + \frac{200}{3} = 200.$$

Next, suppose we are given the variances within each section. Let

$$\frac{1}{10} \sum_{i \in \text{sec } 1} (x_i - 90)^2 = 10, \qquad \frac{1}{20} \sum_{i \in \text{sec } 2} (x_i - 60)^2 = 20.$$

The first expression is the variance of the scores within section 1 (the variance of the score of a randomly selected student, given $Y = 1$). Thus

$$\mathrm{Var}(X \mid Y = 1) = 10.$$

Similarly, the second expression is the variance within section 2, so

$$\mathrm{Var}(X \mid Y = 2) = 20.$$

The abstract conditional variance $\mathrm{Var}(X \mid Y)$ is a random variable that takes the value 10 if $Y = 1$ and 20 if $Y = 2$. Using the section probabilities, we find its expectation:

$$\mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] = \frac{1}{3} \cdot 10 + \frac{2}{3} \cdot 20 = \frac{10}{3} + \frac{40}{3} = \frac{50}{3}.$$

Applying the law of total variance

We now have the two components needed to use the law of total variance,

$$\mathrm{Var}(X) = \mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big).$$

Substituting the values obtained above gives

$$\mathrm{Var}(X) = \frac{50}{3} + 200.$$

This is the variance of the scores of a randomly selected student from the entire class.

In this context, the law of total variance has a natural interpretation. The overall variability of scores in the class has two components. The first term, $\mathbb{E}[\mathrm{Var}(X \mid Y)]$, measures the average variability *within* a typical section, averaged over sections. The second term, $\mathrm{Var}(\mathbb{E}[X \mid Y])$, measures how much the section means themselves differ from the overall class mean, that is, the variability *between* sections. Together, these two sources of randomness add up to the total variance of student scores in the class.

### 1.6.23 Mean of the sum of a random number of random variables

Consider a shopping experiment in which you visit a random number of stores. Let $N$ be the (integer-valued) random variable denoting how many stores you visit. At each store $i$, you spend a random amount $X_i$.
Assume that the $X_i$ are identically distributed and independent of each other, and that they are also independent of $N$. Let

$$\mu = \mathbb{E}[X_i]$$

denote their common mean. Define the total amount of money spent as

$$Y = \sum_{i=1}^{N} X_i.$$

To find $\mathbb{E}[Y]$, we condition on the value of $N$. For a fixed numerical value $n$, the conditional total becomes

$$Y \mid (N = n) = \sum_{i=1}^{n} X_i,$$

which is now a sum of a fixed number $n$ of random variables. Using linearity of expectation and the fact that conditioning on $N = n$ does not change the distribution of the $X_i$ (by independence of $N$ and the $X_i$), we obtain

$$\mathbb{E}[Y \mid N = n] = \mathbb{E}\Big[\sum_{i=1}^{n} X_i \,\Big|\, N = n\Big] = \sum_{i=1}^{n} \mathbb{E}[X_i \mid N = n] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n\mu.$$

This is an equality between numbers, valid for each specific $n$.
We can now apply the Total Expectation Theorem:

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid N]\big] = \sum_{n} \mathbb{E}[Y \mid N = n]\,\mathbb{P}(N = n) = \sum_{n} n\mu\,\mathbb{P}(N = n).$$

Since $\mu$ is a constant, we factor it out:

$$\mathbb{E}[Y] = \mu \sum_{n} n\,\mathbb{P}(N = n) = \mu\,\mathbb{E}[N].$$

Thus the expected total spending is

$$\mathbb{E}[Y] = \mathbb{E}[N]\,\mathbb{E}[X_i].$$

We can arrive at the same result using the Law of Iterated Expectations in a more abstract form. From the calculation above, for each $n$ we have $\mathbb{E}[Y \mid N = n] = n\mu$. The abstract conditional expectation $\mathbb{E}[Y \mid N]$ is therefore the random variable that takes the value $n\mu$ whenever $N = n$, that is,

$$\mathbb{E}[Y \mid N] = N\mu.$$

Applying the law of iterated expectations,

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid N]\big] = \mathbb{E}[N\mu] = \mu\,\mathbb{E}[N],$$

since $\mu$ is a constant and can be pulled outside the expectation. This confirms that the mean of the sum of a random number of i.i.d. terms equals the mean number of terms times the mean of each term.

### 1.6.24 Variance of the sum of a random number of random variables

We continue with the shopping model in which you visit a random number $N$ of stores and spend an amount $X_i$ in store $i$. The total amount spent is

$$Y = \sum_{i=1}^{N} X_i.$$

We assume that the $X_i$ are i.i.d., with

$$\mu = \mathbb{E}[X_i], \qquad \sigma^2 = \mathrm{Var}(X_i),$$

and that they are independent of $N$.
To find $\mathrm{Var}(Y)$, we use the law of total variance:

$$\mathrm{Var}(Y) = \mathbb{E}\big[\,\mathrm{Var}(Y \mid N)\big] + \mathrm{Var}\,\big(\mathbb{E}[Y \mid N]\big).$$

We already know that, for each fixed $n$,

$$\mathbb{E}[Y \mid N = n] = n\mu.$$

Therefore, the abstract conditional expectation is

$$\mathbb{E}[Y \mid N] = N\mu.$$

The second term in the law of total variance is then

$$\operatorname{Var}\big(\mathbb{E}[Y \mid N]\big) = \operatorname{Var}(N\mu) = \mu^2 \operatorname{Var}(N),$$

because multiplying a random variable by a constant multiplies its variance by the square of that constant.
For the first term, condition again on $N = n$. Given $N = n$, the total is a sum of $n$ i.i.d. variables:

$$Y \mid (N = n) = \sum_{i=1}^{n} X_i.$$

Using independence and identical distributions, we have

$$\operatorname{Var}(Y \mid N = n) = \operatorname{Var}\Big(\sum_{i=1}^{n} X_i\Big) = n\sigma^2.$$

This is an equality between numbers, valid for each fixed $n$. As an equality between random variables, it becomes

$$\operatorname{Var}(Y \mid N) = N\sigma^2,$$

since $\operatorname{Var}(Y \mid N)$ takes value $n\sigma^2$ whenever $N = n$.
We can now compute the first term in the law of total variance:

$$\mathbb{E}\big[\operatorname{Var}(Y \mid N)\big] = \mathbb{E}[N\sigma^2] = \sigma^2\mathbb{E}[N],$$

where $\sigma^2$ is a constant and comes outside the expectation.
Putting the two pieces together, we obtain

$$\operatorname{Var}(Y) = \underbrace{\sigma^2\mathbb{E}[N]}_{\mathbb{E}[\operatorname{Var}(Y|N)]} + \underbrace{\mu^2 \operatorname{Var}(N)}_{\operatorname{Var}(\mathbb{E}[Y|N])}.$$

Equivalently,

$$\operatorname{Var}\Big(\sum_{i=1}^{N} X_i\Big) = \mathbb{E}[N]\ \operatorname{Var}(X_1) + \operatorname{Var}(N)\ \big(\mathbb{E}[X_1]\big)^2.$$

Interpretation

The variance of the total amount spent $Y$ has two distinct components. The term $\mathbb{E}[N]\sigma^2$ captures the variability coming from how much you spend in a typical store, averaged over the random number of stores. The term $\mu^2 \operatorname{Var}(N)$ captures the extra variability caused by the randomness in the number of stores itself. Together, these two sources of randomness determine the total variance of $Y$.

## 1.7 Bayesian Inference

In this section, we outline several important domains in which probabilistic modeling and statistical inference play a central role. So far, we have concentrated on analyzing abstract probability models, but in practice these models are useful only to the extent that they are relevant to, and informed by, real data. The field of inference and statistics is precisely concerned with using data generated by the real world to construct, select, and validate such models.

Traditionally, statisticians would often be asked to analyze relatively small data sets, for example a limited number of medical records in order to evaluate whether a treatment is effective. In contrast, modern applications generate massive data sets and call for complex models involving a large number of parameters. At the same time, advances in computing power allow us to handle these large models, so that the range of applications and opportunities has expanded dramatically.

One classical setting involves the deliberate design of a data collection procedure, followed by a relatively simple prediction task. A basic example is polling, where we collect responses from a sample of individuals and use them to predict the outcome of an election. A closely related area is marketing and advertising, where the goal is to make predictions at the level of individual consumers rather than an entire population.

A particularly prominent application in this context is that of recommendation systems. One collects data such as ratings that users assign to movies or other items, which can be organized in a large table with users indexing the rows and items indexing the columns. Using these partially observed data, one then aims to predict missing entries in the table, such as whether a given user would like a particular movie that they have not yet rated.

Finance is another major application domain. Financial markets are inherently uncertain, but they also produce extensive historical data sets. A central question is how to use these data to build models that can capture market behavior and support tasks such as risk assessment or pricing of financial instruments.

In the natural sciences, there has been a revolution driven by the availability of large-scale data. In the life sciences, for instance, genomic data are analyzed in order to understand how combinations of genes relate to diseases or to map out complex biochemical pathways inside cells. An emerging frontier is neuroscience, where increasingly detailed measurements of brain activity raise the possibility of building models that describe how neural systems process information.

Similar themes arise across many other scientific disciplines. In climate and environmental science, large and intricate models are calibrated using extensive observational data to understand and predict phenomena such as climate change. In physics, sophisticated inference techniques are used to detect rare events or objects, such as elusive particles or distant exoplanets, amidst large volumes of noisy measurements.

Engineering applications also provide numerous examples. Engineers are often concerned with designing systems that perform reliably in uncertain or noisy environments, and probabilistic models guide both analysis and design. Signal processing is a prototypical area, where the goal is to recover an underlying signal from corrupted observations, such as reconstructing the content of a radio transmission received in the presence of noise.

The range of application domains is enormous, and it continues to grow as new types of data become available. Despite the diversity of these settings, the fundamental probabilistic and statistical concepts that arise are remarkably similar. Our goal in the subsequent sections is to develop core methodologies that can be applied across this wide spectrum of problems.

Illustrative example

Consider the setting of movie recommendation systems. We can represent the data by a large table, where each row corresponds to a user and each column corresponds to a movie, and the entry in row $i$ and column $j$ is the rating that user $i$ assigned to movie $j$ whenever such a rating is available. The inference task is then to use the observed entries of this table to predict missing ratings, for example to decide whether user $i$ is likely to enjoy movie $j$ and should therefore be recommended that movie.

### 1.7.1 Types of inference problems

In the field of inference, it is useful to distinguish between different types of problems that arise. One broad distinction is between building models of a system and using a given model to make inferences about unknown quantities. In some situations, constructing or calibrating the model itself is the primary goal, while in others the model is taken as given and the focus is on unobserved variables that appear in that model.

To illustrate this distinction, consider a simple communication setting. A transmitter sends a signal $S$ through a medium, which attenuates the signal by a factor $a$, and the signal is additionally corrupted by random noise $W$. The receiver observes a signal $X$ that can be described by the linear relation

$$X = aS + W.$$

Here, $W$ is modeled as random noise with some known distribution, while $a$ and $S$ may be unknown.

In one version of the problem, our goal is to characterize the medium. We transmit a known pilot signal $S$, we observe the corresponding $X$, and we use the relation $X = aS + W$ to infer the value of the attenuation factor $a$. In this case, we are effectively estimating a parameter of the system and thereby constructing or refining a model of the medium.

In a different version of the problem, we assume that the medium has already been characterized. The attenuation factor $a$ is known, but now the transmitted signal $S$ is unknown to the receiver. Given the observation $X$ and the model $X = aS + W$, the task is to infer the value of $S$.
From a mathematical standpoint, these two versions have the same structural form. In both cases, we work with the same linear relation between $X$, $S$, and $a$, but we treat different quantities as known or unknown. This means that similar methodologies can often be applied to both types of problems, even though their interpretations are quite different.
Another important distinction in inference is between hypothesis testing problems and estimation problems. In hypothesis testing, the unknown takes values in a finite or small discrete set, corresponding to a few candidate models of the world. The goal is to decide which of these candidate models is the correct one, based on the available data, while keeping the probability of making an incorrect decision small.
A classical example is a radar detection problem. We receive a radar return signal and wish to decide between two hypotheses: either there is an airplane present in the range of the radar, or there is no airplane. The inference task is to choose between these two discrete alternatives, using the observed data and an appropriate decision rule.
In estimation problems, the unknown is typically numerical and may vary over a continuum of possible values. We aim to produce an estimate that is close to the true value of the quantity of interest, according to some distance or loss function. Here, instead of choosing among a few discrete models, we select an estimate from an infinite set of possible values, such as the real line.
These two types of problems, hypothesis testing and estimation, often require different solution strategies and performance criteria. In hypothesis testing, performance is usually quantified in terms of error probabilities associated with incorrect decisions between the competing hypotheses. In estimation, the emphasis is on the accuracy of the estimate, measured for example by mean squared error or another notion of distance from the true value.
Despite these differences, both hypothesis testing and estimation are central components of statistical inference. Many practical problems involve elements of both, such as first deciding which model or scenario is likely to hold, and then estimating numerical parameters within the chosen model. In the following sections, we will develop methods tailored to each of these problem types and explore how they can be combined in more complex applications.

Example: detection versus estimation
Consider again the radar setting. In a pure detection problem, we use the observed signal to decide between the hypotheses *airplane present* and *airplane absent*. In a related estimation problem, assuming that an airplane is present, we might further use the radar data to estimate a continuous quantity such as the distance or velocity of the airplane, aiming for an estimate that is close to the true value.

### 1.7.2 The Bayesian inference framework

In the Bayesian inference framework, the unknown quantity of interest is modeled as a random variable, denoted by $\Theta$. Unlike other approaches in which $\Theta$ is treated as an unknown but fixed constant, here $\Theta$ has a probability distribution that captures our uncertainty about its value. This initial distribution is called the *prior* distribution of $\Theta$, and it encodes our beliefs about $\Theta$ before any data are observed.
We then obtain data in the form of an observation $X$. The observation $X$ is itself a random variable, and we model it probabilistically by specifying the conditional distribution of $X$ given $\Theta$. This conditional distribution describes how the data would be generated under each possible value of $\Theta$, and it is sometimes referred to as the *likelihood model*.
Once we observe a specific numerical value $x$ of $X$, we can use Bayes' rule to update our prior beliefs. The result is the *posterior* distribution of $\Theta$ given the observation $X = x$. In a discrete setting, this posterior is a conditional PMF, whereas in a continuous setting, it is a conditional PDF.
The posterior distribution represents a complete Bayesian solution to the inference problem. It summarizes all the information about $\Theta$ that is available after combining the prior with the observed data through the probabilistic model. In particular, it allows us to quantify the relative plausibility of different values of $\Theta$ in light of the data.
A natural question is how to choose the prior distribution. In some situations, symmetry or lack of prior information suggests that all values in a certain range should be regarded as equally likely, leading to a uniform prior over that range. In other cases, prior information may come from previous experiments or studies, and the new data can then be used to refine and update these earlier conclusions.
The choice of prior can also be subjective, reflecting personal or expert beliefs about the parameter $\Theta$. Regardless of its origin, the prior should respect any known constraints, such as the possible range of $\Theta$. Values outside that range are assigned zero prior probability, ensuring that the posterior also assigns zero probability to impossible parameter values.
To appreciate the idea of a posterior distribution, consider an election forecasting problem. Let $\Theta$ denote the number of electoral votes that a candidate will receive. Instead of making a single yes/no prediction about the winner, a Bayesian analysis produces a full distribution over the possible values of $\Theta$, providing a detailed picture of the uncertainty about the outcome.

In such a setting, the results of a poll might be presented as a histogram or curve showing the probabilities of different possible vote totals. This graphical representation corresponds to the posterior distribution of $\Theta$ given the polling data. It is much more informative than a single statement of which candidate is more likely to win, because it reveals how confident or uncertain the prediction really is.
While the posterior distribution is a complete answer to the Bayesian inference problem, we often wish to summarize it by a single number called an *estimate*. One common choice is the *maximum a posteriori* (MAP) estimate. In the discrete case, the MAP estimate is the value of $\Theta$ that has the largest posterior probability, and in the continuous case it is the point at which the posterior density is maximized.
Another important summary is the *conditional expectation* estimator. Here we report the posterior mean, that is, the conditional expectation $\mathbb{E}[\Theta \mid X = x]$. This estimator is also known as the least mean squares (LMS) estimator, because it minimizes the mean squared error between the estimate and the true value of $\Theta$.
It is useful to distinguish between an *estimator* and an *estimate*. An estimator is a rule or function, typically denoted by $g$, that maps observed data to a numerical value; in abstract terms, it is a random variable $\hat{\Theta} = g(X)$. An estimate is the numerical value that results when we apply this rule to a particular observed value $x$, namely $\hat{\theta} = g(x)$.
Because $X$ is random, the estimator $\hat{\Theta}$ is also a random variable and can be studied using probabilistic tools. Different estimators, such as the MAP estimator or the conditional expectation estimator, have different properties and performance measures. In subsequent discussions, we will analyze these properties and see how they guide the choice of an appropriate estimator for a given inference problem.

Example: Bayesian estimation in an election
Let $\Theta$ be the (unknown) number of electoral votes that candidate A will receive in an election. Based on past elections and prior knowledge, we assign a prior distribution to $\Theta$, reflecting which totals we consider more or less plausible before any new polling data are collected. After running a poll and observing data $X = x$, we update our beliefs using Bayes' rule to obtain the posterior distribution of $\Theta$, which can be visualized as a probability mass function over the possible vote totals and used to report quantities such as the MAP estimate or the posterior mean.

### 1.7.3 Discrete parameter, discrete observation

We now consider Bayesian inference in the case where both the unknown parameter $\Theta$ and the observation $X$ are discrete random variables. The parameter $\Theta$ takes values in a finite or countable set, and we can interpret each possible value as a different hypothesis about the underlying model. Given a prior PMF for $\Theta$ and a conditional PMF for $X$ given $\Theta$, Bayes' rule allows us to compute the posterior PMF of $\Theta$ once an observation has been made.
Suppose that $\Theta$ can take values in the set $\{1, 2, 3\}$. After observing a particular value $X = x$, we apply Bayes' rule and obtain the conditional PMF $p_{\Theta|X}(\theta \mid x)$. For concreteness, imagine that this posterior distribution assigns probabilities 0.1, 0.6, and 0.3 to the values $\theta = 1$, $\theta = 2$, and $\theta = 3$, respectively.
At this stage, the posterior distribution provides a complete probabilistic description of our uncertainty about $\Theta$ given the observation. However, we may wish to summarize this information by reporting a single numerical estimate of $\Theta$. One natural way to do this is to use the maximum a posteriori (MAP) rule, which selects the value of $\theta$ that has the largest posterior probability.
Formally, the MAP estimate $\hat{\theta}_{\text{MAP}}$ is defined by

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p_{\Theta|X}(\theta \mid x).$$

In the example above, the posterior probability is largest at $\theta = 2$, so the MAP estimate would be $\hat{\theta} = 2$. This estimate can be viewed as the most likely value of $\Theta$ given the observed data.
Another common way to construct an estimate is to use the least mean squares (LMS) rule. In this approach, we choose as our estimate the conditional expectation of $\Theta$ given the observed value,

$$\hat{\theta}_{\text{LMS}} = \mathbb{E}[\Theta \mid X = x] = \sum_{\theta} \theta \, p_{\Theta|X}(\theta \mid x).$$

In the numerical example with probabilities 0.1, 0.6, and 0.3, this gives

$$\hat{\theta}_{\text{LMS}} = 1 \cdot 0.1 + 2 \cdot 0.6 + 3 \cdot 0.3 = 2.2.$$

We now turn to the question of how to assess the quality of an estimate. When the values of $\Theta$ are interpreted as distinct hypotheses, a natural performance criterion is the probability of error. Given that we have observed $X = x$ and produced an estimate $\hat{\theta}$, we define the conditional probability of error as

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta \mid X = x\big).$$

Once the observation $x$ has been fixed, the estimate $\hat{\theta}$ is just a number, while $\Theta$ remains random, described by the posterior distribution. The event of making an error is $\{\Theta \neq \hat{\theta}\}$, and its conditional probability is

$$\mathbb{P}\big(\Theta \neq \hat{\theta} \mid X = x\big) = 1 - p_{\Theta|X}(\hat{\theta} \mid x).$$

Equivalently, this is the sum of the posterior probabilities of all values of $\theta$ different from $\hat{\theta}$.
In the numerical example, if we use the MAP estimate $\hat{\theta} = 2$, the conditional probability of error becomes

$$\mathbb{P}\big(\Theta \neq 2 \mid X = x\big) = 0.1 + 0.3 = 0.4.$$

If instead we had chosen $\hat{\theta} = 3$, the conditional probability of error would be

$$\mathbb{P}\big(\Theta \neq 3 \mid X = x\big) = 0.1 + 0.6 = 0.7,$$

which is larger.
More generally, for any fixed observation $x$, choosing $\hat{\theta}$ to maximize the posterior probability $p_{\Theta|X}(\hat{\theta} \mid x)$ is equivalent to minimizing the conditional probability of error. Indeed, minimizing $\mathbb{P}(\Theta \neq \hat{\theta} \mid X = x)$ is the same as maximizing $\mathbb{P}(\Theta = \hat{\theta} \mid X = x)$. Thus, the MAP rule is optimal in the sense that it produces the smallest possible conditional probability of error for each given value of the observation.
Often, we are also interested in the overall probability of error, not conditioned on a specific observation. This overall probability is

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta\big),$$

which averages the conditional error probabilities over all possible observations. Using the law of total probability and conditioning on the observation, we can write

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta\big) = \sum_x \mathbb{P}\big(\hat{\Theta} \neq \Theta \mid X = x\big)\mathbb{P}(X = x).$$

There is also an alternative expression obtained by conditioning on the value of $\Theta$ instead of $X$. In that case, the overall probability of error becomes

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta\big) = \sum_\theta \mathbb{P}\big(\hat{\Theta} \neq \theta \mid \Theta = \theta\big)\mathbb{P}(\Theta = \theta).$$

Either of these formulas can be used for calculations, and the choice between them depends on which conditioning leads to simpler expressions in a given problem.
The optimality of the MAP rule for the overall probability of error follows from its optimality at the conditional level. For each value of $x$, the MAP estimate minimizes $\mathbb{P}(\hat{\Theta} \neq \Theta \mid X = x)$. Since the overall probability of error is a weighted sum of these conditional probabilities, it must also be smallest when we use the MAP rule.
In summary, when $\Theta$ is discrete and interpreted as a set of alternative hypotheses, the MAP rule is the preferred estimator when the objective is to minimize the probability of making an incorrect decision. The LMS estimator, on the other hand, is more appropriate when the parameter is numerical and performance is measured in terms of mean squared error. Both estimators arise naturally from the posterior distribution, which encapsulates all the information provided by the data and the prior model.

Example: three-hypothesis decision
Suppose that $\Theta$ takes values in $\{1, 2, 3\}$ and that, after observing $X = x$, the posterior PMF is given by

$$p_{\Theta|X}(\theta \mid x) = \begin{cases} 0.1, & \theta = 1, \\ 0.6, & \theta = 2, \\ 0.3, & \theta = 3. \end{cases}$$

The MAP estimate is $\hat{\theta}_{\text{MAP}} = 2$, because this value has the largest posterior probability. The corresponding conditional probability of error is

$$\mathbb{P}\big(\Theta \neq 2 \mid X = x\big) = 0.1 + 0.3 = 0.4,$$

which is smaller than the error probability that would result from choosing either $\hat{\theta} = 1$ or $\hat{\theta} = 3$.

### 1.7.4 Discrete parameter, continuous observation

We now consider Bayesian inference in the case where the parameter $\Theta$ is discrete, while the observation $X$ is continuous. As before, $\Theta$ takes values in a finite or countable set, and we can interpret each possible value as a different hypothesis about the underlying model. The key difference is that the distribution of $X$ is now described by probability density functions rather than probability mass functions.
The prior information about $\Theta$ is captured by a PMF $p_\Theta(\theta)$. For the observation $X$, we specify, for each possible value of $\Theta$, a conditional probability density function $f_{X|\Theta}(x \mid \theta)$. These conditional densities describe how the continuous observation is generated under each hypothesis.
Bayes' rule still provides the link between the prior and the posterior, with the only change being that PDFs now appear in place of PMFs for the observation. For a given observation $X = x$, the posterior PMF of $\Theta$ is given by

$$p_{\Theta|X}(\theta \mid x) = \frac{f_{X|\Theta}(x \mid \theta)\, p_\Theta(\theta)}{\sum_{\theta'} f_{X|\Theta}(x \mid \theta')\, p_\Theta(\theta')},$$

where the denominator is the unconditional PDF of $X$ evaluated at $x$.
A standard example arises in communication systems. Suppose that $\Theta$ can take one of three possible values, corresponding to three different signals that might be transmitted. The transmitted signal is corrupted by additive noise, and the receiver observes a random variable

$$X = \Theta + W,$$

where $W$ represents noise.
A typical assumption is that the noise $W$ has mean zero, some known variance, and is independent of $\Theta$. If we know the PDF $f_W(w)$, then conditioning on $\Theta = \theta$ yields

$$X \mid (\Theta = \theta) = \theta + W.$$

Adding the constant $\theta$ simply shifts the distribution of $W$, so the conditional density of $X$ given $\Theta = \theta$ is

$$f_{X|\Theta}(x \mid \theta) = f_W(x - \theta).$$

Using these conditional densities together with the prior $p_\Theta(\theta)$, we can apply Bayes' rule and compute the posterior probabilities $p_{\Theta|X}(\theta \mid x)$. Once this posterior PMF has been obtained, the situation is analogous to the discrete–discrete case. In particular, we can again apply the maximum a posteriori rule to construct an estimate of $\Theta$.
The MAP estimate $\hat{\theta}_{\text{MAP}}$ is defined, for the observed value $x$, by

$$\hat{\theta}_{\text{MAP}} = \arg\max_\theta\, p_{\Theta|X}(\theta \mid x).$$

That is, we choose the value of $\theta$ that has the largest posterior probability given the continuous observation. This value is interpreted as the most likely hypothesis in light of the data.
Once an estimate has been chosen, we may again be interested in the probability of error. For a fixed observation $x$, the conditional probability of error is

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta \mid X = x\big) = 1 - p_{\Theta|X}(\hat{\theta} \mid x),$$

since an error occurs whenever $\Theta$ differs from the chosen estimate $\hat{\theta}$. As in the purely discrete case, minimizing this conditional probability of error is equivalent to maximizing the posterior probability of the chosen hypothesis.
Thus, for each fixed $x$, the MAP rule produces the smallest possible conditional probability of error. The fact that $X$ is continuous does not change this conclusion. The posterior PMF $p_{\Theta|X}(\theta \mid x)$ plays exactly the same conceptual role as in the discrete–discrete setting.
We can also consider the overall, or unconditional, probability of error,

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta\big),$$

which averages the conditional probabilities of error over all possible observations and parameters. One way to express this quantity is to condition on the observation $X$ and integrate over all possible values of $x$.
In this form, the law of total probability becomes

$$\mathbb{P}\big(\hat{\Theta} \neq \Theta\big) = \int \mathbb{P}\big(\hat{\Theta} \neq \Theta \mid X = x\big) f_X(x)\, dx,$$

where $f_X(x)$ is the unconditional PDF of $X$. This expression is the continuous analogue of the sum that appears in the discrete case.

An alternative expression is obtained by conditioning on the parameter $\Theta$. We can write

$$\mathbb{P}(\hat{\Theta} \neq \Theta) = \sum_{\theta} \mathbb{P}(\hat{\Theta} \neq \theta \mid \Theta = \theta) p_{\Theta}(\theta).$$

In practical calculations, it may be more convenient to use either the integral representation or this summation representation, depending on which is easier to evaluate for the given model.
The optimality of the MAP rule for the overall probability of error follows the same reasoning as in the discrete–discrete case. For each fixed observation $x$, the MAP rule minimizes $\mathbb{P}(\hat{\Theta} \neq \Theta \mid X = x)$. Since the overall probability of error is obtained by integrating these conditional probabilities with respect to the distribution of $X$, the MAP rule also minimizes the unconditional probability of error.
Therefore, when $\Theta$ represents a finite set of hypotheses and the performance criterion is the probability of making an incorrect decision, the MAP rule is optimal whether the observation $X$ is discrete or continuous. The role of the continuous observation is to determine, through the likelihood functions and Bayes' rule, the posterior probabilities on the discrete set of hypotheses. Once those posterior probabilities are known, the decision problem and its solution proceed exactly as in the purely discrete setting.

Example: three-level signal in noise
Suppose that a transmitter sends a signal that can take three possible levels $\{-1, 0, 1\}$, and let $\Theta$ denote the transmitted level. The received signal is

$$X = \Theta + W,$$

where $W$ is zero-mean Gaussian noise with variance $\sigma^2$ and is independent of $\Theta$. Given an observed value $x$, the conditional density $f_{X|\Theta}(x \mid \theta)$ is a Gaussian PDF centered at $\theta$, and Bayes' rule combines these densities with the prior probabilities $p_{\Theta}(\theta)$ to produce the posterior probabilities $p_{\Theta|X}(\theta \mid x)$; the MAP estimate then selects the value of $\theta \in \{-1, 0, 1\}$ with the largest posterior probability, yielding the decision rule that minimizes the probability of symbol detection error.

### 1.7.5 Continuous parameter, continuous observation

We now turn to the case where both the parameter $\Theta$ and the observation $X$ are continuous random variables. As before, the Bayesian framework specifies a prior distribution for $\Theta$ and a conditional distribution for $X$ given $\Theta$. Bayes' rule then provides the posterior distribution of $\Theta$ given the observed value of $X$.
In this setting, all distributions are described by probability density functions. The prior information about the parameter is encoded in a prior density $f_{\Theta}(\theta)$. The observation model is given by the conditional density $f_{X|\Theta}(x \mid \theta)$, which describes how $X$ would be distributed if the parameter took the value $\theta$.
Once we observe a particular numerical value $x$, Bayes' rule takes the form

$$f_{\Theta|X}(\theta \mid x) = \frac{f_{X|\Theta}(x \mid \theta)\, f_{\Theta}(\theta)}{\int f_{X|\Theta}(x \mid \theta')\, f_{\Theta}(\theta')\, d\theta'},$$

which defines the posterior density of $\Theta$ given $X = x$. This posterior density summarizes our updated knowledge about $\Theta$ after taking into account both the prior and the observed data.
A particularly important and widely used class of continuous Bayesian models is the family of linear normal models. In such models, the variables of interest are combined through linear relationships, and all of them are modeled as Gaussian random variables. Because Gaussian distributions are closed under linear transformations, these models often allow closed-form expressions for the posterior distribution.
As a simple example, consider a continuous-valued signal $\Theta$ that we wish to estimate. We observe a noisy version of this signal,

$$X = \Theta + W,$$

where $W$ is additive noise that is independent of $\Theta$ and has a known distribution, often taken to be normal with mean zero and some variance. Given the observation $X = x$, our task is to infer the underlying signal value $\Theta$.
More elaborate versions of this problem involve vector-valued parameters and multiple observations. In such cases, $\Theta$ may be a vector with several components, and we may obtain a collection of measurements $X$ that depend linearly on $\Theta$ plus noise. These multivariate linear normal models form an important part of the study of Bayesian inference and will be examined in more detail elsewhere.
Another example of a continuous model arises when estimating the parameter of a uniform distribution. Suppose that $X$ is uniformly distributed over an interval whose endpoints depend on an unknown parameter $\Theta$. The parameter $\Theta$ is then given a prior distribution, and the goal is to use one or more observations of $X$ to infer the true value of $\Theta$.
In the continuous–continuous setting, we again wish to construct estimators of $\Theta$ based on the observation $X$. Two principal Bayesian estimators appear here, just as in the discrete case. The first is the maximum a posteriori estimator, and the second is the least mean squares estimator.

Given the posterior density $f_{\Theta|X}(\theta \mid x)$, the maximum a posteriori (MAP) estimator chooses the value of $\theta$ at which this density is largest. For a specific observed value $x$, the MAP estimate is defined by

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} f_{\Theta|X}(\theta \mid x).$$

This provides a single value that is regarded as the most plausible parameter value in light of the data and the prior. The least mean squares (LMS) estimator is based on the conditional expectation of $\Theta$ given $X$. For each observed value $x$, the LMS estimate is

$$\hat{\theta}_{\text{LMS}} = \mathbb{E}[\Theta \mid X = x] = \int \theta\, f_{\Theta|X}(\theta \mid x)\, d\theta.$$

This estimator is also known as the conditional expectation estimator or the posterior mean.
To evaluate and compare estimators in this setting, a natural performance measure is the squared distance between the estimate and the true parameter value. For a given estimator $\hat{\Theta} = g(X)$ and a given observation $X = x$, the conditional mean squared error is defined as

$$\mathbb{E}\big[(\hat{\Theta} - \Theta)^2 \mid X = x\big].$$

This quantity measures, on average, how far the estimator is from the true parameter, conditioned on the observed data.
We can also consider the unconditional, or overall, mean squared error of the estimator. This is obtained by averaging over both the randomness in $\Theta$ and the randomness in $X$,

$$\mathbb{E}\big[(\hat{\Theta} - \Theta)^2\big],$$

and it provides a single number summarizing the overall accuracy of the estimator. Both conditional and unconditional mean squared error play an important role in analyzing the performance of estimators.
The LMS estimator has a particularly important optimality property with respect to mean squared error. Among all possible estimators that are functions of $X$, the conditional expectation $\mathbb{E}[\Theta \mid X]$ achieves the smallest mean squared error. This fact justifies the interpretation of the LMS estimator as the best, in the mean squared error sense, within the Bayesian framework.
In practice, the choice between the MAP estimator and the LMS estimator depends on the goals of the application. If we are interested in the most probable parameter value, the MAP estimate is a natural choice. If instead we wish to minimize mean squared error, the LMS estimator is the appropriate one.
In summary, the continuous–continuous Bayesian setting extends the same ideas we have seen in discrete models to the realm of probability densities. We specify a prior, a likelihood model, and then apply Bayes' rule to obtain the posterior distribution. From this posterior, we can construct estimators such as the MAP and LMS estimators and evaluate their performance using criteria such as mean squared error.

Example: Gaussian signal in Gaussian noise
Let $\Theta$ be a Gaussian random variable with mean $\mu_\Theta$ and variance $\sigma_\Theta^2$. Suppose we observe

$$X = \Theta + W,$$

where $W$ is independent Gaussian noise with mean 0 and variance $\sigma_W^2$. In this linear normal model, the posterior distribution of $\Theta$ given $X = x$ is also Gaussian, and both the MAP and LMS estimators coincide, yielding a linear function of $x$ that minimizes the mean squared error between the estimate and the true value of $\Theta$.

### 1.7.6 Inferring the unknown bias of a coin and the Beta distribution

We consider a coin with an unknown bias $\Theta$, where $\Theta$ takes values in the interval $[0, 1]$. Consistent with the Bayesian viewpoint, we treat $\Theta$ as a random variable and assign to it a prior distribution. We then flip the coin $n$ times independently and record the number of heads obtained.
Let $K$ denote the number of heads in these $n$ tosses. Given a particular value $\Theta = \theta$, the random variable $K$ has a binomial distribution with parameters $n$ and $\theta$. Our goal is to use the observed value of $K$ to infer information about the unknown parameter $\Theta$.
We begin with the case where the prior distribution of $\Theta$ is uniform on the unit interval. This prior can be interpreted as expressing a state of ignorance, in the sense that all values of $\theta \in [0, 1]$ are considered equally plausible a priori. The corresponding prior density is

$$f_\Theta(\theta) = 1, \qquad 0 \le \theta \le 1.$$

Once we observe a specific value $K = k$, Bayes' rule in the mixed continuous–discrete setting takes the form

$$f_{\Theta|K}(\theta \mid k) = \frac{p_{K|\Theta}(k \mid \theta)\, f_\Theta(\theta)}{\int_0^1 p_{K|\Theta}(k \mid \theta')\, f_\Theta(\theta')\, d\theta'},$$

for $0 \leq \theta \leq 1$. Here, $f_{\Theta|K}(\theta \mid k)$ is the posterior density of $\Theta$ given that $K = k$.
Given $\Theta = \theta$, the number of heads $K$ follows a binomial distribution, so

$$p_{K|\Theta}(k \mid \theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Substituting this into Bayes' rule and using the uniform prior, we obtain

$$f_{\Theta|K}(\theta \mid k) = \frac{\binom{n}{k}\theta^k(1-\theta)^{n-k} \cdot 1}{\int_0^1 \binom{n}{k}\theta'^k(1-\theta')^{n-k}\, d\theta'}, \qquad 0 \leq \theta \leq 1.$$

The binomial coefficient $\binom{n}{k}$ does not depend on $\theta$ and can be absorbed into the normalizing constant in the denominator. Thus, up to a constant factor, the dependence on $\theta$ in the posterior density is

$$f_{\Theta|K}(\theta \mid k) \propto \theta^k(1-\theta)^{n-k}, \qquad 0 \leq \theta \leq 1.$$

Outside the interval $[0, 1]$, both the prior and the posterior densities are equal to zero.
Densities of the form

$$f(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \leq \theta \leq 1,$$

are known as *Beta* distributions with parameters $\alpha$ and $\beta$. Comparing this with the expression above, we see that the posterior distribution of $\Theta$ given $K = k$ is a Beta distribution with parameters $\alpha = k + 1$ and $\beta = n - k + 1$.
Next, we consider the more general case in which the prior distribution of $\Theta$ is itself a Beta distribution. Specifically, suppose that

$$f_\Theta(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \leq \theta \leq 1,$$

for some $\alpha > 0$ and $\beta > 0$. This prior allows us to encode prior beliefs that favor certain regions of the interval $[0, 1]$.
Using the same likelihood as before, the joint dependence on $\theta$ in the numerator of Bayes' rule is

$$p_{K|\Theta}(k \mid \theta)\, f_\Theta(\theta) \propto \left[\theta^k(1-\theta)^{n-k}\right]\left[\theta^{\alpha-1}(1-\theta)^{\beta-1}\right].$$

Multiplying these terms, we obtain

$$p_{K|\Theta}(k \mid \theta)\, f_\Theta(\theta) \propto \theta^{\alpha+k-1}(1-\theta)^{\beta+n-k-1}, \qquad 0 \leq \theta \leq 1.$$

After normalization, the posterior density takes the form

$$f_{\Theta|K}(\theta \mid k) \propto \theta^{\alpha+k-1}(1-\theta)^{\beta+n-k-1}, \qquad 0 \leq \theta \leq 1,$$

and is equal to zero outside this interval. This shows that the posterior distribution of $\Theta$ is again a Beta distribution, now with updated parameters $\alpha' = \alpha + k$ and $\beta' = \beta + n - k$.
We have thus discovered an important property of the Beta family. If the prior distribution of $\Theta$ is Beta with parameters $\alpha$ and $\beta$, and the data consist of $n$ Bernoulli trials with $k$ heads, then the posterior distribution of $\Theta$ is also Beta, with parameters $\alpha + k$ and $\beta + n - k$. In other words, the Beta distribution is *conjugate* to the Bernoulli and binomial likelihood in the Bayesian framework.
This conjugacy has several practical advantages. It leads to simple update rules for the parameters $\alpha$ and $\beta$ as more data are observed. In particular, each observed head increases the first parameter $\alpha$ by one, and each observed tail increases the second parameter $\beta$ by one.
Because of this structure, it is convenient to update the posterior distribution recursively as new coin tosses are recorded. Starting from an initial Beta prior, we can process each observation in turn, updating the parameters after each toss instead of recomputing the posterior from scratch. This makes the Beta–binomial model a powerful and flexible tool for modeling and learning about unknown probabilities in Bernoulli-type experiments.

Example: updating a Beta prior
Suppose that the prior distribution of the coin bias $\Theta$ is Beta with parameters $\alpha = 2$ and $\beta = 3$, reflecting prior information that favors values of $\theta$ near 0.4. We then flip the coin $n = 5$ times and observe $k = 4$ heads. Using the update rule, the posterior distribution of $\Theta$ is Beta with parameters $\alpha' = 2 + 4 = 6$ and $\beta' = 3 + 5 - 4 = 4$, which concentrates more mass near higher values of $\theta$ and reflects the evidence from the observed data.

### 1.7.7 Inferring the unknown bias of a coin: point estimates

We continue with the problem of estimating the unknown bias $\Theta$ of a coin, where $\Theta \in [0,1]$. We assume a uniform prior on $[0,1]$ and observe the number of heads $K$ in $n$ independent tosses of the coin. The posterior density of $\Theta$ given $K = k$ is a Beta distribution of the form

$$f_{\Theta|K}(\theta \mid k) = d(n,k)\,\theta^k(1-\theta)^{n-k}, \qquad 0 \le \theta \le 1,$$

where $d(n,k)$ is a normalizing constant that depends on $n$ and $k$ but not on $\theta$.
We first derive the maximum a posteriori (MAP) estimate of $\Theta$. By definition, the MAP estimate is the value of $\theta$ at which the posterior density $f_{\Theta|K}(\theta \mid k)$ attains its maximum. Since $d(n,k)$ does not depend on $\theta$, maximizing the posterior is equivalent to maximizing the function

$$g(\theta) = \theta^k(1-\theta)^{n-k}, \qquad 0 < \theta < 1.$$

It is more convenient to maximize the logarithm of this function. Define

$$\ell(\theta) = \log g(\theta) = k\log\theta + (n-k)\log(1-\theta).$$

To find the maximizer, we differentiate $\ell(\theta)$ with respect to $\theta$ and set the derivative equal to zero.
The derivative is

$$\ell'(\theta) = \frac{k}{\theta} - \frac{n-k}{1-\theta}.$$

Setting $\ell'(\theta) = 0$ gives

$$\frac{k}{\theta} = \frac{n-k}{1-\theta},$$

which implies

$$k(1-\theta) = \theta(n-k).$$

Solving for $\theta$, we obtain

$$k - k\theta = n\theta - k\theta \quad \Rightarrow \quad k = n\theta \quad \Rightarrow \quad \theta = \frac{k}{n}.$$

Thus, for the observed value $K = k$, the MAP estimate of the bias is

$$\hat{\theta}_{\text{MAP}} = \frac{k}{n}.$$

This expression is quite natural: it coincides with the empirical frequency of heads in the $n$ tosses. The corresponding estimator, viewed as a function of the random variable $K$, is

$$\hat{\Theta}_{\text{MAP}} = \frac{K}{n},$$

which is a random variable whose realized value is the MAP estimate once $K$ is observed.
We now turn to the least mean squares (LMS) estimator, which is the conditional expectation of $\Theta$ given $K = k$. By definition,

$$\hat{\theta}_{\text{LMS}} = \mathbb{E}[\Theta \mid K = k] = \int_0^1 \theta\, f_{\Theta|K}(\theta \mid k)\, d\theta.$$

Substituting the expression for the posterior density, we obtain

$$\hat{\theta}_{\text{LMS}} = \int_0^1 \theta\, d(n,k)\,\theta^k(1-\theta)^{n-k}\, d\theta = d(n,k)\int_0^1 \theta^{k+1}(1-\theta)^{n-k}\, d\theta.$$

To proceed, we need an explicit expression for the normalizing constant $d(n,k)$. This constant is determined by the requirement that $f_{\Theta|K}(\theta \mid k)$ integrates to 1 over the interval $[0,1]$. Thus,

$$1 = \int_0^1 f_{\Theta|K}(\theta \mid k)\, d\theta = d(n,k)\int_0^1 \theta^k(1-\theta)^{n-k}\, d\theta,$$

so

$$d(n,k) = \frac{1}{\int_0^1 \theta^k(1-\theta)^{n-k}\, d\theta}.$$

We will make use of the following integral identity. For non-negative integers $\alpha$ and $\beta$,

$$\int_0^1 \theta^\alpha(1-\theta)^\beta\, d\theta = \frac{\alpha!\,\beta!}{(\alpha+\beta+1)!}.$$

Applying this formula with $\alpha = k$ and $\beta = n - k$, we find

$$\int_0^1 \theta^k (1-\theta)^{n-k}\, d\theta = \frac{k!\,(n-k)!}{(n+1)!},$$

and therefore

$$d(n,k) = \frac{(n+1)!}{k!\,(n-k)!}.$$

We now evaluate the integral appearing in the expression for $\hat{\theta}_{\text{LMS}}$. Using the same identity with $\alpha = k+1$ and $\beta = n-k$, we obtain

$$\int_0^1 \theta^{k+1} (1-\theta)^{n-k}\, d\theta = \frac{(k+1)!\,(n-k)!}{(n+2)!}.$$

Substituting into the expression for $\hat{\theta}_{\text{LMS}}$, we get

$$\hat{\theta}_{\text{LMS}} = d(n,k)\,\frac{(k+1)!\,(n-k)!}{(n+2)!} = \frac{(n+1)!}{k!\,(n-k)!}\,\frac{(k+1)!\,(n-k)!}{(n+2)!}.$$

We simplify this expression step by step. First, we cancel the factors $(n-k)!$ in the numerator and denominator, obtaining

$$\hat{\theta}_{\text{LMS}} = \frac{(n+1)!}{k!}\,\frac{(k+1)!}{(n+2)!}.$$

Next, we use the identities

$$(k+1)! = (k+1)\,k!, \qquad (n+2)! = (n+2)\,(n+1)!,$$

which lead to

$$\hat{\theta}_{\text{LMS}} = \frac{(n+1)!}{k!}\,\frac{(k+1)\,k!}{(n+2)\,(n+1)!} = \frac{k+1}{n+2}.$$

Thus, the LMS estimate of $\Theta$ given $K = k$ is

$$\hat{\theta}_{\text{LMS}} = \frac{k+1}{n+2}.$$

The corresponding estimator, as a function of the random variable $K$, is

$$\hat{\Theta}_{\text{LMS}} = \frac{K+1}{n+2}.$$

We can now compare the MAP and LMS estimates. The MAP estimate for the observed value $K = k$ is

$$\hat{\theta}_{\text{MAP}} = \frac{k}{n},$$

while the LMS estimate is

$$\hat{\theta}_{\text{LMS}} = \frac{k+1}{n+2}.$$

These two expressions are similar but not identical, which reflects the fact that the mode and the mean of a Beta distribution are generally different.
However, when the number of tosses $n$ is large, the two estimates are close to each other. Indeed, for large $n$, the difference between $k/n$ and $(k+1)/(n+2)$ becomes negligible. In this sense, as the amount of data increases, the MAP and LMS estimators for the coin bias nearly coincide.

Example: numerical comparison of MAP and LMS estimates
Suppose we flip a coin $n = 8$ times and observe $k = 6$ heads. Under a uniform prior, the MAP estimate of the bias is

$$\hat{\theta}_{\text{MAP}} = \frac{6}{8} = 0.75,$$

while the LMS estimate is

$$\hat{\theta}_{\text{LMS}} = \frac{6+1}{8+2} = \frac{7}{10} = 0.7.$$

In this moderate-sample regime, the two estimates are close but not identical, with the LMS estimate being slightly more conservative and closer to the center of the interval.

### 1.7.8 Recognizing normal PDFs

Recall that a normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ has a PDF of the form

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big( - \frac{(x-\mu)^2}{2\sigma^2}\Big).$$

Sometimes, we are given a density that already looks like a normal density, and we want to recognize its mean and variance by matching it to this standard form.
Suppose we are told that a random variable $X$ has PDF

$$f_X(x) = c \exp\Big( - 8(x-3)^2\Big),$$

for some normalizing constant $c$. To identify the mean and variance, we compare the exponent with the canonical normal form. We can write

$$-8(x-3)^2 = -\frac{(x-3)^2}{2\sigma^2},$$

which implies

$$\frac{1}{2\sigma^2} = 8 \quad \Rightarrow \quad \sigma^2 = \frac{1}{16}.$$

The term $x - 3$ in the exponent shows that the distribution is centered at 3, so $\mu = 3$. Finally, to find $c$, we note that the PDF must integrate to 1 and that the normalizing constant for a normal distribution is $1/(\sqrt{2\pi\sigma^2})$. Since $\sigma^2 = 1/16$, we have $\sigma = 1/4$, and hence

$$c = \frac{1}{\sqrt{2\pi\sigma^2}} = \frac{1}{\sigma\sqrt{2\pi}} = \frac{1}{\frac{1}{4}\sqrt{2\pi}} = \frac{4}{\sqrt{2\pi}}.$$

Recognizing a general quadratic exponent
Consider now a more general PDF of the form

$$f_X(x) = c \exp\big( - \alpha x^2 - \beta x - \gamma\big),$$

where $\alpha > 0$ and $c$ is a normalizing constant. We will show that this is also a normal PDF and identify its mean and variance.
The key step is to rewrite the quadratic term in the exponent by *completing the square*. We start from

$$\alpha x^2 + \beta x + \gamma.$$

Factor out $\alpha$ from the first two terms:

$$\alpha x^2 + \beta x + \gamma = \alpha\Big(x^2 + \frac{\beta}{\alpha}x\Big) + \gamma.$$

We now complete the square inside the parentheses. Observe that

$$x^2 + \frac{\beta}{\alpha}x = \Big(x + \frac{\beta}{2\alpha}\Big)^2 - \Big(\frac{\beta}{2\alpha}\Big)^2.$$

Substituting this identity, we obtain

$$\alpha x^2 + \beta x + \gamma = \alpha\Big(x + \frac{\beta}{2\alpha}\Big)^2 - \alpha\Big(\frac{\beta}{2\alpha}\Big)^2 + \gamma = \alpha\Big(x + \frac{\beta}{2\alpha}\Big)^2 - \frac{\beta^2}{4\alpha} + \gamma.$$

Therefore, the exponent in the PDF can be rewritten as

$$-\alpha x^2 - \beta x - \gamma = -\alpha\Big(x + \frac{\beta}{2\alpha}\Big)^2 + \frac{\beta^2}{4\alpha} - \gamma.$$

This leads to the expression

$$f_X(x) = c \exp\Big( - \alpha\Big(x + \frac{\beta}{2\alpha}\Big)^2\Big) \exp\Big(\frac{\beta^2}{4\alpha} - \gamma\Big).$$

The second exponential factor does not depend on $x$, so it can be absorbed into the constant by defining

$$c' = c \exp\Big(\frac{\beta^2}{4\alpha} - \gamma\Big).$$

Thus, the PDF becomes

$$f_X(x) = c' \exp\Big( - \alpha\Big(x + \frac{\beta}{2\alpha}\Big)^2\Big).$$

We can now match this with the standard normal form. We want to write the exponent as

$$-\frac{(x-\mu)^2}{2\sigma^2}.$$

Comparing this expression with

$$-\alpha\Big(x + \frac{\beta}{2\alpha}\Big)^2,$$

we see that

$$x - \mu = x + \frac{\beta}{2\alpha},$$

so that

$$\mu = -\frac{\beta}{2\alpha}.$$

For the variance, we equate the coefficients of the squared term in the exponent:

$$\alpha = \frac{1}{2\sigma^2} \quad \Rightarrow \quad \sigma^2 = \frac{1}{2\alpha}.$$

Therefore, $X$ is normally distributed with mean $\mu = -\beta/(2\alpha)$ and variance $\sigma^2 = 1/(2\alpha)$. The normalizing constant $c'$ must then equal $1/\sqrt{2\pi\sigma^2}$. From this, if desired, one can solve for the original constant $c$.

Using the peak to identify the mean

Once we know that a density is normal, there is a faster way to identify its mean. The mean of a normal random variable coincides with the location of the peak of its PDF. Thus, rather than completing the square in full, we can instead find the point $x$ where the exponent is maximized (or, equivalently, where the quadratic inside the minus sign is minimized).

For the PDF

$$f_X(x) = c \exp\big( - \alpha x^2 - \beta x - \gamma\big),$$

the exponent is

$$-\alpha x^2 - \beta x - \gamma.$$

Maximizing this expression is equivalent to minimizing the quadratic

$$\alpha x^2 + \beta x + \gamma.$$

To find the minimizing point, we differentiate with respect to $x$ and set the derivative equal to zero:

$$\frac{d}{dx}\big(\alpha x^2 + \beta x + \gamma\big) = 2\alpha x + \beta = 0.$$

Solving for $x$ gives

$$x = -\frac{\beta}{2\alpha}.$$

This is the location of the minimum of the quadratic, hence the location of the maximum of the exponent and the peak of the PDF. Since the distribution is normal, this point is exactly the mean:

$$\mu = -\frac{\beta}{2\alpha}.$$

The variance can still be recovered by matching the quadratic coefficient $\alpha$ to the standard form, yielding again

$$\sigma^2 = \frac{1}{2\alpha}.$$

### 1.7.9 Normal unknown and additive noise

We consider a simple linear model with additive noise. There is an unknown parameter, modeled as a random variable $\Theta$, that we wish to estimate. What we observe is a noisy version of $\Theta$, given by

$$X = \Theta + W,$$

where $W$ is an additive noise term.
We assume that $\Theta$ and $W$ are independent standard normal random variables, that is,

$$\Theta \sim \mathcal{N}(0,1), \qquad W \sim \mathcal{N}(0,1), \qquad \Theta \perp W.$$

According to the Bayesian approach, inference about $\Theta$ is based on the posterior distribution of $\Theta$ given the observation $X = x$. To obtain this posterior, we first write down the prior and the likelihood.
The prior density of $\Theta$ is the standard normal density

$$f_\Theta(\theta) = \frac{1}{\sqrt{2\pi}} \exp\Big(-\frac{\theta^2}{2}\Big).$$

Next, we determine the conditional distribution of $X$ given $\Theta$. Condition on the event $\{\Theta = \theta\}$. In this conditional universe, the observation satisfies

$$X \mid (\Theta = \theta) = \theta + W.$$

Since $W$ is independent of $\Theta$, conditioning on $\Theta = \theta$ does not change the distribution of $W$. Thus $W$ remains standard normal, and $X$ is the sum of a constant $\theta$ and a standard normal random variable. This implies

$$X \mid (\Theta = \theta) \sim \mathcal{N}(\theta, 1).$$

The corresponding conditional density of $X$ given $\Theta = \theta$ is

$$f_{X|\Theta}(x \mid \theta) = \frac{1}{\sqrt{2\pi}} \exp\Big(-\frac{(x-\theta)^2}{2}\Big).$$

Bayes' rule gives the posterior density of $\Theta$ given $X = x$ as

$$f_{\Theta|X}(\theta \mid x) = \frac{f_{X|\Theta}(x \mid \theta)\, f_\Theta(\theta)}{f_X(x)},$$

where $f_X(x)$ is the marginal density of $X$. For purposes of identifying the posterior form, we can ignore the denominator $f_X(x)$, which is just a normalizing constant depending on $x$ but not on $\theta$. Thus, up to a proportionality constant,

$$f_{\Theta|X}(\theta \mid x) \propto f_{X|\Theta}(x \mid \theta)\, f_\Theta(\theta).$$

Substituting the expressions for the prior and the likelihood, we obtain

$$f_{\Theta|X}(\theta \mid x) \propto \exp\Big(-\frac{(x-\theta)^2}{2}\Big) \exp\Big(-\frac{\theta^2}{2}\Big).$$

Combining the exponents, we get

$$f_{\Theta|X}(\theta \mid x) \propto \exp\Big(-\frac{(x-\theta)^2 + \theta^2}{2}\Big).$$

We now expand the quadratic in $\theta$:

$$(x-\theta)^2 + \theta^2 = (x^2 - 2x\theta + \theta^2) + \theta^2 = x^2 - 2x\theta + 2\theta^2.$$

Thus

$$f_{\Theta|X}(\theta \mid x) \propto \exp\Big(-\frac{x^2 - 2x\theta + 2\theta^2}{2}\Big).$$

We rearrange the exponent by separating the terms that depend on $\theta$ from the ones that depend only on $x$:

$$-\frac{x^2 - 2x\theta + 2\theta^2}{2} = -\frac{x^2}{2} + x\theta - \theta^2.$$

Hence

$$f_{\Theta|X}(\theta \mid x) \propto \exp\Big(-\frac{x^2}{2}\Big) \exp\big(x\theta - \theta^2\big).$$

The factor $\exp(-x^2/2)$ does not involve $\theta$ and can be absorbed into the normalizing constant. Therefore, as a function of $\theta$,

$$f_{\Theta|X}(\theta \mid x) \propto \exp\big(x\theta - \theta^2\big).$$

We rewrite the exponent $x\theta - \theta^2$ by completing the square in $\theta$. Observe that

$$x\theta - \theta^2 = -\theta^2 + x\theta = -\Big(\theta^2 - x\theta\Big) = -\Big[\theta^2 - x\theta + \Big(\frac{x}{2}\Big)^2 - \Big(\frac{x}{2}\Big)^2\Big] = -\Big(\theta - \frac{x}{2}\Big)^2 + \Big(\frac{x}{2}\Big)^2.$$

Substituting into the exponent gives

$$f_{\Theta|X}(\theta \mid x) \propto \exp\Big(-\Big(\theta - \frac{x}{2}\Big)^2 + \Big(\frac{x}{2}\Big)^2\Big) = \exp\Big(\Big(\frac{x}{2}\Big)^2\Big)\exp\Big(-\Big(\theta - \frac{x}{2}\Big)^2\Big).$$

Again, the factor $\exp\big((x/2)^2\big)$ is a constant with respect to $\theta$ and can be absorbed into the normalization. Thus the posterior density has the form

$$f_{\Theta|X}(\theta \mid x) \propto \exp\Big(-\Big(\theta - \frac{x}{2}\Big)^2\Big).$$

We recognize this as the kernel of a normal density with mean $x/2$ and variance $1/2$. Indeed, a normal density with mean $m$ and variance $\sigma^2$ has exponent $-(\theta - m)^2/(2\sigma^2)$. In our case, the exponent is $-(\theta - x/2)^2$, which corresponds to

$$\frac{1}{2\sigma^2} = 1 \quad \Rightarrow \quad \sigma^2 = \frac{1}{2}.$$

Therefore, the posterior distribution is

$$\Theta \mid (X = x) \sim \mathcal{N}\Big(\frac{x}{2}, \frac{1}{2}\Big).$$

Since the posterior is normal, the conditional expectation and the mode coincide. Thus, the least mean squares estimator (posterior mean) and the maximum a posteriori (MAP) estimator are the same:

$$\mathbb{E}[\Theta \mid X = x] = \frac{x}{2}.$$

Equivalently, we can obtain this value by locating the peak of the posterior density. Because the posterior kernel is

$$\exp\big(x\theta - \theta^2\big),$$

we can find the peak by minimizing the quadratic term inside the negative exponent. Ignoring constants that do not depend on $\theta$, the relevant function is

$$q(\theta) = \theta^2 - x\theta.$$

To minimize $q(\theta)$, we take the derivative with respect to $\theta$ and set it equal to zero:

$$\frac{d}{d\theta}q(\theta) = 2\theta - x = 0.$$

Solving for $\theta$ yields

$$\theta = \frac{x}{2}.$$

This is the location of the maximum of the posterior density and therefore the posterior mean and MAP estimate. From the perspective of estimators (random variables), we define the estimator $\hat{\Theta}(X)$ as

$$\hat{\Theta}(X) = \mathbb{E}[\Theta \mid X] = \frac{X}{2}.$$

When we observe a particular numerical value $X = x$, the corresponding estimate is the number $x/2$. This estimator balances prior information (that $\Theta$ is centered at 0) with the observed data $X$, producing an estimate that lies halfway between the prior mean 0 and the observation $x$.

Generalizations

The same structure arises in more general settings where $\Theta$ and $W$ are independent normal random variables with arbitrary means and variances. In such cases, one can show, by repeating the same type of calculation, that the posterior distribution of $\Theta$ given $X$ is still normal. Because the posterior is normal, its mean coincides with its mode, so the least mean squares estimator and the MAP estimator are again identical. Moreover, the resulting estimator of $\Theta$ turns out to be a linear function of the observation $X$, reflecting a general feature of linear models with Gaussian priors and Gaussian noise.

### 1.7.10 The case of multiple observations

We consider a model in which an unknown parameter $\Theta$ is observed through multiple noisy measurements. For $i = 1, \ldots, n$, each observation satisfies

$$X_i = \Theta + W_i,$$

where $W_i$ is an additive noise term.
We assume that $\Theta$ is a normal random variable with mean $x_0$ and variance $\sigma_0^2$:

$$\Theta \sim \mathcal{N}(x_0, \sigma_0^2).$$

For each $i$, the noise term is also normal with mean 0 and variance $\sigma_i^2$:

$$W_i \sim \mathcal{N}(0, \sigma_i^2),$$

and we assume that $\Theta, W_1, \ldots, W_n$ are all mutually independent. Our goal is to estimate $\Theta$ on the basis of the observations $X_1, \ldots, X_n$.
In the Bayesian framework, estimation is based on the posterior distribution of $\Theta$ given the observations. Let $\mathbf{X} = (X_1, \ldots, X_n)$ and $\mathbf{x} = (x_1, \ldots, x_n)$. Bayes' rule for continuous random variables gives

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) = \frac{f_{\mathbf{X}|\Theta}(\mathbf{x} \mid \theta)\, f_\Theta(\theta)}{f_{\mathbf{X}}(\mathbf{x})},$$

where $f_\Theta$ is the prior density of $\Theta$, $f_{\mathbf{X}|\Theta}$ is the conditional joint density of the measurements given $\Theta$, and $f_{\mathbf{X}}$ is the marginal density of $\mathbf{X}$.
We start by determining the conditional distribution of a single observation $X_i$ given $\Theta$. Condition on $\{\Theta = \theta\}$. In this conditional universe, we have

$$X_i \mid (\Theta = \theta) = \theta + W_i.$$

Since $W_i$ is independent of $\Theta$, its distribution is unaffected by conditioning and remains $\mathcal{N}(0, \sigma_i^2)$. Thus

$$X_i \mid (\Theta = \theta) \sim \mathcal{N}(\theta, \sigma_i^2).$$

The conditional density of $X_i$ given $\Theta = \theta$ is therefore

$$f_{X_i|\Theta}(x_i \mid \theta) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\Big( - \frac{(x_i - \theta)^2}{2\sigma_i^2}\Big).$$

We now turn to the conditional joint density $f_{\mathbf{X}|\Theta}(\mathbf{x} \mid \theta)$. Given $\Theta = \theta$, the measurement equations are

$$X_i = \theta + W_i, \quad i = 1, \ldots, n.$$

The noise variables $W_1, \ldots, W_n$ are independent, and conditioning on $\Theta = \theta$ does not change this independence. Since each $X_i$ is obtained from $W_i$ by adding the constant $\theta$, the random variables $X_1, \ldots, X_n$ are also independent in the conditional universe. Therefore, the conditional joint density factors as

$$f_{\mathbf{X}|\Theta}(\mathbf{x} \mid \theta) = \prod_{i=1}^n f_{X_i|\Theta}(x_i \mid \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\Big( - \frac{(x_i - \theta)^2}{2\sigma_i^2}\Big).$$

The prior density of $\Theta$ is

$$f_\Theta(\theta) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\Big( - \frac{(\theta - x_0)^2}{2\sigma_0^2}\Big).$$

Plugging these expressions into Bayes' rule, we obtain, up to a normalizing constant,

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto f_{\mathbf{X}|\Theta}(\mathbf{x} \mid \theta)\, f_\Theta(\theta) = \left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\Big( - \frac{(x_i - \theta)^2}{2\sigma_i^2}\Big)\right] \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\Big( - \frac{(\theta - x_0)^2}{2\sigma_0^2}\Big).$$

The prefactors involving $1/\sqrt{2\pi\sigma_i^2}$ and $1/\sqrt{2\pi\sigma_0^2}$ do not depend on $\theta$ and can be absorbed into a constant. Thus

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto \exp\left( - \frac{(\theta - x_0)^2}{2\sigma_0^2} - \sum_{i=1}^n \frac{(x_i - \theta)^2}{2\sigma_i^2}\right).$$

To identify the posterior distribution, we focus on the exponent as a function of $\theta$. Let

$$Q(\theta) = \frac{(\theta - x_0)^2}{2\sigma_0^2} + \sum_{i=1}^n \frac{(x_i - \theta)^2}{2\sigma_i^2}.$$

Then

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto \exp\big(-Q(\theta)\big).$$

We expand $Q(\theta)$ and collect terms that are quadratic, linear, and constant in $\theta$. First, for the prior term,

$$\frac{(\theta - x_0)^2}{2\sigma_0^2} = \frac{1}{2\sigma_0^2}\big(\theta^2 - 2x_0\theta + x_0^2\big) = \frac{\theta^2}{2\sigma_0^2} - \frac{x_0}{\sigma_0^2}\theta + \frac{x_0^2}{2\sigma_0^2}.$$

Similarly, each measurement term expands as

$$\frac{(x_i - \theta)^2}{2\sigma_i^2} = \frac{1}{2\sigma_i^2}\big(x_i^2 - 2x_i\theta + \theta^2\big) = \frac{\theta^2}{2\sigma_i^2} - \frac{x_i}{\sigma_i^2}\theta + \frac{x_i^2}{2\sigma_i^2}.$$

Summing over $i$, we obtain

$$\sum_{i=1}^{n} \frac{(x_i - \theta)^2}{2\sigma_i^2} = \theta^2 \sum_{i=1}^{n} \frac{1}{2\sigma_i^2} - \theta \sum_{i=1}^{n} \frac{x_i}{\sigma_i^2} + \sum_{i=1}^{n} \frac{x_i^2}{2\sigma_i^2}.$$

Combining the prior term and the measurement terms, the quadratic function $Q(\theta)$ takes the form

$$Q(\theta) = \frac{1}{2}\left(\Big(\frac{1}{\sigma_0^2} + \sum_{i=1}^{n} \frac{1}{\sigma_i^2}\Big)\theta^2 - 2\Big(\frac{x_0}{\sigma_0^2} + \sum_{i=1}^{n} \frac{x_i}{\sigma_i^2}\Big)\theta\right) + (\text{constant in } \theta).$$

The constant term does not affect the shape of the posterior density as a function of $\theta$, so we omit it. Thus, up to an additive constant,

$$Q(\theta) = \frac{1}{2}A\theta^2 - B\theta,$$

where

$$A = \frac{1}{\sigma_0^2} + \sum_{i=1}^{n} \frac{1}{\sigma_i^2}, \qquad B = \frac{x_0}{\sigma_0^2} + \sum_{i=1}^{n} \frac{x_i}{\sigma_i^2}.$$

Hence,

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto \exp\Big(-\frac{1}{2}A\theta^2 + B\theta\Big).$$

We now complete the square in $\theta$. Consider the exponent

$$-\frac{1}{2}A\theta^2 + B\theta = -\frac{A}{2}\theta^2 + B\theta.$$

Factor out $-\frac{A}{2}$:

$$-\frac{A}{2}\theta^2 + B\theta = -\frac{A}{2}\Big(\theta^2 - \frac{2B}{A}\theta\Big).$$

We write the quadratic inside the parentheses as a complete square:

$$\theta^2 - \frac{2B}{A}\theta = \Big(\theta - \frac{B}{A}\Big)^2 - \Big(\frac{B}{A}\Big)^2.$$

Substituting,

$$-\frac{A}{2}\theta^2 + B\theta = -\frac{A}{2}\Big(\Big(\theta - \frac{B}{A}\Big)^2 - \Big(\frac{B}{A}\Big)^2\Big) = -\frac{A}{2}\Big(\theta - \frac{B}{A}\Big)^2 + \frac{A}{2}\Big(\frac{B}{A}\Big)^2.$$

Therefore,

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto \exp\Big(-\frac{A}{2}\Big(\theta - \frac{B}{A}\Big)^2\Big)\exp\Big(\frac{A}{2}\Big(\frac{B}{A}\Big)^2\Big).$$

The second exponential factor does not depend on $\theta$ and can be absorbed into the normalizing constant. Thus the posterior density has the form

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto \exp\Big(-\frac{A}{2}\Big(\theta - \frac{B}{A}\Big)^2\Big).$$

Comparing with the standard normal kernel

$$\exp\Big(-\frac{(\theta - m)^2}{2\sigma^2}\Big),$$

we identify

$$\frac{1}{2\sigma^2} = \frac{A}{2} \quad \Rightarrow \quad \sigma^2 = \frac{1}{A},$$

and

$$m = \frac{B}{A}.$$

Therefore, the posterior distribution is normal:

$$\Theta \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}\Big(\frac{B}{A}, \frac{1}{A}\Big),$$

that is,

$$\Theta \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}\left(\frac{\dfrac{x_0}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{x_i}{\sigma_i^2}}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{1}{\sigma_i^2}}, \frac{1}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{1}{\sigma_i^2}}\right).$$

Because the posterior is normal, the conditional expectation and the mode coincide. Thus, the least mean squares estimator (posterior mean) and the maximum a posteriori (MAP) estimator are identical. The posterior mean is

$$\mathbb{E}[\Theta \mid \mathbf{X} = \mathbf{x}] = \frac{\dfrac{x_0}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{x_i}{\sigma_i^2}}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{1}{\sigma_i^2}}.$$

Interpreted as an estimator (a random variable), we write

$$\hat{\Theta}(\mathbf{X}) = \mathbb{E}[\Theta \mid \mathbf{X}] = \frac{\dfrac{x_0}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{X_i}{\sigma_i^2}}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^n \frac{1}{\sigma_i^2}}.$$

This expression has a natural interpretation as a weighted average. Each term $X_i$ is multiplied by the weight $1/\sigma_i^2$, and the prior mean $x_0$ is multiplied by the weight $1/\sigma_0^2$. The denominator is the sum of all these weights, so we can write

$$\hat{\Theta}(\mathbf{X}) = \sum_{i=0}^n w_i Z_i,$$

where

$$Z_0 = x_0, \quad Z_i = X_i \text{ for } i \geq 1,$$

and

$$w_i = \frac{1/\sigma_i^2}{\displaystyle\sum_{j=0}^n 1/\sigma_j^2}, \quad i = 0, \ldots, n,$$

with the convention that $\sigma_0^2$ is the prior variance. The prior mean $x_0$ thus acts as an additional observation, and all pieces of information are combined in a symmetric and linear fashion.
The dependence of the weights on the variances matches our intuition. If a particular variance $\sigma_i^2$ is large, the corresponding observation $X_i$ is very noisy and should receive a small weight. Conversely, observations with small noise variance contribute more to the final estimate. In summary, the Bayesian estimator in this normal model is a linear function of the measurements and the prior mean, and the posterior distribution of $\Theta$ remains normal.

### 1.7.11 The mean squared error

We continue with the model in which an unknown parameter $\Theta$ is observed through multiple noisy measurements. For $i = 1, \ldots, n$,

$$X_i = \Theta + W_i,$$

where $\Theta \sim \mathcal{N}(x_0, \sigma_0^2)$, $W_i \sim \mathcal{N}(0, \sigma_i^2)$, and $\Theta, W_1, \ldots, W_n$ are independent. We have seen that the posterior distribution of $\Theta$ given the observations $\mathbf{X} = (X_1, \ldots, X_n)$ is normal with density

$$f_{\Theta|\mathbf{X}}(\theta \mid \mathbf{x}) \propto \exp\left(-\frac{(\theta - x_0)^2}{2\sigma_0^2} - \sum_{i=1}^n \frac{(x_i - \theta)^2}{2\sigma_i^2}\right).$$

The posterior mean, which is also the maximum a posteriori (MAP) estimate, is

$$\hat{\theta} = \mathbb{E}[\Theta \mid \mathbf{X} = \mathbf{x}] = \frac{\dfrac{x_0}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{x_i}{\sigma_i^2}}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}.$$

As an estimator (a function of the random vector $\mathbf{X}$), we write

$$\hat{\Theta}(\mathbf{X}) = \frac{\dfrac{x_0}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{X_i}{\sigma_i^2}}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}.$$

We now assess the performance of this estimator in terms of the mean squared error (MSE). Given that we have observed $\mathbf{X} = \mathbf{x}$, the estimator $\hat{\Theta}(\mathbf{X})$ takes the numerical value $\hat{\theta}$, and the remaining uncertainty about $\Theta$ is described by the posterior distribution. The conditional mean squared error is defined as

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2 \,\big|\, \mathbf{X} = \mathbf{x}\big].$$

In the conditional universe where $\mathbf{X} = \mathbf{x}$ is fixed, $\hat{\Theta}(\mathbf{X}) = \hat{\theta}$ is a constant. Moreover, $\hat{\theta}$ is the mean of the posterior distribution of $\Theta$ given $\mathbf{X} = \mathbf{x}$. Thus we are taking the expectation of the squared distance from the mean:

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2 \,\big|\, \mathbf{X} = \mathbf{x}\big] = \mathbb{E}\big[(\Theta - \hat{\theta})^2 \,\big|\, \mathbf{X} = \mathbf{x}\big],$$

which is exactly the variance of the posterior distribution:

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2 \,\big|\, \mathbf{X} = \mathbf{x}\big] = \mathrm{Var}(\Theta \mid \mathbf{X} = \mathbf{x}).$$

Therefore, the conditional mean squared error is the posterior variance of $\Theta$.
We now compute this posterior variance. Recall that the exponent in the posterior density can be written as a quadratic function of $\theta$:

$$-\frac{(\theta - x_0)^2}{2\sigma_0^2} - \sum_{i=1}^{n} \frac{(x_i - \theta)^2}{2\sigma_i^2} = -\frac{1}{2}A\theta^2 + B\theta + (\text{constant}),$$

where

$$A = \frac{1}{\sigma_0^2} + \sum_{i=1}^{n} \frac{1}{\sigma_i^2}.$$

From the previous derivation, we know that

$$\Theta \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}\Big(\frac{B}{A}, \frac{1}{A}\Big).$$

Thus the posterior variance is

$$\mathrm{Var}(\Theta \mid \mathbf{X} = \mathbf{x}) = \frac{1}{A} = \frac{1}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}.$$

This is the conditional mean squared error of the estimator $\hat{\Theta}(\mathbf{X})$ given the specific observations $\mathbf{X} = \mathbf{x}$:

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2 \,\big|\, \mathbf{X} = \mathbf{x}\big] = \frac{1}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}.$$

A remarkable property of this model is that this quantity does not depend on the particular observed values $x_1, \ldots, x_n$; it depends only on the variances. Thus, regardless of what data we observe, the remaining uncertainty about $\Theta$, measured by the posterior variance, is always the same.
We now consider the unconditional mean squared error, that is, the MSE before performing any measurements. This is

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2\big].$$

Using the law of total expectation,

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2\big] = \mathbb{E}\Big[\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2 \mid \mathbf{X}\big]\Big].$$

For each realization $\mathbf{X} = \mathbf{x}$, the inner expectation is the conditional MSE, which we have found to be equal to the constant

$$\frac{1}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}.$$

Since this does not depend on $\mathbf{X}$, we can pull it out of the expectation:

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2\big] = \frac{1}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}} \cdot \mathbb{E}[statswithr.github](https://statswithr.github.io/book/inference-and-decision-making-$$

Thus the unconditional mean squared error equals the conditional mean squared error. In other words, the expected performance of the estimator, before making any measurements, is equal to the posterior variance obtained after the measurements, and this value is the same for every possible outcome of the measurements.
Let us now interpret this expression and examine some special cases. First, suppose that at least one of the noise variances $\sigma_i^2$ is very small. Then the corresponding term $1/\sigma_i^2$ is very large. Consequently, the sum

$$\frac{1}{\sigma_0^2} + \sum_{i=1}^{n} \frac{1}{\sigma_i^2}$$

is large, and its reciprocal, the mean squared error, is small. This reflects the fact that if we have even a single high-quality (low-noise) measurement, we can estimate $\Theta$ quite accurately.
On the other hand, if all noise variances $\sigma_i^2$ are large, then each term $1/\sigma_i^2$ is small. The sum of these small terms is also small, so its reciprocal is large. In this case, all measurements are very noisy, and we cannot expect to estimate $\Theta$ well; the mean squared error is large.
Next, consider the special case where all variances, including the prior variance, are equal:

$$\sigma_0^2 = \sigma_1^2 = \cdots = \sigma_n^2 = \sigma^2.$$

Then

$$\frac{1}{\sigma_0^2} + \sum_{i=1}^{n} \frac{1}{\sigma_i^2} = \frac{1}{\sigma^2} + \sum_{i=1}^{n} \frac{1}{\sigma^2} = \frac{n+1}{\sigma^2}.$$

Therefore, the mean squared error simplifies to

$$\mathbb{E}\big[(\Theta - \hat{\Theta}(\mathbf{X}))^2\big] = \frac{1}{(n+1)/\sigma^2} = \frac{\sigma^2}{n+1}.$$

This expression shows explicitly how the estimation error decreases as the number of observations $n$ increases. Each additional measurement contributes an extra term $1/\sigma^2$ to the denominator, reducing the overall variance and improving the quality of the estimate.

A concrete example: single observation and standard normals

To better appreciate the invariance of the conditional variance with respect to the data, let us revisit the simple model with a single observation and standard normals. Assume

$$\Theta \sim \mathcal{N}(0,1), \qquad W \sim \mathcal{N}(0,1), \qquad \Theta \perp W,$$

and

$$X = \Theta + W.$$

In this case, $n = 1$, $\sigma_0^2 = 1$, and $\sigma_1^2 = 1$. The general formula gives

$$\mathrm{Var}(\Theta \mid X = x) = \frac{1}{\dfrac{1}{1} + \dfrac{1}{1}} = \frac{1}{2}.$$

Thus, starting from a prior variance of 1, the variance after observing $X$ is reduced to 1/2. This reduction in variance reflects the information gained from the measurement.

Importantly, the conditional variance is equal to $1/2$ for every value of $x$. Whether the observed value $x$ is close to 0 or far away from 0, the remaining uncertainty about $\Theta$, as measured by the posterior variance, is the same. If we observe $x = 0$, our estimate is $\hat{\theta} = x/2 = 0$, and the posterior distribution of $\Theta$ is $\mathcal{N}(0, 1/2)$, which is narrower than the prior $\mathcal{N}(0, 1)$. If instead we observe, say, $x$ far from 0, then the estimate $\hat{\theta} = x/2$ will be correspondingly far from 0, and the posterior distribution will be $\mathcal{N}(x/2, 1/2)$. In this latter case, the observation is surprising in light of the prior, but once it is observed, the remaining uncertainty about $\Theta$ is again characterized by a normal distribution with variance $1/2$.
This behavior is a distinctive feature of linear Gaussian models. The combination of normal priors with normal likelihoods leads to posterior distributions that are also normal, with a variance that depends only on the prior and noise variances, and not on the particular observed values. As a result, the performance of the MAP or least mean squares estimator can be fully characterized by a single number, the posterior variance

$$\frac{1}{\dfrac{1}{\sigma_0^2} + \displaystyle\sum_{i=1}^{n} \frac{1}{\sigma_i^2}},$$

which serves both as the conditional and unconditional mean squared error.

### 1.7.12 Multiple parameters; trajectory estimation

We now consider a model with multiple parameters and multiple observations, motivated by a simple physical scenario. A ball is thrown upwards and follows a trajectory determined by Newton's laws. The vertical displacement $x(t)$ of the ball at time $t$ is modeled by a quadratic function

$$x(t) = \Theta_0 + \Theta_1 t + \Theta_2 t^2,$$

where:

$\Theta_0$ is the initial height, $\Theta_1$ is the initial vertical velocity, $\Theta_2$ is related to the gravitational acceleration (typically negative)

Suppose that the parameters $\Theta_0, \Theta_1, \Theta_2$ are unknown. We model them as random variables and treat the actual physical values as realizations of these random variables. To apply Bayesian inference, we specify a joint prior distribution for the parameter vector

$$\mathbf{\Theta} = (\Theta_0, \Theta_1, \Theta_2)^\top.$$

A common assumption is that the parameters are independent and normally distributed:

$$\Theta_k \sim \mathcal{N}(0, \sigma_{\Theta_k}^2), \quad k = 0, 1, 2,$$

with independence across $k$. The prior variances $\sigma_{\Theta_k}^2$ are chosen to reflect our initial uncertainty about the parameters (for example, how well we know the initial height or velocity).
We observe the ball at times $t_1, \ldots, t_n$. At each time $t_i$, we measure the vertical position with noise, obtaining

$$X_i = \Theta_0 + \Theta_1 t_i + \Theta_2 t_i^2 + W_i, \quad i = 1, \ldots, n,$$

where $W_i$ is the measurement noise. We assume that

$$W_i \sim \mathcal{N}(0, \sigma_i^2),$$

and that the noise variables $W_1, \ldots, W_n$ are independent of each other and independent of $\mathbf{\Theta}$. This is often a realistic model for repeated measurements made by a sensor.
Let

$$\mathbf{X} = (X_1, \ldots, X_n)^\top, \quad \mathbf{x} = (x_1, \ldots, x_n)^\top,$$

and

$$\mathbf{\Theta} = (\Theta_0, \Theta_1, \Theta_2)^\top, \quad \boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)^\top.$$

Our goal is to estimate $\mathbf{\Theta}$ from the observations $\mathbf{X}$ using Bayesian inference, that is, to compute the posterior density $f_{\mathbf{\Theta}|\mathbf{X}}(\boldsymbol{\theta} \mid \mathbf{x})$. Bayes' rule for continuous random vectors gives

$$f_{\mathbf{\Theta}|\mathbf{X}}(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{f_{\mathbf{X}|\mathbf{\Theta}}(\mathbf{x} \mid \boldsymbol{\theta})\, f_{\mathbf{\Theta}}(\boldsymbol{\theta})}{f_{\mathbf{X}}(\mathbf{x})}.$$

Here:

- $f_{\mathbf{\Theta}}(\boldsymbol{\theta})$ is the prior density of the parameter vector,

- $f_{\mathbf{X}|\mathbf{\Theta}}(\mathbf{x} \mid \boldsymbol{\theta})$ is the conditional joint density of the measurements given the parameters,
- $f_{\mathbf{X}}(\mathbf{x})$ is the marginal density of the observations.

We first determine the conditional distribution of a single observation $X_i$ given $\mathbf{\Theta} = \boldsymbol{\theta}$. Conditioning on $\mathbf{\Theta} = \boldsymbol{\theta}$, the measurement model becomes

$$X_i \mid (\mathbf{\Theta} = \boldsymbol{\theta}) = \theta_0 + \theta_1 t_i + \theta_2 t_i^2 + W_i.$$

The noise term $W_i$ remains $\mathcal{N}(0, \sigma_i^2)$ under this conditioning because $W_i$ is independent of $\mathbf{\Theta}$. Therefore,

$$X_i \mid (\mathbf{\Theta} = \boldsymbol{\theta}) \sim \mathcal{N}\big(\theta_0 + \theta_1 t_i + \theta_2 t_i^2,\ \sigma_i^2\big),$$

and the conditional density is

$$f_{X_i|\mathbf{\Theta}}(x_i \mid \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left( - \frac{\big(x_i - (\theta_0 + \theta_1 t_i + \theta_2 t_i^2)\big)^2}{2\sigma_i^2} \right).$$

Next, we determine the conditional joint density $f_{\mathbf{X}|\mathbf{\Theta}}(\mathbf{x} \mid \boldsymbol{\theta})$. Given $\mathbf{\Theta} = \boldsymbol{\theta}$, each $X_i$ is obtained from the corresponding noise variable $W_i$ by adding the deterministic quantity $\theta_0 + \theta_1 t_i + \theta_2 t_i^2$. The noise variables $W_1, \ldots, W_n$ are independent, so the measurements $X_1, \ldots, X_n$ are also independent in the conditional universe where $\mathbf{\Theta}$ is fixed. It follows that

$$f_{\mathbf{X}|\mathbf{\Theta}}(\mathbf{x} \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} f_{X_i|\mathbf{\Theta}}(x_i \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left( - \frac{\big(x_i - (\theta_0 + \theta_1 t_i + \theta_2 t_i^2)\big)^2}{2\sigma_i^2} \right).$$

For the prior, assuming independent normal components with zero means, we have

$$f_{\mathbf{\Theta}}(\boldsymbol{\theta}) = \prod_{k=0}^{2} \frac{1}{\sqrt{2\pi\sigma_{\Theta_k}^2}} \exp\Big( - \frac{\theta_k^2}{2\sigma_{\Theta_k}^2} \Big).$$

Putting these pieces into Bayes' rule and ignoring the marginal density $f_{\mathbf{X}}(\mathbf{x})$, which does not depend on $\boldsymbol{\theta}$, we obtain, up to a normalizing constant,

$$\begin{aligned} f_{\mathbf{\Theta}|\mathbf{X}}(\boldsymbol{\theta} \mid \mathbf{x}) &\propto f_{\mathbf{X}|\mathbf{\Theta}}(\mathbf{x} \mid \boldsymbol{\theta})\, f_{\mathbf{\Theta}}(\boldsymbol{\theta}) \\ &\propto \left[ \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left( - \frac{\big(x_i - (\theta_0 + \theta_1 t_i + \theta_2 t_i^2)\big)^2}{2\sigma_i^2} \right) \right] \left[ \prod_{k=0}^{2} \frac{1}{\sqrt{2\pi\sigma_{\Theta_k}^2}} \exp\Big( - \frac{\theta_k^2}{2\sigma_{\Theta_k}^2} \Big) \right]. \end{aligned}$$

The multiplicative constants in front of the exponentials do not depend on $\boldsymbol{\theta}$ and can be absorbed into a single constant. Therefore, the posterior density is proportional to

$$\exp\left( - \sum_{k=0}^{2} \frac{\theta_k^2}{2\sigma_{\Theta_k}^2} - \sum_{i=1}^{n} \frac{\big(x_i - (\theta_0 + \theta_1 t_i + \theta_2 t_i^2)\big)^2}{2\sigma_i^2} \right).$$

This is the exponential of the negative of a quadratic function of $\boldsymbol{\theta}$. Because the exponent is a quadratic form in the parameters, the posterior distribution of $\mathbf{\Theta}$ is a multivariate normal distribution. Although we do not explicitly write its mean vector and covariance matrix here, the structure matches the general result: a normal prior combined with a linear-Gaussian observation model produces a normal posterior.
To obtain a point estimate of the parameters, a natural choice is the maximum a posteriori (MAP) estimate. The MAP estimate

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \arg\max_{\boldsymbol{\theta}} f_{\mathbf{\Theta}|\mathbf{X}}(\boldsymbol{\theta} \mid \mathbf{x})$$

is the value of $\boldsymbol{\theta}$ that maximizes the posterior density. Since the exponential function is strictly increasing, maximizing the posterior is equivalent to minimizing the negative of the exponent. Thus we consider the quadratic cost function

$$J(\boldsymbol{\theta}) = \sum_{k=0}^{2} \frac{\theta_k^2}{2\sigma_{\Theta_k}^2} + \sum_{i=1}^{n} \frac{\big(x_i - (\theta_0 + \theta_1 t_i + \theta_2 t_i^2)\big)^2}{2\sigma_i^2}.$$

The MAP estimate is

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \arg\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}).$$

The function $J(\boldsymbol{\theta})$ is a quadratic function of $\theta_0, \theta_1, \theta_2$. To find its minimum, we take the partial derivatives with respect to each parameter and set them equal to zero:

$$\frac{\partial J}{\partial \theta_k}(\boldsymbol{\theta}) = 0, \quad k = 0, 1, 2.$$

Each partial derivative is a linear function of $\theta_0, \theta_1, \theta_2$, because the derivative of a quadratic function is linear. Therefore, we obtain a system of three linear equations in the three unknowns $\theta_0, \theta_1, \theta_2$. In matrix form, this can be written as

$$\mathbf{A}\boldsymbol{\theta} = \mathbf{b},$$

for an appropriate $3 \times 3$ matrix $\mathbf{A}$ and vector $\mathbf{b}$ determined by the times $t_i$, the observations $x_i$, the noise variances $\sigma_i^2$, and the prior variances $\sigma_{\Theta_k}^2$. Solving this linear system yields the MAP estimate $\hat{\boldsymbol{\theta}}_{\text{MAP}}$.
In practice, this solution is computed numerically using standard linear algebra methods. The resulting estimator has several important properties that extend those seen in the single-parameter case. The posterior distribution remains Gaussian, the MAP estimate coincides with the posterior mean, and the estimator is a linear function of the observations $\mathbf{X}$. These features make such linear-Gaussian models particularly attractive for applications such as trajectory estimation and tracking.

### 1.7.13 LMS estimation without any observations

Suppose that we wish to estimate a random variable $\Theta$ and that we know its probability distribution, given either by a probability mass function (if $\Theta$ is discrete) or by a probability density function (if $\Theta$ is continuous). We are interested in a point estimate $\hat{\theta}$ of $\Theta$, in the special case where there are no observations available: all we have is the prior distribution of $\Theta$.
As a concrete example, assume that $\Theta$ is uniformly distributed over the interval $[4, 10]$. We can consider different estimation rules that use only this prior. One possibility is the maximum a posteriori (MAP) rule, which selects a value $\hat{\theta}$ at which the posterior density of $\Theta$ is maximized. Since there are no observations, the posterior coincides with the prior, which is flat on $[4, 10]$, so every point in $[4, 10]$ is a maximizer and the MAP rule does not yield a unique estimate.
Another possibility is to use the conditional expectation estimator. Again, with no observations, the conditional expectation reduces to the ordinary expectation, so in this example we obtain

$$\hat{\theta} = \mathbb{E}[\Theta] = \frac{4+10}{2} = 7.$$

In contrast to the MAP rule, this rule produces a unique numerical estimate.
To decide systematically between such candidates, we introduce a performance criterion. The criterion we adopt is the mean squared error (MSE) of an estimator $\hat{\theta}$, defined by

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\big[(\Theta - \hat{\theta})^2\big],$$

where the expectation is with respect to the distribution of $\Theta$. Our goal is to find the value of $\hat{\theta}$ that minimizes this quantity.
We view $\text{MSE}(\hat{\theta})$ as a function of the scalar variable $\hat{\theta}$. Expanding the square inside the expectation gives

$$(\Theta - \hat{\theta})^2 = \Theta^2 - 2\Theta\hat{\theta} + \hat{\theta}^2.$$

Taking expectations and using linearity, we obtain

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[\Theta^2] - 2\hat{\theta}\,\mathbb{E}[\Theta] + \hat{\theta}^2,$$

because $\hat{\theta}$ is a constant and can be taken outside the expectation. Thus the mean squared error is a quadratic function of $\hat{\theta}$.
To minimize this quadratic function, we differentiate it with respect to $\hat{\theta}$ and set the derivative equal to zero:

$$\frac{d}{d\hat{\theta}}\,\text{MSE}(\hat{\theta}) = -2\,\mathbb{E}[\Theta] + 2\hat{\theta} = 0.$$

Solving for $\hat{\theta}$ yields

$$\hat{\theta}^* = \mathbb{E}[\Theta].$$

Therefore, the estimator that minimizes the mean squared error is the expected value of the random variable $\Theta$.
It is useful to derive the same result in a second, more probabilistic way. We start from the defining criterion

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\big[(\Theta - \hat{\theta})^2\big].$$

Using the general identity

$$\mathbb{E}\big[Z^2\big] = \text{Var}(Z) + \big(\mathbb{E}[Z]\big)^2$$

for any random variable $Z$, we apply it to the random variable $Z = \Theta - \hat{\theta}$ and obtain

$$\mathbb{E}\big[(\Theta - \hat{\theta})^2\big] = \mathrm{Var}(\Theta - \hat{\theta}) + \big(\mathbb{E}[\Theta - \hat{\theta}]\big)^2.$$

Since subtracting a constant does not change the variance, we have

$$\mathrm{Var}(\Theta - \hat{\theta}) = \mathrm{Var}(\Theta),$$

which does not depend on $\hat{\theta}$. Furthermore,

$$\mathbb{E}[\Theta - \hat{\theta}] = \mathbb{E}[\Theta] - \hat{\theta}.$$

Combining these facts, we rewrite the mean squared error as

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\Theta) + \big(\mathbb{E}[\Theta] - \hat{\theta}\big)^2.$$

The first term, $\mathrm{Var}(\Theta)$, is independent of $\hat{\theta}$, so there is nothing we can do about it. To minimize $\mathrm{MSE}(\hat{\theta})$, we only need to minimize the second term, which is a square of a real quantity. A square is minimized by making its argument equal to zero, so we choose $\hat{\theta}$ so that

$$\mathbb{E}[\Theta] - \hat{\theta} = 0,$$

that is,

$$\hat{\theta}^* = \mathbb{E}[\Theta].$$

This confirms, without using calculus, that the mean-square optimal estimate of $\Theta$ is its expectation.
Once we adopt the estimate $\hat{\theta}^* = \mathbb{E}[\Theta]$, the corresponding minimum mean squared error is obtained by substituting this value into the expression above. In that case, the squared term vanishes and we obtain

$$\mathrm{MSE}(\hat{\theta}^*) = \mathrm{Var}(\Theta).$$

Thus, when we estimate $\Theta$ by its mean, the least possible value of the mean squared error, over all constant estimators, is equal to the variance of $\Theta$.

Summary of the LMS rule without observations
In the special case in which no observations are available and only the prior distribution of $\Theta$ is known, the least mean squares estimator is obtained by setting the estimate equal to the expected value of $\Theta$. Among all constant estimates $\hat{\theta}$, the choice $\hat{\theta} = \mathbb{E}[\Theta]$ uniquely minimizes the mean squared error, and the minimum attainable mean squared error is equal to $\mathrm{Var}(\Theta)$.

### 1.7.14 LMS estimation with a single observation

We now consider a random variable $\Theta$ with a known prior distribution, together with an observation $X$ whose conditional distribution given $\Theta$ is known. After observing a particular numerical value $x$ of $X$, we want to produce a point estimate $\hat{\theta}$ of $\Theta$ based on this information.
Bayes' rule allows us to combine the prior distribution of $\Theta$ with the likelihood of the observation $X = x$ and obtain the posterior distribution of $\Theta$ given $X = x$. This posterior can be discrete or continuous, depending on the model. Once we have this posterior distribution, we wish to choose a point estimate of $\Theta$ that is optimal under a mean squared error criterion, within the *conditional* universe determined by the event $X = x$.
Recall that in the absence of any observations, the estimator that minimizes the mean squared error is the unconditional expectation $\mathbb{E}[\Theta]$. In the present setting, for a fixed observation $X = x$, we work within the conditional universe determined by that event and consider the conditional mean squared error

$$\mathbb{E}\big[(\Theta - \hat{\theta})^2 \mid X = x\big]$$

as a function of the scalar $\hat{\theta}$. By repeating the same algebraic steps as in the unconditional case, but now in this conditional universe, we find that the minimizing choice is

$$\hat{\theta}^*(x) = \mathbb{E}[\Theta \mid X = x].$$

More abstractly, we can view the LMS estimator as the random variable

$$\hat{\Theta}_{\mathrm{LMS}} = \mathbb{E}[\Theta \mid X],$$

which is a function of the random observation $X$. For each realized value $x$, the estimator outputs the numerical value $\mathbb{E}[\Theta \mid X = x]$. Thus, the optimal estimate in the sense of minimizing conditional mean squared error is the conditional expectation of $\Theta$ given the observed data.

We can make this optimality statement precise using inequalities. In the unconditional setting, the fact that $\mathbb{E}[\Theta]$ minimizes the mean squared error can be written as

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta])^2\big] \leq \mathbb{E}\big[(\Theta - c)^2\big], \qquad \text{for all constants } c.$$

In the conditional setting, for a fixed observation $X = x$, the analogous statement is

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X = x])^2 \mid X = x\big] \leq \mathbb{E}\big[(\Theta - g(x))^2 \mid X = x\big],$$

for any alternative numerical estimate $g(x)$.
We usually think of an estimator as a function $g$ that maps each possible observation $x$ to an estimate $g(x)$. In this language, the inequality above says that, for each fixed $x$, the choice $g(x) = \mathbb{E}[\Theta \mid X = x]$ yields the smallest possible conditional mean squared error. Equivalently, for every $x$,

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X = x])^2 \mid X = x\big] \leq \mathbb{E}\big[(\Theta - g(x))^2 \mid X = x\big].$$

To interpret this more abstractly, notice that both sides of the inequality can be viewed as random variables. The left-hand side is the random variable that takes the value

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X = x])^2 \mid X = x\big]$$

whenever $X = x$, while the right-hand side is the random variable that takes the value

$$\mathbb{E}\big[(\Theta - g(x))^2 \mid X = x\big]$$

whenever $X = x$. Thus we have an inequality between two conditional expectations, viewed as random variables indexed by $X$.
Since these are random variables, we can take expectations on both sides. Using the law of iterated expectations, the expectation of a conditional expectation becomes an unconditional expectation, so we obtain

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X])^2\big] \leq \mathbb{E}\big[(\Theta - g(X))^2\big],$$

for any estimator of the form $g(X)$. The left-hand side is the overall mean squared error of the LMS estimator $\mathbb{E}[\Theta \mid X]$, and the right-hand side is the mean squared error of an arbitrary estimator $g(X)$.
This inequality expresses the global optimality of the LMS estimator. It states that among all estimators that are functions of the observation $X$, the estimator

$$\hat{\Theta}_{\text{LMS}} = \mathbb{E}[\Theta \mid X]$$

achieves the smallest possible mean squared error. Any other estimator $g(X)$ can do no better (and typically does worse) in terms of mean squared error.

LMS estimator with one observation
In summary, when a single observation $X$ is available and the joint distribution of $\Theta$ and $X$ is known, the least mean squares estimator of $\Theta$ is the conditional expectation $\mathbb{E}[\Theta \mid X]$, viewed as a function of the observed value of $X$. This estimator minimizes the mean squared error over all possible estimators of the form $g(X)$, both in each conditional universe where $X = x$ and in terms of the overall, unconditional mean squared error.

### 1.7.15 LMS performance evaluation

Least mean squares (LMS) estimation provides a particularly simple and useful rule: if we wish to keep the mean squared error small, we should use the conditional expectation of the quantity of interest given the available data as our estimate. Concretely, if $\Theta$ is the random variable to be estimated and $X$ is the observation, then the LMS estimator is given by $\mathbb{E}[\Theta \mid X]$, which becomes a numerical value once a specific observation is available.
Suppose that we have already observed a particular value $X = x$, and we use the estimate

$$\hat{\theta}(x) = \mathbb{E}[\Theta \mid X = x].$$

The corresponding conditional mean squared error in this conditional universe is

$$\mathbb{E}\big[(\Theta - \hat{\theta}(x))^2 \mid X = x\big].$$

By the definition of variance, this quantity is equal to the conditional variance of $\Theta$ given $X = x$, that is,

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X = x])^2 \mid X = x\big] = \text{Var}(\Theta \mid X = x).$$

Thus, for each specific measurement $x$, the *optimal* mean squared error achieved by the LMS estimator is precisely $\mathrm{Var}(\Theta \mid X = x)$. If we have already observed $x$ and are asked to quantify the quality of our estimate, this conditional variance is the appropriate performance measure to report, since it describes the remaining uncertainty about $\Theta$ after using the observation.
Now consider the situation before any measurement is taken. At this stage, $X$ is still a random variable, and therefore the conditional variance $\mathrm{Var}(\Theta \mid X)$ is itself a random variable that depends on the eventual value of $X$. If we want to evaluate in advance the overall performance of the LMS estimator, we should average the conditional mean squared error over all possible observations.
Formally, the overall mean squared error of the LMS estimator is

$$\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X])^2\big].$$

Using the law of iterated expectations, this can be written as

$$\mathbb{E}\Big[\,\mathbb{E}\big[(\Theta - \mathbb{E}[\Theta \mid X])^2 \,\big|\, X\big]\Big] = \mathbb{E}\big[\,\mathrm{Var}(\Theta \mid X)\big].$$

In words, the overall mean squared error of the LMS estimator equals the expectation of the conditional variance, averaged over the distribution of $X$.
We can interpret $\mathrm{Var}(\Theta \mid X)$ as the *abstract* conditional variance: it is a random variable that takes the numerical value $\mathrm{Var}(\Theta \mid X = x)$ whenever $X$ happens to be equal to $x$. The overall performance of the LMS estimator, prior to observing $X$, is then given by $\mathbb{E}[\mathrm{Var}(\Theta \mid X)]$, the average of these conditional variances over all possible outcomes of $X$.
It is also important to emphasize when LMS estimation is applicable. LMS methods are designed for estimation problems in which the quality of an estimator is measured by mean squared error. In contrast, in hypothesis testing problems, the primary performance metric is usually the probability of error, rather than mean squared error, so LMS considerations are generally not central in that context.
There are interesting cases in which the LMS estimator and the maximum a posteriori (MAP) estimator coincide. If the posterior distribution of $\Theta$ given $X$ has a single mode and is symmetric around some point, then the mode (used by the MAP rule) and the center of symmetry (the mean, used by LMS) coincide, so both estimators produce the same numerical value. An important example arises in linear-normal models, in which the posterior distribution of $\Theta$ given $X$ is normal.

LMS PERFORMANCE AND MAP COINCIDENCE
For each realized observation, the LMS estimator achieves a conditional mean squared error equal to the conditional variance $\mathrm{Var}(\Theta \mid X = x)$, and the overall mean squared error is the expectation $\mathbb{E}[\mathrm{Var}(\Theta \mid X)]$. In symmetric, unimodal posterior distributions such as those arising in linear-normal models, the LMS and MAP estimators coincide, since they both select the common point where the posterior is centered and maximized.

### 1.7.16 The multidimensional case

In our discussion so far, we have focused on the case of a single unknown random variable $\Theta$ and a single observation $X$, and we have seen that the LMS estimator that minimizes the mean squared error is the conditional expectation $\mathbb{E}[\Theta \mid X]$. We now consider what happens when we have multiple observations and possibly multiple parameters.
Suppose that instead of a single observation, we observe a vector $X = (X_1, \ldots, X_n)$, for which we have a probabilistic model relating $X$ to the unknowns. Once a numerical vector $x = (x_1, \ldots, x_n)$ is observed, we move to the conditional universe corresponding to $X = x$. The arguments used in the one-dimensional case do not rely on the fact that $X$ is scalar, so they extend directly to this vector setting.
In particular, if $\Theta$ is still a single unknown parameter (possibly random), and $X$ is a vector of observations, the LMS estimator is again given by the conditional expectation

$$\hat{\Theta}_{\mathrm{LMS}} = \mathbb{E}[\Theta \mid X].$$

Thus, in the presence of multiple observations, the optimal estimate (in the mean squared error sense) of the scalar parameter $\Theta$ is the conditional expectation of $\Theta$ given the entire observation vector.
Next, consider the case in which the unknown parameter itself is a vector

$$\Theta = (\Theta_1, \ldots, \Theta_m).$$

We are interested in estimating each component $\Theta_j$ on the basis of the observed data $X$. By applying the same LMS argument component-wise, we find that the LMS estimate of the $j$-th component is

$$\hat{\Theta}_{j,\mathrm{LMS}} = \mathbb{E}[\Theta_j \mid X], \qquad j = 1, \ldots, m.$$

Equivalently, the LMS estimator of the entire vector $\Theta$ is the conditional expectation vector $\mathbb{E}[\Theta \mid X]$, whose components are these conditional expectations.
In summary, the least mean squares principle yields a simple and unified prescription: regardless of how many observations or parameters are involved, the LMS estimate of any particular parameter (or component) is the conditional expectation of that parameter given all available observations. This provides a general solution to LMS estimation with multiple parameters and multiple observations.
However, despite its conceptual simplicity and generality, LMS estimation can be difficult to carry out in practice. The first challenge is modeling: in many real-world applications, we may not have a reliable or tractable probabilistic model for the joint behavior of $\Theta$ and the observation vector $X$, especially when these are high-dimensional.
Even when an accurate model is available, there are often serious computational obstacles. The LMS estimator $\mathbb{E}[\Theta \mid X]$ is, in general, a nonlinear function of the observations, which can make both analysis and numerical evaluation difficult. Moreover, to obtain the posterior distribution needed for this conditional expectation, we must apply Bayes' rule. Bayes' rule expresses the posterior density as

$$f_{\Theta|X}(\theta \mid x) = \frac{f_{X|\Theta}(x \mid \theta)\, f_\Theta(\theta)}{f_X(x)},$$

where the denominator

$$f_X(x) = \int f_{X|\Theta}(x \mid \theta)\, f_\Theta(\theta)\, d\theta$$

is obtained by integrating over all possible parameter values. If $\Theta$ is multi-dimensional, this integral becomes a multi-dimensional integral, which may be very challenging or practically impossible to compute.
Furthermore, once the posterior $f_{\Theta|X}(\theta \mid x)$ is known, computing the LMS estimator for a component $\Theta_j$ requires evaluating

$$\mathbb{E}[\Theta_j \mid X = x] = \int \theta_j\, f_{\Theta|X}(\theta \mid x)\, d\theta,$$

which is again a multi-dimensional integral over the parameter space. Therefore, in high-dimensional problems, both the normalization step in Bayes' rule and the computation of the conditional expectations can be computationally demanding.
These modeling and computational difficulties motivate the search for alternative estimation methods that are easier to apply in practice. In particular, it is often desirable to use approaches that require less detailed probabilistic modeling or that are computationally simpler than full LMS estimation, even if they may not achieve the absolute minimum possible mean squared error.

### 1.7.17 Properties of the LMS estimation error

Let $\Theta$ be the random variable that we wish to estimate, and let $X$ denote the observation (or collection of observations). The least mean squares estimator of $\Theta$ based on $X$ is

$$\hat{\Theta} = \mathbb{E}[\Theta \mid X].$$

We define the estimation error as

$$\tilde{\Theta} = \hat{\Theta} - \Theta.$$

We begin by examining the expectation of the estimator and of the estimation error. Using the law of iterated expectations,

$$\mathbb{E}[\hat{\Theta}] = \mathbb{E}\big[\mathbb{E}[\Theta \mid X]\big] = \mathbb{E}[\Theta].$$

Thus, the LMS estimator has the same mean as the random variable $\Theta$. It follows that

$$\mathbb{E}[\tilde{\Theta}] = \mathbb{E}[\hat{\Theta} - \Theta] = \mathbb{E}[\hat{\Theta}] - \mathbb{E}[\Theta] = 0,$$

so the estimation error has zero mean.
In fact, a stronger statement holds: the conditional expectation of the estimation error given the observation is also zero. Indeed,

$$\mathbb{E}[\tilde{\Theta} \mid X] = \mathbb{E}[\hat{\Theta} - \Theta \mid X] = \mathbb{E}[\hat{\Theta} \mid X] - \mathbb{E}[\Theta \mid X].$$

In the conditional universe where $X$ is known, $\hat{\Theta} = \mathbb{E}[\Theta \mid X]$ is a deterministic function of $X$, so

$$\mathbb{E}[\hat{\Theta} \mid X] = \hat{\Theta} = \mathbb{E}[\Theta \mid X],$$

and therefore

$$\mathbb{E}[\tilde{\Theta} \mid X] = \mathbb{E}[\Theta \mid X] - \mathbb{E}[\Theta \mid X] = 0.$$

Next, we study the covariance between the estimation error $\tilde{\Theta}$ and the estimator $\hat{\Theta}$. By definition,

$$\mathrm{Cov}(\tilde{\Theta}, \hat{\Theta}) = \mathbb{E}[\tilde{\Theta}\,\hat{\Theta}] - \mathbb{E}[\tilde{\Theta}]\,\mathbb{E}[\hat{\Theta}].$$

We already know that $\mathbb{E}[\tilde{\Theta}] = 0$, so the second term is zero, and we only need to evaluate $\mathbb{E}[\tilde{\Theta}\,\hat{\Theta}]$.
To do this, we use conditioning. Consider the conditional expectation

$$\mathbb{E}[\tilde{\Theta}\,\hat{\Theta} \mid X].$$

In the conditional universe determined by $X$, the estimator $\hat{\Theta}$ is a constant (a deterministic function of $X$), so it can be factored out:

$$\mathbb{E}[\tilde{\Theta}\,\hat{\Theta} \mid X] = \hat{\Theta}\,\mathbb{E}[\tilde{\Theta} \mid X].$$

But we have already established that $\mathbb{E}[\tilde{\Theta} \mid X] = 0$, and hence

$$\mathbb{E}[\tilde{\Theta}\,\hat{\Theta} \mid X] = 0.$$

Applying the law of iterated expectations, we obtain

$$\mathbb{E}[\tilde{\Theta}\,\hat{\Theta}] = \mathbb{E}\big[\mathbb{E}[\tilde{\Theta}\,\hat{\Theta} \mid X]\big] = \mathbb{E} = 0.$$

Therefore,

$$\mathrm{Cov}(\tilde{\Theta}, \hat{\Theta}) = 0,$$

so the estimation error and the LMS estimator are uncorrelated.
We can now use this fact to relate variances. From the definition of $\tilde{\Theta}$, we can write

$$\Theta = \hat{\Theta} - \tilde{\Theta}.$$

Thus, $\Theta$ is the difference of two random variables that have zero covariance. For any two random variables $A$ and $B$ with $\mathrm{Cov}(A, B) = 0$, we have

$$\mathrm{Var}(A \pm B) = \mathrm{Var}(A) + \mathrm{Var}(B).$$

Applying this to $A = \hat{\Theta}$ and $B = -\tilde{\Theta}$, we obtain

$$\mathrm{Var}(\Theta) = \mathrm{Var}(\hat{\Theta} - \tilde{\Theta}) = \mathrm{Var}(\hat{\Theta}) + \mathrm{Var}(\tilde{\Theta}).$$

This identity shows that the total variance of $\Theta$ decomposes into two parts: the variance of the estimator $\hat{\Theta}$ and the variance of the estimation error $\tilde{\Theta}$. It can also be viewed as a particular manifestation of the law of total variance, expressed here in terms of the LMS estimator and its error. These properties form an important theoretical foundation for more advanced developments in least-squares estimation, even if we do not exploit them further in simple applications.

Key properties of the LMS error

The LMS estimator $\hat{\Theta} = \mathbb{E}[\Theta \mid X]$ has the same mean as $\Theta$, and the corresponding estimation error $\tilde{\Theta} = \hat{\Theta} - \Theta$ has zero mean, both unconditionally and conditionally on the observation. Moreover, the error is uncorrelated with the estimator, and the variance of $\Theta$ decomposes as $\mathrm{Var}(\Theta) = \mathrm{Var}(\hat{\Theta}) + \mathrm{Var}(\tilde{\Theta})$, highlighting how the total uncertainty splits between the variability of the estimate and the residual estimation error.

## 1.8 Limit Theorems and classical statistics

### 1.8.1 Markov inequality and tail bounds

Let $X$ be a non-negative random variable with finite expectation $\mathbb{E}[X]$ and let $a > 0$. We are interested in bounding the probability of the *extreme* event $\{X \geq a\}$, that is, the event that $X$ takes a very large value compared to its typical scale.
The Markov inequality provides such a bound:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Intuitively, if the average value $\mathbb{E}[X]$ is small, then the random variable cannot be very large too often, so the probability of exceeding a large threshold $a$ must be small.
We now give two derivations of this inequality.

### 1.8.2 Integral-based derivation

Assume that $X$ is a continuous, non-negative random variable with probability density function $f_X(x)$. Then the expectation of $X$ is

$$\mathbb{E}[X] = \int_0^\infty x f_X(x)\, dx.$$

We are interested in the event $\{X \geq a\}$, so we consider the restricted integral

$$\int_a^\infty x f_X(x)\, dx.$$

Because we are integrating a non-negative function over a smaller region, we have

$$\int_a^\infty x f_X(x)\, dx \leq \int_0^\infty x f_X(x)\, dx = \mathbb{E}[X].$$

On the other hand, for every $x \geq a$ we have $x \geq a$, so on the interval $[a, \infty)$ we obtain the pointwise inequality

$$x f_X(x) \geq a f_X(x).$$

Hence

$$\int_a^\infty x f_X(x)\, dx \geq \int_a^\infty a f_X(x)\, dx = a \int_a^\infty f_X(x)\, dx = a\, \mathbb{P}(X \geq a).$$

Combining the two inequalities yields

$$a\, \mathbb{P}(X \geq a) \leq \mathbb{E}[X],$$

and therefore

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

The same argument applies in the discrete case by replacing the integral with a sum, and in full generality by using the language of expectations of non-negative random variables.

Indicator-based derivation
We now present a more abstract derivation that does not rely on integrals or sums. Fix $a > 0$ and define a new random variable $Y$ by

$$Y = \begin{cases} 0, & \text{if } X < a, \\ a, & \text{if } X \geq a. \end{cases}$$

By construction, $Y$ only takes values 0 and $a$. We claim that

$$Y \leq X \quad \text{for all outcomes.}$$

Indeed, if $X < a$, then $Y = 0 \leq X$. If $X \geq a$, then $Y = a \leq X$. Thus $Y(\omega) \leq X(\omega)$ for every outcome $\omega$, which implies

$$\mathbb{E}[Y] \leq \mathbb{E}[X].$$

On the other hand, since $Y$ is equal to $a$ on the event $\{X \geq a\}$ and equal to zero otherwise, its expectation is

$$\mathbb{E}[Y] = a\, \mathbb{P}(X \geq a).$$

Combining the two expressions for $\mathbb{E}[Y]$ yields

$$a\,\mathbb{P}(X \geq a) \leq \mathbb{E}[X],$$

and therefore

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

which is again the Markov inequality.

Example: Exponential distribution
Suppose $X$ is exponentially distributed with parameter $\lambda = 1$. Then

$$\mathbb{E}[X] = 1 \quad \text{and} \quad \mathbb{P}(X \geq a) = e^{-a} \quad \text{for } a \geq 0.$$

The Markov inequality gives

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a} = \frac{1}{a}, \quad a > 0.$$

In this case the true tail probability $e^{-a}$ decays exponentially in $a$, whereas the Markov bound $1/a$ decays only polynomially, so the bound is valid but quite loose for large $a$. This illustrates that Markov inequality often provides a conservative tail bound when limited information about the distribution is available.

Example: Uniform distribution and symmetry trick
Let $X$ be uniformly distributed on the interval $[-4, 4]$. The density of $X$ is constant and equal to $1/8$ on this interval, so the true probability of the event $\{X \geq 3\}$ is

$$\mathbb{P}(X \geq 3) = \frac{\text{length of } [3,4]}{\text{length of } [-4,4]} = \frac{1}{8}.$$

We cannot directly apply Markov inequality to $X$ because it takes negative values. However,

$$\{X \geq 3\} \subseteq \{|X| \geq 3\},$$

so

$$\mathbb{P}(X \geq 3) \leq \mathbb{P}(|X| \geq 3).$$

The random variable $|X|$ is non-negative, so we can apply Markov inequality to $|X|$. The distribution of $|X|$ is uniform on $[0, 4]$, hence

$$\mathbb{E}[|X|] = 2.$$

Using Markov inequality with $a = 3$ we obtain

$$\mathbb{P}(|X| \geq 3) \leq \frac{\mathbb{E}[|X|]}{3} = \frac{2}{3},$$

and therefore

$$\mathbb{P}(X \geq 3) \leq \frac{2}{3}.$$

This is a correct but very loose upper bound compared to the true value $1/8$.
We can improve the bound by exploiting the symmetry of the uniform distribution on $[-4, 4]$. By symmetry around zero,

$$\mathbb{P}(X \geq 3) = \mathbb{P}(X \leq -3),$$

and the event $\{|X| \geq 3\}$ is the union of $\{X \geq 3\}$ and $\{X \leq -3\}$, which are disjoint and have equal probability. Hence

$$\mathbb{P}(|X| \geq 3) = \mathbb{P}(X \geq 3) + \mathbb{P}(X \leq -3) = 2\,\mathbb{P}(X \geq 3),$$

so

$$\mathbb{P}(X \geq 3) = \frac{1}{2}\,\mathbb{P}(|X| \geq 3) \leq \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

The improved bound $1/3$ is still loose but noticeably closer to the true value $1/8$ than the original $2/3$. This example shows that Markov inequality can sometimes be sharpened by combining it with additional structural information, such as symmetry.

### 1.8.3 Chebyshev inequality

Let $X$ be a random variable with finite mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathrm{Var}(X)$. The Chebyshev inequality states that for any $c > 0$,

$$\mathbb{P}\big(|X - \mu| \geq c\big) \leq \frac{\sigma^2}{c^2}.$$

This expresses the idea that if the variance $\sigma^2$ is small, then the random variable $X$ is unlikely to deviate far from its mean, and that the probability of being at distance at least $c$ from the mean decays at least on the order of $1/c^2$.

### 1.8.4 Derivation from the Markov inequality

To prove Chebyshev inequality, we apply the Markov inequality to a suitable non-negative random variable. Consider the squared deviation from the mean,

$$Y = (X - \mu)^2,$$

which is always non-negative. The event that $|X - \mu|$ is at least $c$ can be written in terms of $Y$ as

$$\{|X - \mu| \geq c\} = \{(X - \mu)^2 \geq c^2\} = \{Y \geq c^2\}.$$

Thus

$$\mathbb{P}\big(|X - \mu| \geq c\big) = \mathbb{P}\big(Y \geq c^2\big).$$

Since $Y \geq 0$, we may apply the Markov inequality to $Y$ with threshold $a = c^2$,

$$\mathbb{P}\big(Y \geq c^2\big) \leq \frac{\mathbb{E}[Y]}{c^2}.$$

But

$$\mathbb{E}[Y] = \mathbb{E}\big[(X - \mu)^2\big] = \mathrm{Var}(X) = \sigma^2,$$

so we obtain

$$\mathbb{P}\big(|X - \mu| \geq c\big) \leq \frac{\sigma^2}{c^2},$$

which is precisely Chebyshev inequality.

Deviation in standard deviation units

A common way to interpret Chebyshev inequality is in terms of the number of standard deviations away from the mean. Let $\sigma = \sqrt{\mathrm{Var}(X)}$ be the standard deviation of $X$, and fix $k > 0$. We consider the event that $X$ lies at least $k$ standard deviations away from its mean,

$$\{|X - \mu| \geq k\sigma\}.$$

Applying Chebyshev inequality with $c = k\sigma$, we obtain

$$\mathbb{P}\big(|X - \mu| \geq k\sigma\big) \leq \frac{\sigma^2}{(k\sigma)^2} = \frac{1}{k^2}.$$

For example, if $k = 3$, then

$$\mathbb{P}\big(|X - \mu| \geq 3\sigma\big) \leq \frac{1}{9}.$$

This bound holds for *any* distribution with finite variance, regardless of its shape.

Exponential example and comparison with Markov

We now revisit the exponential example from the Markov inequality discussion. Suppose that $X$ is an exponential random variable with parameter 1, so that

$$\mathbb{E}[X] = 1, \qquad \mathrm{Var}(X) = 1, \qquad \sigma = 1.$$

We are interested in bounding the probability

$$\mathbb{P}(X \geq a),$$

where $a > 1$.

First, recall the Markov inequality bound for this probability. Writing

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a} = \frac{1}{a},$$

we see that Markov inequality gives a tail bound that decays like $1/a$.

We now use Chebyshev inequality to obtain a different bound. Since $\mu = 1$, the event $\{X \geq a\}$ corresponds to being at least a distance $a - 1$ above the mean:

$$\{X \geq a\} = \{X - \mu \geq a - 1\}.$$

Therefore,

$$\{X \geq a\} \subseteq \{|X - \mu| \geq a - 1\},$$

and hence

$$\mathbb{P}(X \geq a) \leq \mathbb{P}\big(|X - \mu| \geq a - 1\big).$$

Applying Chebyshev inequality with $c = a - 1$, we obtain

$$\mathbb{P}\big(|X - \mu| \geq a - 1\big) \leq \frac{\sigma^2}{(a-1)^2} = \frac{1}{(a-1)^2}.$$

Thus

$$\mathbb{P}(X \geq a) \leq \frac{1}{(a-1)^2}, \qquad a > 1.$$

For large $a$, the Markov bound behaves like $1/a$, while the Chebyshev bound behaves like $1/a^2$. Since $1/a^2$ tends to zero faster than $1/a$, the Chebyshev inequality gives a smaller and therefore more informative upper bound in the tail for large $a$.
This improvement reflects the fact that Chebyshev inequality uses more information about the distribution of $X$: it depends not only on the mean $\mu$ but also on the variance $\sigma^2$.

### 1.8.5 Weak law of large numbers

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables drawn from a common distribution with finite mean $\mu = \mathbb{E}[X_i]$ and finite variance $\sigma^2 = \mathrm{Var}(X_i)$. We define the sample mean

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

The sample mean $\overline{X}_n$ is itself a random variable, since it is a function of the random variables $X_1, \ldots, X_n$. It is important to distinguish it from the true mean $\mu$, which is a fixed number determined by the underlying distribution.
We begin by computing the expectation of the sample mean. Using linearity of expectation,

$$\mathbb{E}[\overline{X}_n] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\mu = \frac{n\mu}{n} = \mu.$$

Thus, the expected value of the sample mean is equal to the true mean, so $\overline{X}_n$ is an *unbiased* estimator of $\mu$.
Next, we compute the variance of the sample mean. Recall that for a constant $a$ and a random variable $Z$,

$$\mathrm{Var}(aZ) = a^2\,\mathrm{Var}(Z).$$

We have

$$\mathrm{Var}(\overline{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\,\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right).$$

Because the $X_i$ are independent, the variance of their sum is the sum of their variances,

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\mathrm{Var}(X_i) = \sum_{i=1}^{n}\sigma^2 = n\sigma^2.$$

Therefore,

$$\mathrm{Var}(\overline{X}_n) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

As $n$ increases, the variance of the sample mean decreases proportionally to $1/n$, which means that the distribution of $\overline{X}_n$ becomes more and more concentrated around $\mu$.
We now apply Chebyshev inequality to the random variable $\overline{X}_n$. For any $\varepsilon > 0$, Chebyshev inequality tells us that

$$\mathbb{P}\big(|\overline{X}_n - \mathbb{E}[\overline{X}_n]| \geq \varepsilon\big) \leq \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2}.$$

Using $\mathbb{E}[\overline{X}_n] = \mu$ and $\mathrm{Var}(\overline{X}_n) = \sigma^2/n$, we obtain

$$\mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

For fixed $\varepsilon > 0$, as $n \to \infty$ the right-hand side converges to zero,

$$\lim_{n\to\infty} \frac{\sigma^2}{n\varepsilon^2} = 0.$$

Hence

$$\lim_{n\to\infty} \mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon\big) = 0.$$

This statement is the *weak law of large numbers*: for any fixed tolerance $\varepsilon > 0$, the probability that the sample mean differs from the true mean by more than $\varepsilon$ goes to zero as the number of samples $n$ grows.
Equivalently, we can rewrite the inequality in a more suggestive form. For each $n$,

$$\mathbb{P}\big(|\overline{X}_n - \mu| < \varepsilon\big) = 1 - \mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon\big) \geq 1 - \frac{\sigma^2}{n\varepsilon^2}.$$

Taking limits as $n \to \infty$, we obtain

$$\lim_{n\to\infty} \mathbb{P}\big(|\overline{X}_n - \mu| < \varepsilon\big) = 1,$$

which says that the sample mean $\overline{X}_n$ converges in probability to the true mean $\mu$.

Interpretation and Bernoulli model
We can interpret the weak law of large numbers in terms of measurement and noise. Each observation $X_i$ can be decomposed as

$$X_i = \mu + W_i,$$

where $W_i$ represents the random *noise* in the $i$-th measurement, with $\mathbb{E}[W_i] = 0$ and $\mathrm{Var}(W_i) = \sigma^2$. The sample mean becomes

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}\sum_{i=1}^{n}(\mu + W_i) = \mu + \frac{1}{n}\sum_{i=1}^{n} W_i.$$

The average of the noise terms, $\frac{1}{n}\sum_{i=1}^{n} W_i$, has mean zero and variance $\sigma^2/n$, so as $n$ grows these fluctuations become smaller and the sample mean $\overline{X}_n$ becomes increasingly close to $\mu$ with high probability.
A particularly important special case is the Bernoulli model for repeated experiments. Suppose we repeat the same experiment $n$ times, and let $A$ be an event with probability $\mathbb{P}(A) = p$. For each trial $i$, define the indicator random variable

$$X_i = \begin{cases} 1, & \text{if event } A \text{ occurs on trial } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\mathbb{E}[X_i] = p$, and the sample mean

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is the *empirical frequency* of event $A$, that is, the fraction of trials on which $A$ occurs. By the weak law of large numbers, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(|\overline{X}_n - p| \geq \varepsilon\big) = 0,$$

which means that the empirical frequency of event $A$ will be close to its probability $p$ with high probability when the number of trials is large. In this sense, the weak law of large numbers formalizes and justifies the interpretation of probabilities as long-run relative frequencies.

### 1.8.6 Polling and the weak law of large numbers

Consider a referendum in which each voter intends either to vote *yes* or *no*. Let $p$ denote the (unknown) fraction of the population that intends to vote *yes*.
We model a poll as follows. We select $n$ individuals independently and uniformly at random from the population, and for each person $i$ we define a Bernoulli random variable

$$X_i = \begin{cases} 1, & \text{if person } i \text{ intends to vote yes,} \\ 0, & \text{if person } i \text{ intends to vote no.} \end{cases}$$

Then

$$\mathbb{P}(X_i = 1) = p, \qquad \mathbb{P}(X_i = 0) = 1 - p.$$

Hence

$$\mathbb{E}[X_i] = p, \qquad \mathrm{Var}(X_i) = p(1-p).$$

The empirical fraction of *yes* answers in the poll is the sample mean

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

This is a natural estimator for the unknown parameter $p$. Using the general formulas from the weak law of large numbers, we have

$$\mathbb{E}[\overline{X}_n] = p, \qquad \mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}.$$

Suppose we want to understand how accurate the poll is as an estimator of $p$. Fix an accuracy level $\varepsilon > 0$, for example $\varepsilon = 0.01$ corresponding to $\pm 1$ percentage point. We are interested in the probability that the estimation error exceeds $\varepsilon$,

$$\mathbb{P}\big(|\overline{X}_n - p| \geq \varepsilon\big).$$

By Chebyshev inequality applied to $\overline{X}_n$, we have

$$\mathbb{P}\big(|\overline{X}_n - p| \geq \varepsilon\big) \leq \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2}.$$

Bounding the variance without knowing $p$

The bound above still depends on the unknown parameter $p$. However, the variance of a Bernoulli random variable is maximized at $p = 1/2$. Indeed, the function $p(1-p)$ is a concave parabola on $[0,1]$ with maximum

$$p(1-p) \leq \frac{1}{4} \quad \text{for all } p \in [0,1].$$

Therefore

$$\mathbb{P}\big(|\overline{X}_n - p| \geq \varepsilon\big) \leq \frac{1}{4n\varepsilon^2}.$$

This gives a worst-case bound that does not require knowledge of $p$.
Now take $\varepsilon = 0.01$ (one percentage point) and $n = 10{,}000$. Then

$$\varepsilon^2 = 10^{-4}, \quad n\varepsilon^2 = 10{,}000 \cdot 10^{-4} = 1,$$

so

$$\mathbb{P}\big(|\overline{X}_{10000} - p| \geq 0.01\big) \leq \frac{1}{4}.$$

Thus, with a sample size of 10,000, the probability that the poll is off by more than 1 percentage point is at most 25% according to this Chebyshev-based bound. Equivalently,

$$\mathbb{P}\big(|\overline{X}_{10000} - p| < 0.01\big) \geq 1 - \frac{1}{4} = \frac{3}{4},$$

so the confidence level is at least 75%.

Choosing the sample size for a desired accuracy and confidence

Suppose now that we want to design the poll so that the probability of a large error is at most some target $\delta$, for a given accuracy $\varepsilon$. Starting from

$$\mathbb{P}\big(|\overline{X}_n - p| \geq \varepsilon\big) \leq \frac{1}{4n\varepsilon^2},$$

it is enough to choose $n$ such that

$$\frac{1}{4n\varepsilon^2} \leq \delta.$$

Solving for $n$, we get

$$n \geq \frac{1}{4\delta\varepsilon^2}.$$

In the numerical example from the video, we take

$$\varepsilon = 0.01, \qquad \delta = 0.05.$$

Then

$$\varepsilon^2 = 10^{-4}, \quad 4\delta\varepsilon^2 = 4 \cdot 0.05 \cdot 10^{-4} = 0.2 \cdot 10^{-4} = 2 \cdot 10^{-5},$$

so

$$n \geq \frac{1}{2 \cdot 10^{-5}} = \frac{10^5}{2} = 50{,}000.$$

Thus, taking

$$n = 50{,}000$$

is sufficient to ensure that

$$\mathbb{P}\big(|\overline{X}_n - p| \geq 0.01\big) \leq 0.05,$$

that is, the probability that the estimation error exceeds one percentage point is at most 5%.
The poll specifications involve two parameters. *Accuracy* is determined by $\varepsilon$, the allowed absolute error in estimating $p$. *Confidence* is determined by $1 - \delta$, the probability with which the error is guaranteed to be within $\varepsilon$. Using the Chebyshev bound, we see that achieving higher accuracy (smaller $\varepsilon$) or higher confidence (smaller $\delta$) requires a larger sample size $n$.
In practice, the Chebyshev-based sample sizes are quite conservative because Chebyshev inequality is not very tight. Real-world polling often uses more refined approximations (for example, based on the normal distribution and the central limit theorem), which yield substantially smaller required sample sizes for the same nominal accuracy and confidence levels.

### 1.8.7 Convergence in probability

Let $\{Y_n\}_{n\geq 1}$ be a sequence of random variables (not necessarily independent) and let $a$ be a real number. We say that $Y_n$ *converges in probability* to $a$ if, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(|Y_n - a| \geq \varepsilon\big) = 0.$$

Equivalently, for every fixed $\varepsilon > 0$, the probability that $Y_n$ lies within the band $(a-\varepsilon, a+\varepsilon)$ tends to 1 as $n$ increases.
In terms of the weak law of large numbers, if $\overline{X}_n$ is the sample mean of i.i.d. random variables with mean $\mu$, then

$$\lim_{n\to\infty} \mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon\big) = 0 \quad \text{for all } \varepsilon > 0,$$

so $\overline{X}_n$ converges in probability to $\mu$.

### 1.8.8 Comparison with convergence of numbers

For sequences of real numbers $\{x_n\}$, the statement $x_n \to a$ means that for every $\varepsilon > 0$ there exists an index $n_0$ such that for all $n \geq n_0$,

$$|x_n - a| < \varepsilon.$$

Geometrically, if we draw a band $(a-\varepsilon, a+\varepsilon)$ around the limit $a$, then beyond some index $n_0$ all terms $x_n$ stay inside this band.
In the random-variable setting, for each $n$ the quantity $Y_n$ is not a fixed number but a random outcome. Instead of requiring that $Y_n$ always lies inside the band, we require that the *probability* of being outside the band becomes arbitrarily small as $n$ grows. In other words, for large $n$, almost all of the probability mass of $Y_n$ is concentrated inside any prescribed band around the limit $a$.

Intuitive picture
Fix $\varepsilon > 0$, and consider the interval $(a-\varepsilon, a+\varepsilon)$. For each $n$, the distribution of $Y_n$ may place some probability mass outside this band. Convergence in probability requires that

$$\mathbb{P}\big(|Y_n - a| \geq \varepsilon\big)$$

become smaller and smaller as $n$ increases, eventually getting arbitrarily close to zero.
Equivalently, if we draw the probability density (or mass function) of $Y_n$, then as $n$ grows, the bulk of that distribution must migrate into the band around $a$, leaving only a vanishingly small tail outside. This must happen for *every* choice of $\varepsilon > 0$, no matter how small, although for smaller $\varepsilon$ we may need to go to larger $n$ to see this concentration.

Basic properties
Convergence in probability behaves similarly to ordinary convergence of numbers in many respects. Suppose that $X_n \to a$ in probability and $Y_n \to b$ in probability, where $a$ and $b$ are constants.

First, if $g$ is a continuous function, then

$$X_n \to a \text{ in probability} \quad \Longrightarrow \quad g(X_n) \to g(a) \text{ in probability.}$$

For example, if $g(x) = x^2$, then $X_n^2$ converges in probability to $a^2$.
Second, sums behave well under convergence in probability. If $X_n \to a$ in probability and $Y_n \to b$ in probability, then

$$X_n + Y_n \to a + b \text{ in probability.}$$

These properties are direct analogues of the familiar facts for deterministic sequences.
However, there is an important caveat. If $X_n$ converges in probability to some number $a$, it does *not* necessarily follow that $\mathbb{E}[X_n]$ converges to $a$. Convergence in probability is a statement about the distributions of $X_n$ concentrating near $a$, but it does not by itself guarantee convergence of expectations, and additional conditions are often required to relate the two.

### 1.8.9 Related topics: stronger inequalities and modes of convergence

The tools developed so far, such as Markov and Chebyshev inequalities and convergence in probability, can be extended in several directions. We briefly discuss stronger tail bounds and other notions of convergence for random variables.

Sharper tail bounds: Chernoff and the central limit theorem
One limitation of Markov and Chebyshev inequalities is that they often produce loose bounds on tail probabilities, and these bounds typically decay only on the order of $1/n$ as the number of samples $n$ increases. For example, Chebyshev inequality tells us that for the sample mean $\overline{X}_n$ of i.i.d. random variables with mean $\mu$ and variance $\sigma^2$,

$$\mathbb{P}\big(|\overline{X}_n - \mu| \geq a\big) \leq \frac{\sigma^2}{na^2},$$

so the upper bound decreases like $1/n$.
Under additional assumptions, one can obtain much stronger, exponentially decaying bounds. Chernoff bounds are a family of inequalities that state that for suitable random variables (for example, sums of bounded or sub-Gaussian i.i.d. random variables),

$$\mathbb{P}\big(|\overline{X}_n - \mu| \geq a\big) \leq \exp\big(-c(a)\,n\big),$$

where $c(a) > 0$ depends on $a$ and on the underlying distribution. Because the right-hand side decays exponentially fast in $n$, these bounds can show that large deviations of the sample mean are far less likely than Chebyshev inequality suggests.
Another type of approximation for tail probabilities is provided by the central limit theorem. Very loosely, the central limit theorem says that, under mild conditions, the sample mean $\overline{X}_n$ behaves approximately like a normal random variable with the same mean and variance as $\overline{X}_n$. Concretely, after centering at $\mu$ and scaling by $\sigma/\sqrt{n}$, the resulting normalized sample mean has a distribution that approaches the standard normal distribution as $n$ becomes large. This allows us to approximate probabilities involving $\overline{X}_n$ using the normal distribution, leading to more accurate estimates than those given by Chebyshev inequality.

Almost sure convergence and the strong law
Convergence in probability is not the only way in which random variables can converge. Another important notion is *convergence with probability one*, also called *almost sure convergence*.
Let $\{Y_n\}$ be a sequence of random variables defined on a common probability space, and let $Y$ be another random variable. We say that $Y_n$ converges to $Y$ *with probability one* if

$$\mathbb{P}\Big(\big\{\omega : \lim_{n\to\infty} Y_n(\omega) = Y(\omega)\big\}\Big) = 1.$$

In words, for almost every outcome $\omega$ of the underlying experiment, the sequence of real numbers $Y_1(\omega), Y_2(\omega), \ldots$ converges to the real number $Y(\omega)$. Convergence with probability one is a stronger requirement than convergence in probability: whenever $Y_n$ converges to $Y$ with probability one, it also converges to $Y$ in probability.
The law of large numbers also holds under this stronger notion of convergence. Specifically, the *strong law of large numbers* states that for a sequence of i.i.d. random variables with finite mean $\mu$, the sample mean $\overline{X}_n$ converges to $\mu$ with probability one. Because almost sure convergence is more demanding than convergence in probability, this result is called the strong law of large numbers.

Convergence in distribution

A third notion of convergence concerns the distributions of random variables rather than their pointwise behavior. Let $\{X_n\}$ be a sequence of random variables with cumulative distribution functions (CDFs) $F_{X_n}$, and let $X$ be a random variable with CDF $F_X$. We say that $X_n$ *converges in distribution* to $X$ if, for all points $x$ at which $F_X$ is continuous,

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x).$$

This means that the distributions of the $X_n$ approach the distribution of $X$ in the sense that their CDFs converge pointwise at continuity points of the limiting CDF.
Convergence in distribution is weaker than convergence in probability: if $X_n$ converges to $X$ in probability, then it also converges to $X$ in distribution, but not conversely in general. This type of convergence is central in the central limit theorem, where the statement is that the distribution of the appropriately normalized sample mean converges to the standard normal distribution.

### 1.8.10 Sums and averages of i.i.d. random variables

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables with common mean $\mu$ and variance $\sigma^2$. Define the sum

$$S_n = \sum_{i=1}^{n} X_i$$

and the sample mean

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

By linearity of expectation,

$$\mathbb{E}[S_n] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n\mu$$

and

$$\mathbb{E}[\overline{X}_n] = \mathbb{E}\Big[\frac{1}{n}\sum_{i=1}^{n} X_i\Big] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mu.$$

Thus the expected value of the sum grows linearly with $n$ while the sample mean keeps the same expectation $\mu$.
For the variances, independence implies that the covariance terms vanish and the variances add up. Therefore,

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n}\mathrm{Var}(X_i) = n\sigma^2$$

and

$$\mathrm{Var}(\overline{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\,\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\,n\sigma^2 = \frac{\sigma^2}{n}.$$

Hence the standard deviation of the sum is $\sqrt{n}\sigma$ while the standard deviation of the sample mean is $\sigma/\sqrt{n}$.

Three ways to scale the sum
The same underlying random variables can be viewed in three different ways. Each view corresponds to a different scaling of the sum and leads to a different limiting behavior as $n$ becomes large.
*1. The raw sum.* Consider the unscaled sum $S_n$. Its variance is $n\sigma^2$ and its standard deviation is $\sqrt{n}\sigma$. As $n$ increases, the distribution of $S_n$ spreads out more and more. If the $X_i$ take values in some fixed interval, say between $-1$ and 1, then $S_n$ takes values between $-n$ and $n$ and this range grows linearly with $n$. Thus, in this picture, the total variability increases without bound.
*2. The sample mean.* Now consider the sample mean $\overline{X}_n = S_n/n$. Its variance is $\sigma^2/n$ and its standard deviation is $\sigma/\sqrt{n}$. As $n$ grows, the distribution of $\overline{X}_n$ becomes more and more concentrated around $\mu$. In fact, the variance tends to zero and $\overline{X}_n$ converges in probability to $\mu$ by the weak law of large numbers. The limiting distribution in this scaling is degenerate: all the probability mass accumulates at the single point $\mu$.
*3. The centered and $\sqrt{n}$-scaled sum.* Between these two extremes, there is a more interesting normalization obtained by centering the sum by its mean and scaling by $\sqrt{n}$. Define

$$T_n = \frac{S_n - n\mu}{\sqrt{n}}.$$

The expectation of $T_n$ is zero and the variance is

$$\mathrm{Var}(T_n) = \frac{1}{n}\,\mathrm{Var}(S_n) = \frac{1}{n}\,n\sigma^2 = \sigma^2.$$

Thus the distribution of $T_n$ stays centered at zero and its spread, as measured by the variance, remains constant as $n$ increases. The detailed shape of the distribution may change with $n$, but its location and overall width do not blow up or collapse.

### From sums to standardized sums

The normalization in terms of $T_n$ can be pushed one step further by scaling also by the standard deviation $\sigma$. Assuming $\sigma > 0$, define the standardized sum

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\,\sigma}.$$

Equivalently, since $\overline{X}_n = S_n/n$, we can write

$$Z_n = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}.$$

By construction, the random variable $Z_n$ has mean 0 and variance 1. Its distribution is centered at zero, it does not drift as $n$ changes, and its spread remains fixed at unit variance. The question is whether there is a limiting shape for the distribution of $Z_n$ as $n$ tends to infinity, and if so, what that shape is.

### Statement of the Central Limit Theorem

The Central Limit Theorem (CLT) answers this question. Under the assumptions above, it asserts that the standardized sum $Z_n$ converges in distribution to a standard normal random variable. Formally, if $X_1, X_2, \ldots, X_n$ are i.i.d. with mean $\mu$ and variance $\sigma^2$ where $0 < \sigma^2 < \infty$, then for every real number $z$,

$$\lim_{n\to\infty} \mathbb{P}(Z_n \le z) = \Phi(z),$$

where $\Phi$ is the cumulative distribution function of a standard normal random variable. In words, for large $n$ the distribution of $Z_n$ is approximately standard normal.

The CLT provides a powerful approximation tool. When $n$ is large, probabilities involving sums or averages of i.i.d. random variables can be approximated using the normal distribution. For example, to approximate

$$\mathbb{P}(a \le S_n \le b),$$

we rewrite this probability in terms of $Z_n$. Specifically,

$$\mathbb{P}(a \le S_n \le b) = \mathbb{P}\Big(\frac{a - n\mu}{\sqrt{n}\,\sigma} \le Z_n \le \frac{b - n\mu}{\sqrt{n}\,\sigma}\Big),$$

and for large $n$ this is approximately

$$\Phi\Big(\frac{b - n\mu}{\sqrt{n}\,\sigma}\Big) - \Phi\Big(\frac{a - n\mu}{\sqrt{n}\,\sigma}\Big).$$

The approximation improves as the sample size $n$ increases, provided that the underlying distribution of the $X_i$ has finite variance and is not too heavy-tailed.

### Interpretation and relation to the law of large numbers

The law of large numbers and the central limit theorem describe complementary aspects of the behavior of sums of i.i.d. random variables. The weak law of large numbers states that $\overline{X}_n$ converges in probability to $\mu$. Equivalently, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(|\overline{X}_n - \mu| > \varepsilon\big) = 0.$$

This tells us that for large $n$, $\overline{X}_n$ is very likely to be close to $\mu$, but it does not describe the detailed shape of the distribution of the fluctuations around $\mu$.

The central limit theorem refines this picture by zooming in on fluctuations of order $1/\sqrt{n}$ around the mean. By considering the scaled variable

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}},$$

the CLT tells us that these normalized fluctuations converge in distribution to a standard normal random variable. Thus, while the law of large numbers gives a qualitative statement about convergence to a constant, the CLT gives a quantitative description of how the distribution concentrates and what the limiting shape of that concentration is.

Together, these results explain why the normal distribution appears so frequently when dealing with sums, averages, and aggregate quantities built from many small, independent contributions.

### 1.8.11 Discussion of the Central Limit Theorem

The central limit theorem is a deep and highly nonintuitive result. At first sight, there is no obvious reason why a standardized sum of many random variables should have an approximately normal distribution. What makes the theorem especially remarkable is its *universality*. In its classical form, it does not require us to know the full distribution of the individual $X_i$. As long as the $X_i$ are i.i.d. with finite mean $\mu$ and variance $\sigma^2$, the standardized sum behaves, for large $n$, like a normal random variable. When applying the theorem in practice, the only properties of the underlying distribution that we actually need are the mean and variance.

From a computational point of view, the CLT is an extremely convenient shortcut. Even if we know the exact distribution of each $X_i$, computing the distribution of $S_n$ exactly would involve convolving that distribution with itself $n$ times, which quickly becomes intractable as $n$ grows. In contrast, using the CLT reduces the calculation to a normal approximation, which involves only standard normal tables or built-in functions. Despite being an approximation, the CLT is typically very accurate for reasonably large $n$, as numerous examples show.

On a more philosophical level, the CLT explains why models based on normal random variables are so natural. Many real-world quantities are affected by the combined influence of many small, independent noise sources. Each individual noise source may be far from normal, but their cumulative effect is well modeled by a normal random variable. This explains the ubiquity of the normal distribution across very different fields and applications, from physics and engineering to finance and biology.

What the CLT actually says

The mathematical content of the CLT is a statement about convergence of distribution functions. We consider the standardized sum

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\,\sigma},$$

where $S_n = X_1 + \cdots + X_n$. The theorem states that for every real number $z$,

$$\lim_{n\to\infty} \mathbb{P}(Z_n \le z) = \Phi(z),$$

where $\Phi$ is the CDF of a standard normal random variable. Equivalently, the CDF of $Z_n$ converges pointwise (at all continuity points of $\Phi$) to the CDF of a standard normal random variable.

It is important to notice what the theorem *does not* say. The basic statement is about CDFs and does not itself assert convergence of PDFs or PMFs. If the $X_i$ are continuous, so that $Z_n$ has a PDF $f_{Z_n}$, one might ask whether $f_{Z_n}$ converges to the standard normal density. Such results do exist, but they require additional regularity assumptions on the distribution of the $X_i$. In practice, however, it is common and convenient to visualize the CLT through plots of PDFs or histograms, even though the underlying theorem is formulated at the level of CDFs.

Generalizations beyond the i.i.d. case

The classical CLT assumes that the $X_i$ are independent and identically distributed with finite variance. It is natural to ask how far these assumptions can be relaxed. There are versions of the CLT that apply when the $X_i$ are not identically distributed. In such cases, one typically requires conditions on the sequence of means and variances, such as Lindeberg or Lyapunov conditions, that prevent a small number of terms from dominating the sum. Under suitable assumptions, a suitably standardized sum still converges in distribution to a normal random variable.

The assumption of complete independence can also be weakened. There are central limit theorems for sequences with *weak dependence*, where random variables that are far apart in the index are nearly independent, even though nearby ones may be correlated. In many time series or spatial processes, this type of weak dependence is a more realistic assumption than exact independence. Under appropriate mixing or dependence conditions, one can still obtain a CLT for partial sums.

Finally, one may ask how the theorem is proved. For simple special cases, such as sums of Bernoulli random variables (leading to binomial distributions), the CLT can be derived by careful manipulations of binomial coefficients and limits. However, such derivations do not extend to general distributions. The modern general proofs rely on transform methods: one studies expectations of the form $\mathbb{E}[e^{sZ_n}]$ (moment generating functions) or $\mathbb{E}[e^{itZ_n}]$ (characteristic functions) and shows that they converge to the corresponding quantity for the standard normal. Deep results from analysis then imply that convergence of these transforms entails convergence of the associated CDFs.

Practical use of the CLT

From a practical standpoint, the CLT allows us to treat the standardized sum $Z_n$ as if it were exactly standard normal when $n$ is large. Recalling the definition,

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\,\sigma},$$

we can solve for $S_n$ to obtain

$$S_n = n\mu + \sqrt{n}\,\sigma\, Z_n.$$

Thus $S_n$ is an affine (linear plus constant) function of $Z_n$. If we approximate $Z_n$ as standard normal, then $S_n$ is approximated as normal with mean $n\mu$ and variance $n\sigma^2$. In practice, we therefore often proceed by simply assuming that $S_n \sim \mathcal{N}(n\mu, n\sigma^2)$ when $n$ is large enough.
This approximation is surprisingly good even for moderate values of $n$. Values of $n$ on the order of 30 can already yield useful accuracy, though the precise quality of the approximation depends on the distribution of the individual $X_i$. If the $X_i$ are themselves normal, then $S_n$ is exactly normal for all $n$, and no approximation is involved. If the $X_i$ have distributions that are roughly symmetric around their mean and unimodal (a single peak), then the CLT approximation tends to become accurate for smaller $n$. On the other hand, very heavy tails, strong skewness, or multimodality can slow down the convergence to the normal shape and may require larger sample sizes before the approximation is reliable.

### 1.8.12 Illustration of the Central Limit Theorem

We now look at numerical examples that illustrate how the central limit theorem behaves in practice. Consider first a discrete random variable that is uniformly distributed on the set $\{1, 2, \ldots, 8\}$. Let $X_1, X_2, \ldots$ be i.i.d. with this distribution and consider the sum

$$S_n = X_1 + X_2 + \cdots + X_n.$$

For $n = 2$, the PMF of $S_2$ is the convolution of the PMF of $X_1$ with itself and can be computed explicitly by hand. The resulting PMF has a triangular shape: small probabilities at the extremes and larger probabilities near the middle. As $n$ increases, we repeatedly convolve the original PMF with itself. For $n = 4$, the PMF of $S_4$ already starts to resemble the familiar bell shape of a normal distribution. By the time we take $n = 32$, the PMF of $S_{32}$ is visually almost indistinguishable from a normal curve with the same mean and variance.
This convergence is so fast that one can use it in practice to generate samples that are approximately normal. One simple idea is to draw 32 independent samples from the discrete uniform on $\{1, \ldots, 8\}$, add them up, and then (if desired) apply an affine transformation so that the resulting random variable has the desired mean and variance. The distribution of this sum is then very close to normal, and the approximation improves further if we increase $n$.

Effect of asymmetry: truncated geometric example
The previous example started from a symmetric distribution, which is particularly favorable for fast convergence to the normal shape. Next, consider a different starting point: a truncated geometric distribution supported on $\{1, 2, \ldots, k\}$ for some integer $k$. This distribution is markedly asymmetric, with a heavy concentration of probability near the lower values and a long tail towards the higher values.
Let $X_1, X_2, \ldots$ be i.i.d. with this truncated geometric distribution and again consider the sums $S_n$. If we add $n = 8$ such random variables, the PMF of $S_8$ does *not* look very close to a normal shape. The distribution remains visibly skewed: one tail is much heavier than the other. Increasing to $n = 16$ yields a PMF that is somewhat closer to symmetric and has a smoother shape, but the asymmetry is still clearly noticeable. Even at $n = 32$, the PMF of $S_{32}$ continues to show a difference between the two tails, and the fit to a normal curve is not nearly as good as in the symmetric discrete-uniform example.
In this case, the slow convergence is due to the strong asymmetry of the original distribution. The central limit theorem still holds, but more terms are needed before the distribution of the sum becomes well-approximated by a normal distribution. This example highlights an important practical lesson: while the CLT guarantees convergence in distribution as $n \to \infty$, the rate at which the approximation becomes accurate for finite $n$ depends strongly on properties of the underlying distribution, such as symmetry, skewness, and tail behavior.

### 1.8.13 Normal approximation to the binomial

Consider independent Bernoulli random variables $X_1, X_2, \ldots, X_n$ with parameter $p$. The sum

$$S_n = X_1 + X_2 + \cdots + X_n$$

has a binomial distribution with parameters $n$ and $p$. We know that

$$\mathbb{E}[S_n] = np, \qquad \mathrm{Var}(S_n) = np(1-p).$$

The central limit theorem tells us that the standardized version of $S_n$,

$$Z_n = \frac{S_n - np}{\sqrt{np(1-p)}},$$

has a distribution function that, as $n \to \infty$, approaches the CDF $\Phi$ of a standard normal random variable. In practice, for moderately large $n$, we approximate $Z_n$ as standard normal and hence $S_n$ as normal with mean $np$ and variance $np(1-p)$.

Approximating a binomial tail probability

Let us see how this works for a concrete numerical example. Take $n = 36$ and $p = 0.5$, so

$$\mathbb{E}[S_n] = np = 18, \qquad \sqrt{\mathrm{Var}(S_n)} = \sqrt{np(1-p)} = \sqrt{9} = 3.$$

We wish to approximate

$$\mathbb{P}(S_n \le 21).$$

*First attempt (direct CLT with cutoff at 21).* We write

$$\mathbb{P}(S_n \le 21) = \mathbb{P}\Big(\frac{S_n - 18}{3} \le \frac{21-18}{3}\Big) = \mathbb{P}\Big(Z_n \le 1\Big).$$

Approximating $Z_n$ by a standard normal $Z$, we get

$$\mathbb{P}(S_n \le 21) \approx \mathbb{P}(Z \le 1) = \Phi(1) \approx 0.8413.$$

This is a reasonable but not very accurate approximation of the exact binomial probability.

*Second attempt (cutoff at 22).* Observe that, because $S_n$ takes only integer values,

$$\{S_n \le 21\} = \{S_n < 22\}.$$

We can therefore approximate instead

$$\mathbb{P}(S_n < 22) = \mathbb{P}\Big(\frac{S_n - 18}{3} < \frac{22-18}{3}\Big) = \mathbb{P}(Z_n < 1.33).$$

Using again the standard normal approximation, we get

$$\mathbb{P}(S_n \le 21) = \mathbb{P}(S_n < 22) \approx \Phi(1.33) \approx 0.9082.$$

This second approximation overshoots the exact value, whereas the first approximation undershoots it. This suggests that a more accurate answer may lie between these two numbers.

Continuity correction

To understand how to improve the approximation, it is useful to compare the discrete binomial distribution with its continuous normal approximation. The binomial PMF can be visualized as a collection of bars centered at integer points, while the normal approximation replaces these bars by a smooth bell curve with the same mean and variance. When we compute a probability such as $\mathbb{P}(S_n \le 21)$ exactly, we are summing the probabilities of the bars at all integer points $k \le 21$. In the first approximation above, we used the normal area from $-\infty$ up to 21. In the second, we used the area from $-\infty$ up to 22. Neither choice fully aligns the area under the normal curve with the discrete mass of the binomial.

A more sensible choice is to separate the bar at 21 from the bar at 22 by using the midpoint 21.5. In other words, instead of approximating $\mathbb{P}(S_n \le 21)$ by the area under the normal curve up to 21 or up to 22, we use the area up to 21.5. Formally, we write

$$\mathbb{P}(S_n \le 21) = \mathbb{P}(S_n < 21.5),$$

and approximate

$$\mathbb{P}(S_n \le 21) \approx \mathbb{P}\Big(Z_n \le \frac{21.5-18}{3}\Big) = \mathbb{P}(Z_n \le 1.17) \approx \Phi(1.17) \approx 0.8790.$$

This value is much closer to the exact binomial probability than either of the previous two approximations. The adjustment of $\pm 0.5$ in the argument of the normal CDF is called a *continuity correction*. It systematically improves normal approximations to binomial probabilities, especially when $n$ is not extremely large or when the probability mass changes noticeably from one integer point to the next.

Approximating individual binomial probabilities

The continuity correction idea can also be used to approximate individual probabilities of the binomial PMF. For example, to approximate

$$\mathbb{P}(S_n = 19),$$

we view the event $\{S_n = 19\}$ as corresponding to the interval between midpoints 18.5 and 19.5. In terms of the normal approximation, we write

$$\mathbb{P}(S_n = 19) = \mathbb{P}(18.5 \le S_n \le 19.5) \approx \mathbb{P}(18.5 \le S_n \le 19.5 \text{ under the normal model}).$$

Standardizing, we subtract the mean 18 and divide by the standard deviation 3:

$$\mathbb{P}(S_n = 19) \approx \mathbb{P}\Big(\frac{18.5-18}{3} \le Z \le \frac{19.5-18}{3}\Big) = \mathbb{P}(0.17 \le Z \le 0.5),$$

where $Z$ is a standard normal random variable. Using the CDF $\Phi$,

$$\mathbb{P}(0.17 \le Z \le 0.5) = \Phi(0.5) - \Phi(0.17) \approx 0.6915 - 0.5675 = 0.1240$$

(rounded to three decimal places). The exact binomial probability $\mathbb{P}(S_n = 19)$ is very close to this approximate value. In general, to approximate $\mathbb{P}(S_n = k)$ for a binomial random variable $S_n$, we use

$$\mathbb{P}(S_n = k) \approx \mathbb{P}(k - 0.5 \le S_n \le k + 0.5) \approx \Phi\Big(\frac{k+0.5-np}{\sqrt{np(1-p)}}\Big) - \Phi\Big(\frac{k-0.5-np}{\sqrt{np(1-p)}}\Big).$$

Under suitable conditions on $n$ and $p$, there are theorems that guarantee these approximations become very accurate as $n$ grows.

### 1.8.14 Polling revisited: CLT approximation

We return to the polling problem and now use the central limit theorem to obtain more accurate and informative results than those provided by Chebyshev's inequality. We want to estimate an unknown proportion $p$, representing the fraction of the population that will vote *yes* in a referendum. We select $n$ people independently and uniformly at random from the population. For each selected person $i$, we define a Bernoulli random variable $X_i$ that is equal to 1 if the response is *yes* and 0 otherwise. Under the sampling assumptions, the random variables $X_1, X_2, \ldots, X_n$ are independent, identically distributed Bernoulli with parameter $p$.
We let

$$S_n = \sum_{i=1}^{n} X_i$$

denote the total number of *yes* responses in the sample and define the sample mean

$$M_n = \frac{S_n}{n}.$$

The sample mean $M_n$ is our estimator for $p$. We are interested in the accuracy of this estimator and, in particular, in controlling the probability that the estimation error exceeds a given tolerance. For concreteness, suppose that we want the error to be at most one percentage point, that is,

$$|M_n - p| \le 0.01,$$

with high probability. Since we cannot guarantee this with certainty (unless we survey essentially the entire population), we instead require that the probability of violating the specification be small:

$$\mathbb{P}\big(|M_n - p| \ge 0.01\big) \text{ is small.}$$

Our goal is to approximate this probability and then choose $n$ so that it is below a prescribed level, such as 5%.

Expressing the error event in standardized form
To apply the CLT we rewrite the error event in terms of the standardized sum. The mean and variance of a Bernoulli($p$) random variable are

$$\mu = \mathbb{E}[X_i] = p, \qquad \sigma^2 = \mathrm{Var}(X_i) = p(1-p), \qquad \sigma = \sqrt{p(1-p)}.$$

Recall that $M_n = S_n/n$. Consider the event

$$|M_n - p| \ge 0.01.$$

Since $M_n = S_n/n$, we can rewrite this as

$$\Big|\frac{S_n}{n} - p\Big| \ge 0.01.$$

Equivalently,

$$\Big|\frac{S_n - np}{n}\Big| \ge 0.01.$$

Next, multiply both sides by $\sqrt{n}$:

$$\Big|\frac{S_n - np}{n}\Big|\sqrt{n} = \Big|\frac{S_n - np}{\sqrt{n}}\Big| \ge 0.01\sqrt{n}.$$

We then divide both sides by $\sigma$ and obtain

$$\Big|\frac{S_n - np}{\sqrt{n}\,\sigma}\Big| \geq \frac{0.01\sqrt{n}}{\sigma}.$$

The quantity

$$Z_n = \frac{S_n - np}{\sqrt{n}\,\sigma}$$

is precisely the standardized sum that appears in the CLT for Bernoulli random variables. Thus we have expressed the error event in terms of $Z_n$:

$$\big\{|M_n - p| \geq 0.01\big\} = \big\{|Z_n| \geq 0.01\sqrt{n}/\sigma\big\}.$$

Consequently,

$$\mathbb{P}\big(|M_n - p| \geq 0.01\big) = \mathbb{P}\Big(|Z_n| \geq \frac{0.01\sqrt{n}}{\sigma}\Big).$$

By the central limit theorem, $Z_n$ is approximately standard normal for large $n$. We therefore approximate

$$\mathbb{P}\Big(|Z_n| \geq \frac{0.01\sqrt{n}}{\sigma}\Big) \approx \mathbb{P}\Big(|Z| \geq \frac{0.01\sqrt{n}}{\sigma}\Big),$$

where $Z$ is a standard normal random variable.

Worst-case bound and reduction to the standard normal

The preceding expression still involves $\sigma$, which depends on the unknown $p$. However, for a Bernoulli($p$) variable we have

$$\sigma^2 = p(1-p) \leq \frac{1}{4}, \qquad \text{hence} \qquad \sigma \leq \frac{1}{2}.$$

Since $\sigma \leq 1/2$, we have

$$\frac{0.01\sqrt{n}}{\sigma} \geq 0.01\sqrt{n} \cdot 2 = 0.02\sqrt{n}.$$

Thus

$$\mathbb{P}\Big(|Z| \geq \frac{0.01\sqrt{n}}{\sigma}\Big) \leq \mathbb{P}\big(|Z| \geq 0.02\sqrt{n}\big),$$

because the event $\{|Z| \geq 0.02\sqrt{n}\}$ corresponds to being in the outer part of the tails, which is contained in the event $\{|Z| \geq 0.01\sqrt{n}/\sigma\}$ when $0.01\sqrt{n}/\sigma$ is larger. Therefore, a conservative (worst-case) bound is

$$\mathbb{P}\big(|M_n - p| \geq 0.01\big) \lesssim \mathbb{P}\big(|Z| \geq 0.02\sqrt{n}\big).$$

Using symmetry of the standard normal distribution, we can express the right-hand side as

$$\mathbb{P}\big(|Z| \geq a\big) = \mathbb{P}(Z \leq -a) + \mathbb{P}(Z \geq a) = 2\mathbb{P}(Z \geq a) = 2\big(1 - \Phi(a)\big),$$

where $a = 0.02\sqrt{n}$. Thus,

$$\mathbb{P}\big(|M_n - p| \geq 0.01\big) \lesssim 2\big(1 - \Phi(0.02\sqrt{n})\big).$$

Numerical example: $n = 10{,}000$

Suppose we take a sample of size $n = 10{,}000$. Then

$$\sqrt{n} = 100, \qquad 0.02\sqrt{n} = 2.$$

Our bound becomes

$$\mathbb{P}\big(|M_{10{,}000} - p| \geq 0.01\big) \lesssim 2\big(1 - \Phi(2)\big).$$

From standard normal tables,

$$\Phi(2) \approx 0.9772,$$

so

$$2\big(1 - \Phi(2)\big) \approx 2 \cdot (1 - 0.9772) = 2 \cdot 0.0228 = 0.0456.$$

Thus, with a sample size of 10,000, the probability that the polling error exceeds one percentage point is about 4.6%, which is slightly below the target level of 5%. This means that using 10,000 respondents is more than sufficient to achieve the desired accuracy with high probability.

Choosing the sample size for a desired error probability

We can also invert the above reasoning to determine $n$ given a target probability. Suppose we want

$$\mathbb{P}\big(|M_n - p| \geq 0.01\big) \approx 2\big(1 - \Phi(0.02\sqrt{n})\big) = 0.05.$$

Solving for $0.02\sqrt{n}$, we get

$$2\big(1 - \Phi(0.02\sqrt{n})\big) = 0.05 \quad \Longrightarrow \quad 1 - \Phi(0.02\sqrt{n}) = 0.025 \quad \Longrightarrow \quad \Phi(0.02\sqrt{n}) = 0.975.$$

Looking up $\Phi^{-1}(0.975)$ in the standard normal table, we find

$$\Phi^{-1}(0.975) \approx 1.96.$$

Therefore we require

$$0.02\sqrt{n} = 1.96 \quad \Longrightarrow \quad \sqrt{n} = \frac{1.96}{0.02} = 98 \quad \Longrightarrow \quad n = 98^2 = 9604.$$

So a sample size of approximately $n = 9604$ is sufficient to ensure that the probability of an error larger than one percentage point is about 5% in the worst case over all possible values of $p$. This is slightly smaller than 10,000, reflecting the fact that our earlier choice of $n = 10{,}000$ was somewhat conservative.

Practical implications and looser specifications

In real-world polling, one rarely sees surveys with sample sizes as large as 10,000. Typical national polls often use sample sizes on the order of 1,000. This discrepancy is explained by the fact that practical polling specifications are usually looser than $\pm 1\%$. For example, pollsters commonly report a margin of error of $\pm 3\%$ at a confidence level of about 95%. If we replace 0.01 in our derivations by 0.03 and repeat the calculations, the required sample size scales roughly like $1/(0.03)^2$, which is about one-ninth of what is needed for 0.01. Starting from $n \approx 9604$ for a 1% margin of error and 95% confidence, dividing by 9 leads to a sample size of a little over 1000, which matches what is observed in typical polling practice.

In summary, the CLT-based analysis shows that with $n$ on the order of a few thousand, one can obtain very accurate estimates of population proportions with small margins of error. It also provides a transparent formula that relates the desired accuracy and confidence level to the required sample size.